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2023-2025

Portfolio

CCNP &
Cybersecurity



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Lab 1: Multi-Area OSPF

Purpose

The primary purpose of this lab was to review OSPF, one of the biggest IP routing protocols we learned as CCNA. In order to achieve this purpose, we were asked to create a multi-area OSPF network using 5 routers and 3 OSPF areas that utilized both IPv4 and IPv6.

Background Information on Lab Concepts

Open Shortest Path First (OSPF) is a link-state routing protocol that works by exchanging routing information between routers in a network to determine the best paths to reach various destinations. OSPF operates by exchanging link-state information between routers to build a consistent view of the network topology. It then calculates the shortest path to each destination and uses this information to populate routing tables, enabling efficient packet forwarding within the network.

Multi-Area OSPF is a more advanced form of OSPF, where there are multiple OSPF areas, as the name suggests. It offers several significant benefits in large and complex network environments. One of the primary advantages is enhanced scalability. By dividing the OSPF domain into multiple areas, it reduces the size of the Link-State Database (LSDB) and the overhead associated with exchanging routing information. This results in faster convergence times and reduced resource utilization. Additionally, multi-area OSPF improves network stability and fault tolerance. If a problem occurs within one area, it typically doesn't affect routers in other areas, minimizing the scope of network disruptions. Furthermore, it allows for more efficient network management, as administrators can apply different routing policies and controls within each area to suit specific requirements.

For example, if you were in a library searching for a book, the difference between a library with lots of different sections (kind of like various OSPF areas) and one library with just one big section is clear. In the multi-section library, you've got different librarians for each area, like fiction, non-fiction, and reference books. It's like having experts for every book category, making it super easy to find what you're looking for and

getting help from someone who knows their stuff. But in the one-section library, there's only one librarian trying to handle all the books. It can take longer to find what you want, and you might not get the specialized help you need.

Lab Summary

Before starting with the physical lab, we first had to create a topology scheme on Packet Tracer, with each device and OSPF area listed out. After creating the topology scheme, we then created a table with IPv4 and IPv6 addresses along with their respective subnet masks for each device port. After confirming that our subnetting and router topology would function with OSPF, we started to set up the physical routers with basic configurations like a new device name, IP addresses, and enabling IPv6 routing. After setting up basic configurations, we started to set up the OSPFv2 areas by enabling OSPF on the router itself, assigning a router-id, and advertising the networks connected to each port. After setting up OSPFv2, we set up OSPFv3 by setting another router-id, and then applying it to each router's port interface. To verify that we set up OSPF properly, we viewed both the IPv4 and IPv6 routes and checked if we could see every network and if we could successfully ping across the network. We managed to both ping and traceroute using IPv4 and IPv6 from R1 to R5, showing that we had indeed created a multi-area OSPF network that supported both IPv4 and IPv6.

Lab Commands

Ipv6 unicast-routing: This command enables ipv6 on that device and sets up a ipv6 routing table

Interface [Interface]: Goes into the interface you type in

Ip/ipv6 address [Address]: Sets an ipv4 or ipv6 address (you need to type the address in)

Ipv6 enable: Enables ipv6 processing on a interface

Ipv6 OSPF [Process-id] area [Area Number]: Assigns that port to a OSPFv3 area

No Shutdown: Turns a port on

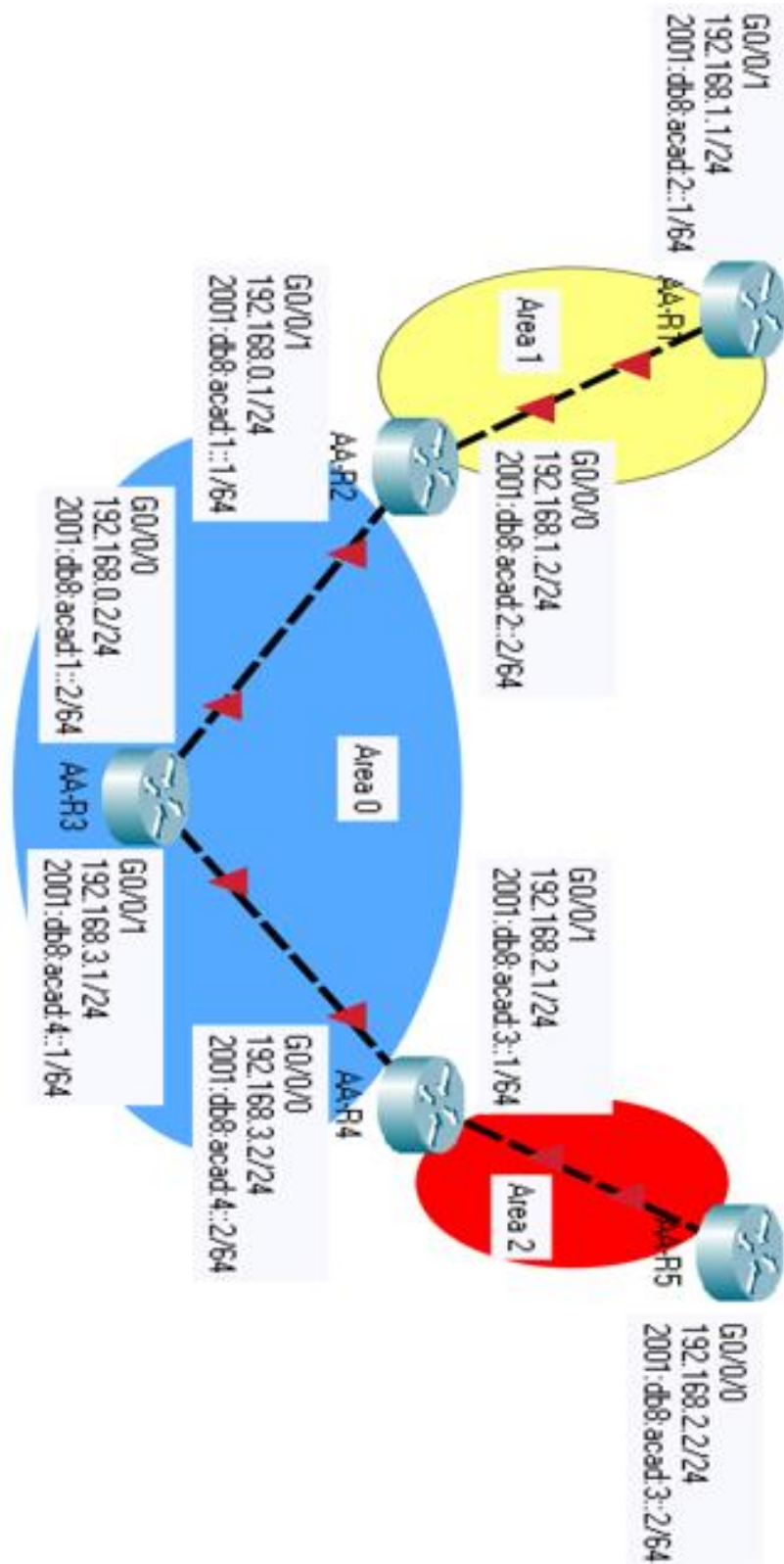
Router OSPF [Process-id]: Enables OSPFv2 routing on your router

Router-id [router-id]: Puts a router id on your router used for OSPF elections

Network [Network Number][Wildcard Mask] area [Area number]:
Sets a network address on the link in the area you select

Ipv6 router OSPF [Process-id]: Enables OSPFv3 on your router

Network Diagram with IPs



Configurations

Pings/Trace Route:

```
AA-R1#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 192.168.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 192.168.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 192.168.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 192.168.3.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 192.168.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
AA-R1#ping 2001:db8:acad:2::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2::1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 2001:db8:acad:2::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2::2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/16 ms
AA-R1#ad:1::1
Trying AD:1::1 ...
% Destination unreachable; gateway or host down

AA-R1#ping 2001:db8:acad:2::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2::2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
AA-R1#ping 2001:db8:acad:4::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:4::1, timeout is 2 seconds:
!!!!
```

```
AA-R1#traceroute 192.168.2.2
Type escape sequence to abort.
Tracing the route to 192.168.2.2
VRF info: (vrf in name/id, vrf out name/id)
 1 192.168.1.2 0 msec 0 msec 1 msec
 2 192.168.0.2 0 msec 0 msec 1 msec
 3 192.168.3.2 0 msec 1 msec 0 msec
 4 192.168.2.2 1 msec 1 msec *
```

```
AA-R1#
```

Router 1:

Current configuration : 3820 bytes

Last configuration change at 21:25:37 UTC Fri Sep 8 2023

version 16.9

service timestamps debug datetime msec

service timestamps log datetime msec

platform qfp utilization monitor load 80

platform punt-keepalive disable-kernel-core

hostname AA-R1

boot-start-marker

boot-end-marker

vrf definition Mgmt-intf

address-family ipv4

exit-address-family

address-family ipv6

exit-address-family

no aaa new-model

login on-success log

subscriber templating

ipv6 unicast-routing

multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-3571946550

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-3571946550

revocation-check none

rsa-keypair TP-self-signed-3571946550

crypto pki certificate chain TP-self-signed-3571946550

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030

31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274

69666963 6174652D 33353731 39343635 3530301E 170D3233 30393038
32313232
30365A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D33
35373139
34363535 30308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 0100ED47 38BBA43B 6AFC5F45 5D7210FB 1600701A 7A10844D
3D3AC12F
54466586 D0369449 6E1299D1 1BC00360 A75A79F5 5D4B2001 E8075648
BFA3D4FF
E3DE2D0D C11D3374 981D1BD3 F873EFF1 28C4C8DB E33C083E 5180C2C3
09D3C632
73B3E435 49D03E8B 94618338 F2CA4F24 EF6C292E 01C747E3 F8B8636F
445C935E
E19D4E1D 1DA78C81 5A1D7B3C F7FD172C 2DDC3868 5991278C DAB937ED
C6CE25C5
289BD560 00B03BF5 2B6858B8 D9EE3F7E 2E7488F6 B9A16ED0 F83D6507
D116A1F2
21B053A1 C8701737 7A07EA0D 240FFCBC 8A5A91DD 73CFA379 E8F0E3D2
640A4575
722C7EE9 B05E4ADB 35BC019E 33890A1A 4220A614 CCEE91D6 C6910527
7342C1CF
B096B219 61630203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 1498D94D CFCA177F DBACD4A2 1EE973F3
41F24A94
C7301D06 03551D0E 04160414 98D94DCF CA177FDB ACD4A21E E973F341
F24A94C7
300D0609 2A864886 F70D0101 05050003 82010100 C662AAA9 8B211D02
BAAD17A4
9ED95436 BA4BD735 129DCC87 FE7741A3 F7EA2533 7EE377F3 6131EDA5
34B91A48
273EC786 FD3B7CFD 7B29CE9B EE54922E C36A7CFC A0428FE0 5065702E
4014D58D
5E48B266 BC38CE0D F0708088 A3ECF4F7 BFBE634C 67A478F5 D26A7932
6C68BA10
8F35CC78 4F4F6FBB 4BFD7811 34E05895 BCBC4F5D D89CFBAF C2265F4E
0F6F36A0
2017E9D8 4188DE2E 8DB5C7F0 912D25A7 AC3AC0B8 C37F1AC6 E792F7D3
8A8AAC76
4A713E11 E3949EB5 69A13255 5339FBF7 B754CE8B 9B3ED2C1 9539F25A
61CB3AD8
CA3897A5 398AF6F2 86B5E12F 8029EBA2 D1C35110 FE164985 D6956786
E5B67854
E6C9C2AB CE487149 ED5FE4DF 38F7D364 2E868094

```
quit

license udi pid ISR4321/K9 sn FDO214420HS
no license smart enable
diagnostic bootup level minimal

spanning-tree extend system-id

redundancy
mode none

interface GigabitEthernet0/0/0
no ip address
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.1.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::1/64
ipv6 enable
ipv6 ospf 10 area 1

interface Serial0/1/0
no ip address

interface Serial0/1/1
no ip address

interface GigabitEthernet0/2/0
no ip address
negotiation auto

interface GigabitEthernet0/2/1
no ip address
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto

router ospf 10
router-id 1.1.1.1
network 192.168.1.0 0.0.0.255 area 1
```



```
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
ipv6 router ospf 10
router-id 1.1.1.1
```

```
control-plane
```

```
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
```

```
end
```

```

AA-R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA  192.168.0.0/24 [110/2] via 192.168.1.2, 01:03:22, GigabitEthernet0/0/1
       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
O IA  192.168.2.0/24 [110/4] via 192.168.1.2, 00:52:24, GigabitEthernet0/0/1
O IA  192.168.3.0/24 [110/3] via 192.168.1.2, 01:02:24, GigabitEthernet0/0/1
AA-R1#show ipv6 route
IPv6 Routing Table - default - 6 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
OI  2001:DB8:ACAD:1::/64 [110/2]
       via FE80::B6A8:B9FF:FE01:B2D0, GigabitEthernet0/0/1
C  2001:DB8:ACAD:2::/64 [0/0]
       via GigabitEthernet0/0/1, directly connected
L  2001:DB8:ACAD:2::1/128 [0/0]
       via GigabitEthernet0/0/1, receive
OI  2001:DB8:ACAD:3::/64 [110/4]
       via FE80::B6A8:B9FF:FE01:B2D0, GigabitEthernet0/0/1
OI  2001:DB8:ACAD:4::/64 [110/3]
       via FE80::B6A8:B9FF:FE01:B2D0, GigabitEthernet0/0/1
L  FF00::/8 [0/0]
       via Null0, receive

```

Router 2:

Current configuration : 3914 bytes

Last configuration change at 21:18:41 UTC Fri Sep 8 2023

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core

hostname AA-R2

boot-start-marker
boot-end-marker

vrf definition Mgmt-intf

address-family ipv4
exit-address-family

address-family ipv6
exit-address-family

no aaa new-model

login on-success log

subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-3937601207
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-3937601207
revocation-check none
rsa-keypair TP-self-signed-3937601207

crypto pki certificate chain TP-self-signed-3937601207
certificate self-signed 01
```

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 33393337 36303132 3037301E 170D3233 30393038
32313134
30325A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D33
39333736
30313230 37308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 01009C3B A624F264 22F3F291 F1EBE21C 5EF4A0B1 DC30647F
4979DE80
940D6490 88A72F94 00B530E7 6754C77E 114B0AEA D75D6018 14842FC8
0300F035
D401C0F5 8B520237 D7C1747A 3CE2F836 35542EA3 758B84E8 23BEDC9C
7E3EF895
8BBE7C03 53AF34F0 66C50C0C 049A8C4F 1401A92B AF079E01 0C12947C
38EDBD17
5E1EC5A9 569E0CEC 9182F5D9 DDD82571 2292D3A8 12D0D691 E9C72E6D
8AE70EEC
7A113197 A51D77BB 870428DC 0DCDABA3 09E04D97 3F5073FB AAFE6932
AC033C4C
F33DEC76 E6F05121 614EB2BC 5721C63E 88F2BB5B E9FC545C 923D6D82
E68D7615
4F3ED89E 015410D1 545EA171 20E1B4D0 4CD2DA0C CC1FD779 06C5D648
C40710FB
BF643D56 3A3B0203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 14B9DACD 87EE185D 183E7821 971BA65C
6A3A5136
EE301D06 03551D0E 04160414 B9DACD87 EE185D18 3E782197 1BA65C6A
3A5136EE
300D0609 2A864886 F70D0101 05050003 82010100 2E6FB4CC 78800096
83F968D8
BD7B5E42 3F879AD5 777EB00F 5D409540 63004263 29E79FD0 BCF6D439
FA659B10
286765C9 D38D950C 3143AF13 4B612D8B 8F2687FC 11B5CF5C BB7059AB
8FD2DDF3
AD125D00 78244912 19D3BE9B 9BAE1E97 0F386ED3 6E961D9F 210986AE
61A5D8A8
E83A46AA 84C74B84 B10F6BEC 69A42517 FDAE35D8 6323CC7C D2D3B399
80CDAD21
090CE63E B9EE473D 4D222059 1D52D573 66658481 5035C534 18F39D61
E539FAF1

```

B9DFDBC4 315023BD 29017EC1 D73CA563 CAC8D65E 721C9A19 43D05E65
4AFE9DBA
6FC528F4 B618BD55 50210DC9 445479E1 64B9BC82 0DBC01D 1CAAED4
0D7556E8
4F868E3C 8FC5B095 EE1098D1 86457B32 08B09F90
quit

```

```

license udi pid ISR4321/K9 sn FDO214421BX
no license smart enable
diagnostic bootup level minimal

```

```
spanning-tree extend system-id
```

```

redundancy
mode none

```

```

interface GigabitEthernet0/0/0
ip address 192.168.1.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::2/64
ipv6 enable
ipv6 ospf 10 area 1

```

```

interface GigabitEthernet0/0/1
ip address 192.168.0.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:1::1/64
ipv6 enable
ipv6 ospf 10 area 0

```

```

interface Serial0/1/0
no ip address
shutdown

```

```

interface Serial0/1/1
no ip address
shutdown

```

```

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

```

```

router ospf 10
router-id 2.2.2.2

```

```
network 192.168.1.0 0.0.0.255 area 1
network 192.168.0.0 0.0.0.255 area 0

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 10
router-id 2.2.2.2

control-plane

line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login

end
```

```

AA-R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.2/32 is directly connected, GigabitEthernet0/0/0
O IA   192.168.2.0/24 [110/3] via 192.168.0.2, 00:51:07, GigabitEthernet0/0/1
O       192.168.3.0/24 [110/2] via 192.168.0.2, 01:00:27, GigabitEthernet0/0/1
AA-R2#show ipv
AA-R2#show ipv6 r
AA-R2#show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
C   2001:DB8:ACAD:1::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:1::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:2::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:2::2/128 [0/0]
    via GigabitEthernet0/0/0, receive
OI  2001:DB8:ACAD:3::/64 [110/3]
    via FE80::521C:B0FF:FE63:3200, GigabitEthernet0/0/1
O   2001:DB8:ACAD:4::/64 [110/2]
    via FE80::521C:B0FF:FE63:3200, GigabitEthernet0/0/1
L   FF00::/8 [0/0]
    via Null0, receive

```

Router 3:

Current configuration : 4066 bytes

Last configuration change at 21:20:04 UTC Fri Sep 8 2023

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core

hostname AA-R3

boot-start-marker
boot-end-marker

vrf definition Mgmt-intf

address-family ipv4
exit-address-family

address-family ipv6
exit-address-family

no aaa new-model

login on-success log

subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-1411542050
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1411542050
revocation-check none
rsa-keypair TP-self-signed-1411542050

crypto pki certificate chain TP-self-signed-1411542050
certificate self-signed 01
```


30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31343131 35343230 3530301E 170D3233 30393038
32313136
32345A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31
34313135
34323035 30308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 01009D91 242FD337 90599CCD 2FDCA4BE 22594CD8 0121391D
806E8904
89AF962E 670EDA31 954E6992 F978E5C4 25F543AA 24E422D0 FE41CFC9
EE33B583
4AE99C8D 192B5879 2BC401DC D9643C48 5AEF0023 C8C98D45 7AB64C62
0CAB710B
709572FD B88F2785 7BC5EECA FD4F4975 323D673A 9125C7F2 74C37A96
FF3050FF
6F11587E 5B2BA493 A87BC4C0 B195931A 8EF05026 BA110FA9 DEBDC11D
88B8A9ED
5A632C59 E48C58D3 353A643F 29896765 C5F03237 406B9F61 6CAA3ED1
FB41DCB8
80FFD280 96681862 C4F520F6 DEA45EF5 59CAE010 F0A8F7D3 2C61FE4B
4450BBBB
8AE621E6 D00E32AF F92EA703 AF2A037E C0F7B23E D072A3EB 670126EA
5AE1389E
459B43FD F3E70203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 140C45AA 391585AB 2179FB4C E5DA957F
62D64A97
E3301D06 03551D0E 04160414 0C45AA39 1585AB21 79FB4CE5 DA957F62
D64A97E3
300D0609 2A864886 F70D0101 05050003 82010100 1DE5C7E9 125C8E84
D23D322C
F3B71591 F391BB7F 64C79D7C 7C89E125 90F4480D BD0B3CC1 CA058D2A
6BBD91CE
9C17BE3D 76CB13B0 888FF84F 57B238EB 174C5198 9033AFB0 6DFF72F8
ECEE593A
6F2C23AF D7FEF735 1A77F61E BE2251F5 2BC27E5D 0D358E5E 53EC075C
F82E3681
FC76B173 00ABE7BF 806DD600 325A56A9 88E77CE3 6E73A69A 14BF976A
98E462BD
EE2073A5 98041267 25E69ACC AA89C44E 7505C951 9C86374E 67DA5FE2
3F8876C2

```
AB4BBDFE 7F6C45E5 2A24F5D4 F620A3E1 FE4A488F DEB1412A DC6B2A19
71B0FF18
9F5F0D82 FBA076C0 1F05AC0E BF4A3F29 2184164F E646784E 52A0DF6F
6B8F4CA2
```

```
ACAF8598 7B07EABF 02316694 B9270930 BB20188D
quit
```

```
license udi pid ISR4321/K9 sn FDO215009PN
no license smart enable
diagnostic bootup level minimal
```

```
spanning-tree extend system-id
```

```
redundancy
mode none
```

```
interface GigabitEthernet0/0/0
ip address 192.168.0.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:1::2/64
ipv6 enable
ipv6 ospf 10 area 0
```

```
interface GigabitEthernet0/0/1
ip address 192.168.3.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:4::1/64
ipv6 enable
ipv6 ospf 10 area 0
```

```
interface Serial0/1/0
no ip address
shutdown
```

```
interface Serial0/1/1
no ip address
shutdown
```

```
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
```

```
interface GigabitEthernet0/2/1
no ip address
shutdown
```

```
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

router ospf 10
router-id 3.3.3.3
network 192.168.0.0 0.0.0.255 area 0
network 192.168.3.0 0.0.0.255 area 0

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 10
router-id 3.3.3.3

control-plane

line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login

end
```

```

AA-R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.0.2/32 is directly connected, GigabitEthernet0/0/0
O IA    192.168.1.0/24 [110/2] via 192.168.0.1, 01:03:39, GigabitEthernet0/0/0
O IA    192.168.2.0/24 [110/2] via 192.168.3.2, 00:53:26, GigabitEthernet0/0/1
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0/1
AA-R3#show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
C    2001:DB8:ACAD:1::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L    2001:DB8:ACAD:1::2/128 [0/0]
    via GigabitEthernet0/0/0, receive
OI   2001:DB8:ACAD:2::/64 [110/2]
    via FE80::B6A8:B9FF:FE01:B2D1, GigabitEthernet0/0/0
OI   2001:DB8:ACAD:3::/64 [110/2]
    via FE80::227:90FF:FED5:FB60, GigabitEthernet0/0/1
C    2001:DB8:ACAD:4::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L    2001:DB8:ACAD:4::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
L    FF00::/8 [0/0]
    via Null0, receive

```

Router 4:

Current configuration : 3914 bytes

Last configuration change at 21:18:41 UTC Fri Sep 8 2023

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
```

```
hostname AA-R4
```

```
boot-start-marker
boot-end-marker
```

```
vrf definition Mgmt-intf
```

```
address-family ipv4
exit-address-family
```

```
address-family ipv6
exit-address-family
```

```
no aaa new-model
```

```
login on-success log
```

```
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
```

```
crypto pki trustpoint TP-self-signed-3937601207
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-3937601207
revocation-check none
rsa-keypair TP-self-signed-3937601207
```

```
crypto pki certificate chain TP-self-signed-3937601207
certificate self-signed 01
```

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 33393337 36303132 3037301E 170D3233 30393038
32313134
30325A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D33
39333736
30313230 37308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 01009C3B A624F264 22F3F291 F1EBE21C 5EF4A0B1 DC30647F
4979DE80
940D6490 88A72F94 00B530E7 6754C77E 114B0AEA D75D6018 14842FC8
0300F035
D401C0F5 8B520237 D7C1747A 3CE2F836 35542EA3 758B84E8 23BEDC9C
7E3EF895
8BBE7C03 53AF34F0 66C50C0C 049A8C4F 1401A92B AF079E01 0C12947C
38EDBD17
5E1EC5A9 569E0CEC 9182F5D9 DDD82571 2292D3A8 12D0D691 E9C72E6D
8AE70EEC
7A113197 A51D77BB 870428DC 0DCDABA3 09E04D97 3F5073FB AAFE6932
AC033C4C
F33DEC76 E6F05121 614EB2BC 5721C63E 88F2BB5B E9FC545C 923D6D82
E68D7615
4F3ED89E 015410D1 545EA171 20E1B4D0 4CD2DA0C CC1FD779 06C5D648
C40710FB
BF643D56 3A3B0203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 14B9DACD 87EE185D 183E7821 971BA65C
6A3A5136
EE301D06 03551D0E 04160414 B9DACD87 EE185D18 3E782197 1BA65C6A
3A5136EE
300D0609 2A864886 F70D0101 05050003 82010100 2E6FB4CC 78800096
83F968D8
BD7B5E42 3F879AD5 777EB00F 5D409540 63004263 29E79FD0 BCF6D439
FA659B10
286765C9 D38D950C 3143AF13 4B612D8B 8F2687FC 11B5CF5C BB7059AB
8FD2DDF3
AD125D00 78244912 19D3BE9B 9BAE1E97 0F386ED3 6E961D9F 210986AE
61A5D8A8
E83A46AA 84C74B84 B10F6BEC 69A42517 FDAE35D8 6323CC7C D2D3B399
80CDAD21
090CE63E B9EE473D 4D222059 1D52D573 66658481 5035C534 18F39D61
E539FAF1

```

B9DFDBC4 315023BD 29017EC1 D73CA563 CAC8D65E 721C9A19 43D05E65
4AFE9DBA
6FC528F4 B618BD55 50210DC9 445479E1 64B9BC82 0DBC01D 1CAAED4
0D7556E8
4F868E3C 8FC5B095 EE1098D1 86457B32 08B09F90
quit

```

```

license udi pid ISR4321/K9 sn FDO214421BX
no license smart enable
diagnostic bootup level minimal

```

```

spanning-tree extend system-id

```

```

redundancy
mode none

```

```

interface GigabitEthernet0/0/0
ip address 192.168.3.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:4::2/64
ipv6 enable
ipv6 ospf 10 area 0

```

```

interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:3::1/64
ipv6 enable
ipv6 ospf 10 area 2

```

```

interface Serial0/1/0
no ip address
shutdown

```

```

interface Serial0/1/1
no ip address
shutdown

```

```

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

```

```

router ospf 10

```

```
router-id 4.4.4.4
network 192.168.2.0 0.0.0.255 area 2
network 192.168.3.0 0.0.0.255 area 0

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 10
router-id 4.4.4.4

control-plane

line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login

end
```



```

AA-R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O      192.168.0.0/24 [110/2] via 192.168.3.1, 00:52:51, GigabitEthernet0/0/0
O IA   192.168.1.0/24 [110/3] via 192.168.3.1, 00:52:51, GigabitEthernet0/0/0
       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L      192.168.2.1/32 is directly connected, GigabitEthernet0/0/1
       192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.3.0/24 is directly connected, GigabitEthernet0/0/0
L      192.168.3.2/32 is directly connected, GigabitEthernet0/0/0
AA-R4#show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
O      2001:DB8:ACAD:1::/64 [110/2]
       via FE80::521C:B0FF:FE63:3201, GigabitEthernet0/0/0
OI     2001:DB8:ACAD:2::/64 [110/3]
       via FE80::521C:B0FF:FE63:3201, GigabitEthernet0/0/0
C      2001:DB8:ACAD:3::/64 [0/0]
       via GigabitEthernet0/0/1, directly connected
L      2001:DB8:ACAD:3::1/128 [0/0]
       via GigabitEthernet0/0/1, receive
C      2001:DB8:ACAD:4::/64 [0/0]
       via GigabitEthernet0/0/0, directly connected
L      2001:DB8:ACAD:4::2/128 [0/0]
       via GigabitEthernet0/0/0, receive
L      FF00::/8 [0/0]
       via Null0, receive

```

Router 5:

Current configuration : 3793 bytes

Last configuration change at 21:40:29 UTC Fri Sep 8 2023

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
```

```
hostname AA-R5
```

```
boot-start-marker
boot-end-marker
```

```
vrf definition Mgmt-intf
```

```
address-family ipv4
exit-address-family
```

```
address-family ipv6
exit-address-family
```

```
no aaa new-model
```

```
login on-success log
```

```
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
```

```
crypto pki trustpoint TP-self-signed-2100162634
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-2100162634
revocation-check none
rsa-keypair TP-self-signed-2100162634
```

```
crypto pki certificate chain TP-self-signed-2100162634
certificate self-signed 01
```

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 32313030 31363236 3334301E 170D3233 30393038
32313335
31325A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32
31303031
36323633 34308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 01009955 E050D2D1 D680D1A0 38846E12 BDD46C32 6F15762C
C8E6CF00
4E2CF7B4 0AD9326B 9D3FA2D8 DCD6B2FD 0049012A 49A78389 54877080
5109F2CB
9D9A9727 27C22E5E 0C9364AE EBD874B9 47F78459 9FEF4029 49E39860
A2E9A55E
98009A57 74DF9C87 6C044675 D5E9B088 85B948E5 009B85B0 63E92098
ACB5E034
758F42D8 70213087 BD235494 9AC658F8 A1A6659D C37E7B9C 4113FE18
27DA8992
9227CDF6 D3D605BF 37BB61A0 529576E3 058B5913 78FF12DD DCB92345
E6DB1C53
F8FBE92F F374EA54 F1D0ADB4 1EABA454 EDFE0EC7 988E1EDD D9A50869
9622A45B
69229F17 27D576EF 6C8B01F1 74DBC8EF 6053D0BA E5AE57CE 59999579
D4F9EF39
D65ADC8E 0B8F0203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 14FC2E3B 8EB769F1 61200CB8 D253023E
871AA5FD
C6301D06 03551D0E 04160414 FC2E3B8E B769F161 200CB8D2 53023E87
1AA5FDC6
300D0609 2A864886 F70D0101 05050003 82010100 80151111 A1B19DCE
10EA6741
48A35C51 A2404F83 D2C0682F F1C65AA4 D1B37CFE 34FC76AA D12528FA
46311A82
69530F59 25E8F00D F38969AB D93ACBBC F1D20CD1 E36B541B 6C568524
6CBC5FFB
E8CD6A66 63E6439A D90145B1 8B7ECC74 5CE052A8 0B72CD8B 5DCC190E
892BAA59
C03A0806 D2ACC0B6 4A10A8DC A0BD9DFF F77CD384 0E8136F0 CCA69D25
0783BCB0
16D8F7D7 2236BFB4 E1D90090 4352D862 D923C7FF 9D93EB85 FB26548D
E0E712DE

```
ED824E4E 6F0C63F0 A4633326 2D5E6208 2A58EA75 401A8CFE 9DC9450B
7577E52C
```

```
B1C4E2BB CA2B341B 7AC834AE 0EB79FBF 1CD30E14 32BAA608 514DC045
9BD66930
```

```
12B53F1A 07143566 3DA05AC7 DAA25FC5 71CDC389
```

```
quit
```

```
license udi pid ISR4321/K9 sn FDO21491FBC
```

```
no license smart enable
```

```
diagnostic bootup level minimal
```

```
spanning-tree extend system-id
```

```
redundancy
```

```
mode none
```

```
interface GigabitEthernet0/0/0
```

```
ip address 192.168.2.2 255.255.255.0
```

```
negotiation auto
```

```
ipv6 address 2001:DB8:ACAD:3::2/64
```

```
ipv6 enable
```

```
ipv6 ospf 10 area 2
```

```
interface GigabitEthernet0/0/1
```

```
no ip address
```

```
shutdown
```

```
negotiation auto
```

```
interface Serial0/1/0
```

```
no ip address
```

```
shutdown
```

```
interface Serial0/1/1
```

```
no ip address
```

```
shutdown
```

```
interface GigabitEthernet0
```

```
vrf forwarding Mgmt-intf
```

```
no ip address
```

```
shutdown
```

```
negotiation auto
```

```
router ospf 10
```

```
router-id 5.5.5.5
```

```
network 192.168.2.0 0.0.0.255 area 2
```

```
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
ipv6 router ospf 10
router-id 5.5.5.5
```

```
control-plane
```

```
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
```

```
end
```

```

AA-R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA  192.168.0.0/24 [110/3] via 192.168.2.1, 00:53:39, GigabitEthernet0/0/0
O IA  192.168.1.0/24 [110/4] via 192.168.2.1, 00:53:39, GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.2.0/24 is directly connected, GigabitEthernet0/0/0
L      192.168.2.2/32 is directly connected, GigabitEthernet0/0/0
O IA  192.168.3.0/24 [110/2] via 192.168.2.1, 00:53:39, GigabitEthernet0/0/0
AA-R5#show ipv6 route
IPv6 Routing Table - default - 6 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDR - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
OI  2001:DB8:ACAD:1::/64 [110/3]
    via FE80::227:90FF:FED5:FB61, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:2::/64 [110/4]
    via FE80::227:90FF:FED5:FB61, GigabitEthernet0/0/0
C   2001:DB8:ACAD:3::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:3::2/128 [0/0]
    via GigabitEthernet0/0/0, receive
OI  2001:DB8:ACAD:4::/64 [110/2]
    via FE80::227:90FF:FED5:FB61, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
    via Null0, receive

```

Problems

One problem we faced was accidentally assigning both interfaces of a router to the same subnet. We forgot that two interfaces of a router cannot be assigned to the same subnet, due to the risk of broadcasts creating an infinite loop, taking up bandwidth and potentially crashing the devices in that network if more devices were added. Through this, we learned that we have to assign every port of a router to a different subnet, even if there wasn't a current way for broadcasts to create a loop.

Conclusion

In this lab, we created a multiple-area OSPF network with 5 routers and 3 areas that supported both IPv4 and IPv6. We enabled OSPFv2 on the routers themselves, and enabled OSPFv3 on the router interfaces. Overall, the lab went relatively smoothly, with a few hiccups regarding the subnetting of router interfaces. The entire lab was a review to refresh ourselves on how to set up routers and OSPF.

Lab 2: Stubby OSPF Areas and OSPF LSA's

Purpose

The primary purpose of the lab was to learn how to setup different types of stubby OSPF areas, configuring one of each type, and combining them all to create a functional network. After the network was setup, we used WireShark to capture packets sent by the different OSPF areas to prove that we had indeed set up the areas correctly and that certain types of LSAs were getting blocked.

Background Information on Lab Concepts

Open Shortest Path First (OSPF) is a link-state routing protocol that works by exchanging routing information between routers in a network to determine the best paths to reach various destinations. OSPF operates by exchanging link-state information between routers to build a consistent view of the network topology. It then calculates the shortest path to each destination and uses this information to populate routing tables, enabling efficient packet forwarding within the network. There are different types of OSPF areas that each have their own benefits, allowing for more efficient usage of network resources.

Stub Areas are similar to normal OSPF areas except for the fact that they block Type 4 Summary ASBR LSAs and Type 5 External LSAs. The lack of Type 5 LSAs means that Stub Areas cannot be connected to another network that uses a different routing protocol. Having no ASBR in the network also means that there is no point in generating Type 4 LSAs, which would tell other routers in the area where the nonexistent ASBR is. Routers in Stub Areas receive information about routers in the same OSPF domain, but always route to the ABR of that area.

Totally Stub Areas are like normal Stub Areas, except they also block Type 3 Summary LSAs. The lack of Type 3 LSAs forces all routers in a totally stubby area to create a default route to the ABR of the area, routing all packets to it. This uses less network resources and bandwidth, allowing for more efficient usage of networks resources.

Not So Stubby Areas are similar to stub areas, in that they both use a default route to the ABR in scenarios where the destination IP is not explicitly known in the routing table. However, NSSA can still accept external routes by converting Type 5 LSAs into Type 7 LSAs, which they use if instead of the default route if it matches their destination IP. NSSAs are a middle ground between Stub areas and normal areas, allowing the introduction of external routes, but preventing the full propagation of them throughout the domain.

Lab Summary

First, we created a topology of the lab in Packet Tracer, including a map of all the Areas, which port each wire is connected into, and the IPv4 addresses and subnets. After creating the topology, we made a list of all the commands we would type into the routers in a text file, confirming that they would create the desired result by pasting them into routers in Packet Tracer. After creating a list of all the commands, we created the physical network following the topology we made in packet tracer and pasted the text from the text file into the global configuration mode of each router. Once we finished pasting in the configurations, we plugged computers into a switch in each special OSPF area and captured the LSA packets in that area using Wireshark. By checking the LSA packets in each area, we could indeed confirm that the correct LSA messages were getting blocked, showing that we had indeed set up each type of area correctly.

Lab Commands

Interface [Interface]: Goes into the interface you type in

Ip address [Address]: Sets an ipv4 or ipv6 address (you need to type the address in)

No Shutdown: Turns a port on

Router OSPF [Process-id]: Enables OSPFv2 routing on your router

Router-id [router-id]: Puts a router id on your router used for OSPF elections

Network [Network Number][Wildcard Mask] area [Area number]:

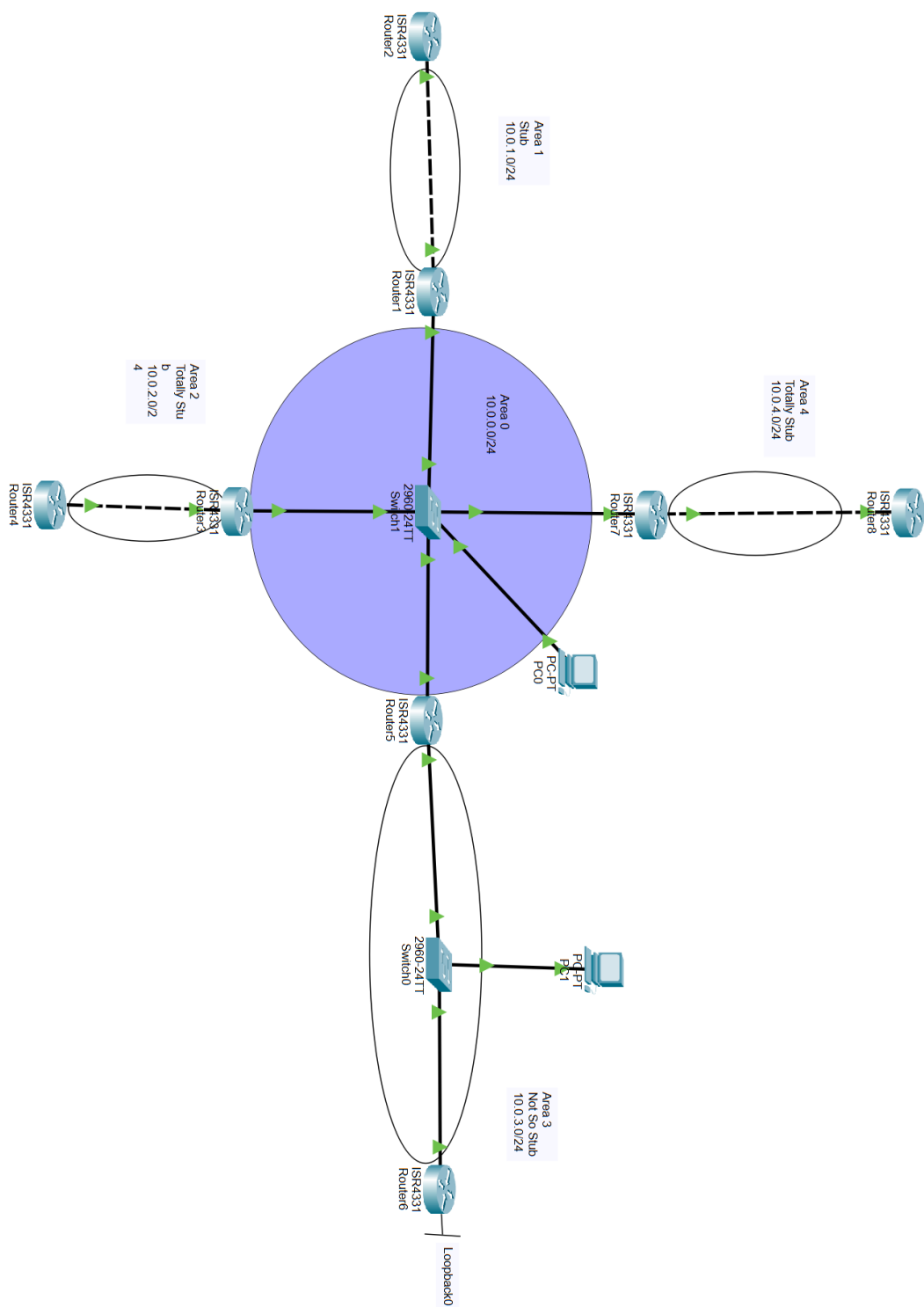
Sets a network address on the link in the area you select

Area xx stub: Turns that OSPF area into a Stub Area

Area xx stub no-summary: Turns that OSPF area into a Totally Stub Area

Area xx nssa: Turns that OSPF area into a NSSA

Network Diagram with IP's



Configurations

Router 1:

```
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname JacobAaronAidenR1

ip cef
no ipv6 cef

spanning-tree mode pvst

interface GigabitEthernet0/0/0
 ip address 10.0.0.1 255.255.255.0
 ip ospf 1 area 0
 duplex auto
 speed auto

interface GigabitEthernet0/0/1
 ip address 10.0.1.1 255.255.255.0
 ip ospf 1 area 1
 duplex auto
 speed auto

interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown

interface Vlan1
 no ip address
 shutdown

router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 area 1 stub

ip classless

ip flow-export version 9
```

```
line con 0

line aux 0

line vty 0 4
  login
```

```
end
```

Router 2:

```
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
hostname JacobAaronAidenR2
```

```
ip cef
no ipv6 cef
```

```
spanning-tree mode pvst
```

```
interface GigabitEthernet0/0/0
  no ip address
  duplex auto
  speed auto
  shutdown
```

```
interface GigabitEthernet0/0/1
  ip address 10.0.1.2 255.255.255.0
  ip ospf 1 area 1
  duplex auto
  speed auto
```

```
interface GigabitEthernet0/0/2
  no ip address
  duplex auto
  speed auto
  shutdown
```

```
interface Vlan1
  no ip address
  shutdown
```

```
router ospf 1
  router-id 2.2.2.2
  log-adjacency-changes
```

```
area 1 stub

ip classless

ip flow-export version 9

line con 0

line aux 0

line vty 0 4
login

end
```

Router 3:

```
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname JacobAaronAidenR3

no ip cef
no ipv6 cef

spanning-tree mode pvst

interface GigabitEthernet0/0/0
ip address 10.0.0.2 255.255.255.0
ip ospf 1 area 0
duplex auto
speed auto

interface GigabitEthernet0/0/1
ip address 10.0.2.1 255.255.255.0
ip ospf 1 area 2
duplex auto
speed auto

interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown

interface Vlan1
```

```
no ip address
shutdown

router ospf 1
  router-id 3.3.3.3
  log-adjacency-changes
  area 2 stub no-summary

ip classless

ip flow-export version 9

line con 0

line aux 0

line vty 0 4
  login

end
Router 4:

version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname JacobAaronAidenR4

no ip cef
no ipv6 cef

spanning-tree mode pvst

interface GigabitEthernet0/0/0
  no ip address
  duplex auto
  speed auto
  shutdown

interface GigabitEthernet0/0/1
  ip address 10.0.2.2 255.255.255.0
  ip ospf 1 area 2
  duplex auto
  speed auto

interface GigabitEthernet0/0/2
```

```
no ip address
duplex auto
speed auto
shutdown

interface Vlan1
no ip address
shutdown

router ospf 1
router-id 4.4.4.4
log-adjacency-changes
area 2 stub no-summary

ip classless

ip flow-export version 9

line con 0

line aux 0

line vty 0 4
login

end
```

Router 5:

```
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname JacobAaronAidenR5

no ip cef
no ipv6 cef

spanning-tree mode pvst

interface GigabitEthernet0/0/0
ip address 10.0.0.3 255.255.255.0
ip ospf 1 area 0
duplex auto
speed auto

interface GigabitEthernet0/0/1
```



```
ip address 10.0.3.1 255.255.255.0
ip ospf 1 area 3
duplex auto
speed auto

interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown

interface Vlan1
no ip address
shutdown

router ospf 1
router-id 5.5.5.5
log-adjacency-changes
area 3 nssa

ip classless

ip flow-export version 9

line con 0

line aux 0

line vty 0 4
login

end
```

Router 6:

```
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname JacobAaronAidenR6

no ip cef
no ipv6 cef

spanning-tree mode pvst

interface Loopback0
```

```
ip address 23.23.23.23 255.0.0.0

interface GigabitEthernet0/0/0
no ip address
duplex auto
speed auto
shutdown

interface GigabitEthernet0/0/1
ip address 10.0.3.2 255.255.255.0
ip ospf 1 area 3
duplex auto
speed auto

interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown

interface Vlan1
no ip address
shutdown

router ospf 1
router-id 6.6.6.6
log-adjacency-changes

ip classless
ip route 0.0.0.0 0.0.0.0 Loopback0

ip flow-export version 9

line con 0

line aux 0

line vty 0 4
login
```

end

Router 7:

```
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
hostname JacobAaronAidenR7

no ip cef
no ipv6 cef

spanning-tree mode pvst

interface GigabitEthernet0/0/0
 ip address 10.0.0.4 255.255.255.0
 ip ospf 1 area 0
 duplex auto
 speed auto

interface GigabitEthernet0/0/1
 ip address 10.0.4.1 255.255.255.0
 ip ospf 1 area 4
 duplex auto
 speed auto

interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown

interface Vlan1
 no ip address
 shutdown

router ospf 1
 router-id 7.7.7.7
 log-adjacency-changes

ip classless

ip flow-export version 9

line con 0

line aux 0

line vty 0 4
 login

end
```

Router 8:

```
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname JacobAaronAidenR8

no ip cef
no ipv6 cef

spanning-tree mode pvst

interface Loopback0
 ip address 34.34.34.34 255.0.0.0

interface GigabitEthernet0/0/0
 no ip address
 duplex auto
 speed auto
 shutdown

interface GigabitEthernet0/0/1
 ip address 10.0.4.2 255.255.255.0
 ip ospf 1 area 4
 duplex auto
 speed auto

interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown

interface Vlan1
 no ip address
 shutdown

router ospf 1
 router-id 8.8.8.8
 log-adjacency-changes
 default-information originate

ip classless
ip route 0.0.0.0 0.0.0.0 Loopback0
```

```
ip flow-export version 9

line con 0

line aux 0

line vty 0 4
  login
end
```

LSA Types:

Wireshark packet capture showing OSPF Hello and LS Update packets. The packet list on the left shows a sequence of OSPF packets. The packet details pane on the right shows the structure of an OSPF Hello packet (Type 1) and an LS Update packet (Type 2). The packet bytes pane on the right shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1388	980.318765	10.0.1.1	224.0.0.5	OSPF	94	Hello Packet
1310	980.916410	10.0.1.2	224.0.0.5	OSPF	94	Hello Packet
1318	980.432074	10.0.1.1	224.0.0.5	OSPF	94	Hello Packet
1320	980.270366	10.0.1.2	224.0.0.5	OSPF	94	Hello Packet
1330	980.737558	10.0.1.1	224.0.0.5	OSPF	94	Hello Packet
1331	980.756135	10.0.1.2	224.0.0.5	OSPF	94	Hello Packet
1333	1001.037224	10.0.1.1	224.0.0.5	OSPF	134	LS Update
1334	1001.070430	10.0.1.1	224.0.0.5	OSPF	90	Hello Packet
1335	1001.572045	10.0.1.1	224.0.0.5	OSPF	98	LS Update
1339	1002.072102	10.0.1.1	224.0.0.5	OSPF	98	LS Update
1340	1003.530951	10.0.1.2	224.0.0.5	OSPF	98	LS Acknowledge
1341	1003.574279	10.0.1.1	224.0.0.5	OSPF	158	LS Acknowledge
1346	1007.071415	10.0.1.1	224.0.0.5	OSPF	98	LS Update
1348	1008.406629	10.0.1.2	224.0.0.5	OSPF	98	LS Acknowledge
1350	1009.962132	10.0.1.2	224.0.0.5	OSPF	94	Hello Packet
1351	1009.171855	10.0.1.1	224.0.0.5	OSPF	78	LS Acknowledge
1353	1010.933596	10.0.1.1	224.0.0.5	OSPF	94	Hello Packet
1359	1010.102340	10.0.1.2	224.0.0.5	OSPF	94	Hello Packet
1364	1020.568736	10.0.1.1	224.0.0.5	OSPF	94	Hello Packet
1372	1028.353962	10.0.1.2	224.0.0.5	OSPF	94	Hello Packet
1374	1030.007024	10.0.1.1	224.0.0.5	OSPF	94	Hello Packet
1384	1030.017676	10.0.1.2	224.0.0.5	OSPF	94	Hello Packet
1386	1039.583343	10.0.1.1	224.0.0.5	OSPF	94	Hello Packet
1391	1044.752494	10.0.1.2	224.0.0.5	OSPF	166	LS Update

Frame 1341: 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface VFP_F0R3A638-78F9-4408-9875-C95E55
 Ethernet II, Src: Cisco.81:b7:51 (b4:8b:b0:01:b7:51), Dst: IPmulticast_05 (01:00:5e:00:00:05)
 Internet Protocol Version 4, Src: 10.0.1.1, Dst: 224.0.0.5
 Open Shortest Path First
 OSPF Header
 LSA-type 1 (Router-LSA), len 36
 LSA-type 2 (Network-LSA), len 32
 LSA-type 3 (Summary-LSA (IP network)), len 28
 LSA-type 4 (Summary-LSA (IP network)), len 28
 LSA-type 5 (NSSA AS-External-LSA), len 36

Frame 1346: 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface VFP_F0R3A638-78F9-4408-9875-C95E55
 Ethernet II, Src: Cisco.81:b7:51 (b4:8b:b0:01:b7:51), Dst: IPmulticast_05 (01:00:5e:00:00:05)
 Internet Protocol Version 4, Src: 10.0.1.1, Dst: 224.0.0.5
 Open Shortest Path First
 OSPF Header
 LSA-type 2 (Network-LSA), len 32
 LSA-type 3 (Summary-LSA (IP network)), len 28
 LSA-type 4 (Summary-LSA (IP network)), len 28
 LSA-type 5 (NSSA AS-External-LSA), len 36

Wireshark packet capture showing OSPF LS Update and LS Acknowledge packets. The packet list on the left shows a sequence of OSPF packets. The packet details pane on the right shows the structure of an OSPF LS Update packet (Type 2) and an OSPF LS Acknowledge packet (Type 3). The packet bytes pane on the right shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
4901	1340.189708	10.10.2.2	224.0.0.5	OSPF	90	Hello Packet
4916	1345.261882	10.0.2.2	224.0.0.5	OSPF	94	Hello Packet
4918	1345.097070	10.0.2.1	224.0.0.5	OSPF	94	Hello Packet
4935	1340.827551	10.10.2.2	224.0.0.5	OSPF	90	Hello Packet
4941	1354.918553	10.0.2.2	224.0.0.5	OSPF	94	Hello Packet
4946	1355.879013	10.0.2.1	224.0.0.5	OSPF	94	Hello Packet
4956	1359.755323	10.10.2.2	224.0.0.5	OSPF	90	Hello Packet
4972	1364.141428	10.0.2.2	224.0.0.5	OSPF	94	Hello Packet
4980	1365.559551	10.0.2.1	224.0.0.5	OSPF	94	Hello Packet
5002	1368.945499	10.10.2.2	224.0.0.5	OSPF	90	Hello Packet
5024	1372.360178	10.0.2.1	224.0.0.5	OSPF	126	LS Update
5027	1372.807096	10.0.2.1	224.0.0.5	OSPF	98	LS Update
5028	1373.249801	10.0.2.2	224.0.0.5	OSPF	94	Hello Packet
5038	1374.023023	10.0.2.2	224.0.0.5	OSPF	112	LS Acknowledge
5036	1375.545609	10.0.2.1	224.0.0.5	OSPF	94	Hello Packet
5056	1378.693380	10.10.2.2	224.0.0.5	OSPF	90	Hello Packet
5080	1379.949537	10.0.2.2	224.0.0.5	OSPF	152	LS Update
5085	1382.372128	10.0.2.1	224.0.0.5	OSPF	98	LS Update
5087	1382.498837	10.0.2.1	224.0.0.5	OSPF	90	LS Update
5088	1382.458431	10.0.2.1	224.0.0.5	OSPF	118	LS Acknowledge
5097	1382.945247	10.0.2.2	224.0.0.5	OSPF	94	Hello Packet
5117	1384.877067	10.0.2.2	224.0.0.5	OSPF	98	LS Acknowledge
5120	1385.189123	10.0.2.1	224.0.0.5	OSPF	94	Hello Packet
5135	1387.811281	10.10.2.2	224.0.0.5	OSPF	90	Hello Packet
5145	1400.708462	10.0.2.1	224.0.0.5	OSPF	90	Hello Packet

Frame 5024: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface VFP_F0R3A638-78F9-4408-9875-C95E55
 Ethernet II, Src: Cisco.47:8e:41 (b4:8b:b0:47:8e:41), Dst: IPmulticast_05 (01:00:5e:00:00:05)
 Internet Protocol Version 4, Src: 10.0.2.1, Dst: 224.0.0.5
 Open Shortest Path First
 OSPF Header
 Version: 2
 Message Type: LS Update (4)
 Packet Length: 92
 Source OSPF Router: 1.1.1.1
 Area ID: 0.0.0.0 (Backbone)
 Checksum: 0x0b07 (correct)
 Auth Type: Null (0)
 Auth Data (none): 0000000000000000
 LS Update Packet
 Number of LSAs: 2
 LSA-type 4 (Summary-LSA (ASBR)), len 28
 LSA-type 5 (AS-External-LSA (ASBR)), len 36

Frame 5088: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface VFP_F0R3A638-78F9-4408-9875-C95E55
 Ethernet II, Src: Cisco.47:8e:41 (b4:8b:b0:47:8e:41), Dst: IPmulticast_05 (01:00:5e:00:00:05)
 Internet Protocol Version 4, Src: 10.0.2.1, Dst: 224.0.0.5
 Open Shortest Path First
 OSPF Header
 Version: 2
 Message Type: LS Acknowledge (3)
 Packet Length: 90
 Source OSPF Router: 1.1.1.1
 Area ID: 0.0.0.0 (Backbone)
 Checksum: 0x0b07 (correct)
 Auth Type: Null (0)
 Auth Data (none): 0000000000000000
 LS Acknowledge Packet
 Number of LSAs: 2
 LSA-type 4 (Summary-LSA (ASBR)), len 28
 LSA-type 5 (AS-External-LSA (ASBR)), len 36

Lab 3: Setting Up a BGP Network with 3 Different Routing Protocols

Purpose

The primary purpose of this lab was to learn how to set up BGP, a way for network border routers to route between networks that use different routing protocols. In order to practice setting up BGP, we set up 3 different networks that use OSPF, EIGRP, or RIP as their routing protocol, and connected the border routers using BGP.

Background Information on Lab Concepts

As an inter-domain protocol, BGP, unlike intra-domain IGPs such as OSPF, RIP, and EIGRP do not require communicating routers in the same network to run the same IGP. BGP takes the routes of all these different IGPs and combines them to create its own routing table. Also, BGP doesn't use the link-state advertisements that protocols OSPF uses, and instead uses its own Update Messages. This allows the system resource requirements for routers that use BGP to be much less, unlike the memory and CPU intensive IGPs. This makes BGP much more scalable and flexible than other protocols, perfect for extremely large networks that use different intra-domain protocols like the Internet. However, BGP routes are simpler than the ones created by OSPF, which causes the convergence speed of BGP to be much slower than OSPF.

BGP also has multiple attributes it can assign to a route, which will affect which route a packet would take to get to an end destination. One such attribute is the weight of a path. The weight attribute is generally the most important attribute, and the path with the highest weight will usually be the preferred path for packets to take to a destination. Another attribute is the next hop IP address of the route. This attribute assigns a router to be the next router to send packets to in order to reach a destination, which simplifies routes and prevents security issues. The metric attribute is like the opposite of weight. Instead of affecting how a packet leaves a BGP router, it changes what path a packet from another router takes to reach that specific BGP router. A lower metric has more preference.

Lab Summary

Before starting with the physical lab, we first had to create a topology scheme on Packet Tracer, with each device and routing protocol area listed out. After creating the

topology scheme, we then created a table with IPv4 and IPv6 addresses along with their respective subnet masks for each device port. After confirming that our subnetting and router topology would function with OSPF, EIGRP, RIP, and BGP, we started to set up the physical routers with the same basic configurations such as a new device name, IP addresses, loopback interfaces, and enabling IPv6 routing. After setting up basic configurations, we started to set up each IPv4 network, with things like assigning a router-id, and advertising the networks connected to each port. After setting up basic IPv4 routing, we set up IPv6 routing using a similar process to the IPv4 networks.

We then set up each individual IPv4 OSPF, EIGRP, or RIP area between each set of 2 routers, and did the same for IPv6. After testing and seeing that each set of routers could obtain routes through the protocol, we set up BGP on each of the network border routers. Once BGP was setup and we could ping the loopback interfaces from a different network, we finally added the three attributes to BGP.

Lab Commands

Ipv6 unicast-routing: This command enables ipv6 on that device and sets up a ipv6 routing table

Interface [Interface]: Goes into the interface you type in

IP/IPv6 address [Address]: Sets an ipv4 or ipv6 address (you need to type the address in)

IPv6 enable: Enables ipv6 processing on a interface

IP/IPv6 OSPF [Process-id] area [Area Number]: Assigns that port to a OSPFv3 area

No Shutdown: Turns a port on

Router OSPF [Process-id]: Enables OSPFv2 routing on your router

Router-id [router-id]: Puts a router id on your router used for OSPF elections

Network [Network Number][Wildcard Mask] area [Area number]:
Sets a network address on the link in the area you select for ospf

IPv6 router OSPF [Process-id]: Enables OSPFv3 on your router

Redistribute connected: Redistributes connected routes to other routers in the area

Redistribute ospf/rip/eigrp: Redistributes IPv4 networks learned from OSPF/RIP/EIGRP through BGP

Ipv6 rip [name] enable: Enables IPv6 RIP routing on that interface

Router rip: Enables IPv4 RIP routing on the entire router

Network [Network Number]: Specifies the network for that router's RIP area

Neighbor [Neighbor IP Address]: Specifies any other routers in the same RIP area

Eigrp router-id [router-id]: Puts a router id on your router used for EIGRP elections

Network [IP/IPv6 address] mask [subnet mask]: Advertise this network to all other BGP routers

Neighbor [IP/IPv6 address] remote-as [as-number]: Attempts to create a BGP peer with this ip address

Neighbor [IP/IPv4 address] activate: Establishes BGP adjacency with this ip address

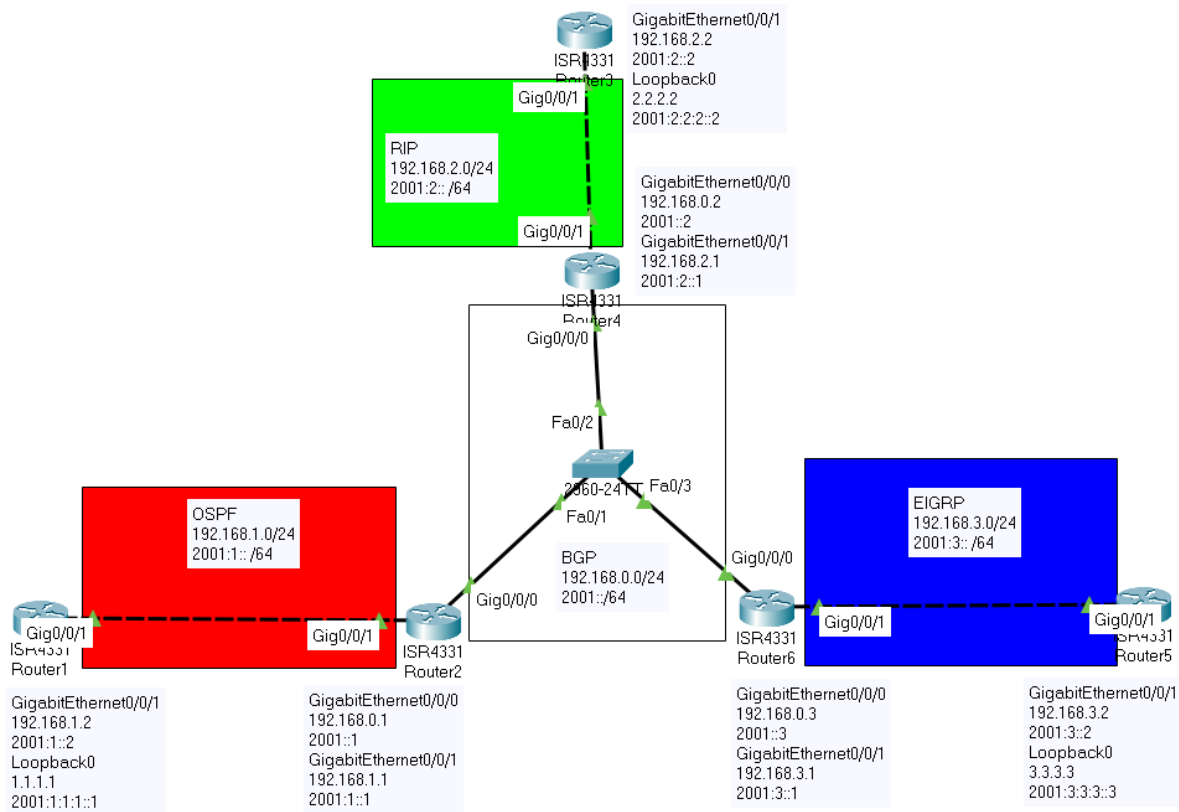
Neighbor [IP/IPv4 address] weight [weight]: Manually changes the weight of the path to the neighbor to a specified value

Neighbor [IP address] next-hop-self: Sets itself as the next hop for any advertised routes

Neighbor [ip-address] route-map [name] out: Advertises this route map to a neighbor's address.

Access-list [number] permit [ip addresses]: Allows all devices with an IP address in the range to go through the specified access list

Network Diagram



Router Configurations

Router 1:

```

!
! Last configuration change at 20:06:21 UTC Wed Nov 8 2023
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname JacobAaronAidenR1
!
boot-start-marker
boot-end-marker
!

```

```
!  
vrf definition Mgmt-intf  
!  
  address-family ipv4  
  exit-address-family  
!  
  address-family ipv6  
  exit-address-family  
!  
!  
no aaa new-model  
  
ipv6 unicast-routing  
  
subscriber templating  
multilink bundle-name authenticated  
  
license udi pid ISR4321/K9 sn FDO214421CF  
  
spanning-tree extend system-id  
  
redundancy  
  mode none  
  
vlan internal allocation policy ascending  
  
interface Loopback0  
  ip address 1.1.1.1 255.255.255.0  
  ip ospf 1 area 0  
  ipv6 address 2001:1:1:1::1/64  
  ipv6 ospf 1 area 0  
  
interface GigabitEthernet0/0/0  
  no ip address  
  negotiation auto  
  
interface GigabitEthernet0/0/1  
  ip address 192.168.1.2 255.255.255.0  
  ip ospf 1 area 0  
  negotiation auto
```

```
ipv6 address 2001:1::2/64
ipv6 ospf 1 area 0

interface Serial0/1/0
no ip address

interface Serial0/1/1
no ip address

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto

interface Vlan1
no ip address

router ospf 1
router-id 1.1.1.1
redistribute connected subnets

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 1
router-id 1.1.1.1
redistribute connected

control-plane

line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login

end
```

Router 2:

Last configuration change at 20:23:14 UTC Wed Nov 8 2023

version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core

hostname JacobAaronAidenR2

boot-start-marker
boot-end-marker

vrf definition Mgmt-intf

address-family ipv4
exit-address-family

address-family ipv6
exit-address-family

no aaa new-model

ipv6 unicast-routing

subscriber templating
multilink bundle-name authenticated

license udi pid ISR4321/K9 sn FDO211216BL

spanning-tree extend system-id

redundancy
mode none

vlan internal allocation policy ascending

interface GigabitEthernet0/0/0
ip address 192.168.0.1 255.255.255.0
negotiation auto
ipv6 address 2001::1/64

```
ipv6 ospf 1 area 0

interface GigabitEthernet0/0/1
 ip address 192.168.1.1 255.255.255.0
 ip ospf 1 area 0
 negotiation auto
 ipv6 address 2001:1::1/64
 ipv6 ospf 1 area 0

interface Serial0/1/0
 no ip address

interface Serial0/1/1
 no ip address

interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 negotiation auto

interface Vlan1
 no ip address

router ospf 1
 router-id 2.2.2.2
 redistribute connected subnets
 redistribute bgp 2 subnets
 passive-interface GigabitEthernet0/0/0

router bgp 2
 bgp log-neighbor-changes
 neighbor 2001::2 remote-as 4
 neighbor 2001::3 remote-as 6
 neighbor 192.168.0.2 remote-as 4
 neighbor 192.168.0.3 remote-as 6

address-family ipv4
 redistribute ospf 1
 no neighbor 2001::2 activate
 no neighbor 2001::3 activate
 neighbor 192.168.0.2 activate
 neighbor 192.168.0.3 activate
 exit-address-family

address-family ipv6
 redistribute ospf 1 include-connected
 neighbor 2001::2 activate
```

```
    neighbor 2001::3 activate
  exit-address-family

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 1
  router-id 2.2.2.2
  passive-interface GigabitEthernet0/0/0
  redistribute connected
  redistribute bgp 2

control-plane

line con 0
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login

end
```

Router 3:

Last configuration change at 20:23:32 UTC Wed Nov 8 2023

```
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core

hostname JacobAaronAidenR3

boot-start-marker
boot-end-marker

vrf definition Mgmt-intf
```



```
address-family ipv4
exit-address-family

address-family ipv6
exit-address-family

no aaa new-model

ipv6 unicast-routing

subscriber templating
multilink bundle-name authenticated

license udi pid ISR4321/K9 sn FDO214420G7

spanning-tree extend system-id

redundancy
mode none

vlan internal allocation policy ascending

interface Loopback0
 ip address 2.2.2.2 255.255.255.0
 ipv6 address 2001:2:2:2::2/64
 ipv6 rip RIPforV6 enable

interface GigabitEthernet0/0/0
 no ip address
 shutdown
 negotiation auto

interface GigabitEthernet0/0/1
 ip address 192.168.2.2 255.255.255.0
 negotiation auto
 ipv6 address 2001:2::2/64
 ipv6 rip RIPforV6 enable

interface Serial0/1/0
 no ip address
 shutdown
```

```
interface Serial0/1/1
  no ip address
  shutdown

interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto

interface Vlan1
  no ip address
  shutdown

router rip
  version 2
  redistribute connected metric transparent
  network 192.168.2.0
  neighbor 192.168.2.1
  no auto-summary

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router rip RIPforV6
  redistribute connected metric 5

control-plane

line con 0
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login

end
```

Router 4:

Last configuration change at 20:29:24 UTC Wed Nov 8 2023

version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core

hostname JacobAaronAidenR4

boot-start-marker
boot-end-marker

vrf definition Mgmt-intf

address-family ipv4
exit-address-family

address-family ipv6
exit-address-family

no aaa new-model

ipv6 unicast-routing

subscriber templating
multilink bundle-name authenticated

license udi pid ISR4321/K9 sn FDO21442B21

spanning-tree extend system-id

redundancy
mode none

vlan internal allocation policy ascending

interface GigabitEthernet0/0/0
ip address 192.168.0.2 255.255.255.0
negotiation auto

```
ipv6 address 2001::2/64
ipv6 rip RIPforV6 enable

interface GigabitEthernet0/0/1
 ip address 192.168.2.1 255.255.255.0
 negotiation auto
 ipv6 address 2001:2::1/64
 ipv6 rip RIPforV6 enable

interface Serial0/1/0
 no ip address

interface Serial0/1/1
 no ip address

interface GigabitEthernet0/2/0
 no ip address
 negotiation auto

interface GigabitEthernet0/2/1
 no ip address
 negotiation auto

interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 negotiation auto

interface Vlan1
 no ip address

router rip
 version 2
 redistribute connected
 redistribute bgp 4 metric 5
 passive-interface GigabitEthernet0/0/0
 network 192.168.2.0
 neighbor 192.168.2.2
 no auto-summary

router bgp 4
 bgp log-neighbor-changes
 neighbor 2001::1 remote-as 2
 neighbor 2001::3 remote-as 6
 neighbor 192.168.0.1 remote-as 2
 neighbor 192.168.0.3 remote-as 6
```

```

address-family ipv4
  redistribute rip
  no neighbor 2001::1 activate
  no neighbor 2001::3 activate
  neighbor 192.168.0.1 activate
  neighbor 192.168.0.3 activate
exit-address-family

```

```

address-family ipv6
  redistribute rip RIPforV6 include-connected
  neighbor 2001::1 activate
  neighbor 2001::3 activate
exit-address-family

```

```

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0

```

```

ipv6 router rip RIPforV6
  redistribute connected
  redistribute bgp 4 metric 5

```

```

control-plane

```

```

line con 0
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login

```

```

end

```

Router 5:

Last configuration change at 20:17:44 UTC Wed Nov 8 2023

```

version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core

```

```
hostname JacobAaronAidenR5

boot-start-marker
boot-end-marker

vrf definition Mgmt-intf

    address-family ipv4
    exit-address-family

    address-family ipv6
    exit-address-family

no aaa new-model

login on-success log

subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated

license udi pid ISR4321/K9 sn FLM24060912
no license smart enable
diagnostic bootup level minimal

spanning-tree extend system-id

redundancy
mode none

interface Loopback0
 ip address 3.3.3.3 255.255.255.0
 ipv6 address 2001:3:3:3::3/64
 ipv6 eigrp 1

interface GigabitEthernet0/0/0
 no ip address
 negotiation auto

interface GigabitEthernet0/0/1
 ip address 192.168.3.2 255.255.255.0
 negotiation auto
```

```
ipv6 address 2001:3::2/64
ipv6 eigrp 1

interface GigabitEthernet0/2/0
no ip address
negotiation auto

interface GigabitEthernet0/2/1
no ip address
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto

router eigrp 1
network 192.168.3.0
redistribute connected metric 10000 100 255 1 1500
eigrp router-id 5.5.5.5

ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router eigrp 1
eigrp router-id 5.5.5.5
redistribute connected metric 10000 100 255 1 1500

control-plane

line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login

end
```

Router 6:

Last configuration change at 20:22:41 UTC Wed Nov 8 2023

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
```

```
hostname JacobAaronAidenR6
```

```
boot-start-marker
boot-end-marker
```

```
vrf definition Mgmt-intf
```

```
    address-family ipv4
    exit-address-family
```

```
    address-family ipv6
    exit-address-family
```

```
no aaa new-model
```

```
login on-success log
```

```
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
```

```
license udi pid ISR4321/K9 sn FLM2408005M
no license smart enable
diagnostic bootup level minimal
```

```
spanning-tree extend system-id
```

```
redundancy
mode none
```

```
interface GigabitEthernet0/0/0
 ip address 192.168.0.3 255.255.255.0
 negotiation auto
```



```
ipv6 address 2001::3/64
ipv6 eigrp 1

interface GigabitEthernet0/0/1
 ip address 192.168.3.1 255.255.255.0
 negotiation auto
 ipv6 address 2001:3::1/64
 ipv6 eigrp 1

interface GigabitEthernet0/2/0
 no ip address
 negotiation auto

interface GigabitEthernet0/2/1
 no ip address
 negotiation auto

interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 negotiation auto

router eigrp 1
 network 192.168.3.0
 redistribute bgp 6 metric 10000 100 255 1 1500
 redistribute connected
 passive-interface GigabitEthernet0/0/0
 eigrp router-id 6.6.6.6

router bgp 6
 bgp log-neighbor-changes
 neighbor 2001::1 remote-as 2
 neighbor 2001::2 remote-as 4
 neighbor 192.168.0.1 remote-as 2
 neighbor 192.168.0.2 remote-as 4

address-family ipv4
 redistribute eigrp 1
 no neighbor 2001::1 activate
 no neighbor 2001::2 activate
 neighbor 192.168.0.1 activate
 neighbor 192.168.0.1 next-hop-self
 neighbor 192.168.0.1 weight 200
 neighbor 192.168.0.2 activate
 neighbor 192.168.0.2 route-map map out
 exit-address-family
```

```

address-family ipv6
  redistribute eigrp 1 include-connected
  neighbor 2001::1 activate
  neighbor 2001::2 activate
exit-address-family

ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0

access-list 1 permit any
ipv6 router eigrp 1
  passive-interface GigabitEthernet0/0/0
  eigrp router-id 6.6.6.6
  redistribute bgp 6 metric 10000 100 255 1 1500
  redistribute connected

route-map map permit 10
  match ip address 1
  set metric 500

route-map map permit 20

control-plane

line con 0
  transport input none
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login

end

```

IP Routes/Traceroutes:

Router 1:

IP Routes:

JacobAaronAidenR1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

      1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.1.1.0/24 is directly connected, Loopback0
L       1.1.1.1/32 is directly connected, Loopback0
      2.0.0.0/24 is subnetted, 1 subnets
O E2    2.2.2.0 [110/1] via 192.168.1.1, 00:42:00,
GigabitEthernet0/0/1
      3.0.0.0/24 is subnetted, 1 subnets
O E2    3.3.3.0 [110/1] via 192.168.1.1, 00:43:32,
GigabitEthernet0/0/1
O E2    192.168.0.0/24 [110/20] via 192.168.1.1, 00:45:54,
GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected,
GigabitEthernet0/0/1
L       192.168.1.2/32 is directly connected,
GigabitEthernet0/0/1
O E2    192.168.2.0/24 [110/1] via 192.168.1.1, 00:42:00,
GigabitEthernet0/0/1
O E2    192.168.3.0/24 [110/1] via 192.168.1.1, 00:43:32,
GigabitEthernet0/0/1

```

JacobAaronAidenR1#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
 IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
 O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
 ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

```

O   2001::/64 [110/2]
    via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/1
C   2001:1::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   2001:1::2/128 [0/0]
    via GigabitEthernet0/0/1, receive
C   2001:1:1:1::/64 [0/0]
    via Loopback0, directly connected
L   2001:1:1:1::1/128 [0/0]
    via Loopback0, receive
OE2 2001:2::/64 [110/1]
    via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/1
OE2 2001:2:2:2::/64 [110/1]
    via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/1
OE2 2001:3::/64 [110/1]
    via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/1
OE2 2001:3:3:3::/64 [110/1]
    via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/1
L   FF00::/8 [0/0]
    via Null0, receive

```

Traceroute:

```

JacobAaronAidenR1#traceroute 1.1.1.1
Type escape sequence to abort.
Tracing the route to 1.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 1.1.1.1 1 msec * 1 msec
JacobAaronAidenR1#traceroute 2.2.2.2
Type escape sequence to abort.
Tracing the route to 2.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.1.1 0 msec 1 msec 0 msec
  2 192.168.0.2 1 msec 1 msec 1 msec
  3 192.168.2.2 1 msec 2 msec *
JacobAaronAidenR1#traceroute 3.3.3.3
Type escape sequence to abort.
Tracing the route to 3.3.3.3
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.1.1 1 msec 0 msec 1 msec
  2 192.168.0.3 1 msec 1 msec 1 msec

```

```
3 192.168.3.2 1 msec 1 msec *
```

```
JacobAaronAidenR1#traceroute 2001:1:1:1::1
Type escape sequence to abort.
Tracing the route to 2001:1:1:1::1
```

```
1 2001:1:1:1::1 1 msec 0 msec 1 msec
JacobAaronAidenR1#traceroute 2001:2:2:2::2
Type escape sequence to abort.
Tracing the route to 2001:2:2:2::2
```

```
1 2001:1::1 6 msec 1 msec 1 msec
2 2001::2 1 msec 1 msec 1 msec
3 2001:2::2 1 msec 1 msec 0 msec
JacobAaronAidenR1#traceroute 2001:3:3:3::3
Type escape sequence to abort.
Tracing the route to 2001:3:3:3::3
```

```
1 2001:1::1 1 msec 1 msec 1 msec
2 2001::3 1 msec 1 msec 1 msec
3 2001:3::2 1 msec 2 msec 1 msec
```

Router 2:

IP Routes:

```
JacobAaronAidenR2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-
user static route
        o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
        a - application route
        + - replicated route, % - next hop override, p -
overrides from PfR
```

```
Gateway of last resort is not set
```

```
1.0.0.0/32 is subnetted, 1 subnets
```

```

O      1.1.1.1 [110/2] via 192.168.1.2, 00:45:16,
GigabitEthernet0/0/1
      2.0.0.0/24 is subnetted, 1 subnets
B      2.2.2.0 [20/1] via 192.168.0.2, 00:41:22
      3.0.0.0/24 is subnetted, 1 subnets
B      3.3.3.0 [20/281856] via 192.168.0.3, 00:42:54
      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.0.0/24 is directly connected,
GigabitEthernet0/0/0
L      192.168.0.1/32 is directly connected,
GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected,
GigabitEthernet0/0/1
L      192.168.1.1/32 is directly connected,
GigabitEthernet0/0/1
B      192.168.2.0/24 [20/0] via 192.168.0.2, 00:41:22
B      192.168.3.0/24 [20/0] via 192.168.0.3, 00:42:54

```

JacobAaronAidenR2#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDR - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a -

Application

```

C      2001::/64 [0/0]
      via GigabitEthernet0/0/0, directly connected
L      2001::1/128 [0/0]
      via GigabitEthernet0/0/0, receive
C      2001:1::/64 [0/0]
      via GigabitEthernet0/0/1, directly connected
L      2001:1::1/128 [0/0]
      via GigabitEthernet0/0/1, receive
O      2001:1:1:1::1/128 [110/1]
      via FE80::B6A8:B9FF:FE47:8E41, GigabitEthernet0/0/1
B      2001:2::/64 [20/0]
      via FE80::B6A8:B9FF:FEA0:2E20, GigabitEthernet0/0/0
B      2001:2:2:2::/64 [20/2]
      via FE80::B6A8:B9FF:FEA0:2E20, GigabitEthernet0/0/0
B      2001:3::/64 [20/0]

```

```

        via FE80::CE7F:76FF:FED1:ADC0, GigabitEthernet0/0/0
B   2001:3:3:3::/64 [20/130816]
        via FE80::CE7F:76FF:FED1:ADC0, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
        via Null0, receive

```

Traceroute:

```

JacobAaronAidenR2#traceroute 1.1.1.1
Type escape sequence to abort.
Tracing the route to 1.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.1.2 1 msec 2 msec *
JacobAaronAidenR2#traceroute 2.2.2.2
Type escape sequence to abort.
Tracing the route to 2.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.0.2 1 msec 0 msec 1 msec
  2 192.168.2.2 [AS 4] 1 msec 1 msec *
JacobAaronAidenR2#traceroute 3.3.3.3
Type escape sequence to abort.
Tracing the route to 3.3.3.3
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.0.3 1 msec 1 msec 1 msec
  2 192.168.3.2 [AS 6] 1 msec 1 msec *

JacobAaronAidenR2#traceroute 2001:1:1:1::1
Type escape sequence to abort.
Tracing the route to 2001:1:1:1::1

  1 2001:1::2 14 msec 1 msec 1 msec
JacobAaronAidenR2#traceroute 2001:2:2:2::2
Type escape sequence to abort.
Tracing the route to 2001:2:2:2::2

  1 2001::2 3 msec 3 msec 3 msec
  2 2001:2::2 [AS 4] 1 msec 2 msec 1 msec
JacobAaronAidenR2#traceroute 2001:3:3:3::3
Type escape sequence to abort.
Tracing the route to 2001:3:3:3::3

  1 2001::3 1 msec 1 msec 1 msec
  2 2001:3::2 [AS 6] 1 msec 1 msec 1 msec

```

Router 3:

IP Routes:

```
JacobAaronAidenR3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-
user static route
        o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
        a - application route
        + - replicated route, % - next hop override, p -
overrides from PfR
```

Gateway of last resort is not set

```
        1.0.0.0/32 is subnetted, 1 subnets
R        1.1.1.1 [120/5] via 192.168.2.1, 00:00:17,
GigabitEthernet0/0/1
        2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        2.2.2.0/24 is directly connected, Loopback0
L        2.2.2.2/32 is directly connected, Loopback0
        3.0.0.0/24 is subnetted, 1 subnets
R        3.3.3.0 [120/5] via 192.168.2.1, 00:00:17,
GigabitEthernet0/0/1
R        192.168.0.0/24 [120/1] via 192.168.2.1, 00:00:17,
GigabitEthernet0/0/1
R        192.168.1.0/24 [120/5] via 192.168.2.1, 00:00:17,
GigabitEthernet0/0/1
        192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected,
GigabitEthernet0/0/1
L        192.168.2.2/32 is directly connected,
GigabitEthernet0/0/1
R        192.168.3.0/24 [120/5] via 192.168.2.1, 00:00:17,
GigabitEthernet0/0/1
```

```
JacobAaronAidenR3#show ipv6 route
IPv6 Routing Table - default - 10 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
        B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
```


IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
 ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
 O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
 ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

```

R 2001::/64 [120/2]
    via FE80::B6A8:B9FF:FEA0:2E21, GigabitEthernet0/0/1
R 2001:1::/64 [120/6]
    via FE80::B6A8:B9FF:FEA0:2E21, GigabitEthernet0/0/1
R 2001:1:1:1::1/128 [120/6]
    via FE80::B6A8:B9FF:FEA0:2E21, GigabitEthernet0/0/1
C 2001:2::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L 2001:2::2/128 [0/0]
    via GigabitEthernet0/0/1, receive
C 2001:2:2:2::/64 [0/0]
    via Loopback0, directly connected
L 2001:2:2:2::2/128 [0/0]
    via Loopback0, receive
R 2001:3::/64 [120/6]
    via FE80::B6A8:B9FF:FEA0:2E21, GigabitEthernet0/0/1
R 2001:3:3:3::/64 [120/6]
    via FE80::B6A8:B9FF:FEA0:2E21, GigabitEthernet0/0/1
L FF00::/8 [0/0]
    via Null0, receive

```

Traceroute:

```

JacobAaronAidenR3#traceroute 1.1.1.1
Type escape sequence to abort.
Tracing the route to 1.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
 1 192.168.2.1 1 msec 1 msec 0 msec
 2 192.168.0.1 1 msec 1 msec 0 msec
 3 192.168.1.2 1 msec 1 msec *
JacobAaronAidenR3#traceroute 2.2.2.2
Type escape sequence to abort.
Tracing the route to 2.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
 1 2.2.2.2 0 msec * 1 msec
JacobAaronAidenR3#traceroute 3.3.3.3
Type escape sequence to abort.
Tracing the route to 3.3.3.3
VRF info: (vrf in name/id, vrf out name/id)

```

```

1 192.168.2.1 0 msec 1 msec 0 msec
2 192.168.0.3 1 msec 1 msec 1 msec
3 192.168.3.2 2 msec 1 msec *
```

```

JacobAaronAidenR3#traceroute 2001:1:1:1::1
Type escape sequence to abort.
Tracing the route to 2001:1:1:1::1
```

```

1 2001:2::1 5 msec 1 msec 1 msec
2 2001::1 1 msec 1 msec 1 msec
3 2001:1::2 1 msec 1 msec 1 msec
```

```

JacobAaronAidenR3#traceroute 2001:2:2:2::2
Type escape sequence to abort.
Tracing the route to 2001:2:2:2::2
```

```

1 2001:2:2:2::2 0 msec 1 msec 0 msec
```

```

JacobAaronAidenR3#traceroute 2001:3:3:3::3
Type escape sequence to abort.
Tracing the route to 2001:3:3:3::3
```

```

1 2001:2::1 1 msec 1 msec 1 msec
2 2001::3 1 msec 2 msec 1 msec
3 2001:3::2 1 msec 1 msec 1 msec
```

Router 4:

IP Routes:

```

JacobAaronAidenR4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-
user static route
        o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
        a - application route
        + - replicated route, % - next hop override, p -
overrides from PfR
```

```

Gateway of last resort is not set
```

```

        1.0.0.0/32 is subnetted, 1 subnets
B       1.1.1.1 [20/2] via 192.168.0.1, 00:38:49
        2.0.0.0/24 is subnetted, 1 subnets
R       2.2.2.0 [120/1] via 192.168.2.2, 00:00:04,
GigabitEthernet0/0/1
        3.0.0.0/24 is subnetted, 1 subnets
B       3.3.3.0 [20/500] via 192.168.0.3, 00:38:49
        192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.0.2/32 is directly connected,
GigabitEthernet0/0/0
B       192.168.1.0/24 [20/0] via 192.168.0.1, 00:38:49
        192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected,
GigabitEthernet0/0/1
L       192.168.2.1/32 is directly connected,
GigabitEthernet0/0/1
B       192.168.3.0/24 [20/500] via 192.168.0.3, 00:38:49

```

JacobAaronAidenR4#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX -
EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr
- Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 -
OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a -
Application

```

C    2001::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L    2001::2/128 [0/0]
    via GigabitEthernet0/0/0, receive
B    2001:1::/64 [20/0]
    via FE80::2F8:2CFF:FE7F:7190, GigabitEthernet0/0/0
B    2001:1:1:1::1/128 [20/1]
    via FE80::2F8:2CFF:FE7F:7190, GigabitEthernet0/0/0
C    2001:2::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L    2001:2::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
R    2001:2:2:2::/64 [120/2]
    via FE80::B6A8:B9FF:FE01:B751, GigabitEthernet0/0/1

```

```

B    2001:3::/64 [20/0]
     via FE80::CE7F:76FF:FED1:ADC0, GigabitEthernet0/0/0
B    2001:3:3:3::/64 [20/130816]
     via FE80::CE7F:76FF:FED1:ADC0, GigabitEthernet0/0/0
L    FF00::/8 [0/0]
     via Null0, receive

```

Traceroute:

```

JacobAaronAidenR4#traceroute 1.1.1.1
Type escape sequence to abort.
Tracing the route to 1.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.0.1 0 msec 1 msec 0 msec
  2 192.168.1.2 [AS 2] 1 msec 1 msec *
JacobAaronAidenR4#traceroute 2.2.2.2
Type escape sequence to abort.
Tracing the route to 2.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.2.2 2 msec 1 msec *
JacobAaronAidenR4#traceroute 3.3.3.3
Type escape sequence to abort.
Tracing the route to 3.3.3.3
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.0.3 1 msec 1 msec 1 msec
  2 192.168.3.2 [AS 6] 1 msec 2 msec *

JacobAaronAidenR4#traceroute 2001:1:1:1::1
Type escape sequence to abort.
Tracing the route to 2001:1:1:1::1

  1 2001::1 2 msec 1 msec 1 msec
  2 2001:1::2 [AS 2] 1 msec 1 msec 1 msec
JacobAaronAidenR4#traceroute 2001:2:2:2::2
Type escape sequence to abort.
Tracing the route to 2001:2:2:2::2

  1 2001:2::2 8 msec 1 msec 1 msec
JacobAaronAidenR4#traceroute 2001:3:3:3::3
Type escape sequence to abort.
Tracing the route to 2001:3:3:3::3

  1 2001::3 1 msec 2 msec 1 msec
  2 2001:3::2 [AS 6] 1 msec 1 msec 1 msec

```

Router 5:

IP Routes:

```
JacobAaronAidenR5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-
user static route
        o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
        a - application route
        + - replicated route, % - next hop override, p -
overrides from PfR
```

Gateway of last resort is not set

```
        1.0.0.0/32 is subnetted, 1 subnets
D EX      1.1.1.1 [170/281856] via 192.168.3.1, 00:38:47,
GigabitEthernet0/0/1
        2.0.0.0/24 is subnetted, 1 subnets
D EX      2.2.2.0 [170/281856] via 192.168.3.1, 00:37:14,
GigabitEthernet0/0/1
        3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C          3.3.3.0/24 is directly connected, Loopback0
L          3.3.3.3/32 is directly connected, Loopback0
D EX 192.168.0.0/24
        [170/28416] via 192.168.3.1, 00:40:06,
GigabitEthernet0/0/1
D EX 192.168.1.0/24
        [170/281856] via 192.168.3.1, 00:38:47,
GigabitEthernet0/0/1
D EX 192.168.2.0/24
        [170/281856] via 192.168.3.1, 00:37:14,
GigabitEthernet0/0/1
        192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C          192.168.3.0/24 is directly connected,
GigabitEthernet0/0/1
L          192.168.3.2/32 is directly connected,
GigabitEthernet0/0/1
```

```
JacobAaronAidenR5#show ipv6 route
IPv6 Routing Table - default - 10 entries
```

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

```
D 2001::/64 [90/28416]
  via FE80::CE7F:76FF:FED1:ADC1, GigabitEthernet0/0/1
EX 2001:1::/64 [170/281856]
  via FE80::CE7F:76FF:FED1:ADC1, GigabitEthernet0/0/1
EX 2001:1:1:1::1/128 [170/281856]
  via FE80::CE7F:76FF:FED1:ADC1, GigabitEthernet0/0/1
EX 2001:2::/64 [170/281856]
  via FE80::CE7F:76FF:FED1:ADC1, GigabitEthernet0/0/1
EX 2001:2:2:2::/64 [170/281856]
  via FE80::CE7F:76FF:FED1:ADC1, GigabitEthernet0/0/1
C 2001:3::/64 [0/0]
  via GigabitEthernet0/0/1, directly connected
L 2001:3::2/128 [0/0]
  via GigabitEthernet0/0/1, receive
C 2001:3:3:3::/64 [0/0]
  via Loopback0, directly connected
L 2001:3:3:3::3/128 [0/0]
  via Loopback0, receive
L FF00::/8 [0/0]
  via Null0, receive
```

Traceroute:

JacobAaronAidenR5#traceroute 1.1.1.1

Type escape sequence to abort.

Tracing the route to 1.1.1.1

VRF info: (vrf in name/id, vrf out name/id)

```
 1 192.168.3.1 1 msec 1 msec 1 msec
 2 192.168.0.1 1 msec 1 msec 1 msec
 3 192.168.1.2 1 msec 1 msec *
```

JacobAaronAidenR5#traceroute 2.2.2.2

Type escape sequence to abort.

Tracing the route to 2.2.2.2

VRF info: (vrf in name/id, vrf out name/id)

```
 1 192.168.3.1 1 msec 1 msec 1 msec
 2 192.168.0.2 0 msec 1 msec 0 msec
```

```

3 192.168.2.2 1 msec 1 msec *
JacobAaronAidenR5#traceroute 3.3.3.3
Type escape sequence to abort.
Tracing the route to 3.3.3.3
VRF info: (vrf in name/id, vrf out name/id)
 1 3.3.3.3 1 msec * 1 msec

```

```

JacobAaronAidenR5#traceroute 2001:1:1:1::1
Type escape sequence to abort.
Tracing the route to 2001:1:1:1::1

```

```

 1 2001:3::1 6 msec 1 msec 1 msec
 2 2001::1 1 msec 1 msec 1 msec
 3 2001:1::2 1 msec 1 msec 1 msec
JacobAaronAidenR5#traceroute 2001:2:2:2::2
Type escape sequence to abort.
Tracing the route to 2001:2:2:2::2

```

```

 1 2001:3::1 1 msec 1 msec 0 msec
 2 2001::2 1 msec 1 msec 1 msec
 3 2001:2::2 1 msec 1 msec 1 msec
JacobAaronAidenR5#traceroute 2001:3:3:3::3
Type escape sequence to abort.
Tracing the route to 2001:3:3:3::3

```

```

 1 2001:3:3:3::3 0 msec 1 msec 0 msec

```

Router 6:

IP Routes:

```

JacobAaronAidenR6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

```

Gateway of last resort is not set

```

    1.0.0.0/32 is subnetted, 1 subnets
B       1.1.1.1 [20/2] via 192.168.0.1, 00:35:07
    2.0.0.0/24 is subnetted, 1 subnets
B       2.2.2.0 [20/0] via 192.168.0.2, 00:33:35
    3.0.0.0/24 is subnetted, 1 subnets
D EX    3.3.3.0 [170/281856] via 192.168.3.2, 00:36:23,
GigabitEthernet0/0/1
    192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.0.3/32 is directly connected,
GigabitEthernet0/0/0
B       192.168.1.0/24 [20/0] via 192.168.0.1, 00:35:07
B       192.168.2.0/24 [20/0] via 192.168.0.2, 00:33:35
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected,
GigabitEthernet0/0/1
L       192.168.3.1/32 is directly connected,
GigabitEthernet0/0/1

```

JacobAaronAidenR6#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
 IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX -
 EIGRP external
 ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr
 - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 -
 OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a -
 Application

```

C    2001::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L    2001::3/128 [0/0]
    via GigabitEthernet0/0/0, receive
B    2001:1::/64 [20/0]
    via FE80::2F8:2CFF:FE7F:7190, GigabitEthernet0/0/0
B    2001:1:1:1::1/128 [20/1]
    via FE80::2F8:2CFF:FE7F:7190, GigabitEthernet0/0/0
B    2001:2::/64 [20/0]
    via FE80::B6A8:B9FF:FEA0:2E20, GigabitEthernet0/0/0
B    2001:2:2:2::/64 [20/2]

```



```

    via FE80::B6A8:B9FF:FEA0:2E20, GigabitEthernet0/0/0
C   2001:3::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   2001:3::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
D   2001:3:3:3::/64 [90/130816]
    via FE80::CE7F:76FF:FE6A:B5E1, GigabitEthernet0/0/1
L   FF00::/8 [0/0]
    via Null0, receive

```

Traceroute:

```

JacobAaronAidenR6#traceroute 1.1.1.1
Type escape sequence to abort.
Tracing the route to 1.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.0.1 0 msec 1 msec 0 msec
  2 192.168.1.2 [AS 2] 1 msec 1 msec *
JacobAaronAidenR6#traceroute 2.2.2.2
Type escape sequence to abort.
Tracing the route to 2.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.0.2 0 msec 1 msec 0 msec
  2 192.168.2.2 [AS 4] 1 msec 1 msec *
JacobAaronAidenR6#traceroute 3.3.3.3
Type escape sequence to abort.
Tracing the route to 3.3.3.3
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.3.2 1 msec 1 msec *
JacobAaronAidenR6#traceroute 2001:1:1:1::1
Type escape sequence to abort.

Tracing the route to 2001:1:1:1::1

  1 2001::1 1 msec 1 msec 0 msec
  2 2001:1::2 [AS 2] 1 msec 1 msec 1 msec
JacobAaronAidenR6#traceroute 2001:2:2:2::2
Type escape sequence to abort.
Tracing the route to 2001:2:2:2::2

  1 2001::2 1 msec 1 msec 0 msec
  2 2001:2::2 [AS 4] 1 msec 1 msec 1 msec
JacobAaronAidenR6#traceroute 2001:3:3:3::3
Type escape sequence to abort.
Tracing the route to 2001:3:3:3::3

  1 2001:3::2 7 msec 1 msec 1 msec

```

Problems:Certain Commands Being Different in Packet Tracer and Physical Rack

In Packet Tracer, you can't set any BGP attributes at all, making the copied configurations from PT into the physical rack incomplete. We needed to add the attribute-changing commands after copying the configs over.

There also isn't the redistribute rip or redistribute eigrp commands in Packet Tracer for some reason, and we needed to add those also.

Set up RIPv1 Instead of RIPv2

RIPv2 supports variable length subnet masking(VLSM) and route summarization, while RIPv1 doesn't. By setting up RIPv1 initially, it made the redistribution of routes much more complicated, making setting up RIPv2 a better option overall.

Network BGP Statements Needing to be Specific:

When adding routes through BGP using network statements, the subnet mask of the network needs to match the exact network you are trying to add to the BGP routing table. If the subnet mask of the network doesn't match, the network isn't added properly to the table.

Conclusion:

Although BGP may be more complicated than other IGPs to set up, the innumerable benefits that it offers over these other protocols is invaluable. BGP utilizes much less system resources, as well as allows for more networks that use different IGPs to communicate with each other. The versatility, flexibility, and scalability of BGP make it the routing protocol of choice for network engineers of large networks such as corporate networks and of course, the Internet, even if it isn't the fastest protocol out there.

Lab 4: Amazon AWS Cloud Foundations Labs 1-3



Purpose

Amazon Web Services, or AWS for short, is one of the most common and robust cloud service providers, along with others like Microsoft Azure and Google Cloud Platform. By renting their platform out through a pay-as-you-go format, these cloud service providers may offer a cheaper, more reliable, and more flexible alternative compared to traditional on-premises infrastructure. Through these three labs, we can start to learn about various fundamental services of AWS and how they combine with other services. Learning about these services will also help us study and prepare for the AWS Cloud Foundations Certification Exam.

Background Information on Lab Concepts

Lab 1: Introduction to AWS IAM

Amazon IAM, or Identity and Access Management, is an integral service for AWS, allowing administrators to centrally manage and assign certain user or user groups specific permissions. Permissions are especially important when working in a company structure, where different positions should have different clearances for certain services. Each employee should only receive the bare minimum permissions in order to carry out their jobs effectively. By using IAM, administrators can share permissions with fine-grain control without sharing login credentials, thus greatly increasing security of the entire organization.

Lab 2: Build your VPC and Launch a Web Server

Virtual Private Clouds, or VPC for short, are virtually isolated networks that the AWS user can create. These networks are highly very similar to traditional networks using your own infrastructure, except you gain the scalability of AWS's infrastructure. With a VPC, you can do everything you can on a traditional network, including subnetting, private/public addressing, and connecting to the internet. VPC's allow you to launch AWS services in them, such as Amazon EC2, Elastic Container Service, and RDS. With Amazon VPCs, the service itself doesn't cost money, but internal services such as NAT gateways, traffic mirroring, or Network Access Analyzer do cost extra.

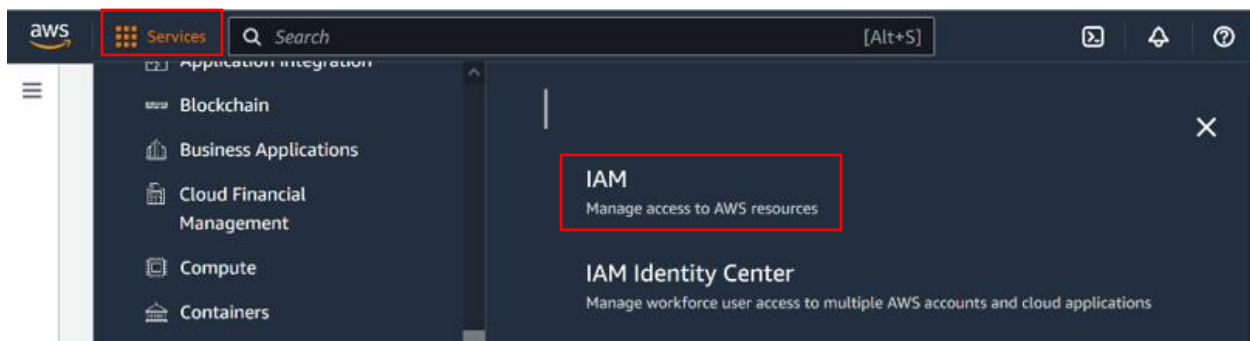
Lab 3: Introduction to Amazon EC2

Amazon Elastic Compute Cloud, or EC2, provides on-demand, scalable virtual machine that users pay for on a pay-as-you-go format. In a traditional networking model, companies are forced to purchase and maintain their own systems. However, with the shift to cloud computing with EC2, companies only need to pay for what resources they use, drastically cutting down on overall costs. Each instance of EC2 is like a virtual computer, where you choose between various configurations of CPU, memory, storage, and graphical processing power. These instances can do most things that a physical computer can do, without needing the user to buy their own machines and taking up the space and electricity required to maintain each machine.

Lab #1 Walkthrough:

Task 1: Exploring the Users and Groups

1. Open the Amazon AWS Management Console, navigate to Services, and choose IAM.



2. In the navigation plane, choose the Users tab.

The screenshot shows the AWS IAM console interface. On the left is the navigation pane for 'Identity and Access Management (IAM)'. It includes a search bar, a 'Dashboard' link, and two main sections: 'Access management' and 'Access reports'. Under 'Access management', the 'Users' link is highlighted with a red box. The main content area on the right is the 'IAM Dashboard'. It features a breadcrumb 'IAM > Dashboard', a 'Security recommendations' section with one warning (Add MFA for root user), and an 'IAM resources' section showing counts for User groups (3), Users (4), and Roles (13).

Identity and Access Management (IAM) ✕

Search IAM

Dashboard

▼ **Access management**

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings

▼ **Access reports**

- Access analyzer

IAM Dashboard

[IAM](#) > Dashboard

Security recommendations 1

Add MFA for root user
Sign in as the root user (or contact your administrator) and register the root user to improve security for this account.

IAM resources
Resources in this AWS Account

User groups	Users	Roles
3	4	13

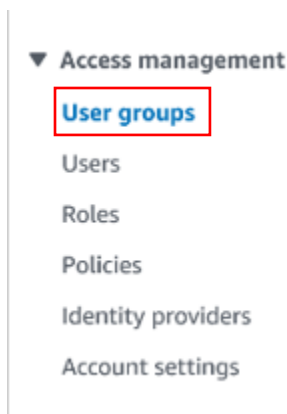
3. When you choose User-1, notice how the user doesn't have any permissions and how its not part of any groups.

The screenshot displays the AWS IAM console interface for a user named 'user-1'. The interface is divided into several sections:

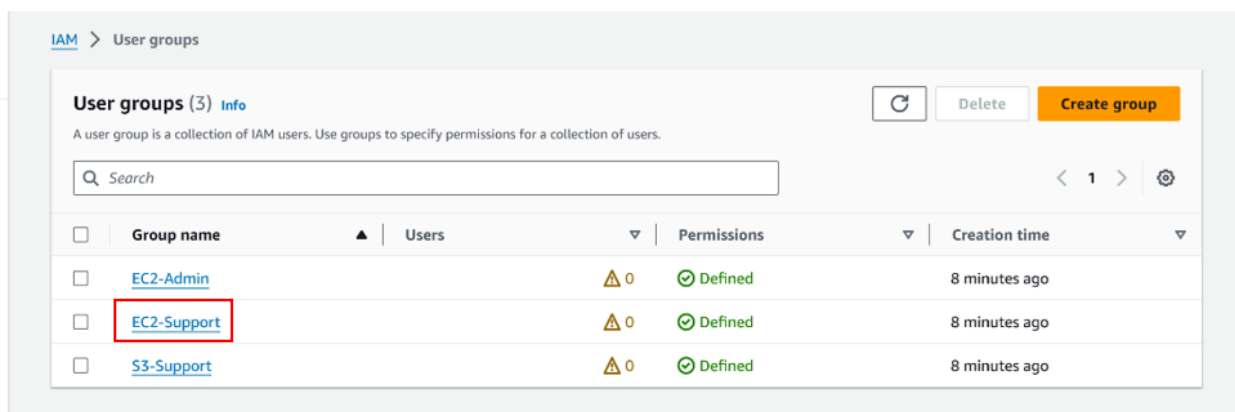
- Summary:** Contains the user's ARN (`arn:aws:iam::056013997774:user/sp166/user-1`), console access status (Enabled without MFA), last console sign-in (Never), and access keys (Access key 1: AKIAQ2CVR3HJUAVWWHP - Active, Access key 2: Create access key).
- Permissions policies (0):** Shows that no permissions policies are attached to the user. A red box highlights the message 'No resources to display'.
- Permissions boundary (not set):** Indicates that no permissions boundary is set for the user. A button 'Set permissions boundary' is available.
- User groups membership (0):** Shows that the user is not a member of any groups. A red box highlights the message 'This user does not belong to any groups.'

The navigation bar at the top shows the user's name 'user-1' and an 'Info' link. The left sidebar contains links for 'Permissions', 'Groups', 'Tags (1)', 'Security credentials', and 'Access Advisor'. The main content area is titled 'user-1' and includes a 'Delete' button.

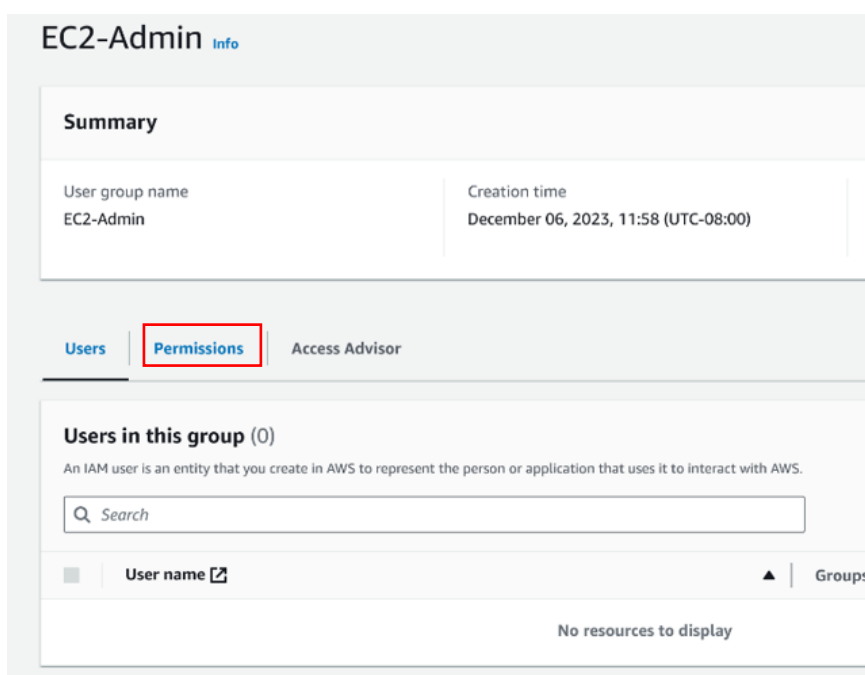
- Navigate to the User Groups tab in the navigations tab.



- Choose the EC2-Support Group



- Choose the Permissions Tab



7. Notice how a managed policy was already created for the User Group called AmazonEC2ReadOnlyAccess. Click the “+” button to view the policy details.

The screenshot shows the AWS IAM console with the 'Permissions' tab selected. Under 'Permissions policies (1)', the 'AmazonEC2ReadOnlyAccess' policy is listed. A red box highlights the policy name and its icon. Another red box highlights the JSON policy document in the editor view below.

Permissions policies (1) Info

You can attach up to 10 managed policies.

Filter by Type: All types

Policy name	Type	Attach
 AmazonEC2ReadOnlyAccess	AWS managed	1

AmazonEC2ReadOnlyAccess

Provides read only access to Amazon EC2 via the AWS Management Console.

```
10  "Effect": "Allow",
11  "Action": "elasticloadbalancing:Describe*",
12  "Resource": "*"
13  },
14  {
15    "Effect": "Allow",
16    "Action": [
17      "cloudwatch:ListMetrics",
18      "cloudwatch:GetMetricStatistics",
19      "cloudwatch:Describe*"
20    ],
21    "Resource": "*"
22  },
23  {
24    "Effect": "Allow",
25    "Action": "autoscaling:Describe*",
26    "Resource": "*"
27  }
28  ]
29 }
```

8. View the policy details for the other two User Groups, S3-Support and EC2-Admin

The image displays two screenshots from the AWS IAM console. The left screenshot shows the 'EC2-Admin' user group details, specifically the 'Permissions policies' tab. It lists the 'EC2-Admin-Policy' with its JSON content:

```
1- {
2  "Version": "2012-10-17",
3  "Statement": [
4    {
5      "Action": [
6        "ec2:Describe*",
7        "ec2:StartInstances",
8        "ec2:StopInstances"
9      ],
10     "Resource": [
11       "*"
12     ]
13   }
14 ]
15 }
```

. The right screenshot shows the 'AmazonS3ReadOnlyAccess' policy details, displaying its JSON content:

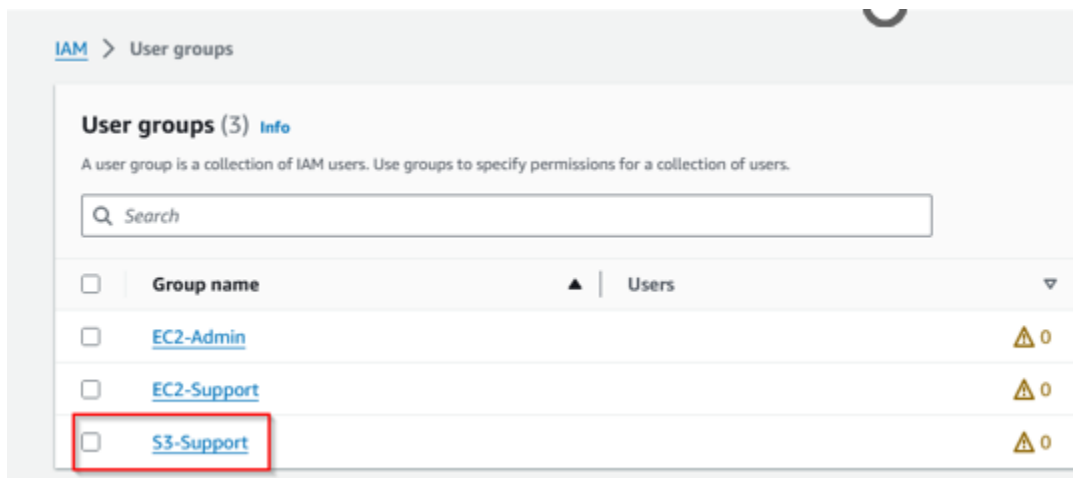
```
1- {
2  "Version": "2012-10-17",
3  "Statement": [
4    {
5      "Effect": "Allow",
6      "Action": [
7        "s3:Get*",
8        "s3:List*",
9        "s3:Describe*",
10       "s3-object-lambda:Get*",
11       "s3-object-lambda:List*"
12     ],
13     "Resource": "*"
14   }
15 ]
16 }
```

. Both screenshots have a red box highlighting the JSON policy content.

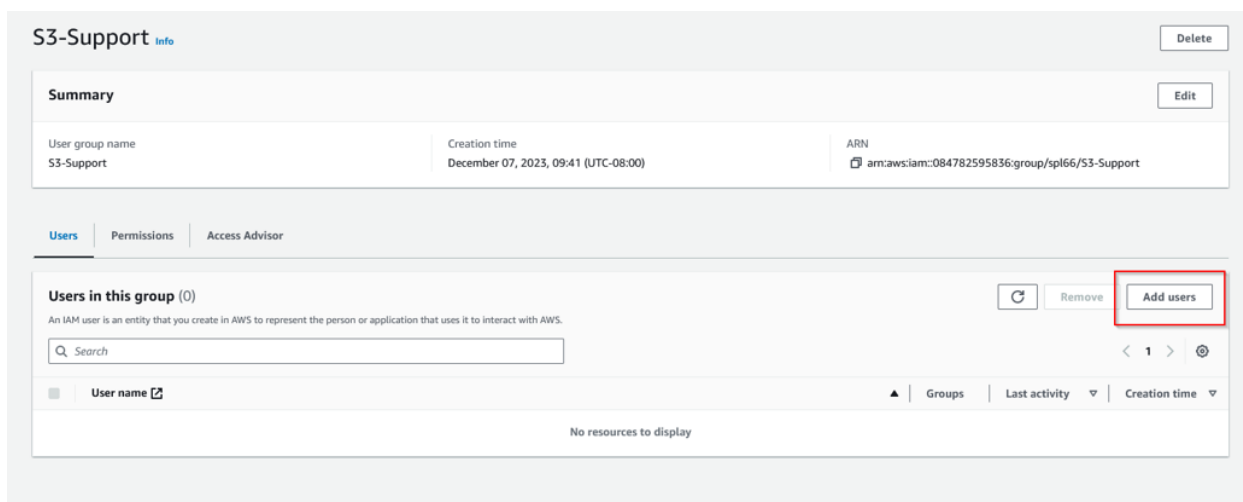
Task #2: Add Users to Groups

1. Navigate to the User Groups tab, and select the S3-Support Group

The image shows the 'Access management' sidebar in the AWS IAM console. The 'User groups' link is highlighted in blue, indicating it is the selected option. Other visible links include 'Users', 'Roles', 'Policies', 'Identity providers', and 'Account settings'.



2. Choose the Users tab, and click Add Users



3. Select user-1 and choose Add users.

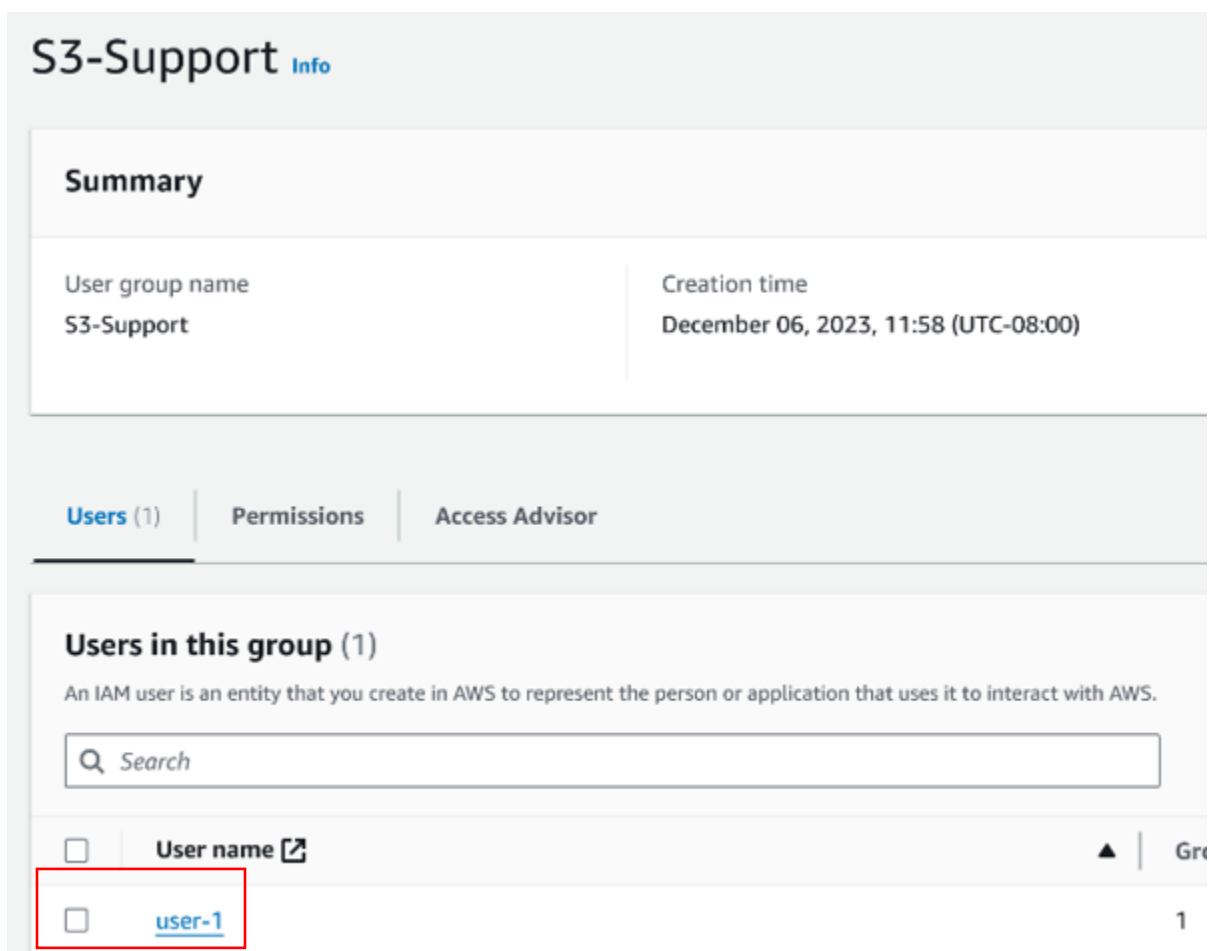
Add users to S3-Support [Info](#)

Other users in this account (1/4)

☐	User name ↗
☐	awsstudent
☒	user-1
☐	user-2
☐	user-3

None	20 minutes ago
None	20 minutes ago
None	20 minutes ago
None	20 minutes ago

4. See that user-1 has been added to the S3-Support Group



S3-Support [Info](#)

Summary

User group name	Creation time
S3-Support	December 06, 2023, 11:58 (UTC-08:00)

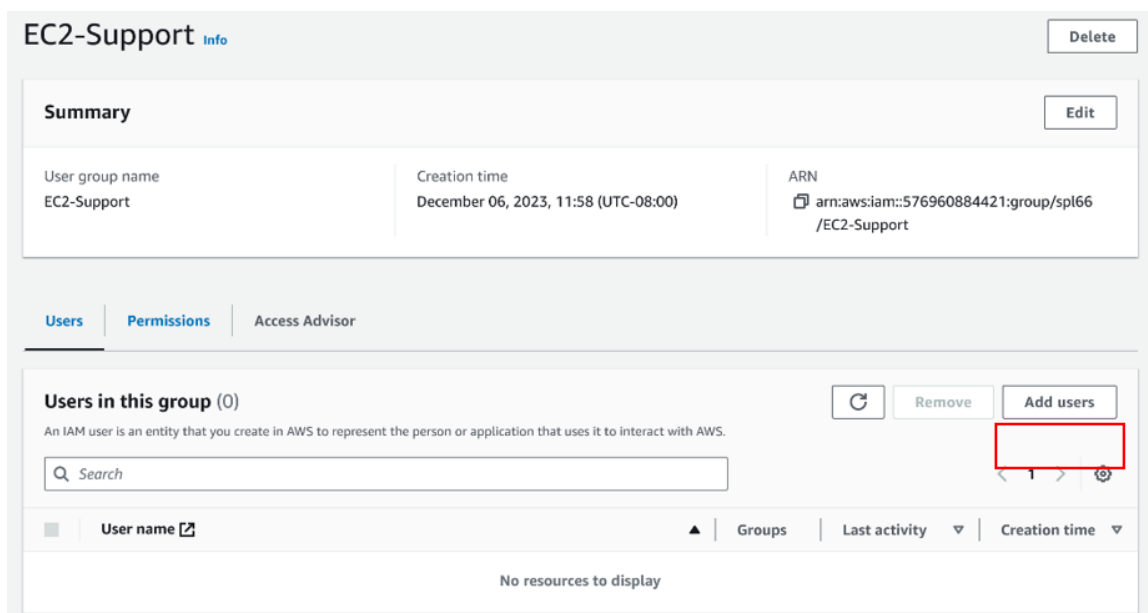
Users (1) | **Permissions** | **Access Advisor**

Users in this group (1)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

☐	User name ↗	▲	Grp
☐	user-1		1

5. Using the previous steps, add user-2 to the EC2-Support Group.



EC2-Support [Info](#) [Delete](#)

Summary [Edit](#)

User group name	Creation time	ARN
EC2-Support	December 06, 2023, 11:58 (UTC-08:00)	arn:aws:iam::576960884421:group/spl66/EC2-Support

Users | **Permissions** | **Access Advisor**

Users in this group (0)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

[↻](#) [Remove](#) [Add users](#)

☐ [User name](#) [↗](#) [▲](#) [Groups](#) [Last activity](#) [▼](#) [Creation time](#) [▼](#)

No resources to display

Add users to EC2-Support Info

Other users in this account (1/4)

☐	User name ↗		
☐	awsstudent	None	20 minutes ago
☐	user-1	None	20 minutes ago
☒	user-2	None	20 minutes ago
☐	user-3	None	20 minutes ago

6. Add user-3 to the EC2-Admin Group.

EC2-Admin Info

Summary

User group name EC2-Admin	Creation time December 06, 2023, 11:58 (UTC-08:00)	ARN arn:aws:iam::576960884421:group/spl66/EC2-Admin
------------------------------	---	--

[Users](#) | [Permissions](#) | [Access Advisor](#)

Users in this group (0)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

☐ [User name](#) [↗](#) [▲](#) [Groups](#) [Last activity](#) [▼](#) [Creation time](#) [▼](#)

No resources to display

Add users to EC2-Admin Info

Other users in this account (1/4)

☐	User name	
☐	awsstudent	None 20 minutes ago
☐	user-1	None 20 minutes ago
☐	user-2	None 20 minutes ago
☒	user-3	

- Navigate to the User Groups tab and verify that each User Group has one User assigned to it.

▼ Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

[IAM](#) > User groups

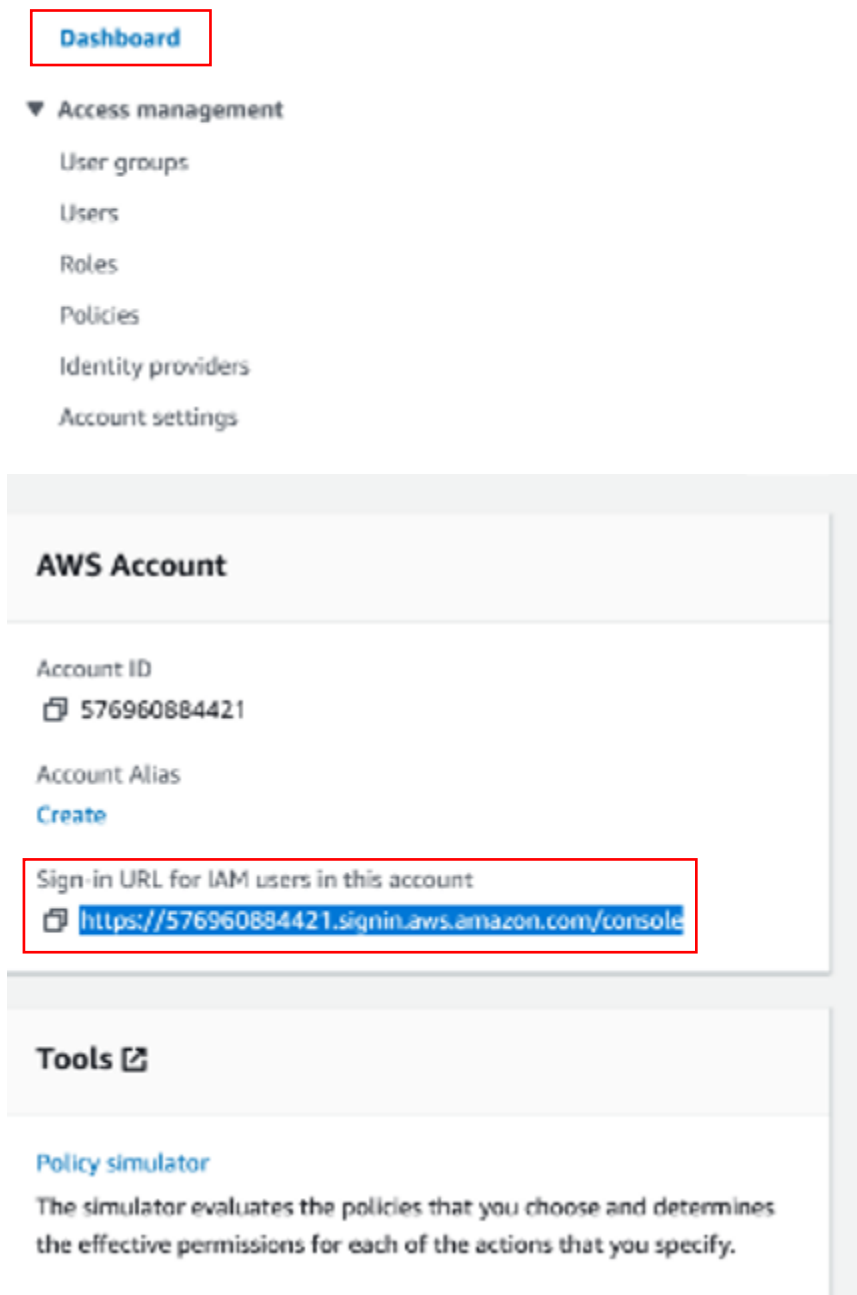
User groups (3) Info

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

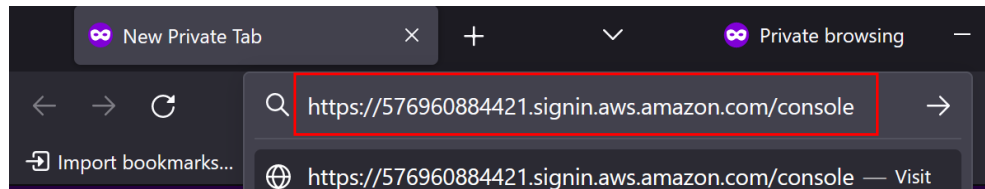
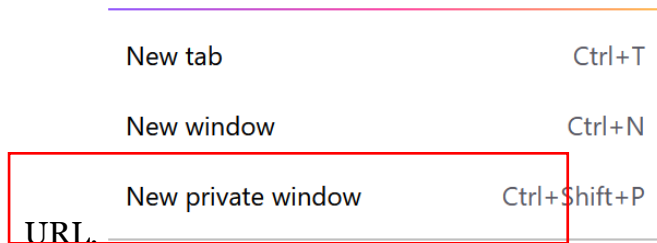
☐	Group name	Users	Permissions	Creation time
☐	EC2-Admin	1	✓ Defined	23 minutes ago
☐	EC2-Support	1	✓ Defined	23 minutes ago
☐	S3-Support	1	✓ Defined	23 minutes ago

Task #3: Sign-In and Test Users

1. Navigate to the dashboard and copy the sign in URL for IAM users in this account.



2. Open a private (incognito) window in a web browser and paste the Sign-in



3. Sign in with IAM username “user-1”, and Password “Lab-Password1”

Aconsole%2Fcanvas&code_challenge=4H9WE2U1cCOLTKBK2GpQ4ezg-o6lXbvTvToBLNrcew&code_challenge_met



Sign in as IAM user

Account ID (12 digits) or account alias

056013997774

IAM user name

user-1

Password

.....|

☐ Remember this account

Sign in

[Sign in using root user email](#)

[Forgot password?](#)

AWS S

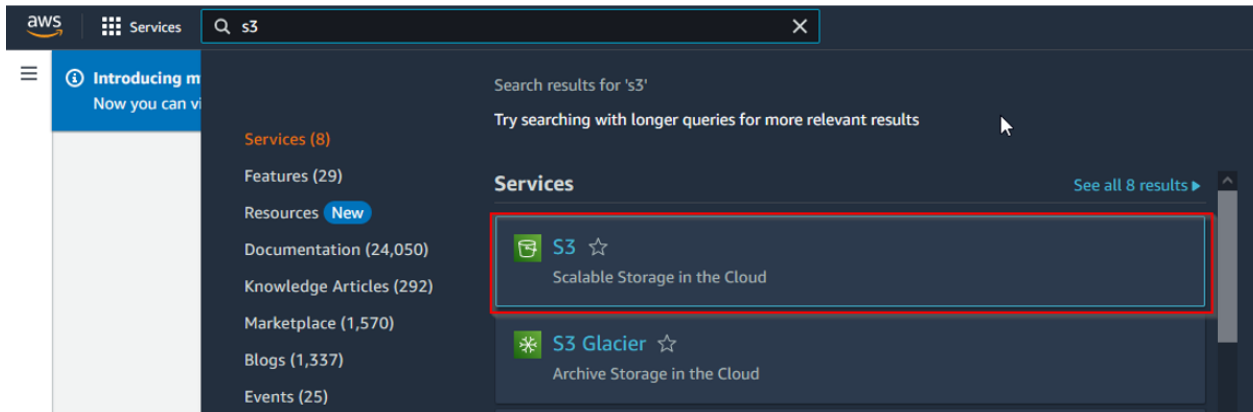
Your new le
500+ free d

GET STARTED

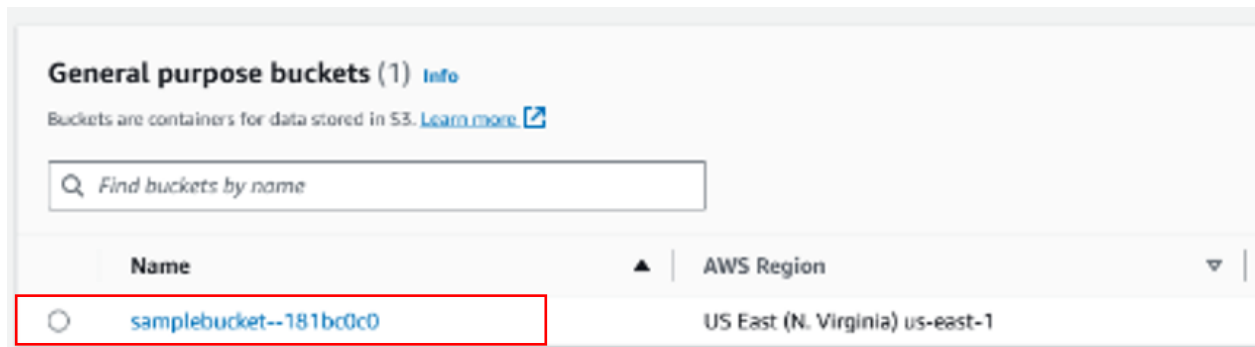
English ▼

[Terms of Use](#) [Privacy Policy](#) © 1996-2023, Amazon Web Services,

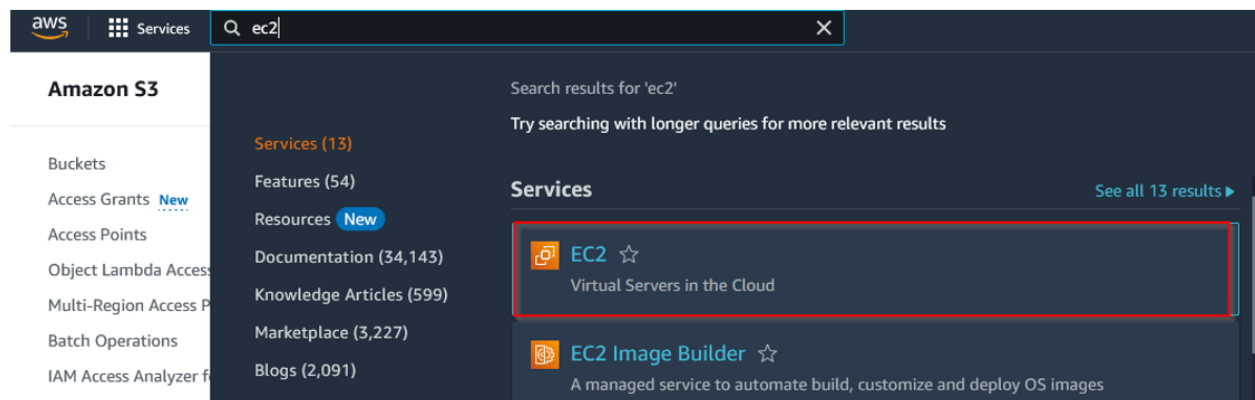
4. In the Services menu, select S3.



5. Choose the bucket that already exists in S3, and notice how we are able to see all the buckets and the contents.



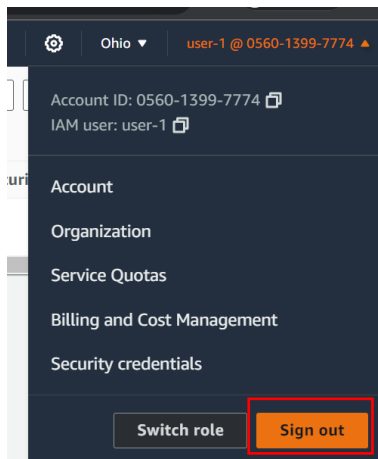
6. Navigate to the services menu and choose EC2.



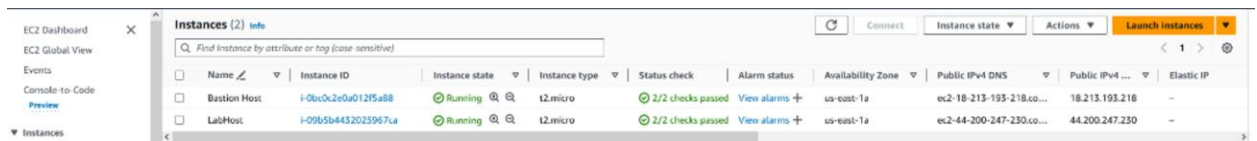
7. Notice how user-1 is unable to see any EC2 instances and is instead greeted with “You are not authorized to perform this operation.” This is because the permissions of the user group that user-1 was put into doesn’t contain these permissions.



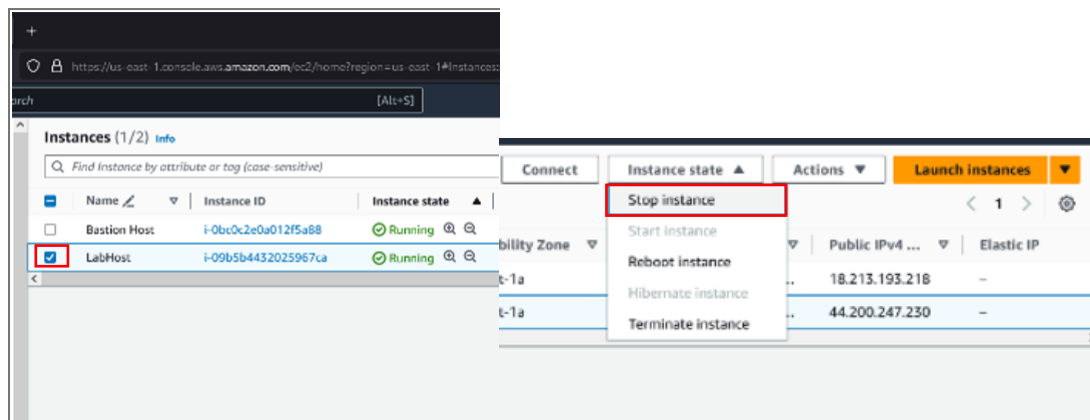
8. Sign out of user-1, and sign in to user-2.



9. Navigate back to the EC2 service. Notice how you can now see the EC2 instances because this user has the permissions to.



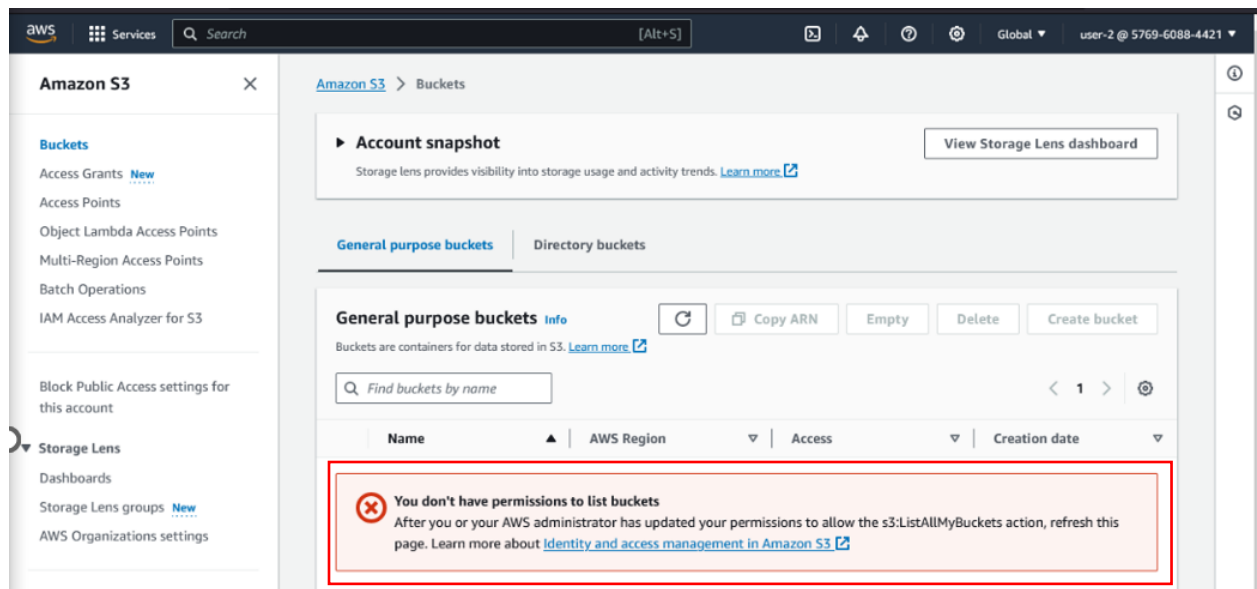
10. Select the instance named "LabHost", and in the "Instance State" menu, choose "Stop Instance".



11. You will receive an error message, demonstrating that while you can see the EC2 Instances, you are unable to edit them.



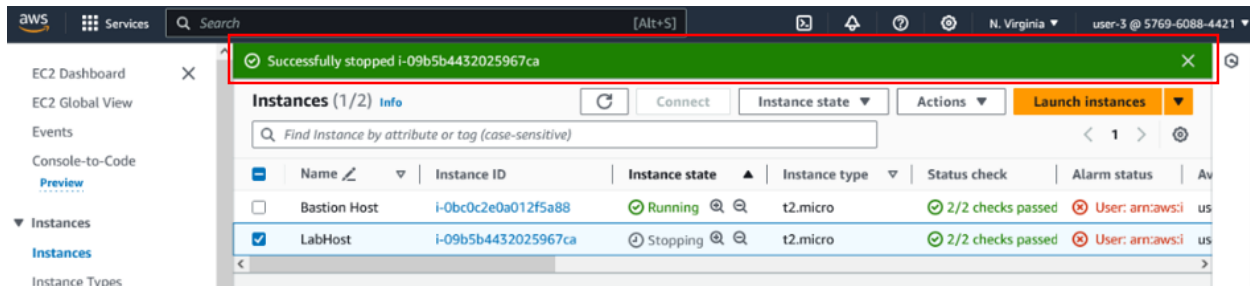
12. Navigate back to S3 and notice how the user is greeted with the “You don’t have permissions to list buckets” message. This is because user-2 doesn’t have access to S3.



13. Sign out of user-2 and log in to user-3. Go back to the LabHost screen and try to stop the instance again.



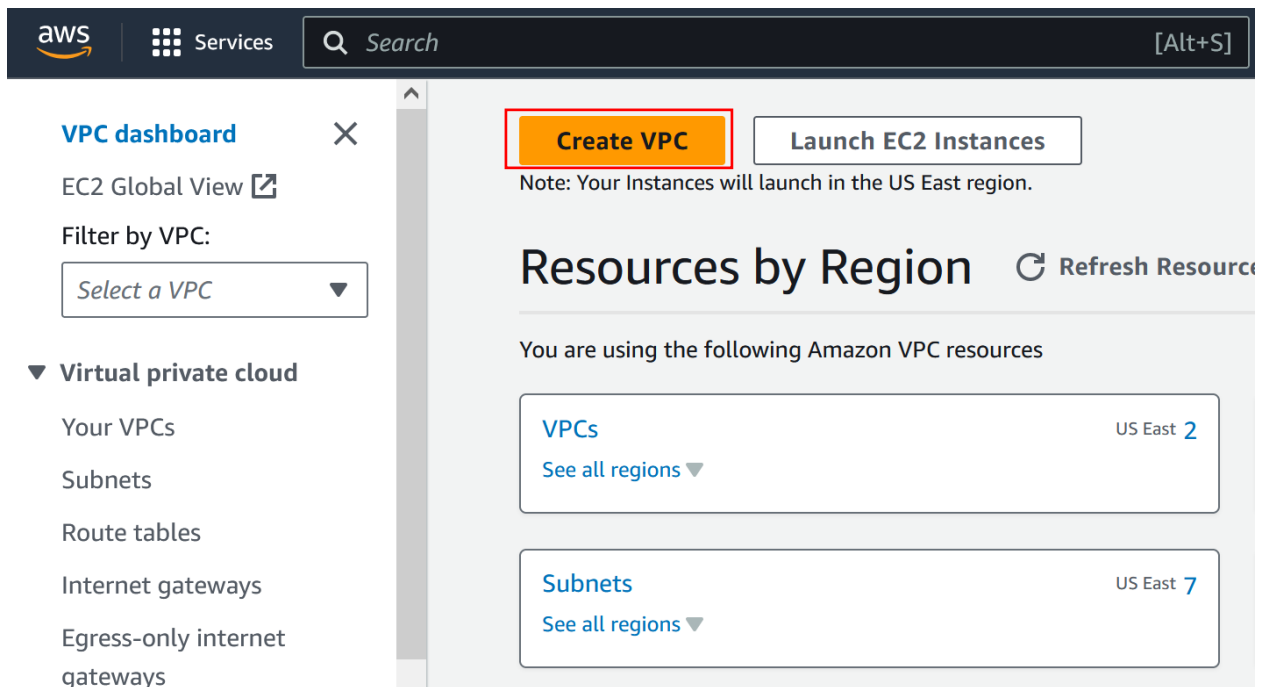
14. Notice how user-3 can successfully stop the instance, due to the permissions assigned by IAM.



Lab #2 Walkthrough

Task #1: Create Your VPC

1. Click the Create VPC button after navigating to the VPC service.



2. Configure the details of the VPC, including number of Availability Zones, public subnets, private subnets, NAT gateways, etc.

[VPC](#) > [Your VPCs](#) > Create VPC

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☐ VPC only

☒ VPC and more

Name tag auto-generation [Info](#)

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

lab

IPv4 CIDR block [Info](#)

Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16

65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)

Default

Number of Availability Zones (AZs) [Info](#)

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1

2

3

Number of public subnets [Info](#)

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0	1
---	---

Number of private subnets [Info](#)

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0	1	2
---	---	---

▼ Customize subnets CIDR blocks**Public subnet CIDR block in us-east-1a**

10.0.0.0/24	256 IPs
-------------	---------

Private subnet CIDR block in us-east-1a

10.0.1.0/24	256 IPs
-------------	---------

NAT gateways (\$) [Info](#)

Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway.

None	In 1 AZ	1 per AZ
------	---------	----------

VPC endpoints [Info](#)

Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None	S3 Gateway
------	------------

DNS options [Info](#)

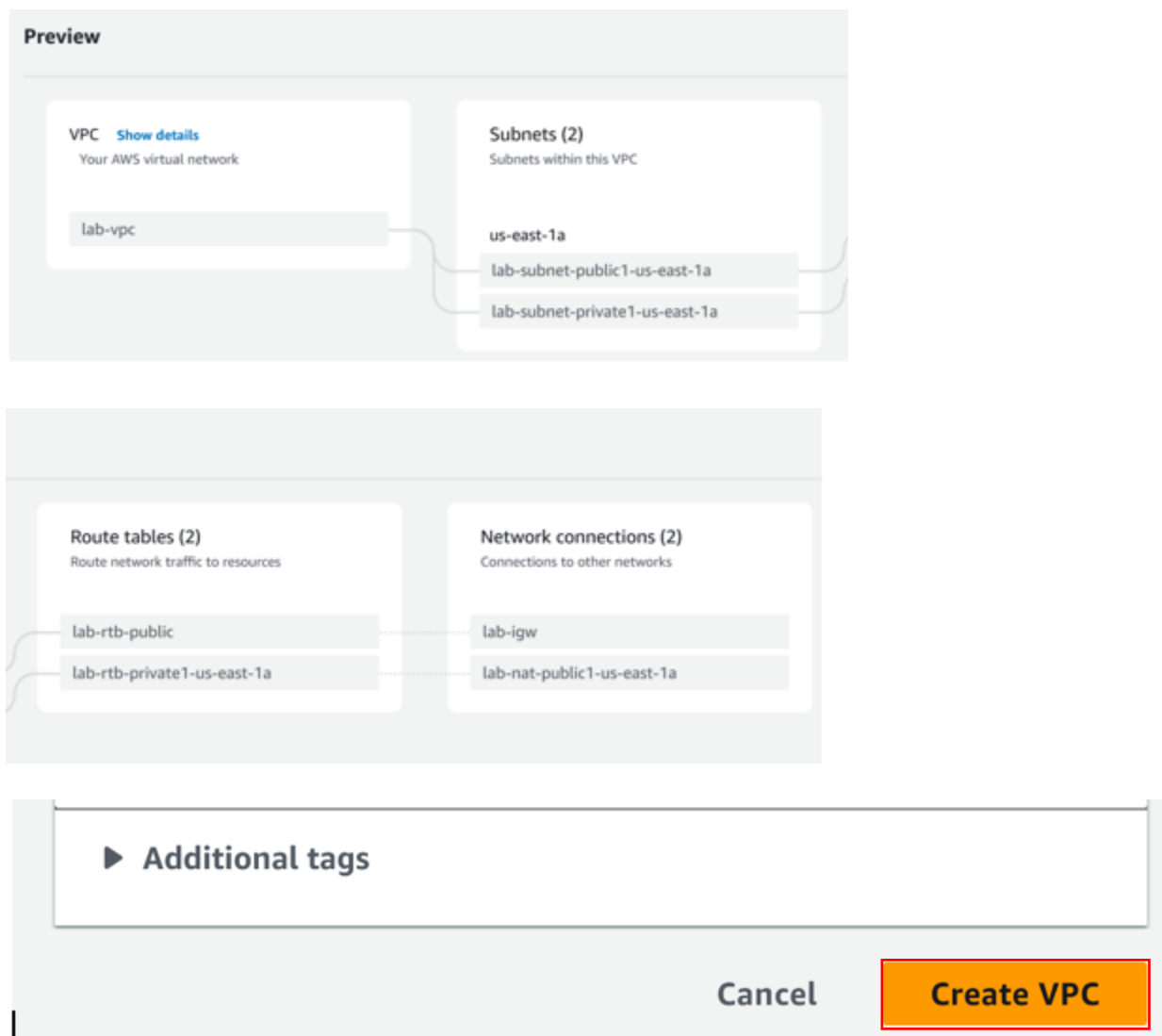
- ☒ Enable DNS hostnames
- ☒ Enable DNS resolution

► Additional tags

Cancel

Create VPC

3. In the Preview tab, confirm the settings you have configured and create the VPC.



4. Once all the resources for the VPC have been created, click View VPC

 Services [Alt+S]

VPC > Your VPCs > Create VPC > Create VPC resources

Create VPC workflow

✔ Success

▼ Details

- ✔ Create VPC: [vpc-001f904f8c3ddb655](#)
- ✔ Enable DNS hostnames
- ✔ Enable DNS resolution
- ✔ Verifying VPC creation: [vpc-001f904f8c3ddb655](#)
- ✔ Create subnet: [subnet-043769d955ccbaea4](#)
- ✔ Create subnet: [subnet-0f4e9505bee25ba6c](#)
- ✔ Create internet gateway: [igw-08f976ddc3c67ebd4](#)
- ✔ Attach internet gateway to the VPC
- ✔ Create route table: [rtb-0e6a75aeddab54c8f](#)
- ✔ Create route
- ✔ Associate route table
- ✔ Allocate elastic IP: [eipalloc-0a84777bd6f3d5ee4](#)
- ✔ Create NAT gateway: [nat-0cf2194c927c458cd](#)
- ✔ Wait for NAT Gateways to activate
- ✔ Create route table: [rtb-03e098212e2717e58](#)
- ✔ Create route
- ✔ Associate route table
- ✔ Verifying route table creation

View VPC

5. Click on “Subnets” and “Route Tables” in order to see more details about them.

The first screenshot shows the AWS Management Console 'Subnets (9)' page. The left sidebar has 'Subnets' highlighted under 'Virtual private cloud'. The main content area displays a table of subnets.

Name	Subnet ID	State	VPC	IPv4 CIDR
lab-subnet-private1-us-east-1a	subnet-0f4e9505bee25ba6c	Available	vpc-001f904f8c3ddb655 lab-vpc	10.0.1.0/24
lab-subnet-public1-us-east-1a	subnet-043769d955ccbaea4	Available	vpc-001f904f8c3ddb655 lab-vpc	10.0.0.0/24
-	subnet-05667fd50e3c4b63	Available	vpc-0c855b95cca507fa2	172.31.80.0/20
-	subnet-02e4cece88b80324f	Available	vpc-0c855b95cca507fa2	172.31.48.0/20
-	subnet-04c78b2cf885abd58	Available	vpc-0c855b95cca507fa2	172.31.16.0/20
-	subnet-0988bcaeeb1190f99	Available	vpc-0c855b95cca507fa2	172.31.32.0/20
Work Public Subnet	subnet-076f01b2d13f82584	Available	vpc-0c753a6831c051993 Wor...	10.0.0.0/24
-	subnet-0cd48c92408bf1b63	Available	vpc-0c855b95cca507fa2	172.31.0.0/20
-	subnet-03d49eeb099faafba	Available	vpc-0c855b95cca507fa2	172.31.64.0/20

The second screenshot shows the AWS Management Console 'Route tables (6)' page. The left sidebar has 'Route tables' highlighted under 'Virtual private cloud'. The main content area displays a table of route tables.

Name	Route table ID	Explicit subnet associati...	Edge associations	Main	VPC
lab-rtb-public	rtb-0e6a75aeddb54c8f	subnet-043769d955ccba...	-	No	vpc-001f904f8c3ddb655 lab-vpc
lab-rtb-private1-us-east-1a	rtb-03e098212e2717e58	subnet-0f4e9505bee25b...	-	No	vpc-001f904f8c3ddb655 lab-vpc
-	rtb-02b471ed22f87d35e	-	-	Yes	vpc-001f904f8c3ddb655 lab-vpc
-	rtb-092c79f23638e780b	-	-	Yes	vpc-0c753a6831c051993 Wor...
Work Public Route Table	rtb-0a3ff42ce0ee83aec	subnet-076f01b2d13f82...	-	No	vpc-0c753a6831c051993 Wor...
-	rtb-0d718f7c746b594bf	-	-	Yes	vpc-0c855b95cca507fa2

Task 2: Create Additional Subnets

1. Click the Subnets tab and click on “Create Subnet”. Configure the subnet with the VPC ID, Subnet Name, Availability Zone, and IPv4 CIDR block.

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet
gateways

Carrier gateways

DHCP option sets

N. Virginia ▼

voclabs/user2949274=Aiden_Yin @ 8438-1426-7431 ▼



Actions ▼

Create subnet

< 1 >

IPv6 CIDR ▼	Available IPv4 address
–	251
–	250
–	4091

VPC

VPC ID

Create subnets in this VPC.

vpc-001f904f8c3ddb655 (lab-vpc)

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

lab-subnet-public2

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US East (N. Virginia) / us-east-1b

IPv4 VPC CIDR block [Info](#)

Choose the IPv4 VPC CIDR block to create a subnet in.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.2.0/24

256 IPs

< > ^ v

► Tags - optional

Remove

Add new subnet

Cancel

Create subnet

 You have successfully created 1 subnet: subnet-0d59b26ed8ba4349e

Subnets (1) [Info](#)

Subnet ID : subnet-0d59b26ed8ba4349e 

Clear filters

☐	Name	Subnet ID	State
☐	lab-subnet-public2	subnet-0d59b26ed8ba4349e	 Available

2. Create a new Subnet with different details.

N. Virginia ▼

voclabs/user2949274=Aiden_Yin @ 8438-1426-7431 ▼

↻

Actions ▼

Create subnet

< 1 >

⚙

ⓘ

🔍

IPv6 CIDR ▼	Available IPv4 address
–	251
–	250
–	4091

VPC

VPC ID

Create subnets in this VPC.

vpc-001f904f8c3ddb655 (lab-vpc) ▼

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

lab-subnet-private2

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US East (N. Virginia) / us-east-1b ▼

IPv4 VPC CIDR block [Info](#)

Choose the IPv4 VPC CIDR block to create a subnet in.

10.0.0.0/16 ▼

IPv4 subnet CIDR block

10.0.3.0/24 256 IPs

< > ^ v

► Tags - optional

Remove

Add new subnet

Cancel

Create subnet

✓ You have successfully created 1 subnet: subnet-05e89e910d4134fe3

Subnets (1) [Info](#)

Find resources by attribute or tag

Subnet ID : subnet-05e89e910d4134fe3 [X](#) [Clear filters](#)

☐	Name	Subnet ID	State	VPC	IPv4 CIDR
☐	lab-subnet-private2	subnet-05e89e910d4134fe3	Available	vpc-001f904f8c3ddb655 lab-vpc	10.0.3.0/24

3. You will now route traffic from the private network to the NAT gateway in order to keep information private from the rest of the network by using a route table. To do so, click the “Route Tables” tab in the navigation panel.

▼ **Virtual private cloud**

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

4. Select the “**lab-rtb-private1-us-east-1a**” route table.

Route tables (1/6) [Info](#)

Find resources by attribute or tag

☐	Name	Route table ID	Explicit subnet associati...
☐	lab-rtb-public	rtb-0e6a75aeddab54c8f	subnet-043769d955ccba...
☒	lab-rtb-private1-us-east-1a	rtb-03e098212e2717e58	subnet-0f4e9505bee25b...
☐	–	rtb-02b471ed22f87d35e	–
☐	–	rtb-092c79f23638e780b	–
☐	Work Public Route Table	rtb-0a3ff42ce0ee83aec	subnet-076f01b2d13f82...
☐	–	rtb-0d718f7c746b594bf	–

- Click the Routes tab and notice that the Destination 0.0.0.0/0 is set to Target nat-xxxxxxx. This means that all traffic bound for the internet will be sent to that specific NAT gateway.

rtb-03e098212e2717e58 / lab-rtb-private1-us-east-1a

Details **Routes** Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Destination	Target
0.0.0.0/0	nat-0cf2194c927c458cd
10.0.0.0/16	local

- Choose the Subnet associations tab. Notice how there is a subnet already associated with this route table. It was associated when we did task 1.

Details Routes **Subnet associations** Edge associations Route propagation Tags

Explicit subnet associations (1)

Find subnet association

Name	Subnet ID	IPv4 CIDR
lab-subnet-private1-us-east-1a	subnet-0f4e9505bee25ba6c	10.0.1.0/24

Subnets without explicit associations (2)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR
lab-subnet-private2	subnet-05e89e910d4134fe3	10.0.3.0/24
lab-subnet-public2	subnet-0d59b26ed8ba4349e	10.0.2.0/24

- In the Explicit Subnet Associations panel, click Edit Subnet Associations. Select the private subnet we just created (lab-subnet-private2). Leave the other subnet selected as well.

Edit subnet associations

< 1 > ⚙

IPv6 CIDR

-

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR
☒	lab-subnet-private1-us-east-1a	subnet-0f4e9505bee25ba6c	10.0.1.0/24
☐	lab-subnet-public1-us-east-1a	subnet-043769d955ccbaea4	10.0.0.0/24
☒	lab-subnet-private2	subnet-05e89e910d4134fe3	10.0.3.0/24
☐	lab-subnet-public2	subnet-0d59b26ed8ba4349e	10.0.2.0/24

Selected subnets

subnet-0f4e9505bee25ba6c / lab-subnet-private1-us-east-1a X subnet-05e89e910d4134fe3 / lab-subnet-private2 X

8. Click Save Associations

< 1 > ⚙

IPv4 CIDR	Route table ID
-	rtb-03e098212e2717e58 / lab-rtb-private1-us-east...
-	rtb-0e6a75aeddab54c8f / lab-rtb-public
-	Main (rtb-02b471ed22f87d35e)
-	Main (rtb-02b471ed22f87d35e)

Cancel **Save associations**

9. Select the lab-rtb-public route table and deselect anything else.

	Name	Route table ID	Explicit subnet associati...
☒	lab-rtb-public	rtb-0e6a75aeddab54c8f	subnet-043769d955ccba...
☐	lab-rtb-private1-us-east-1a	rtb-03e098212e2717e58	2 subnets
☐	-	rtb-02b471ed22f87d35e	-
☐	-	rtb-092c79f23638e780b	-
☐	Work Public Route Table	rtb-0a3ff42ce0ee83aec	subnet-076f01b2d13f82...
☐	-	rtb-0d718f7c746b594bf	-

10. Note how traffic in this route table is directly routed to igw-xxxxxx, which means that there is no NAT gateway, allowing the network to be public.

rtb-0e6a75aeddab54c8f / lab-rtb-public

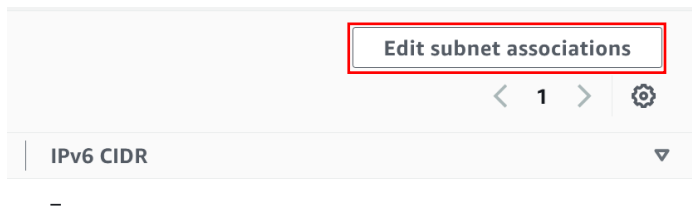
Details	Routes	Subnet associations	Edge associations	Route propagation	Tags
---------	--------	---------------------	-------------------	-------------------	------

Routes (2)

Filter routes

Destination	Target
0.0.0.0/0	igw-08f976ddc3c67ebd4
10.0.0.0/16	local

11. Click the Edit Subnet Associations button again.



12. Leave the “**lab-subnet-public1-us-east-1a**” selected, but also include the “**lab-subnet-public2**” that we just created.

VPC > Route tables > [rtb-0e6a75aeddab54c8f](#) > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

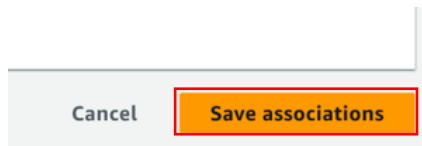
Filter subnet associations

	Name	Subnet ID	IPv4 CIDR
☐	lab-subnet-private1-us-east-1a	subnet-0f4e9505bee25ba6c	10.0.1.0/24
☒	lab-subnet-public1-us-east-1a	subnet-043769d955ccbaea4	10.0.0.0/24
☐	lab-subnet-private2	subnet-05e89e910d4134fe3	10.0.3.0/24
☒	lab-subnet-public2	subnet-0d59b26ed8ba4349e	10.0.2.0/24

Selected subnets

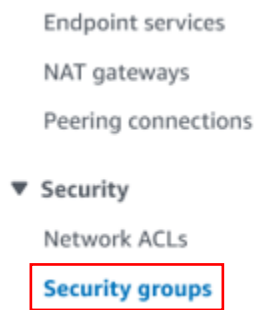
subnet-043769d955ccbaea4 / lab-subnet-public1-us-east-1a X subnet-0d59b26ed8ba4349e / lab-subnet-public2 X

13. Choose Save Associations.

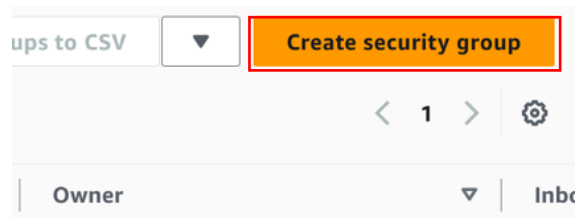


Task 3: Create a VPC Security Group

1. In the left navigation menu, choose Security Groups.



2. Choose Create security group, then configure the group name, description, and VPC.



Basic details

Security group name [Info](#)

Web Security Group

Name cannot be edited after creation.

Description [Info](#)

Enable HTTP Access

VPC [Info](#)

vpc-001f904f8c3ddb655 (lab-vpc)

3. Add an inbound rule with type HTTP, source Anywher-IPv4, and description Permit web requests.

Inbound rules [info](#)

Type info	Protocol info	Port range info	Source info	Description - optional info
HTTP	TCP	80	Anywhere-IPv4	Permit web requests

0.0.0.0/0 X

[Add rule](#)

- After creating the security group, you should receive a message confirming the creation of the group.

🟢 Security group (sg-0a81453fe336862b4 | Web Security Group) was created successfully
[▶ Details](#)

[VPC](#) > [Security Groups](#) > sg-0a81453fe336862b4 - Web Security Group

sg-0a81453fe336862b4 - Web Security Group

Details

Security group name 📄 Web Security Group	Security group ID 📄 sg-0a81453fe336862b4	Description 📄 Enable HTTP Access
Owner 📄 843814267431	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry

[Inbound rules](#) | [Outbound rules](#) | [Tags](#)

Inbound rules (1)

🔍 Search

☐	Name	Security group rule...	IP version	Type	Protocol
☐	-	sgr-0c622319881e412...	IPv4	HTTP	TCP

Task 4: Launch a Web Server Instance

- In the Services menu, choose the EC2 service and create a new instance with the name “Web Server 1”, keeping the default Amazon Machine Image (AMI). Choose “t2.micro” as the instance type, and select “vockey” as the key pair name. Then, configure your network settings accordingly.

EC2 Dashboard

×

EC2 Global View

Events

Console-to-Code

Preview

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

New

▼ Images

AMIs

Resources

You are using the following Amazon EC2 resources in the US East (N. Virginia)

Instances (running)	1	Auto Scaling Groups
Elastic IPs	2	Instances
Load balancers	API Error	Placement groups
Snapshots	0	Volumes

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance ▼

Migrate a server

Note: Your instances will launch in the US East (N. Virginia) Region

Name and tags [Info](#)

Name

Web Server 1

Add additional tags

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUS

Q

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI

Free tier eligible

ami-0230bd60aa48260c6 (64-bit (x86)) / ami-04c97e62cb19d53f1 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.0716 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

Create new key pair

▼ Network settings

Info

VPC - required

Info

vpc-001f904f8c3ddb655 (lab-vpc)

10.0.0.0/16

▼

↻

Subnet

Info

subnet-0d59b26ed8ba4349e

lab-subnet-public2

VPC: vpc-001f904f8c3ddb655

Owner: 843814267431

Availability Zone: us-east-1b

▼

↻

Create new subnet

↗

Auto-assign public IP

Info

Enable

▼

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups

Info

Select security groups

▼

Web Security Group

sg-0a81453fe336862b4

×

VPC: vpc-001f904f8c3ddb655

↻

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

- Configure a script to run on the instance when it launches under Advanced Details and create the instance. You should see a success message.

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

 Choose file

```
#!/bin/bash

# Install Apache Web Server and PHP

dnf install -y httpd wget php mariadb105-server

# Download Lab files

wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip

unzip lab-app.zip -d /var/www/html/

# Turn on web server

chkconfig httpd on
```

Cancel

Launch instance

[Review commands](#)[EC2](#) > [Instances](#) > Launch an instance

✓ **Success**
Successfully initiated launch of instance ([i-016b1e87fe9ba9d59](#))

▶ Launch log

3. Looking at the instances, we can see that the Web Server we just created is functional. It may take some time as the instance needs to pass all of the checks.

Instances (2) Info					
Find Instance by attribute or tag (case-sensitive)					
☐	Name 	Instance ID	Instance state 	Instance type 	Status check
☐	Bastion Host	i-014943a3fad103acd	✓ Running  	t2.micro	✓ 2/2 checks passed
☐	Web Server 1	i-016b1e87fe9ba9d59	✓ Running  	t2.micro	✓ 2/2 checks passed

4. Select the Web Server 1 Instance and copy the Public IPv4 DNS value shown.

Instances (1/2) [Info](#)

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID
☐	Bastion Host	i-0aa505b090af697b9
☒	Web Server 1	i-0c3c99f405a452a96

Instance: i-016b1e87fe9ba9d59 (Web Server 1)

[Details](#) | [Security](#) | [Networking](#) | [Storage](#) | [Status checks](#) | [Monitoring](#) | [Tags](#)

▼ Instance summary [Info](#)

Instance ID i-016b1e87fe9ba9d59 (Web Server 1)	Public IPv4 address 3.82.101.161 open address	Private IPv4 addresses 10.0.2.174
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-3-82-101-161.compute-1.amazonaws.com open address
Hostname type IP name: ip-10-0-2-174.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-2-174.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	

5. Paste the Public IPv4 DNS value into the search bar of any web browser, and you should be taken to the following screen, showing that your web server is functional.

ec2-3-82-101-161.compute-1.amazonaws.com

aws Load Test RDS

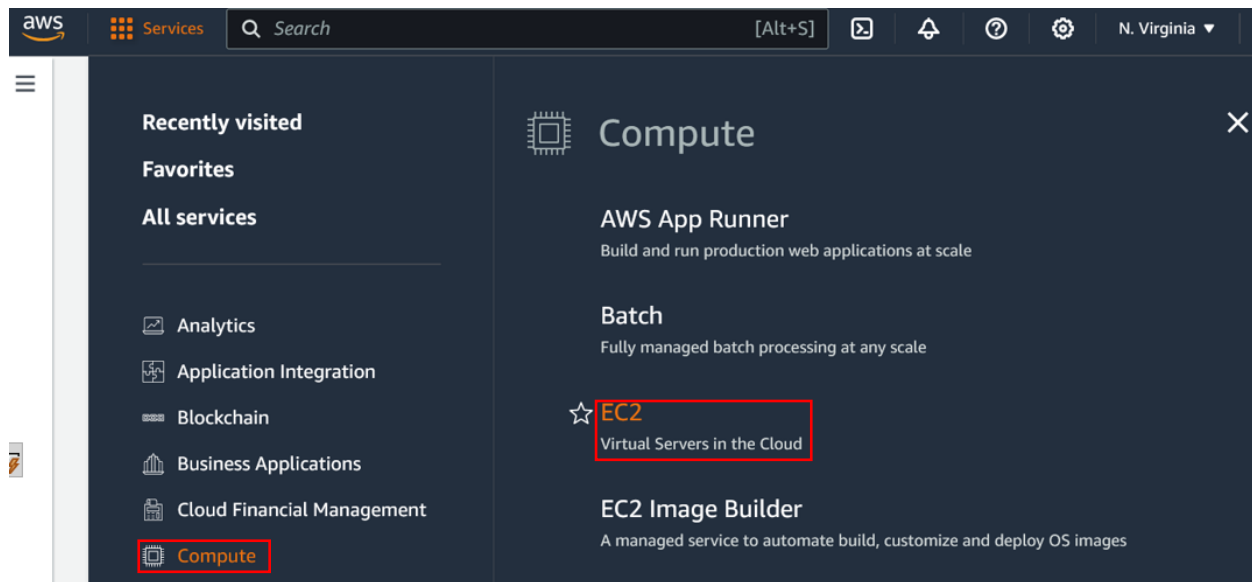
Meta-Data	Value
InstanceId	i-016b1e87fe9ba9d59
Availability Zone	us-east-1b

Current CPU Load: 1%

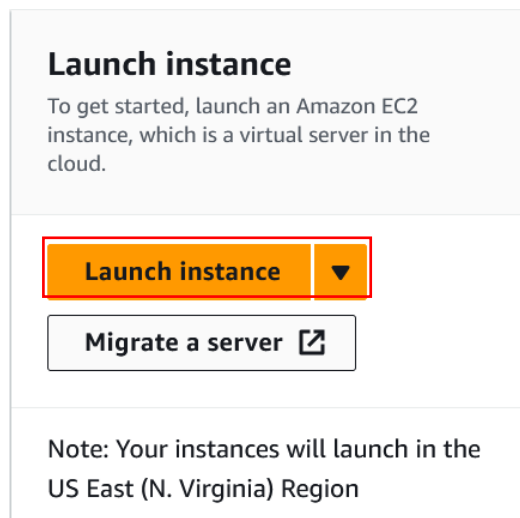
Lab #3 Walkthrough:

Task 1: Launch Your Amazon EC2 Instance

1. In the Services menu, select EC2. Verify that the region your lab is in is N. Virginia (us-east-1).



2. Launch an instance with the name Web Server, Amazon Linux 2023 AMI, t2.micro instance type, vockey key pair login, and the required network settings.



Name and tags [Info](#)

Name

Web Server

Amazon
Linux



macOS



Ubuntu



Windows



Red Hat



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI

Free tier eligible

ami-079db87dc4c10ac91 (64-bit (x86), uefi-preferred) / ami-02cd6549baea35b55 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

▼ Instance type [Info](#) [Get advice](#)

Instance type

t2.micro Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.0716 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance before you launch the instance.

Key pair name - *required*

vockey

▼ Network settings [Info](#)

VPC - *required* [Info](#)

vpc-06df25e906c6de9bf (Lab VPC)
10.0.0.0/16

Subnet [Info](#)

subnet-0c964537828590a58 Public Subnet 2
VPC: vpc-06df25e906c6de9bf Owner: 422987282380
Availability Zone: us-east-1b IP addresses available: 251 CIDR: 10.0.2.0/24

Create new subnet

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - *required*

Web Server Security Group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!\$*

Description - *required* [Info](#)

Security group for my web server

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Remove

Type Info	Protocol Info	Port range Info
ssh	TCP	22

- Under Advanced Details, turn on Termination Protection and paste the code into the user data box.

Termination protection [Info](#)

Enable

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

 Choose file

```
#!/bin/bash

dnf install -y httpd

systemctl enable httpd

systemctl start httpd

echo '<html><h1>Hello From Your Web Server!</h1></html>' > /var/www
/html/index.html|
```

- Launch the Instance.

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

[Review commands](#)

- Click View all instances, select Web Server, and wait until the Instance State is Running and it passes both status checks.

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms [↗](#)

View all instances

☒	Web Server	i-0235d4513419f3e8a	☒ Running	🔍 🔍	t2.micro	☒ 2/2 checks passed
-------------------------------------	------------	---------------------	---	-----	----------	---

Task 2: Monitor your Instance

- In the bottom panel, choose status and alarms. Note that both the system status checks and instance status checks have both been passed.

Instance: i-0235d4513419f3e8a (Web Server)

Details

Status and alarms New

Monitoring

Security

Networking

Storage

Tags

Status checks [Info](#)

Status checks detect problems that may impair i-0235d4513419f3e8a (Web Server) from running your applications.

System status checks

☒ System reachability check passed

Instance status checks

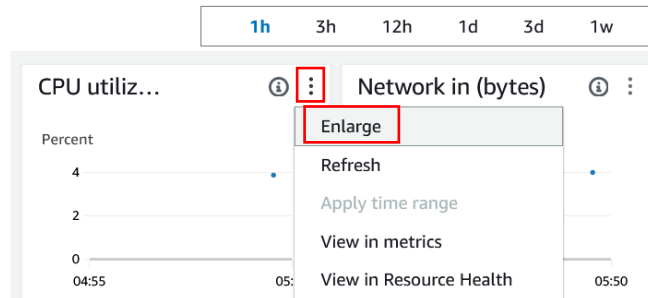
☒ Instance reachability check passed

- Choose the Monitoring tab and note the multiple graphs for different metrics. You can click the three dots on any graph and click enlarge to see a more detailed view about that metric.

Instance: i-0235d4513419f3e8a (Web Server)

 Details | Status and alarms **New** | **Monitoring** | Security | Netv

Alarm recommendations



3. In the Actions menu towards the top of the console, select Monitor and Troubleshoot, then select Get system log. This system log is extremely helpful in debugging problems. Note that the HTTP package we put in the user data was installed here.

The screenshot shows the AWS Management Console interface. At the top, there are navigation icons and a header bar with 'N. Virginia' and 'voclabs/user2949274=Aiden_Yin @'. Below the header, there are buttons for 'Connect', 'Instance state', 'Actions', and 'Launch instances'. The 'Actions' button is highlighted with a red box. A dropdown menu is open under 'Actions', showing options like 'Connect', 'View details', 'Manage instance state', 'Instance settings', 'Networking', 'Security', 'Image and templates', and 'Monitor and troubleshoot'. The 'Monitor and troubleshoot' option is highlighted with a red box. A sub-menu is open under 'Monitor and troubleshoot', showing options like 'Get system log', 'Get instance screenshot', 'Manage detailed monitoring', 'Manage CloudWatch alarms', 'EC2 serial console', 'Replace root volume', and 'Fleet Manager'. The 'Get system log' option is highlighted with a red box.

System log Copy log Download

Review system log for instance i-0235d4513419f3e8a as of Thu Jan 04 2024 21:56:22 GMT-0800 (Pacific Standard Time)

```
[ 33.874751] systemd-sysv-generator[3847]: SysV service '/etc/rc.d/init.d/cfn-hup' lacks a native systemd unit file. Automat
[ 33.897042] zram_generator::config[3849]: zram0: system has too much memory (949MB), limit is 800MB, ignoring.
ci-info: +++++Authorized keys from /home/ec2-user/.ssh/authorized_keys for user ec2-user+++++
ci-info: +-----+
ci-info: | Keytype |                               Fingerprint (sha256)                               | Options
ci-info: +-----+-----+-----+
ci-info: | ssh-rsa | 37:97:63:c1:ac:ff:07:de:85:24:2d:60:32:2e:46:fd:ef:4e:5f:13:34:c0:a2:5a:e5:a9:58:7f:19:8c:2f:6d | -
ci-info: +-----+-----+-----+
<14>Jan 5 05:46:59 cloud-init: #####
<14>Jan 5 05:46:59 cloud-init: ----BEGIN SSH HOST KEY FINGERPRINTS----
<14>Jan 5 05:46:59 cloud-init: 256 SHA256:i3X6W0g20qqL88zPUXIiPxLh7I23y/Cc300/ocoELG4 root@ip-10-0-2-190.ec2.internal (ECDSA)
<14>Jan 5 05:46:59 cloud-init: 256 SHA256:2NbSz0XATvOsQg0LZxFNwYqgWAah96u5BbcJ4PoyKxE root@ip-10-0-2-190.ec2.internal (ED25519)
<14>Jan 5 05:46:59 cloud-init: ----END SSH HOST KEY FINGERPRINTS----
<14>Jan 5 05:46:59 cloud-init: #####
----BEGIN SSH HOST KEY KEYS----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBFuxqGLd5LWhi/AJkLL9DgF/naS+FZy576mylUH+pF9jy3mVR+/mhF
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIFA2j3wkj5boNF8Tz0L7EWz/nUUCoCdeNLgA81Asfko root@ip-10-0-2-190.ec2.internal
----END SSH HOST KEY KEYS----
[ 34.966315] cloud-init[2201]: Cloud-init v. 22.2.2 finished at Fri, 05 Jan 2024 05:46:59 +0000, DataSource DataSourceEc2.
```

4. Ensure Web Server is still selected. Then in the actions menu, choose Monitor and Troubleshoot, and choose Get instance screenshot. This shows you what your EC2 instance would look like if there was a monitor attached to it.

Navigation bar: 🔍 🔔 ? ⚙️ N. Virginia ▼ voclabs/user2949274=Aiden_Yin @ 4229-8728

Buttons: Connect Instance state ▼ Actions ▲ Launch instances ▼

▼	Instance type ▼	Status check	Instance settings ▶	Availability Zone
🔍	t2.micro	🟢 2/2 checks passed	Networking ▶	us-east-1b
🔍	t2.micro	🟢 2/2 checks passed	Security ▶	us-east-1a

Sub-menu for **Actions**:

- Connect
- View details
- Manage instance state
- Instance settings ▶
- Networking ▶
- Security ▶
- Image and templates ▶
- Monitor and troubleshoot ▶**

Sub-menu for **Monitor and troubleshoot**:

- Get system log
- Get instance screenshot**
- Manage detailed monitoring
- Manage CloudWatch alarms
- EC2 serial console
- Replace root volume
- Fleet Manager 📄

Instance screenshot



Download

i-0235d4513419f3e8a (Web Server) on 2024-01-04 at T21:59:21.110 -08:00

```

Amazon Linux 2023
Kernel 6.1.66-91.160.amzn2023.x86_64 on an x86_64 (-)

ip-10-0-2-190 login: [ 32.248423] systemd-sysv-generator[2376]: SysV service '/etc/rc.d/init.d/cfn-hup' lacks a native systemd unit file. Automatically generating a unit file for compatibility. Please update package to include a native systemd unit file, in order to make it more safe and robust.
[ 32.273346] zram_generator::config[2378]: zram0: system has too much memory (949MB), limit is 800MB, ignoring.
[ 33.874751] systemd-sysv-generator[3847]: SysV service '/etc/rc.d/init.d/cfn-hup' lacks a native systemd unit file. Automatically generating a unit file for compatibility. Please update package to include a native systemd unit file, in order to make it more safe and robust.
[ 33.897042] zram_generator::config[3849]: zram0: system has too much memory (949MB), limit is 800MB, ignoring.

```

Task 3: Update Your Security Group and Access the Web Server

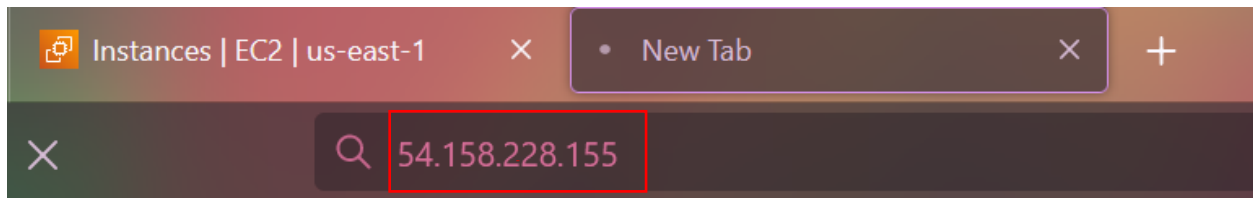
1. Ensure Web Server is still selected and click the Details tab. Copy the public IPv4 Address.

=

Instance: i-0235d4513419f3e8a (Web Server)

Details	Status and alarms New	Monitoring	Security	Networking	Storage								
▼ Instance summary Info <table> <tr> <td>Instance ID</td> <td>Public IPv4 address</td> </tr> <tr> <td> i-0235d4513419f3e8a (Web Server)</td> <td> 54.158.228.155 open address </td> </tr> <tr> <td>IPv6 address</td> <td>Instance state</td> </tr> <tr> <td>—</td> <td> Running</td> </tr> </table>						Instance ID	Public IPv4 address	i-0235d4513419f3e8a (Web Server)	54.158.228.155 open address	IPv6 address	Instance state	—	Running
Instance ID	Public IPv4 address												
i-0235d4513419f3e8a (Web Server)	54.158.228.155 open address												
IPv6 address	Instance state												
—	Running												

2. In a new tab in your web browser, paste the IP address into the search bar. Note how you are currently unable to access the server because the security group is blocking port 80.



3. Keep the browser tab open but navigate back to the EC2 Console. In the left navigation menu, choose Security groups.

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

4. Select the Web Server Security Group, and choose the Inbound Rules tab. Choose Edit inbound rules and add a rule with type HTTP, Source Anywhere-IPv4, and save it.

Security Groups (1/5) [Info](#)

 **Actions** ▾ **Export security groups to**

	Name ▾	Security group ID ▾	Security group name
☐	-	sg-068789ac7bd676237	default
☐	-	sg-05204189dc7efa4f5	Ec2SecurityGroup
☐	-	sg-09189663241b714f9	default
☐	-	sg-0c7775ea5e5547ad3	default
☒	-	sg-056db2bcfd4ce9bae	Web Server Security Group

sg-056db2bcfd4ce9bae - Web Server Security Group

Details **Inbound rules** Outbound rules Tags

Inbound rules  **Manage tags** **Edit inbound rules**

 **1**  

	Name ▾	Security group rule... ▾	IP version ▾	Type ▾	Protocol ▾	Port range ▾
No security group rules found						

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
-	HTTP ▾	TCP	80	Any... ▾	



Add rule

Cancel **Preview changes** **Save rules**

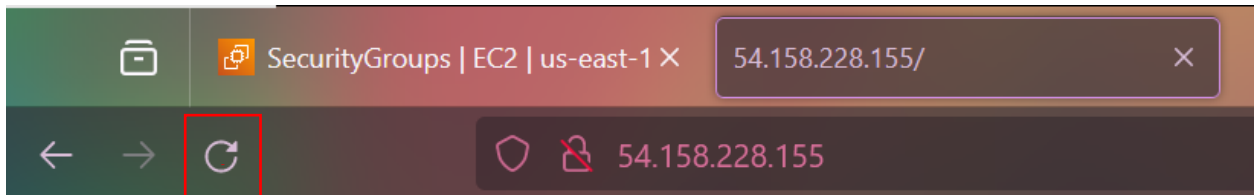
✓ Inbound security group rules successfully modified on security group ([sg-056db2bcfd4ce9bae](#) | [Web Server Secur](#))
 ▶ Details

Security Groups (5) [Info](#) ↻ Actions ▼ Export security groups t

🔍 Find resources by attribute or tag

☐	Name ▼	Security group ID ▼	Security group name
☐	-	sg-068789ac7bd676237	default
☐	-	sg-05204189dc7efa4f5	Ec2SecurityGroup
☐	-	sg-09189663241b714f9	default
☐	-	sg-0c7775ea5e5547ad3	default
☐	-	sg-056db2bcfd4ce9bae	Web Server Security Group

- Refresh the web browser tab with the Web Server on it. You should now see a message from your web server.



Hello From Your Web Server!

Task 4: Resize your Instance: Instance Type and EBS Volume

- Select Instances from the EC2 navigation panel.

▼ Instances**Instances**

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

2. Select Web Server if it isn't already and click stop instance. Proceed through all the prompts and wait till the instance state displays Stopped.

Instances (1/2) [Info](#)   

 Find Instance by attribute or tag (case-sensitive)

	Name 	Instance ID	Instance state 	Instance type
☒	Web Server	i-0235d4513419f3e8a	 Running  	t2.micro
☐	Bastion Host	i-053976705beaa11ad	 Running	

Launch instances

Launch instance from template

Migrate a server

Connect

Stop instance

Stop instance?



Instance IDs

 i-0235d4513419f3e8a (Web Server)

To confirm that you want to stop the instance, choose the *Stop* button below.

Cancel

Stop

Instances (2) [Info](#)  

 Find Instance by attribute or tag (case-sensitive)

☐	Name 	Instance ID	Instance state 
☐	Web Server	i-0235d4513419f3e8a	 Stopped  
☐	Bastion Host	i-053976705beaa11ad	 Running  

3. In the Actions menu near the top, select Instance settings, click change instance type, and change the type to t2.small and click Apply.

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with icons for home, notifications, help, and settings, followed by the region 'N. Virginia' and the user 'voclabs/user2949274=Aiden_Yin @ 42'. Below this, a toolbar contains buttons for 'Connect', 'Instance state', 'Actions', and 'Launch instances'. The 'Actions' dropdown menu is open, showing options like 'Connect', 'View details', 'Manage instance state', 'Instance settings' (highlighted with a red box), 'Networking', 'Security', 'Image and templates', and 'Monitor and troubleshoot'. On the left, a list of instance actions is visible, with 'Change instance type' highlighted by a red box. Below this, a modal window titled 'Change instance type' is displayed. It shows the 'Instance ID' as 'i-0235d4513419f3e8a (Web Server)' and the 'Current instance type' as 't2.micro'. The 'Instance type' dropdown is set to 't2.small' (highlighted with a red box). There is an unchecked checkbox for 'EBS-optimized' with a note that it is not supported for this instance type. At the bottom right of the modal, there are 'Cancel' and 'Apply' buttons, with 'Apply' highlighted by a red box.

Instance ID
i-0235d4513419f3e8a (Web Server)

Current instance type
t2.micro

Instance type
t2.small ▼

☐ EBS-optimized
EBS-optimized is not supported for this instance type

Cancel Apply

4. With the Web Server instance still selected, choose the Storage tab, select the name of the volume id, and select the checkbox next to the name. In the Actions menu, select modify volume and change the size to 10 and proceed through all the prompts.

Instance: i-0235d4513419f3e8a (Web Server)

Details

Status and alarms [New](#)

Monitoring

Security

Networking

Storage

▼ Root device details

Root device name

 /dev/xvda

Root device type

EBS

▼ Block devices

Filter block devices			
Volume ID	Device name	Volume size (GiB)	Attachment status
vol-08a6f88016bb32e6f	/dev/xvda	8	 Attached

Volumes (1/1) [Info](#)Volume ID = vol-08a6f88016bb32e6f

☒	Name ▾	Volume ID ▾	Type ▾	Size ▾	IOPS ▾	Throughput ▾
☒	-	vol-08a6f88016bb32e6f	gp3	8 GiB	3000	125



Actions ▲

[Create volume](#)

Modify volume

Create snapshot

Create snapshot lifecycle policy

Delete volume

Attach volume

Detach volume

Force detach volume

Volume details

Volume ID

 vol-08a6f88016bb32e6fVolume type [Info](#)

General Purpose SSD (gp3) ▾

Size (GiB) [Info](#)

Min: 1 GiB, Max: 16384 GiB. The value must be an integer.

IOPS [Info](#)

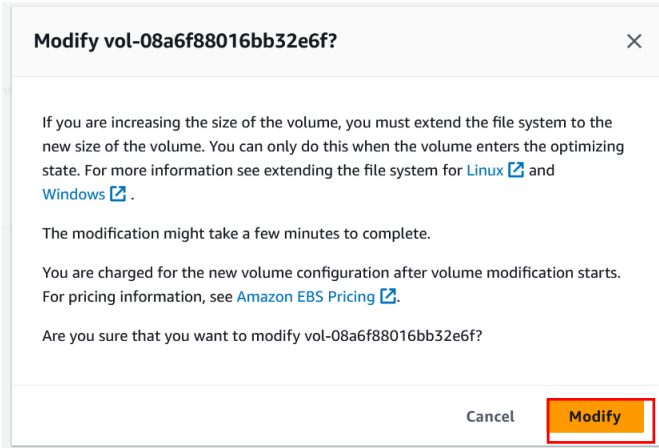
Min: 3000 IOPS, Max: 16000 IOPS. The value must be an integer.

Throughput (MiB/s) [Info](#)

Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Cancel

[Modify](#)



5. In the navigation plane, choose instances again. Select the Web Server Instance, and in the Instance state menu, select Start Instance.

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

	Name	Instance ID	Instance state	Instance type
☒	Web Server	i-0235d4513419f3e8a	Stopped	t2.small
☐	Bastion Host	i-053976705beaa11ad	Runni	

Launch instances

Launch instance from template

Migrate a server

Connect

Stop instance

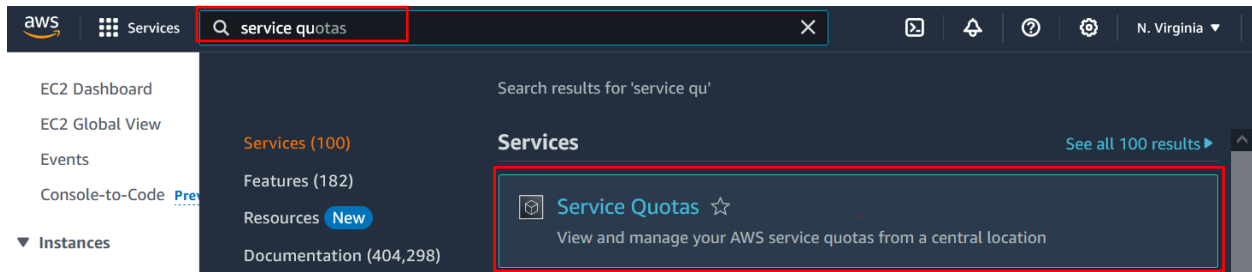
Start instance

Reboot instance

Task 5: Explore EC2 Limits

Amazon EC2 provides different resources that you can use, including images, instances, snapshots, and volumes. When you create an AWS account, there are default limits on these resources on a per-region basis.

1. In the console, use the services search bar to search for service quotas.



2. Choose AWS Services from the navigation menu, search ec2, and choose Amazon Elastic Compute Cloud-

The screenshot shows the AWS Service Quotas console. On the left, a sidebar menu is visible with the following items: Dashboard, AWS services (highlighted with a red box), Quota request history, and a collapsed Organization section containing Quota request template. The main content area has a breadcrumb trail: Service Quotas > AWS services. Below this, the heading 'AWS services' is followed by a search bar containing 'ec2' (highlighted with a red box) and a '5 matches' indicator. A table of results is displayed with the following items:

Service
Amazon EC2 Auto Scaling
Amazon Elastic Compute Cloud (Amazon EC2) (highlighted with a red box)
EC2 Fast Launch
EC2 Image Builder
EC2 VM Import/Export

3. In the search bar, search running on-demand, and observe the filtered list of service quotas. You are unable to go past any of these limits in a region.

Service quotas [info](#) Request increase at account-level

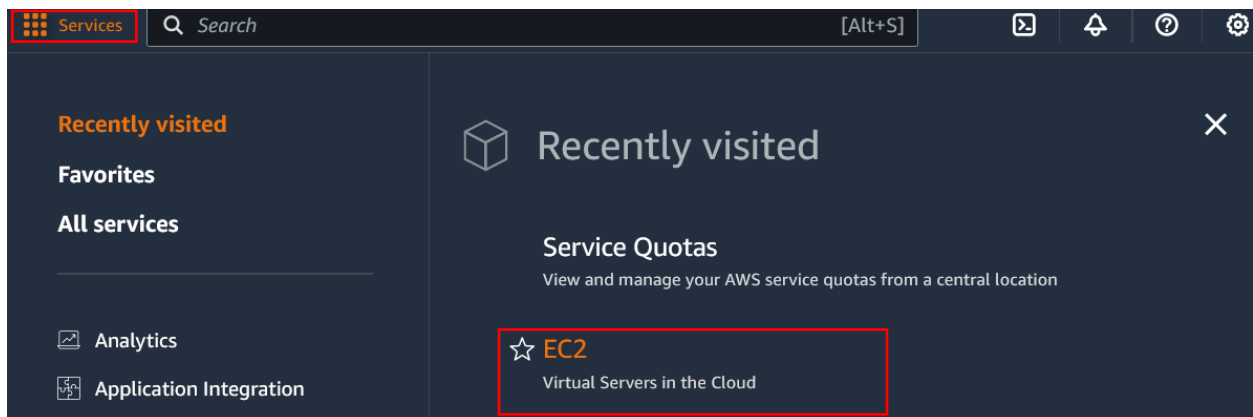
View your applied quota values, default quota values, and request quota increases for quotas. [Learn more](#)

10 matches

	Quota name ▲	Applied quota value ▼	AWS default quota value ▼	Adjustability ▼
☐	Running On-Demand DL instances	96	0	Account-level
☐	Running On-Demand F instances	64	0	Account-level
☐	Running On-Demand G and VT instances	0	0	Account-level
☐	Running On-Demand High Memory instances	0	0	Account-level
☐	Running On-Demand HPC instances	192	0	Account-level
☐	Running On-Demand Inf instances	8	0	Account-level
☐	Running On-Demand P instances	0	0	Account-level
☐	Running On-Demand Standard (A, C, D, H, I, M, R, T, Z) instances	256	5	Account-level
☐	Running On-Demand Trn instances	0	0	Account-level
☐	Running On-Demand X instances	0	0	Account-level

Task 6: Test Termination Protection

1. Navigate back to EC2 through the services menu.



2. Use the Navigation Menu to choose Instances, select Web Server, and choose Terminate Instance from the instance state menu. Follow any prompts that appear. An error message will appear, because termination protection was turned on.

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations [New](#)

▼ Images

☒	Web Server	i-0235d4513419f3e8a	✔ Running	⌵
☐	Bastion Host	i-053976705beaa11ad	✔ Running	⌵

Launch instances

Launch instance from template

Migrate a server

Connect

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate instance

Instance settings ▶



×

Terminate instance?

⚠

On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID	Termination protection
 i-0235d4513419f3e8a (Web Server)	 Enabled

To confirm that you want to terminate the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

Cancel

Terminate

⊗

Failed to terminate an instance: The instance 'i-0235d4513419f3e8a' may not be terminated. Modify its 'disableApiTermination' instance attribute and try again.

- While Web Server is still selected, choose instance settings from the action menu, and choose Change termination protection.

↺

Connect

Instance state ▼

Actions ▲

Launch instances

▼

Instance type ▼

Status check

Attach to Auto Scaling Group

Change termination protection

Change stop protection

Change shutdown behavior

Change auto-recovery behavior

Change instance type

Connect

View details

Manage instance state

Instance settings ▶

Networking ▶

Security ▶

Image and templates ▶

Monitor and troubleshoot ▶

- Uncheck Enable and click save. Select the Web Server instance again and proceed to terminate it. You should now be able to terminate the instance.

Change termination protection



To prevent your instance from being accidentally terminated, you can enable termination protection for the instance. [Learn more](#)

Instance ID

i-0235d4513419f3e8a (Web Server)

Termination protection

☐ Enable



Termination protection disabled.

The instance is no longer protected against accidental termination. If the instance is terminated, data stored on ephemeral storage is lost.

Cancel

Save

	Name	Instance ID	Instance state	Instance type
☒	Web Server	i-0235d4513419f3e8a	Running	t2.small
☐	Bastion Host	i-053976705beaa11ad		

Launch instances

Launch instance from template

Migrate a server

Connect

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate instance

Instance settings

Networking

Security

Image and templates

Monitor and troubleshoot

Terminate instance?

On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID	Termination protection
i-0235d4513419f3e8a (Web Server)	Disabled

To confirm that you want to terminate the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

Cancel

Terminate

Successfully terminated i-0235d4513419f3e8a

Problems:

The labs themselves were quite straightforward, with each step generally being pretty well defined. There were some cases where it took me a bit of time to find what prompt the instructions were talking about, but that is just an issue with design of the instructions. The main problem I experienced was the instances for the EC2 instances and VPC taking much longer than expected to finish starting up. Nothing could be done to solve the issue, except wait for the checks to finish,

In future labs that utilize AWS, I will make sure to keep in mind that these instances may take more time than expected to finish setting up and may require refreshing the page multiple times. The instance does not 100% have an issue until one of the tests explicitly states that it was failed.

Conclusion:

AWS provides a cloud platform for multiple services to run on, providing many benefits over traditional methods of computing. Amazon EC2 provides a simple, yet robust way of cloud compute, while VPC provides a sort of container that can store these EC2 instances. AWS IAM provides a way for administrators to bestow permissions to different accounts to access these services provided by AWS, increasing security of the overall cloud. These services are highly compatible with one another, making the transition from traditional, on-site networks to cloud networking much smoother overall.

Lab 5: Amazon AWS Cloud Foundations Labs 4-6



Purpose

Amazon Web Services, or AWS for short, is one of the most common and robust cloud service providers, along with others like Microsoft Azure and Google Cloud Platform. By renting their platform out through a pay-as-you-go format, these cloud service providers may offer a cheaper, more reliable, and more flexible alternative compared to traditional on-premises infrastructure. Through these three labs, we can start to learn about various fundamental services of AWS and how they combine with other services. Learning about these services will also help us study and prepare for the AWS Cloud Foundations Certification Exam.

Background Information on Lab Concepts

Lab 4: Working with EBS

Amazon Elastic Block Store, or EBS, provides block level storage volumes for use with EC2 instances. Block storage has many differences with object storage. When you edit data stored in block storage, only that specific block that contains the item you want to edit gets changed. However, when you edit data that is store in object storage, the entire file needs to be updated. Block storage is generally faster than object storage, but it is more expensive than object storage. Object storage is generally used for large, unstructured data, while block storage is used in cases such as an application that regularly needs to load data from the cloud. Amazon Elastic Block Storage has many options that users can choose from, such as the size, price, performance, or encryption. EBS volumes can be “snapshotted”, making a copy of that volume that can be used on a different availability zone or recover lost data.

Lab 5: Build your DB Server and Interact with Your DB Using an App

Amazon Relational Database Service, or RDS, is a web service that makes it easier to set up, operate, and scale a relational database in the AWS cloud. Compared to the traditional method of having an on-premises database, using RDS makes Amazon in charge of things like server maintenance, database backups, scaling, and OS patching and installations, making using a database less troublesome. However, there still are user-managed items, meaning that these databases are still prone to user error. Amazon RDS supports many different SQL's such as MySQL. RDS is generally more preferred

than EC2 when just used for databases, as more components are managed by Amazon and less prone to user error.

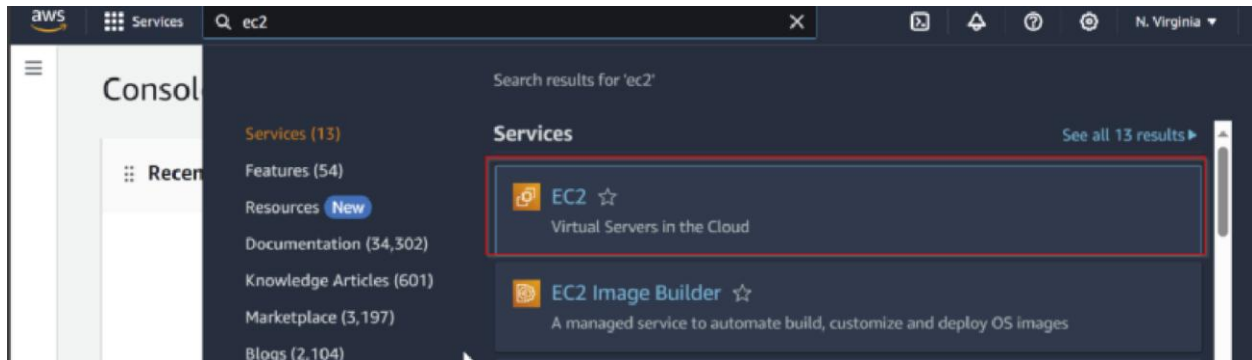
Lab 6: Scale & Load Balance your Architecture

Scaling and Load Balancing are integral parts of virtual machines like EC2. To scale a machine is to increase or decrease its capacity just enough to cover how much it is being used. If a machine isn't being used a lot, the resources that are allocated to it will be scaled down, and vice versa. Load Balancing is the distribution of traffic loads across multiple different EC2 instances. By doing so, it provides increased fault tolerance in case one instance goes down, improving the user experience of the service. AWS load balancing comes in three formats: Application Load Balancers (ALB), Network Load Balancers (NLB), and Classic Load Balancers/Elastic Load Balancer (ELB). These load balancers are based off of the 7 Layer OSI model. Application Load Balancers act on the Application Layer, which is the 7th layer of the model. This type of balancer routes based on things like HTTP headers. ALBs are typically used with web applications. Network Load Balancers, on the other hand, function off the Transport layer, the 4th layer of the OSI model. This balancer routes using TCP/UDP traffic, generally providing high-performance load balancing. Classic Load Balancing, like the name suggests, is the legacy load balancing option offered by AWS. This option can route both HTTP and TCP/UDP, but when compared to using both ALB and NLB, ELB provides less options for load balancing in exchange for a simpler setup process.

Lab #4 Walkthrough:

Task 1: Create a New EBS Volume

1. Navigate to EC2, then click on Instances in the navigation panel on the left.

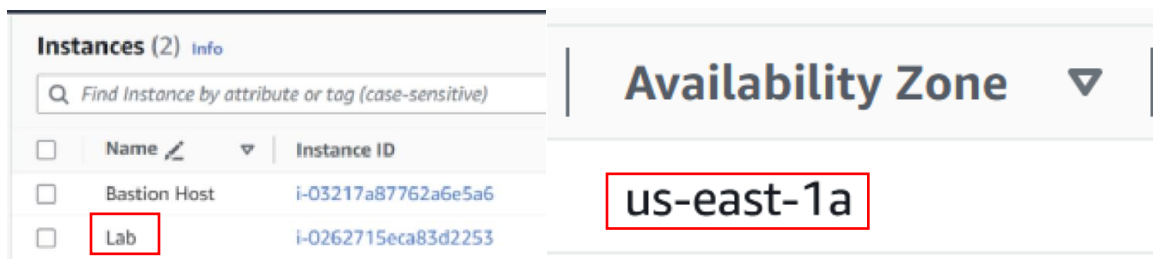


▼ Instances

Instances

- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts

2. An instance named Lab has already been set up for you. Note the availability zone of the instance.



3. In the Navigation Plane, click on Volumes. Then, create a new volume with Volume Type General Purpose SSD (gp2), size 1GiB, the same availability zone as your EC2 instance, and an extra Key tag named My Volume.

▼ Elastic Block Store

[Volumes](#)[Snapshots](#)[Lifecycle Manager](#)

Actions ▼

Create volume



1



Volume settings

Volume type [Info](#)

General Purpose SSD (gp2) ▼

Size (GiB) [Info](#)

1



Min: 1 GiB, Max: 16384 GiB. The value must be an integer.

IOPS [Info](#)

100 / 3000

Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.

Throughput (MiB/s) [Info](#)

Not applicable

Availability Zone [Info](#)

us-east-1a ▼

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and your resources or track your AWS costs.

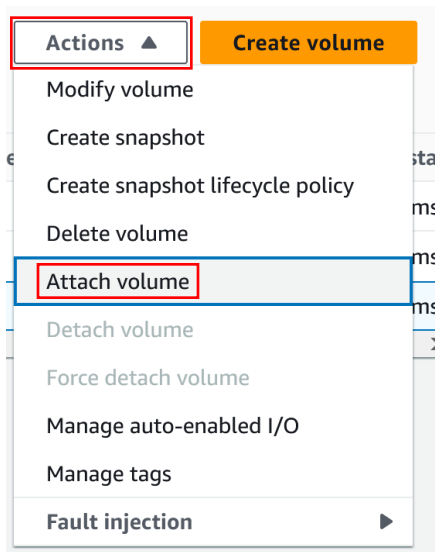
Key

Value - optional

 Task 2: Attach the Volume to an Instance

1. Select My Volume. In the actions menu, choose Attach Volume. Choose the instance field and select the instance that appears.

Volumes (1/3) Info	
Search	
☐	-
☐	-
☒	My Volume
<	



Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

 vol-0d06b393a1087d149 (My Volume)

Availability Zone

us-east-1a

Instance [Info](#)

i-0ecf18399d7d0f2a2 ▼



Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)

/dev/sdf

Recommended device names for Linux: /dev/sda1 for root volume. /dev/sd[f-p] for data volumes.



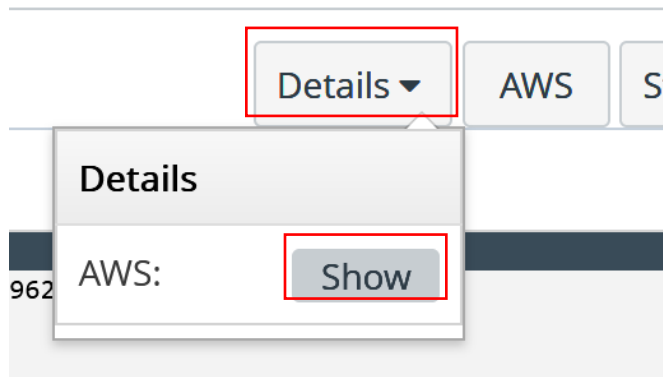
Newer Linux kernels may rename your devices to **/dev/xvdf** through **/dev/xvdp** internally, even when the device name entered here (and shown in the details) is **/dev/sdf** through **/dev/sdp**.

Cancel

Attach volume

Task 3: Connect to Your Amazon EC2 Instance

1. For this lab, we have received an SSH key to connect to the instance that was created for us. In order to access the key, click on the Details button on the Instructure page for the lab. Then click Show next to AWS.



2. Download the ppk file and save the labuser.ppk file somewhere.

Cloud Access

AWS CLI:

Show

Cloud Labs

Remaining session time: 07:45:56 (466 minutes)

Session started at: 2024-01-16T13:21:47-0800

Session to end at: 2024-01-16T21:21:47-0800

Accumulated lab time: 08:31:00 (511 minutes)

(1) ips -- public:3.86.99.12, private:10.1.11.229 (2) i

SSH key

Show

Download PEM

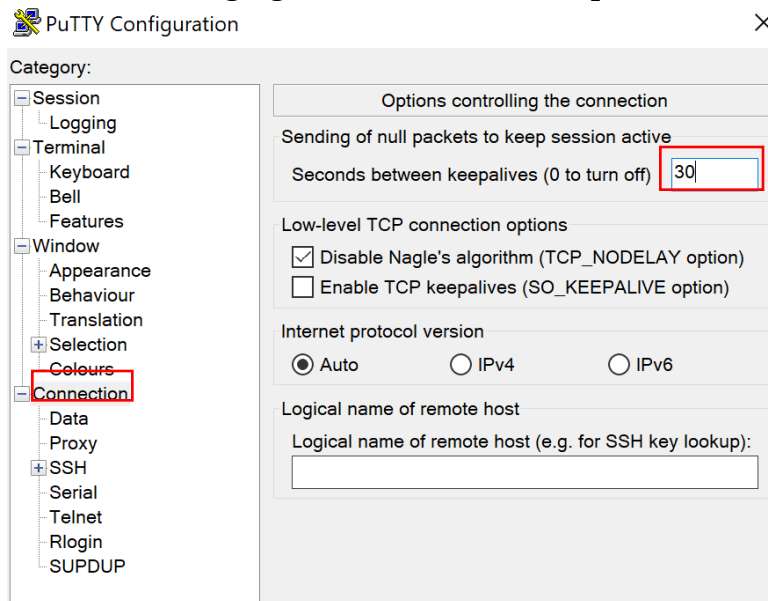
Download PPK

AWS SSO

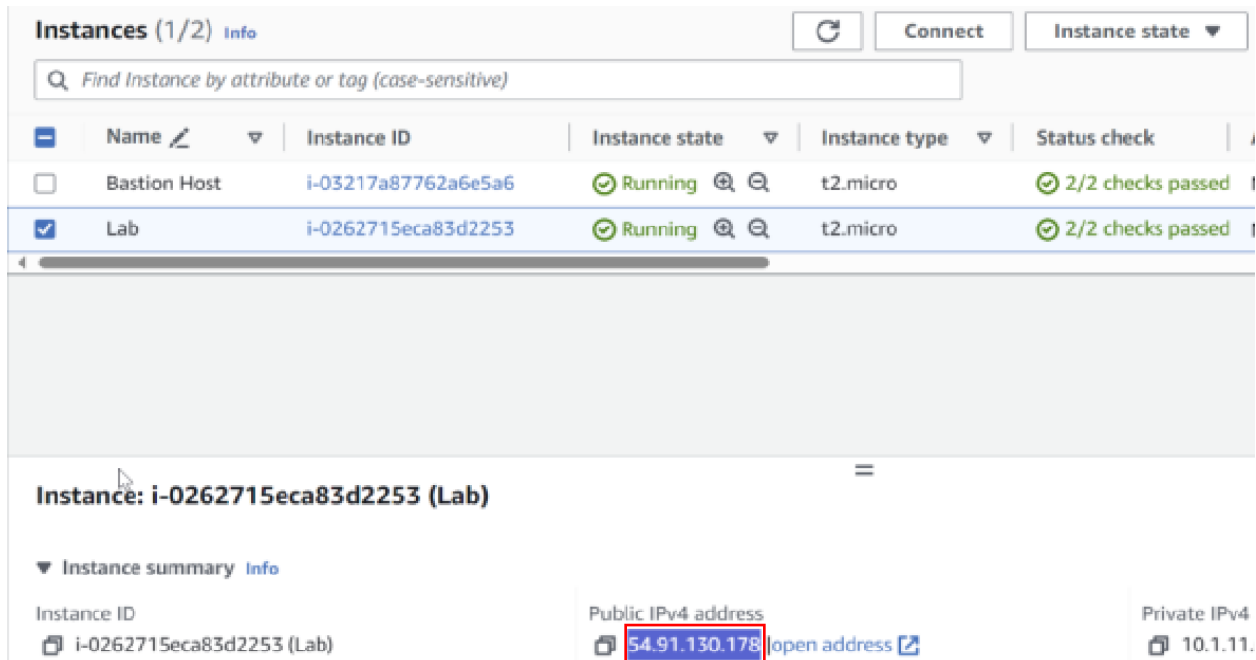
Download URL

SecretKey	8cK/VjoFVeWxAa1la
BastionHost	54.145.167.222
Region	us-east-1
AvailabilityZone	us-east-1a
AccessKey	AKIAZEJVBHR34TLUD
LabInstance	3.86.99.12

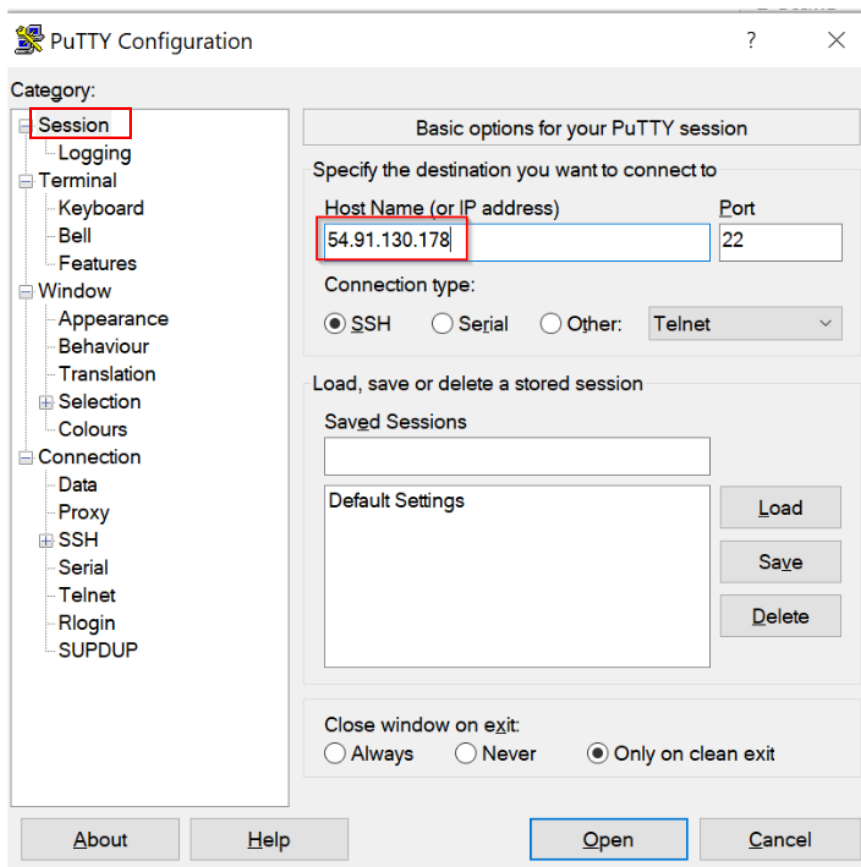
3. Open Putty, and configure Putty to not time out by selecting the Connection menu and changing Seconds between keepalives to 0.



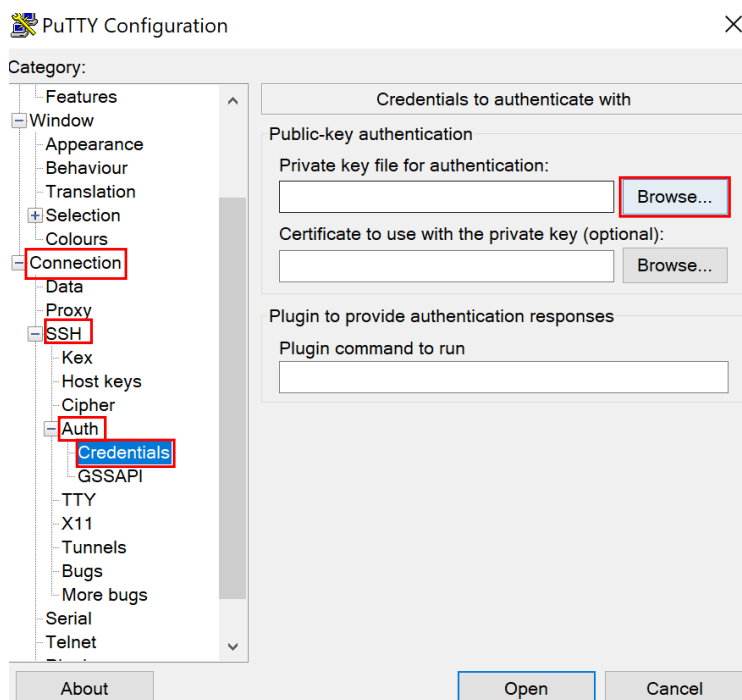
4. Copy the public DNS or IPv4 Address of the EC2 Instance that was already made.



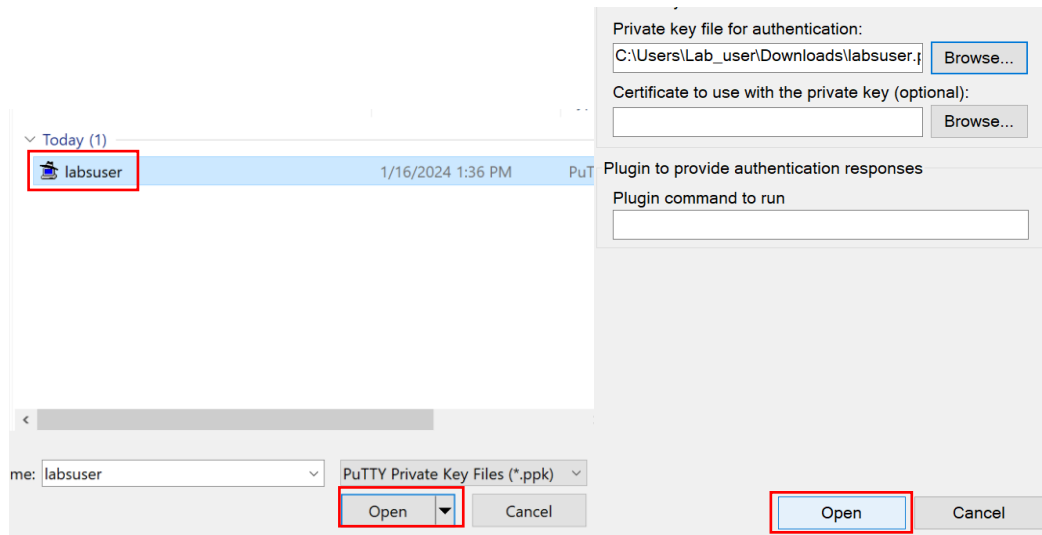
5. In Putty, choose the Session menu, input the IPv4 address we just copied into the Host Name section



6. Navigate back to the Connection menu, expand SSH, expand Auth and choose Credentials. Click on Browse next to Private key file for authentication.



- Choose the labuser.ppk file you recently downloaded, and click open on Putty once it has been selected.



- Choose accept in order to trust and connect to the host.

You have no guarantee that the server is the computer you think it is.

The server's ssh-ed25519 key fingerprint is:

ssh-ed25519 255 SHA256:CZg+mXIJ1qTc702Bg2bdlM3o/czEjx216tE4nUCuc

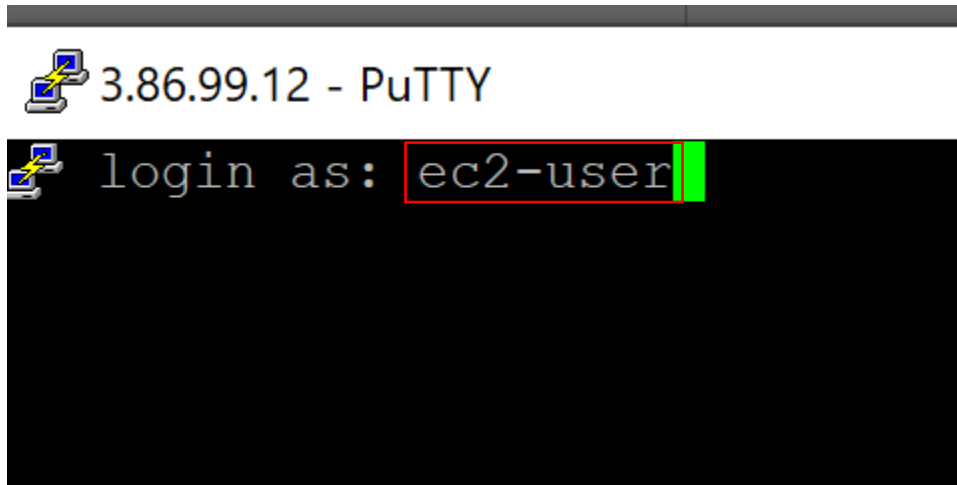
If you trust this host, press "Accept" to add the key to PuTTY's cache and carry on connecting.

If you want to carry on connecting just once, without adding the key to the cache, press "Connect Once".

If you do not trust this host, press "Cancel" to abandon the connection.



9. When prompted, log in as `ec2-user` and press enter. You are now connected to the EC2 instance.



Task 4: Create and Configure Your File System'

1. View the storage available on your device using the `df -h` command.

```
[ec2-user@ip-10-1-11-229 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M  0    4.0M  0%  /dev
tmpfs           475M  0    475M  0%  /dev/shm
tmpfs           190M  2.9M  188M  2%  /run
/dev/xvda1       8.0G  1.5G  6.5G  19%  /
tmpfs           475M  0    475M  0%  /tmp
/dev/xvda128     10M   1.3M  8.7M  13%  /boot/efi
tmpfs           95M   0    95M   0%  /run/user/1000
[ec2-user@ip-10-1-11-229 ~]$
```

2. Create a new ext3 file system on the new volume using the `sudo mkfs -t ext3 /dev/sdf` command.

```
[ec2-user@ip-10-1-11-229 ~]$ sudo mkfs -t ext3 /dev/sdf
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 2764a8c9-ea93-4143-82ab-95d609c73476
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-10-1-11-229 ~]$
```

3. Create a directory for mounting the new storage volume using the `sudo mkdir /mnt/data-store` command.

```
[ec2-user@ip-10-1-11-229 ~]$ sudo mkdir /mnt/data-store
[ec2-user@ip-10-1-11-229 ~]$
```

4. Mount the new volume using the `sudo mount /dev/sdf /mnt/data-store` command.

```
[ec2-user@ip-10-1-11-229 ~]$ sudo mount /dev/sdf /mnt/data-store
[ec2-user@ip-10-1-11-229 ~]$
```

5. Use the `cat /etc/fstab` command to see the setting on the last line.

```
[ec2-user@ip-10-1-11-229 ~]$ cat /etc/fstab
#
UUID=2bebc11-b28e-4409-b1dd-8374c85cc602 / xfs defaults,noatime 1 1
UUID=B34C-BAE4 /boot/efi vfat defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2
[ec2-user@ip-10-1-11-229 ~]$
```

6. View the available storage again and notice the additional line `-/dev/xvdf`.

```
[ec2-user@ip-10-1-11-229 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0    4.0M   0% /dev
tmpfs           475M   0    475M   0% /dev/shm
tmpfs           190M  2.9M   188M   2% /run
/dev/xvda1      8.0G  1.5G   6.5G  19% /
tmpfs           475M   0    475M   0% /tmp
/dev/xvda128    10M   1.3M   8.7M  13% /boot/efi
tmpfs           95M    0     95M   0% /run/user/1000
/dev/xvdf       975M   60K   924M   1% /mnt/data-store
[ec2-user@ip-10-1-11-229 ~]$
```

7. On your mounted volume, create a file and add some text to it.

```
[ec2-user@ip-10-1-11-229 ~]$ sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"
[ec2-user@ip-10-1-11-229 ~]$
```

8. Verify that the text has been written to the volume.

```
[ec2-user@ip-10-1-11-229 ~]$ cat /mnt/data-store/file.txt
some text has been written
[ec2-user@ip-10-1-11-229 ~]$
```

Task 5: Create an Amazon EBS Snapshot

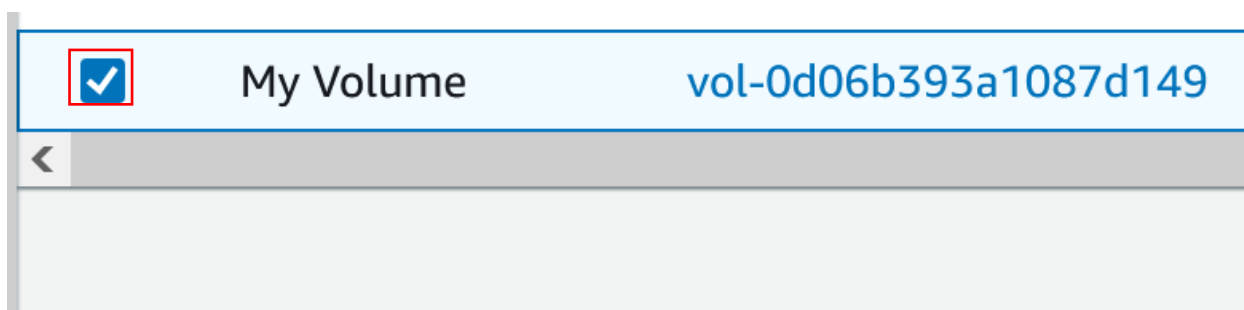
1. Navigate to Volumes and select My Volume.

▼ Elastic Block Store

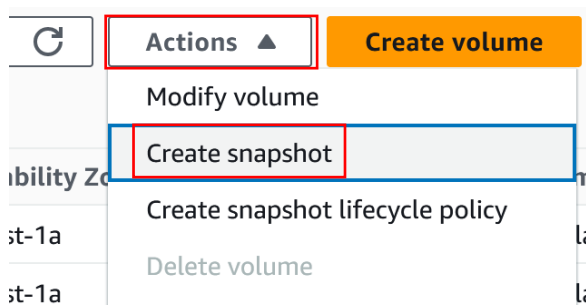
Volumes

Snapshots

Lifecycle Manager



2. In the Actions Menu, click Create snapshot. Add a tag with the key Name, value My Snapshot, and create it.



No tags associated with the resource.

Add tag

You can add 50 more tags.

Tags [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

X

Value - optional

Q My Snapshot

X

Remove

Add tag

You can add 49 more tags.

Cancel

Create snapshot

- Choose Snapshots in the navigation plane on the left. Notice the snapshot that we just took.

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

☐	Name ▾	Snapshot ID ▾	Volume size ▾	Description ▾	Storage tier ▾	Snapshot status ▾
☐	My Snapshot	snap-0721ebe05198a8775	1 GiB	-	Standard	Completed

- Delete the file we created on our volume in Putty using the `sudo rm /mnt/data-store/file.txt` command and verify through the `ls /mnt/data-store/` command.

```
[ec2-user@ip-10-1-11-229 ~]$ ~
sudo rm /mnt/data-store/file.txt
-bash: /home/ec2-user: Is a directory
[ec2-user@ip-10-1-11-229 ~]$ ls /mnt/data-store/
lost+found
```

Task 6: Restore the Amazon EBS Snapshot

1. Select My Snapshot and in the Actions Menu, click Create volume from snapshot. Use the same availability zone as before. Add a tag with the key Name and value Restored Volume.

Snapshots (1/1) [Info](#)

Owned by me ▼

☒	Name ▼	Snapshot ID ▼	Volume size ▼
☒	My Snapshot	snap-0721ebe05198a8775	1 GiB

Actions ▲

Create snapshot

Create volume from snapshot

Create image from snapshot

Copy snapshot



IOPS [Info](#)

3000

Min: 3000 IOPS, Max: 16000 IOPS. The value must be an integer.

Throughput (MiB/s) [Info](#)

125

Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Availability Zone [Info](#)

us-east-1a

Fast snapshot restore [Info](#)

Not enabled for selected snapshot

No tags associated with the resource.

Add tag

You can add 50 more tags.

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name



Value - optional

Q Restored Volume



Remove

Add tag

You can add 49 more tags.

Snapshot summary

Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

Cancel

Create volume

2. Choose Volumes in the left navigation plane, select Restored Volume, and use the Actions menu to attach it to the Lab instance.

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

☐	Name
☐	-
☐	-
☐	My Volume
☒	Restored Volu...

Actions ▲

Create volume

Modify volume

Create snapshot

Create snapshot lifecycle policy

Delete volume

Attach volume

Basic details

Volume ID
vol-0dd2ee6ceb075c563 (Restored Volume)

Availability Zone
us-east-1a

Instance [Info](#)
i-0262715eca83d2253

Device name [Info](#)
/dev/sdg

ⓘ Newer Linux kernels may rename your devices to **/dev/xvdf** through **/dev/xvdp** internally, even when the device name entered here (and shown in the details) is **/dev/sdf** through **/dev/sdp**.

Cancel

Attach volume

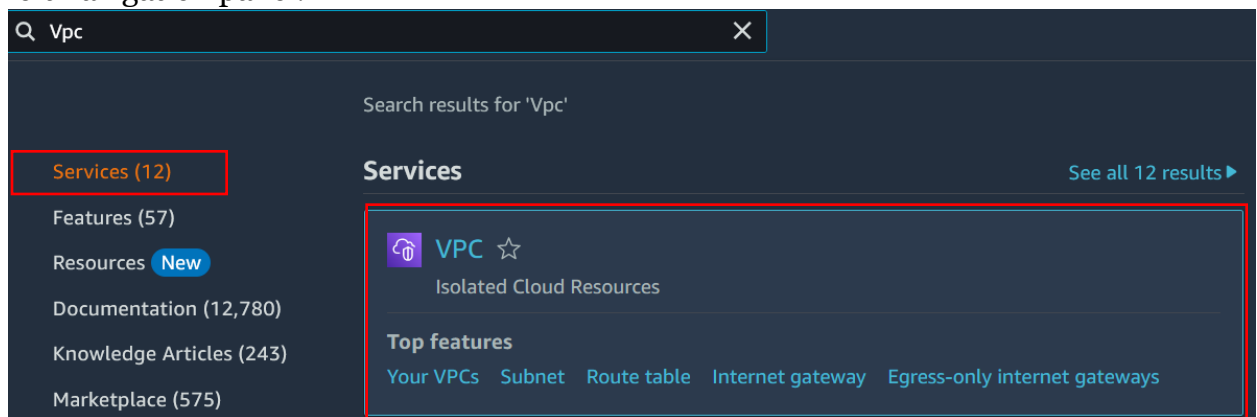
3. Mount the restored volume by creating a new directory, mounting the volume, and verifying that the volume has the file. You should be able to see file.txt after this.

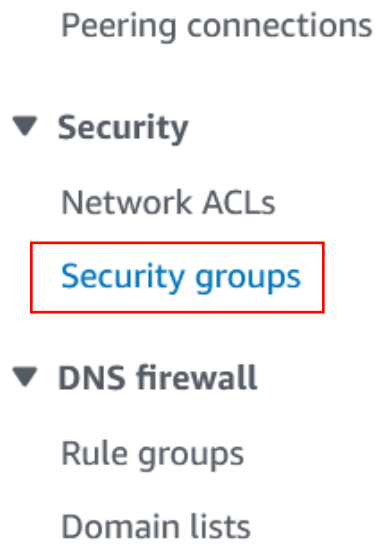
```
[ec2-user@ip-10-1-11-229 ~]$ sudo mkdir /mnt/data-store2
[ec2-user@ip-10-1-11-229 ~]$ 
[ec2-user@ip-10-1-11-229 ~]$ sudo mount /dev/sdg /mnt/data-store2
[ec2-user@ip-10-1-11-229 ~]$ 
[ec2-user@ip-10-1-11-229 ~]$ ls /mnt/data-store2/
file.txt  lost+found
```

Lab #5 Walkthrough

Task 1: Create a Security Group for the RDS DB Instance

1. Use the Services menu to get to the VPC service. Choose security groups from the left navigation panel.





2. Create a new security group with the name DB Security Group, description Permit access from web security group, and VPC Lab VPC. Add an inbound rule with type MySQL/Aurora(3306) and CIDR, IP, Security Group, or Prefix List of Web Security Group.

[Export security groups to CSV](#) 

Create security group

< 1 > 

▼ | Outbound rules count ▼

Basic details

Security group name [Info](#)

DB Security Group

Name cannot be edited after creation.

Description [Info](#)

Permit access from Web Security Group

VPC [Info](#)

vpc-0eb5e1d05831bebe0 (Lab VPC) ▼

Inbound rules [Info](#)

This security group has no inbound rules.

Add rule

Inbound rules [Info](#)

Type [Info](#)

MYSQL/Aurora ▼

Source Info

Custom ▼

Search: sg

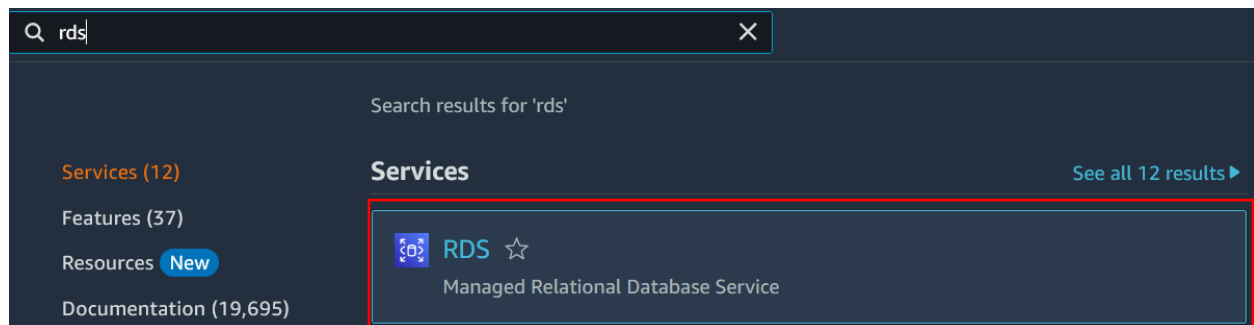
Security Groups

- default | sg-02c43f5b90ecae858
- Web Security Group | sg-0fc6f13bc6c75878a**
Web Security Group

Cancel Create security group

Task 2: Create a DB Subnet Group

1. In the Services Menu, select RDS. Once selected, use the left navigation menu to select subnet groups.



Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations [New](#)

2. Create a new DB subnet group with name DB-subnet-group, description DB subnet group, and VPC Lab VPC. Add a subnet and select the first two availability zones us-east-1a and us-east-1b. Expand the list of values under subnets and select 1.0.1.0/24 and 1.0.3.0/24.



< 1 > 

Name

You won't be able to modify the name after your subnet group has been created.

DB-Subnet-Group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and

Description

DB Subnet Group

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You can't change a different VPC identifier after your subnet group has been created.

Choose a VPC

Lab VPC (vpc-0eb5e1d05831bebe0)

Add subnets

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

☒ us-east-1a

☒ us-east-1b

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Z

Select subnets

us-east-1b

☐ subnet-0312cf989052b8be1 (10.0.2.0/24)

☒ subnet-02fd3594f5a3a744b (10.0.3.0/24)

us-east-1a

☒ subnet-0e7d7a71db31a388d (10.0.1.0/24)

☐ subnet-00186e203365bc7ab (10.0.0.0/24)

Subnets selected (2)

Availability zone	Subnet ID	CIDR block
us-east-1b	subnet-02fd3594f5a3a744b	10.0.3.0/24
us-east-1a	subnet-0e7d7a71db31a388d	10.0.1.0/24

Cancel

Create

Task 3: Create an Amazon RDS DB Instance

1. Choose databases in the left navigation plane.

Amazon RDS[Dashboard](#)[Databases](#)[Query Editor](#)[Performance insights](#)[Snapshots](#)[Exports in Amazon S3](#)[Automated backups](#)

2. Create a new database using MySQL as the engine.

Modify

Actions ▼

Restore from S3

Create database

< 1 >

⚙

Choose a database creation method [Info](#)

☒ **Standard create**
 You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
 Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☒ **MySQL**


☐ MariaDB


3. Choose Dev/Test under templates.

Templates

Choose a sample template to meet your use case.

☐ **Production**
 Use defaults for high availability and fast, consistent performance.

☒ **Dev/Test**
 This instance is intended for development use outside of a production environment.

☐ **Free tier**
 Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

4. Choose Multi-AZ DB instance under Availability and durability.

Availability and durability

Deployment options [Info](#)

The deployment options below are limited to those supported by the engine you selected above.

☐ Single DB instance

Creates a single DB instance with no standby DB instances.

☒ Multi-AZ DB instance

Creates a primary DB instance and a standby DB instance in a different AZ. Provides high availability and data redundancy, but the standby DB instance doesn't support connections for read workloads.

☐ Multi-AZ DB Cluster

Creates a DB cluster with a primary DB instance and two readable standby DB instances, with each DB instance in a different Availability Zone (AZ). Provides high availability, data redundancy and increases capacity to serve read workloads.

- Under Settings, configure the DB instance identifier lab-db, master username main, and master password lab-password.

Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

lab-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

main

1 to 16 alphanumeric characters. The first character must be a letter.

Master password [Info](#)

••••••••

Constraints: At least 8 printable ASCII characters. Can't contain any of the following: / (slash), '(single quote), "(double quote) and @ (at sign).

Confirm master password [Info](#)

••••••••

- Under DB instance class, select burstable classes (includes t classes) and db.t3.micro.

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

- ☒ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
- ☐ Include previous generation classes
- ☐ Standard classes (includes m classes)
- ☐ Memory optimized classes (includes r and x classes)
- ☒ Burstable classes (includes t classes)

db.t3.micro

2 vCPUs 1 GiB RAM Network: 2,085 Mbps

- Under Storage, Select General Purpose SSD and 20 GiB of storage.

Storage

Storage type [Info](#)

General Purpose SSD (gp3)

Performance scales independently from storage

Allocated storage [Info](#)

20

GiB

Minimum: 20 GiB. Maximum: 6,144 GiB

- Under Connectivity, Select Lab VPC.

Connectivity [Info](#)**Compute resource**

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Lab VPC (vpc-0eb5e1d05831bebe0)
4 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

9. Under Existing VPC security groups, choose DB Security Group and deselect default.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups

Choose one or more options

Q

☐ default

☒ DB Security Group

☐ Web Security Group

resiliency, and security.

10. Under Monitoring, uncheck enable enhanced monitoring.

Monitoring

☐ **Enable Enhanced Monitoring**
Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

11. Under Additional Configuration, configure the initial database name to be lab, and uncheck Enable automatic backups and Enable encryption.

▼ **Additional configuration**

Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudW protection turned off.

Database options

Initial database name [Info](#)

lab

If you do not specify a database name, Amazon RDS does not create a database.

Backup

☐ **Enable automated backups**

Creates a point-in-time snapshot of your database

Encryption

☐ **Enable encryption**

Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

12. Create the database, click on the lab-db link, and copy the Endpoint field in the Connectivity & security section.

sary rights for any third-party products or

Cancel **Create database**

Databases (1)

Filter by databases

	DB identifier	Status	Role
	lab-db	Creating	Instance

Connectivity & security

Endpoint & port

Endpoint

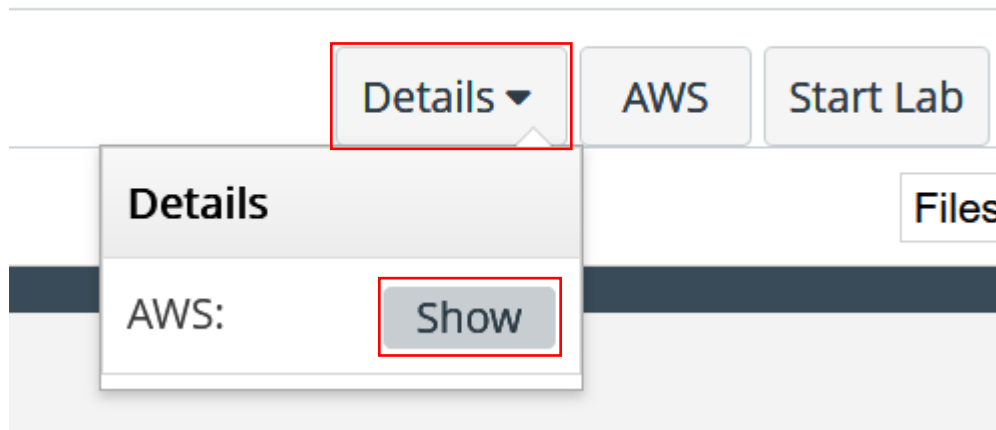
lab-db.creayq6akoyg.us-east-1.rds.amazonaws.com

Port

3306

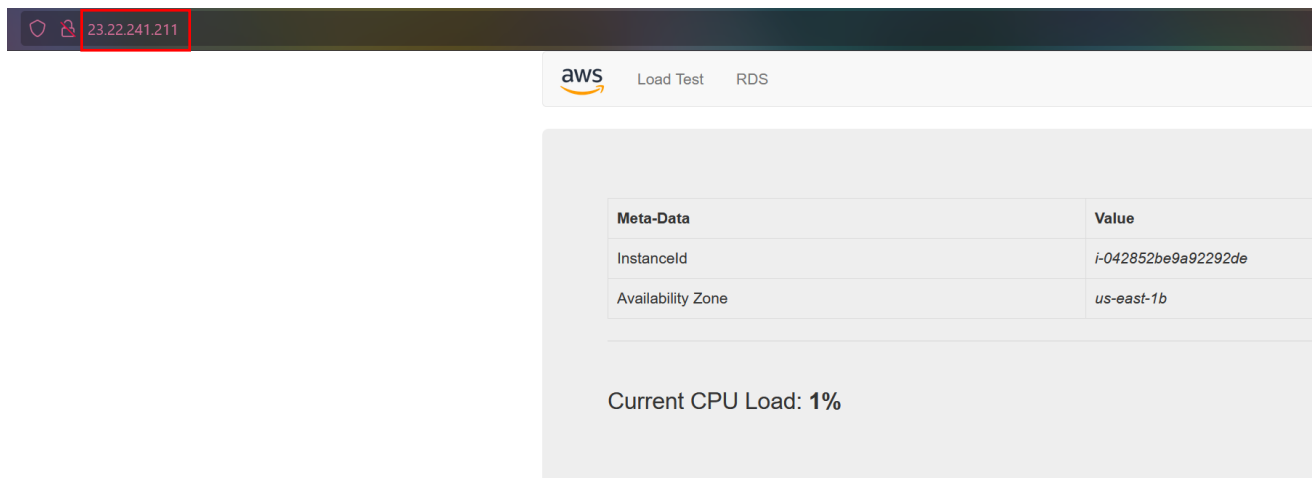
Task 4: Interact with your database

1. Copy the WebServer IP address from the details tab in the instructure website.



SecretKey	qff6zVEVh6d3NqX2E2iC2vvxPJGXp1lQyjdNd2di
WebServer	23.22.241.211
BastionHost	50.19.188.187

2. Paste the IP address into a web browser, and you can now see information about the instance.



3. Choose the RDS link at the top of the page, and configure the Endpoint we copied earlier, database lab, username main, and password lab-password.

 [Load Test](#) [RDS](#)

Meta-Data

Endpoint	lab-db.creayq6akoyg.us-east-1.rds.amazonaws.com
Database	lab
Username	main
Password	••••••••••
	Submit

4. You can now make changes to the address book by adding or removing contacts. An example of adding a contact is shown below.

Address Book

Last name	First name	Phone	Email	Admin	
				Add Contact	
Doe	Jane	010-110-1101	janed@someotheraddress.org	Edit	Remove
Johnson	Roberto	123-456-7890	robertoj@someaddress.com	Edit	Remove

Add Contact

Last Name:

First Name:

Phone:

Email:

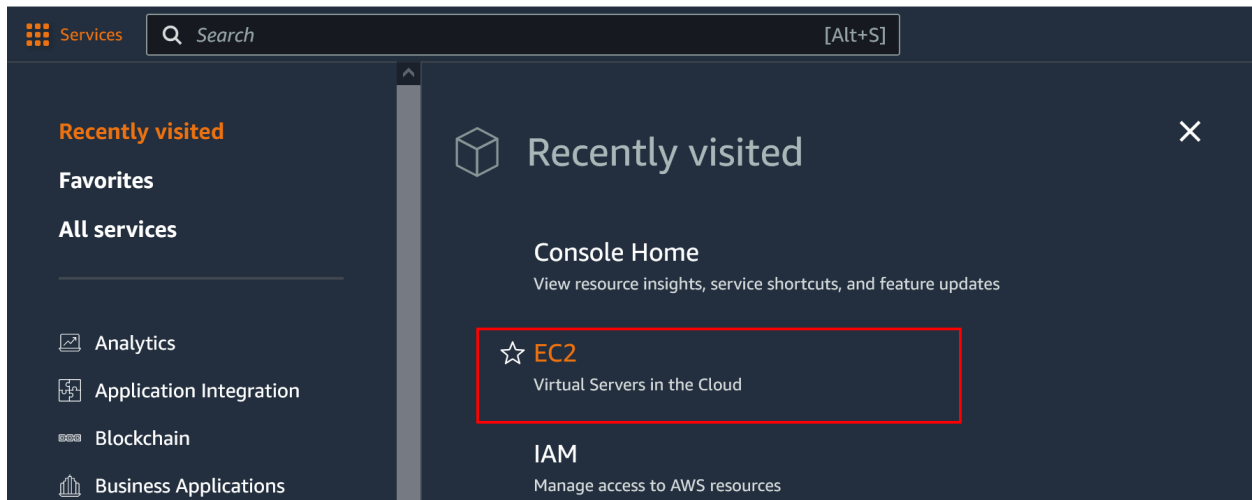
Address Book

Last name	First name	Phone	Email	Admin	
				Add Contact	
bob	jim	098-765-4321	jimbob@address.com	Edit	Remove
Doe	Jane	010-110-1101	janed@someotheraddress.org	Edit	Remove
Johnson	Roberto	123-456-7890	robertoj@someaddress.com	Edit	Remove

Lab #6 Walkthrough

Task 1: Create an AMI for Auto Scaling

1. Using the services menu, select EC2. Choose instances on the left navigation panel.



▼ Instances

[Instances](#)

[Instance Types](#)

[Launch Templates](#)

[Spot Requests](#)

[Savings Plans](#)

[Reserved Instances](#)

[Dedicated Hosts](#)

[Capacity Reservations](#)

[New](#)

2. After verifying that Web Server 1 has passed both checks, select it and go to the action menu. In the action menu, choose Image and templates, then choose Create image. Configure the name to be WebServerAMI and description to be Lab



AMI for Web Server.

	Name 	Instance ID	Instance state 
☐	Bastion Host	i-05f4ce31a7e256bfd	 Terminated  
☐	Bastion Host	i-036072fb6e210ce80	 Running  
☒	Web Server 1	i-0cc9df45ef9c29a4a	 Running  
☐	Web Server 1	i-051fa8a3c879908ba	 Terminated  


Connect
Instance state 
Actions 
Launch instances 

Availability Zone 	Public IPv4 DNS 	
ast-1a	–	
ast-1a	–	
ast-1a	–	
ast-1a	–	

Create image

Create template from instance

Launch more like this

Connect

View details

Manage instance state

Instance settings 

Networking 

Security 

Image and templates 

Monitor and troubleshoot 

Instance ID

 i-0cc9df45ef9c29a4a (Web Server 1)

Image name

WebServerAMI

Maximum 127 characters. Can't be modified after creation.

Image description - *optional*

Lab AMI for Web Server

Maximum 255 characters

Cancel

Create image

Task 2: Create a Load Balancer

1. In the left navigation plane, choose Target Groups.

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores [New](#)

▼ Auto Scaling

Auto Scaling Groups

2. Create a target group with target type Instances, target group name LabGroup, and VPC Lab VPC.

Actions ▼

Create target group

< 1 > ⚙

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

☒ Instances

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

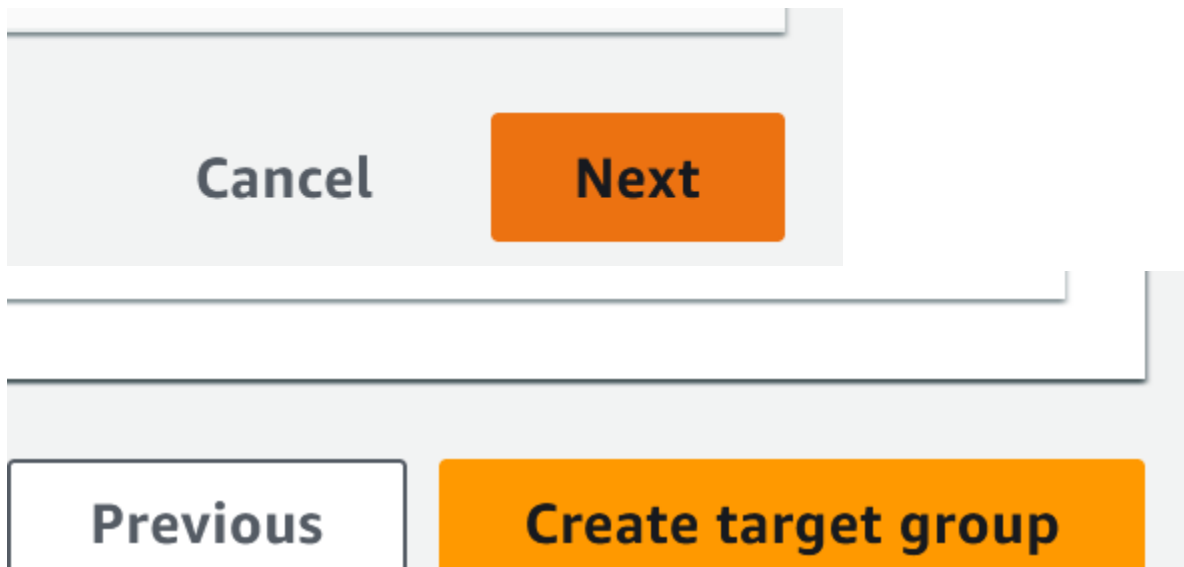
Target group name

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

-	vpc-0096a04ec748e3244	▲
IPv4: 172.31.0.0/16		
Q		
-	vpc-0096a04ec748e3244	✓
IPv4: 172.31.0.0/16		
Lab VPC		
vpc-03930100a6f3b8d67		
IPv4: 10.0.0.0/16		



3. Navigate to the load balancers using the navigation panel.

Network Interfaces

▼ Load Balancing

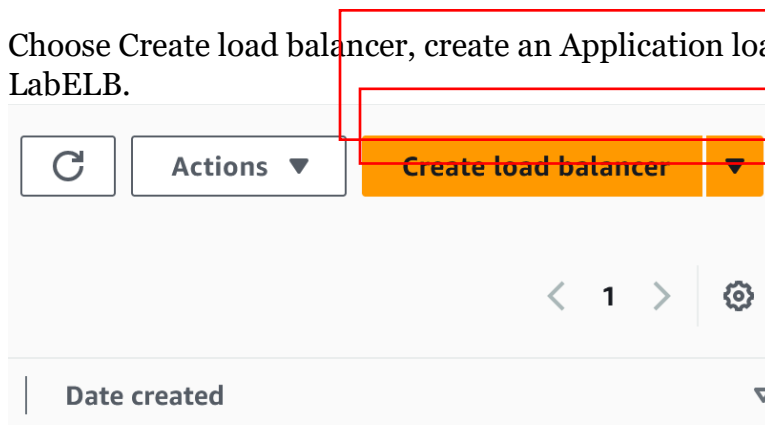
[Load Balancers](#)

Target Groups

Trust Stores [New](#)

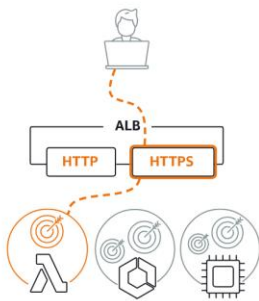
▼ Auto Scaling

4. Choose Create load balancer, create an Application load balancer, and name it LabELB.



Load balancer types

Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the lo

LabELB

A maximum of 32 alphanumeric characters including hyphens are allowed, but th

5. Scroll down to the Network mapping section, choose Lab VPC for VPC, and choose the two public subnets. In the security groups section, choose Web Security Group and deselect the default.

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

Lab VPC

vpc-0ab8e2e179b685957

IPv4: 10.0.0.0/16

Subnet of the VPC are not available for selection.

☒ **us-east-1a (use1-az2)**

Subnet

subnet-0e6004bd017a596ca	Private Subnet 1 ▲
subnet-0e6004bd017a596ca	Private Subnet 1 ✓
subnet-01782562eb7300d63	Public Subnet 1

☒ **us-east-1b (use1-az4)**

Subnet

subnet-0d9657a8e87d35348	Private Subnet 2 ▲
subnet-0d9657a8e87d35348	Private Subnet 2 ✓
subnet-09ca1c5f1b732a47f	Public Subnet 2

Security groups

Select up to 5 security groups ▲

Q

☐ c94925a2128894l5663577t1w587127467423-BastionSecurityGroup-11XKJ1HAC9EHQ
sg-0a67421016b34ceaf VPC: vpc-03930100a6f3b8d67

☐ Web Security Group
sg-0c58d7060d4fb5201 VPC: vpc-03930100a6f3b8d67

☒ default
sg-02ff832345d0b23f2 VPC: vpc-03930100a6f3b8d67

6. For the Listener HTTP:80 Row, set the default action to forward to LabGroup, and proceed to create view the load balancer.

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define to its registered targets.

▼ Listener HTTP:80

Protocol: HTTP Port: 80 (1-65535)

Default action: [Info](#)

Forward to: Select a target group

[Create target](#) Q

LabGroup
Target type: Instance, IPv4

Listener tags - optional

Cancel

Create load balancer



View load balancer

Task 3: Create a Launch Template and an Auto Scaling Group

1. In the left navigation plane, choose Launch templates.

▼ **Instances**

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

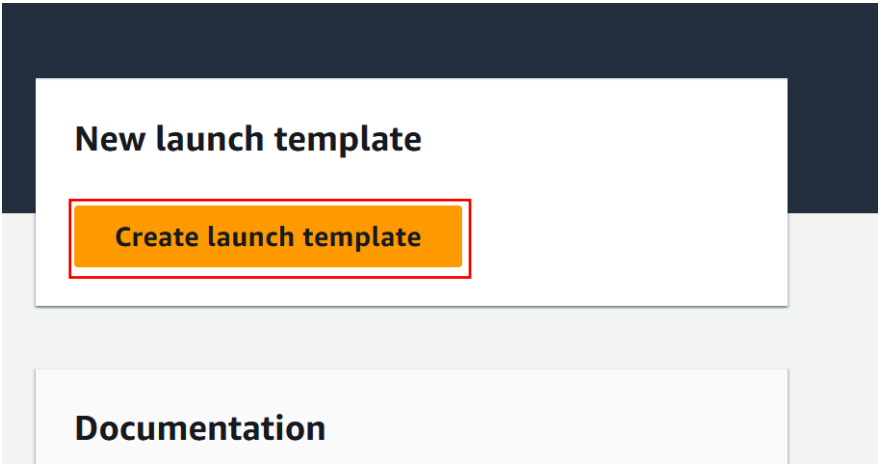
Reserved Instances

Dedicated Hosts

Capacity Reservations

[New](#)

2. Click Create launch template.



New launch template

Create launch template

Documentation

3. Name the Launch Template LabConfig, and select “Provide guidance to help me set up a template that I can use with EC2 Auto Scaling”.

Launch template name and description

Launch template name - *required*

LabConfig

Must be unique to this account. Max 128 chars. No spaces or spe

Auto Scaling guidance [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling



Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

4. Choose Web Server AMI as the AMI type, and choose t2.micro as the instance type.

▼ Application and OS Images (Amazon Machine Image) - required [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Q Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

☒ Owned by me

☐ Shared with me



Browse more AMIs

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

WebServerAMI

ami-088217f2a905baef0

2024-01-24T19:21:54.000Z

Virtualization: hvm ENA enabled: true Root device type: ebs

On-Demand Linux base pricing: 0.0116 USD per Hour

t2.micro Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.0716 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour	Free tier eligible
t2.small Family: t2 1 vCPU 2 GiB Memory Current generation: true On-Demand Windows base pricing: 0.032 USD per Hour On-Demand Linux base pricing: 0.023 USD per Hour On-Demand RHEL base pricing: 0.083 USD per Hour On-Demand SUSE base pricing: 0.053 USD per Hour	
t2.medium Family: t2 2 vCPU 4 GiB Memory Current generation: true	

Don't include in launch template

5. Choose vockey as the key pair name.

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access before you launch the instance.

Key pair name

Don't include in launch template

Q

Specify a custom value...

Don't include in launch template ✓

vockey
Type: rsa

vockey

6. Choose Select existing security group, and select Web Security Group.

▼ Network settings [Info](#)

Subnet [Info](#)

Don't include in launch template ▼

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow spe instance.

☒ Select existing security group

☐ Create security group

☒	Web Security Group VPC: vpc-03930100a6f3b8d67	sg-0c58d7060d4fb5201
☐	default VPC: vpc-0096a04ec748e3244	sg-0cc065eachb5133504
☐	default VPC: vpc-03930100a6f3b8d67	sg-02ff832345d0b23f2
☐	DB Security Group VPC: vpc-03930100a6f3b8d67	sg-044a8713b303b65b0
Select security groups		▲

► Advanced network configuration

7. Navigate to Advanced details and enable Detailed CloudWatch monitoring. Then create the launch template.

▶ Advanced details [Info](#)

Stop protection [Info](#)

Don't include in launch template ▼

Detailed CloudWatch monitoring [Info](#)

Don't include in launch template ▲

Don't include in launch template ✓

Enable

Disable

Storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Create launch template

8. In the success dialog, click the link to LabConfig.



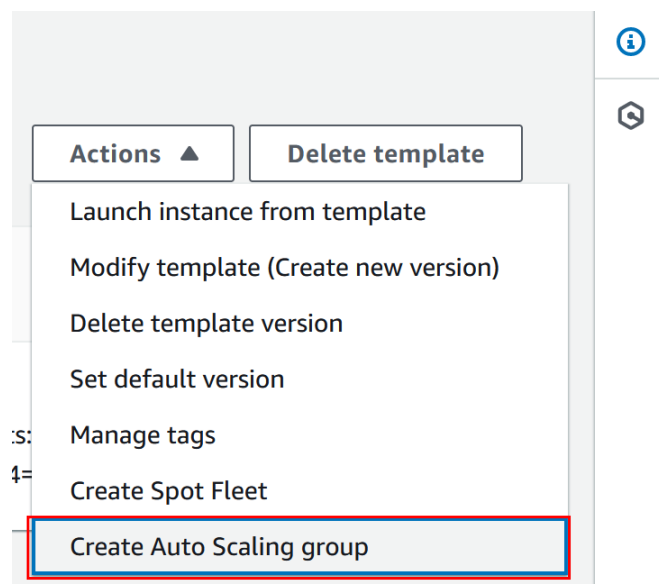
Success

Successfully created **LabConfig(lt-013a2380ddb0e4b96).**

► Actions log

Next Steps

9. From the actions menu, choose Create Auto Scaling Group.



10. Name it Lab Auto Scaling Group, and choose LabConfig as the Launch template.

Name

Auto Scaling group name

Enter a name to identify the group.

Lab Auto Scaling Group

Must be unique to this account in the current Region

Launch template

Choose a launch template that contains the instance security groups.

LabConfig

[Create a launch template](#) 

11. Choose LabVPC as the VPC, and select the two private subnets.

Network [Info](#)

For most applications, you can use multiple Availability Zones across the zones. The default VPC and default subnets are suitable for most applications.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-0096a04ec748e3244

172.31.0.0/16 Default



vpc-03930100a6f3b8d67 (Lab VPC)

10.0.0.0/16



us-east-1a | subnet-0e6004bd017a596ca (Private Subnet 1)

10.0.1.0/24



us-east-1a | subnet-01782562eb7300d63 (Public Subnet 1)

10.0.0.0/24



us-east-1b | subnet-0d9657a8e87d35348 (Private Subnet 2)

10.0.3.0/24



us-east-1b | subnet-09ca1c5f1b732a47f (Public Subnet 2)

10.0.2.0/24

Select Availability Zones and subnets



12. In Advanced options, choose Attach to an existing load balancer and choose LabGroup. Enable group metrics collection within CloudWatch as well.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer you define.



No load balancer

Traffic to your Auto Scaling group will not be fronted by a load balancer.



Attach to an existing load balancer

Choose from your existing load balancers.

Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.



Choose from your load balancer target groups

This option allows you to attach Application, Network, or Gateway Load Balancers.

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are shown.

Select target groups



LabGroup | HTTP

Application Load Balancer: LabELB

Additional settings

Monitoring | [Info](#)

☒ Enable group metrics collection within CloudWatch

13. Under group size, configure desired capacity to 2, minimum capacity to 2, and maximum capacity to 6. Under scaling policies, choose Target tracking scaling policy and configure the name to be LabScalingPolicy, metric type to be Average CPU utilization, and the target value to 60%. Then move on by selecting next.

Group size [Info](#)

Set the initial size of the Auto Scaling group.
automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity.
configured with a set of instance attributes.

Units (number of instances) ▼

Desired capacity

Specify your group size.

2

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changing demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

Equal or less than
desired capacity

Max desired capacity

6

Equal or greater than
desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling [after creating your Auto Scaling group](#)



No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.



Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Metric type [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using better scaling performance.

Target value

Add notifications - *optional* [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

14. When you get prompted to add tags, add a tag with the key Name and value Lab Instance.

Add tags - *optional* [Info](#)

Add tags to help you search, filter, and track your instances. You can add these tags to instances when they are launched.

i You can optionally choose to add tags to instances when you launch them. We recommend caution, as tags added to the launch template will be overridden if there are any tags added to the instances.

Tags (0)

50 remaining

Tags (1)

Key

Value - optional

15. Keep clicking next until you can Create Auto Scaling Group.

The screenshot shows two parts of the AWS console interface. The top part is a navigation bar with three buttons: 'Cancel', 'Previous', and 'Next'. The 'Next' button is highlighted with a red border. The bottom part is a confirmation dialog titled 'Tag new instances' with a 'Yes' button. Below the dialog is another navigation bar with three buttons: 'Cancel', 'Previous', and 'Create Auto Scaling group'. The 'Create Auto Scaling group' button is highlighted with a red border.

Task 4: Verify that Load Balancing is Working

1. Choose Instances in the navigation panel on the left. You will see two instances named Lab Instance. These were made by Auto Scaling.

- ▼ Instances
 - Instances**
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - [New](#)

Instances (6) [Info](#) Find Instance by attribute or tag (case-sensitive)

☐	Name 	Instance ID	Instance state 
☐	Lab Instance	i-00fb231a7efe9fecf	 Running  
☐	Lab Instance	i-0dd1e82de39aac7ab	 Running  

2. Choose Target Groups in the navigation plane and select LabGroup. Choose the Targets tab below, and wait until the status of both Lab Instances are healthy.

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores [New](#)

Target groups (1/1) [Info](#)

Q Filter target groups

☒	Name	ARN
☒	LabGroup	arn:aws:elasticloadbalanci...

Target group: LabGroup

Details

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2) [Info](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets with at least 3 healthy targets.

Q Filter targets

☐	Instance ID	Name	Port	Zone	Health status
☐	i-00fb231a7efe9fecf	Lab Instance	80	us-east-1b	☒ Healthy
☐	i-0dd1e82de39aac7ab	Lab Instance	80	us-east-1b	☒ Healthy

- Choose Load Balancers from the navigation plane and select LabELB. Copy the DNS name of the load balancer under the Details plane.

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores [New](#)

Load balancers (1/1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

☒	Name	DNS name	State
☒	LabELB	LabELB-461398690.us-east-1.elb.amazonaws.com	Active

VPC vpc-03930100a6f3b8d67 Availability Zones subnet-09ca1c5f1b732a47f us-east-1b (use1-az4) subnet-01782562eb7300d63 us-east-1a (use1-az2)	IP address type IPv4 Date created January 24, 2024, 11:30 (UTC-08:00)
---	---

DNS name [Info](#)

[LabELB-461398690.us-east-1.elb.amazonaws.com](#) (A Record)

4. Open a new web browser tab, and paste the DNS name you just copied.

[labelb-461398690.us-east-1.elb.amazonaws.com](#)

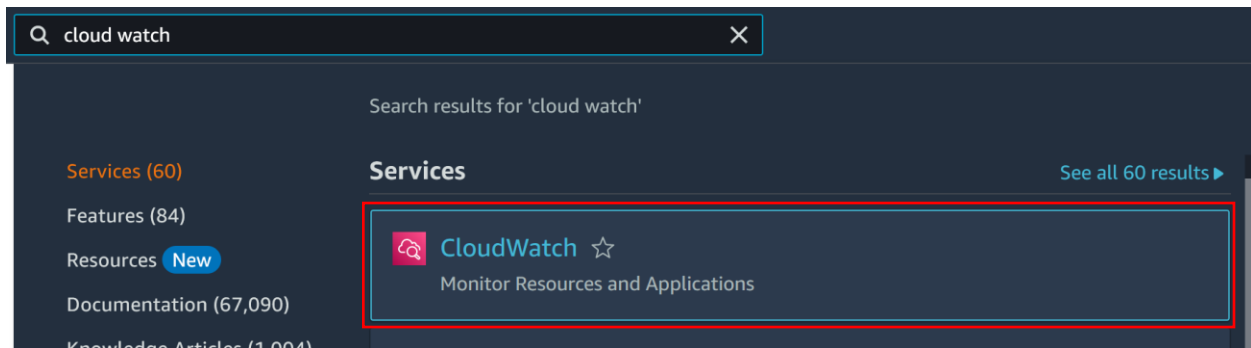
aws Load Test RDS

Meta-Data	Value
InstanceId	i-0dd1e82de39aac7ab
Availability Zone	us-east-1b

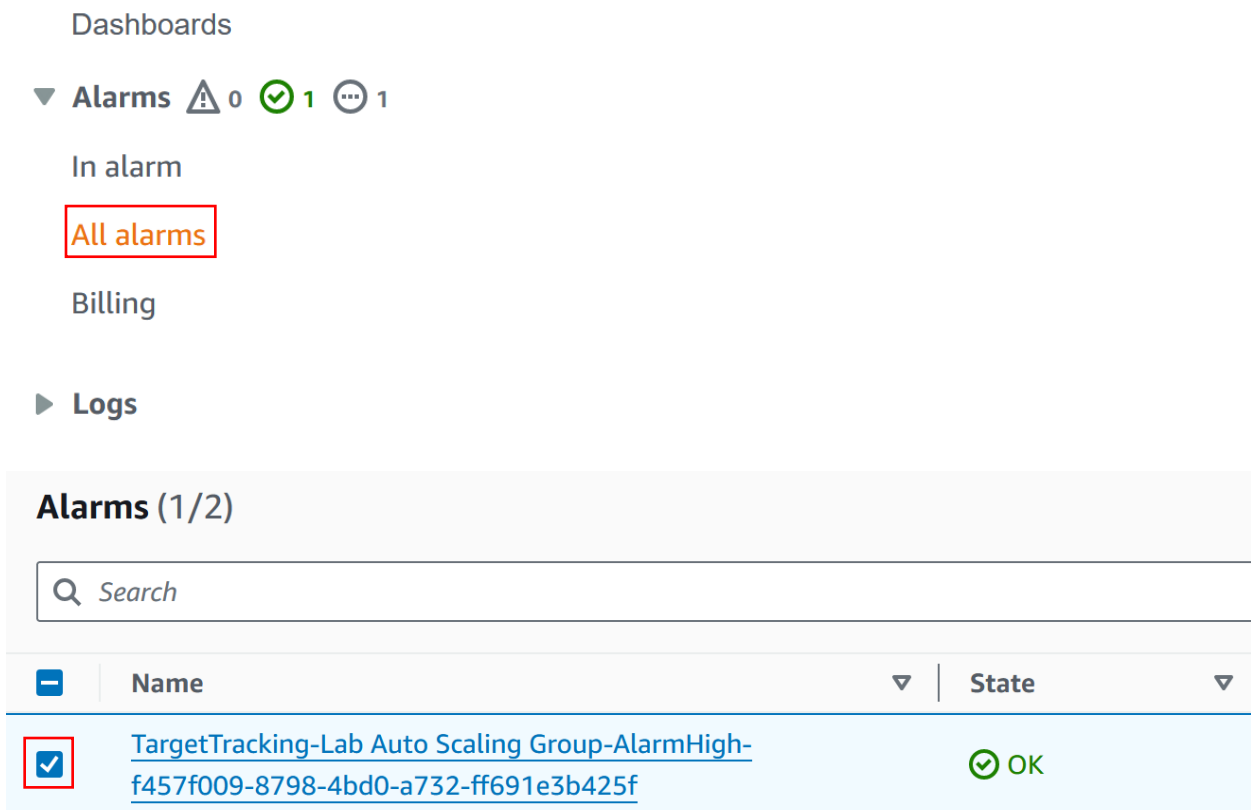
Current CPU Load: 0%

Task 5: Test Auto Scaling

1. Use the services menu to navigate to CloudWatch.



2. In the navigation plane, choose All Alarms. Choose the OK alarm, which has the AlarmHigh in its name.



3. Click on load test on the web browser page. This will test the instance under high load.

 Load Test RDS

Meta-Data	Value
InstanceId	<i>i-0dd1e82de39aac7ab</i>
Availability Zone	<i>us-east-1b</i>

Current CPU Load: **0%**

 Load Test RDS

Generating CPU Load! (auto refresh in 5 seconds)

Current CPU Load: **100%**

- Once the alarm with AlarmHigh enters the In Alarm state, you can now view the extra EC2 instances that were created by Auto Scaling.

Alarms (2)

☐	Name	State
☐	TargetTracking-Lab Auto Scaling Group-AlarmLow-11a6d8a7-7c54-4d1c-92d0-ecdb2d2f97c2	Insufficient data
☐	TargetTracking-Lab Auto Scaling Group-AlarmHigh-f457f009-8798-4bd0-a732-ff691e3b425f	In alarm

Services

Recently visited

Favorites

All services

Recently visited

★ **EC2**
Virtual Servers in the Cloud

Instances

Instances

Instance Types

Launch Templates

☐	Name	Instance ID	Instance state
☐	Lab Instance	i-00fb231a7efe9fecf	Running
☐	Lab Instance	i-0dd1e82de39aac7ab	Running
☐	Lab Instance	i-0f525640c2588d887	Running
☐	Lab Instance	i-08abbefb7c3295323	Running

Task 6: Terminate Web Server 1

1. Select Web Server 1. In the Instance state menu, choose Terminate Instance.

☐	Bastion Host	i-036072fb6e210ce80	Running
☒	Web Server 1	i-0cc9df45ef9c29a4a	Running
☐	Web Server 1	i-051fa8a3c879908ba	Terminated

Connect

Instance state

Ac

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate instance

Zone

Pu

Pul

Termination protection **Disabled**

the terminate button below. Instances with termination
instance cannot be undone.

Cancel**Terminate**

Problems

Once again, the labs themselves were quite straightforward, with each step generally being pretty well defined. There were still some cases where it took me a bit of time to find what prompt the instructions were talking about, but that is just an issue with design of the instructions. The problem with finding the buttons that the instructions were talking about will most likely go away with more practice with different AWS services.

Conclusion

AWS provides a cloud platform for multiple services to run on, providing many benefits over traditional methods of computing and database storage. Amazon EBS provides a fast block level storage system, perfect for web applications that require information as quickly as possible. Amazon RDS allows for the AWS user to use Amazon's own EBS instances as database servers, handing the maintenance and management tasks of the instances to Amazon, allowing them to focus on other tasks in AWS. Scaling and load balancing are integral parts of cloud computing, maximizing fault tolerance within a system and minimizing routing times by splitting up the workload between instances. These labs show how different services in AWS build upon each other and function together to provide a seamless experience for users.

Lab 6: Setting up an iBGP Network



Purpose

The primary purpose of this lab was to learn how to set up iBGP, a way for routers running BGP in the same area to communicate with each other and learn other routes.

Background Information on Lab Concepts

When trying to route traffic inside an autonomous system, network administrations generally use some sort of interior gateway protocol (IGP), such as OSPF, EIGRP, or RIP. These protocols, while being relatively fast, need to store a lot of information in their routing tables. This brings up the problem of scalability; when routing inside a very large autonomous system, the amount of information stores increases exponentially. This leads to a vastly increased amount of resources required to store all that information as the amount of routers go up. Running iBGP over one of these protocols seeks to solve that problem. iBGP, like the previously mentioned IGPs, allows for a full mesh topology, where every router in a network has adjacency to every other router. iBGP uses the loopback addresses of routers, which were learned through the IGP, in order to update its own routing tables and allow adjacency. This provides an increased scalability to the network that BGP provides, while also maintaining most of the speed that an IGP provides. This “full mesh” network allows for increased redundancy within a network, since every router has adjacency with every other router.

iBGP and eBGP, while both being BGP, have many differences between the two. The most basic difference is that routers running iBGP all have the same AS number, while routers running eBGP are all in different autonomous systems. This difference, however, doesn't really affect the commands required to set up adjacency between routers. eBGP also has an administrative distance of 20, while iBGP has an AD of 200, which means that eBGP is preferred when deciding between two routes. However, it is important to note that when only comparing two BGP routes, the BGP attributes are considered first before using administrative distance as a criterion. Because eBGP isn't dependent on another IGP and doesn't require redundancy, it is a more suitable choice for extremely large networks, such as the internet.

One use case of iBGP would be an internet service provider (ISP). An ISP would not use an IGP, simply because they need to route packets to too many different networks. However, they cannot use eBGP because they need to maintain redundancy inside of their network, in case a router goes down. iBGP fills the requirements for an ISP perfectly as it provides increased scalability compared to traditional IGPs, and it provides the redundancy that eBGP doesn't, ensuring their customers get as much internet as they can.

Lab Summary

Before starting with the physical lab, we first had to create a topology scheme on Packet Tracer, with each device and routing protocol area listed out. After creating the topology scheme, we then created a table with IPv4 addresses along with their respective subnet masks for each device port. After confirming that our subnetting and router topology would function with OSPF and BGP, we started to set up the physical routers with the same basic configurations such as a new device name, IP addresses, and loopback interfaces. After setting up basic configurations, we started to set up each IPv4 network, with things like assigning a router-id, and advertising the networks connected to each port. We then set up the IPv4 OSPF area and the iBGP area. After testing and seeing that each set of routers could obtain routes through the protocol, we set up eBGP on each of the network border routers. Once eBGP was setup, we pinged from edge router to edge router to verify all our routes were being redistributed properly.

Lab Commands

IP/ address [Address]: Sets an ipv4 address

IP OSPF [Process-id] area [Area Number]: Assigns that port to a OSPFv2 area

No Shutdown: Turns a port on

Router OSPF [Process-id]: Enables OSPFv2 routing on your router

Router-id [router-id]: Puts a router id on your router used for OSPF elections

Network [Network Number][Wildcard Mask] area [Area number]:

Sets a network address on the link in the area you select for ospf

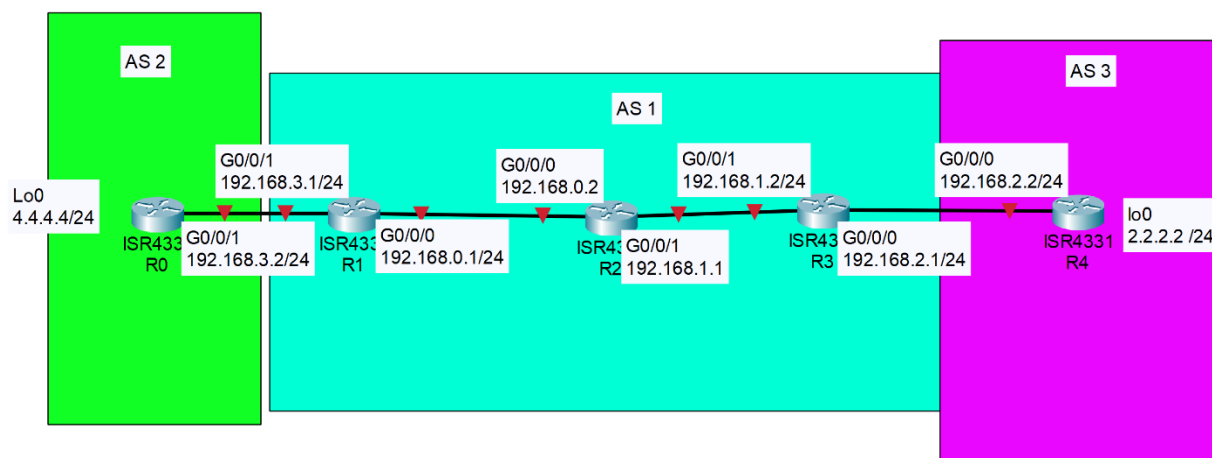
Redistribute connected: Redistributes connected routes to other routers in the area

Redistribute ospf: Redistributes IPv4 networks learned from OSPF through BGP

Network [IP] mask [subnet mask]: Advertise this network to all other BGP routers

Neighbor [IP] remote-as [as-number]: Attempts to create a BGP peer with this ip address

Network Diagram



Router Configurations

Router o:

```

Current configuration : 1532 bytes
!
! Last configuration change at 19:49:23 UTC Wed Feb 14 2024
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname JacobAidenAaronR0
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
  address-family ipv4
  exit-address-family
!
  address-family ipv6
  exit-address-family
!
!
no aaa new-model
!
subscriber templating
multilink bundle-name authenticated
!
license udi pid ISR4321/K9 sn FDO214421CF
!
spanning-tree extend system-id
!
!
redundancy
  mode none
!
!
vlan internal allocation policy ascending
!

```

```

interface Loopback0
  ip address 4.4.4.4 255.255.255.0
!
interface GigabitEthernet0/0/0
  no ip address
  negotiation auto
!
interface GigabitEthernet0/0/1
  ip address 192.168.3.2 255.255.255.0
  negotiation auto
!
interface Serial0/1/0
  no ip address
!
interface Serial0/1/1
  no ip address
!
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  negotiation auto
!
interface Vlan1
  no ip address
!
router bgp 2
  bgp log-neighbor-changes
  neighbor 192.168.3.1 remote-as 1
  !
  address-family ipv4
    network 4.4.4.0 mask 255.255.255.0
    network 192.168.3.0
    redistribute connected
    neighbor 192.168.3.1 activate
  exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
control-plane
!
!
line con 0
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
!
!

```

end

Router 1:

```

Current configuration : 1746 bytes
!
! Last configuration change at 20:06:22 UTC Wed Feb 14 2024
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname JacobAaronAidenR1
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
  address-family ipv4
  exit-address-family
!
  address-family ipv6
  exit-address-family
!
!
no aaa new-model
!
subscriber templating
multilink bundle-name authenticated
!
license udi pid ISR4321/K9 sn FDO211216BL
!
spanning-tree extend system-id
!
!
redundancy
  mode none
!
!
vlan internal allocation policy ascending
!
interface GigabitEthernet0/0/0
  ip address 192.168.0.1 255.255.255.0
  ip ospf 1 area 0

```

```

    negotiation auto
!
interface GigabitEthernet0/0/1
    ip address 192.168.3.1 255.255.255.0
    negotiation auto
!
interface Serial0/1/0
    no ip address
!
interface Serial0/1/1
    no ip address
!
interface GigabitEthernet0
    vrf forwarding Mgmt-intf
    no ip address
    negotiation auto
!
interface Vlan1
    no ip address
!
router ospf 1
    router-id 1.1.1.1
    redistribute bgp 1 subnets
!
router bgp 1
    bgp log-neighbor-changes
    neighbor 192.168.0.2 remote-as 1
    neighbor 192.168.3.2 remote-as 2
    !
    address-family ipv4
        network 2.2.2.0 mask 255.255.255.0
        network 4.4.4.0 mask 255.255.255.0
        network 192.168.0.0
        network 192.168.1.0
        network 192.168.2.0
        network 192.168.3.0
        redistribute ospf 1
        neighbor 192.168.0.2 activate
        neighbor 192.168.3.2 activate
    exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
control-plane
!
line con 0
    stopbits 1
line aux 0

```



```

    stopbits 1
line vty 0 4
    login
!
!
end

```

Router 2:

```

Current configuration : 1729 bytes
!
! Last configuration change at 20:05:28 UTC Wed Feb 14 2024
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname JacobAaronAidenR2
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
    address-family ipv4
    exit-address-family
!
    address-family ipv6
    exit-address-family
!
!
no aaa new-model
!
!
!
subscriber templating
multilink bundle-name authenticated
!
!
license udi pid ISR4321/K9 sn FDO214420G7
!
spanning-tree extend system-id
!
!

```

```

redundancy
  mode none
!
!
vlan internal allocation policy ascending
!
interface GigabitEthernet0/0/0
  ip address 192.168.0.2 255.255.255.0
  ip ospf 1 area 0
  negotiation auto
!
interface GigabitEthernet0/0/1
  ip address 192.168.1.1 255.255.255.0
  ip ospf 1 area 0
  negotiation auto
!
interface Serial0/1/0
!
interface Serial0/1/1
!
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 1
  router-id 2.2.2.2
!
router bgp 1
  bgp log-neighbor-changes
  neighbor 192.168.0.1 remote-as 1
  neighbor 192.168.1.2 remote-as 1
!
address-family ipv4
  network 2.2.2.0 mask 255.255.255.0
  network 4.4.4.0 mask 255.255.255.0
  network 192.168.0.0
  network 192.168.1.0
  network 192.168.2.0
  network 192.168.3.0

```

```

    redistribute connected
    neighbor 192.168.0.1 activate
    neighbor 192.168.1.2 activate
exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
control-plane
!
!
line con 0
    stopbits 1
line aux 0
    stopbits 1
line vty 0 4
    login
!
!
end

```

Router 3:

```

Current configuration : 1878 bytes
!
! Last configuration change at 20:20:57 UTC Wed Feb 14 2024
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname JacobAaronAidenR3
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
    address-family ipv4
    exit-address-family
!

```

```

    address-family ipv6
    exit-address-family
!
!
no aaa new-model
!
subscriber templating
multilink bundle-name authenticated
!
license udi pid ISR4321/K9 sn FDO21442B21
!
spanning-tree extend system-id
!
!
redundancy
    mode none
!
!
vlan internal allocation policy ascending
!
interface GigabitEthernet0/0/0
    ip address 192.168.2.1 255.255.255.0
    negotiation auto
!
interface GigabitEthernet0/0/1
    ip address 192.168.1.2 255.255.255.0
    ip ospf 1 area 0
    negotiation auto
!
interface Serial0/1/0
    no ip address
!
interface Serial0/1/1
    no ip address
!
interface GigabitEthernet0/2/0
    no ip address
    negotiation auto
!
interface GigabitEthernet0/2/1
    no ip address
    negotiation auto
!
interface GigabitEthernet0
    vrf forwarding Mgmt-intf

```

```

no ip address
negotiation auto
!
interface Vlan1
no ip address
!
router ospf 1
router-id 3.3.3.3
redistribute bgp 1 subnets
!
router bgp 1
bgp log-neighbor-changes
neighbor 192.168.1.1 remote-as 1
neighbor 192.168.2.2 remote-as 3
!
address-family ipv4
network 2.2.2.0 mask 255.255.255.0
network 4.4.4.0 mask 255.255.255.0
network 192.168.0.0
network 192.168.1.0
network 192.168.2.0
network 192.168.3.0
redistribute ospf 1
neighbor 192.168.1.1 activate
neighbor 192.168.2.2 activate
exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
control-plane
!
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
end

```

Router 4:

```

Current configuration : 1638 bytes
!
! Last configuration change at 19:47:15 UTC Wed Feb 14 2024
!
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
!
hostname JacobAaronAidenR4
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
  address-family ipv4
  exit-address-family
!
  address-family ipv6
  exit-address-family
!
!
no aaa new-model
!
!
login on-success log
!
subscriber templating
multilink bundle-name authenticated
!
license udi pid ISR4321/K9 sn FLM24060912
no license smart enable
diagnostic bootup level minimal
!
spanning-tree extend system-id

```

```

!
!
redundancy
  mode none
!
!!
interface Loopback0
  ip address 2.2.2.2 255.255.255.0
!
interface GigabitEthernet0/0/0
  ip address 192.168.2.2 255.255.255.0
  negotiation auto
!
interface GigabitEthernet0/0/1
  no ip address
  negotiation auto
!
interface GigabitEthernet0/2/0
  no ip address
  negotiation auto
!
interface GigabitEthernet0/2/1
  no ip address
  negotiation auto
!
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  negotiation auto
!
router bgp 3
  bgp log-neighbor-changes
  neighbor 192.168.2.1 remote-as 1
  !
  address-family ipv4
    network 2.2.2.0 mask 255.255.255.0
    network 192.168.2.0
    redistribute connected
    neighbor 192.168.2.1 activate
  exit-address-family
!
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0

```

```

!
!
control-plane
!
!
line con 0
  transport input none
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
!
!
end

```

IP and BGP Routes

Router o:

```

BGP table version is 14, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, >
best, i - internal,
                r RIB-failure, S Stale, m multipath, b backup-
path, f RT-Filter,
                x best-external, a additional-path, c RIB-
compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight
Path					
*>	2.2.2.0/24	192.168.3.1	1		0 1
i					
*>	4.4.4.0/24	0.0.0.0	0		32768 i
*>	192.168.0.0	192.168.3.1	0		0 1
i					
*>	192.168.1.0	192.168.3.1	2		0 1
i					
*>	192.168.2.0	192.168.3.1	1		0 1
i					
*	192.168.3.0	192.168.3.1	0		0 1
i					
*>		0.0.0.0	0		32768 i

Gateway of last resort is not set

```

      2.0.0.0/24 is subnetted, 1 subnets
B       2.2.2.0 [20/1] via 192.168.3.1, 00:31:39
      4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       4.4.4.0/24 is directly connected, Loopback0
L       4.4.4.4/32 is directly connected, Loopback0
B       192.168.0.0/24 [20/0] via 192.168.3.1, 00:44:57
B       192.168.1.0/24 [20/2] via 192.168.3.1, 00:38:49
B       192.168.2.0/24 [20/1] via 192.168.3.1, 00:36:16
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.3.2/32 is directly connected, GigabitEthernet0/0/1

```

Router 1:

BGP table version is 17, local router ID is 192.168.3.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

 x best-external, a additional-path, c RIB-compressed, Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	2.2.2.0/24	192.168.0.2	1		32768	i
* i		192.168.1.2	1	100	0	i
*>	4.4.4.0/24	192.168.3.2	0		0	2 i
* i	192.168.0.0	192.168.0.2	0	100	0	i
*>		0.0.0.0	0		32768	i
*>	192.168.1.0	192.168.0.2	2		32768	i
* i		192.168.0.2	0	100	0	i
*>	192.168.2.0	192.168.0.2	1		32768	i
* i		192.168.1.2	1	100	0	i
*	192.168.3.0	192.168.3.2	0		0	2 i
*>		0.0.0.0	0		32768	i

Gateway of last resort is not set

```

      2.0.0.0/24 is subnetted, 1 subnets
O E2     2.2.2.0 [110/1] via 192.168.0.2, 00:33:54,
GigabitEthernet0/0/0
      4.0.0.0/24 is subnetted, 1 subnets
B       4.4.4.0 [20/0] via 192.168.3.2, 00:38:59
      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks

```

```

C      192.168.0.0/24 is directly connected, GigabitEthernet0/0/0
L      192.168.0.1/32 is directly connected, GigabitEthernet0/0/0
O      192.168.1.0/24 [110/2] via 192.168.0.2, 00:43:55,
GigabitEthernet0/0/0
O E2   192.168.2.0/24 [110/1] via 192.168.0.2, 00:36:14,
GigabitEthernet0/0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.3.0/24 is directly connected, GigabitEthernet0/0/1
L      192.168.3.1/32 is directly connected, GigabitEthernet0/0/1

```

Router 2:

BGP table version is 29, local router ID is 192.168.1.1
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f
 RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	2.2.2.0/24	192.168.1.2	1		32768	i
* i		192.168.2.2	0	100	0	3 i
*>	4.4.4.0/24	192.168.0.1	1		32768	i
* i		192.168.3.2	0	100	0	2 i
* i	192.168.0.0	192.168.0.1	0	100	0	i
*>		0.0.0.0	0		32768	i
* i	192.168.1.0	192.168.1.2	0	100	0	i
*>		0.0.0.0	0		32768	i
* i	192.168.2.0	192.168.1.2	0	100	0	i
*>		192.168.1.2	1		32768	i
*>	192.168.3.0	192.168.0.1	1		32768	i
* i		192.168.0.1	0	100	0	i

Gateway of last resort is not set

```

      2.0.0.0/24 is subnetted, 1 subnets
O E2   2.2.2.0 [110/1] via 192.168.1.2, 00:32:49,
GigabitEthernet0/0/1
      4.0.0.0/24 is subnetted, 1 subnets
O E2   4.4.4.0 [110/1] via 192.168.0.1, 00:43:32,
GigabitEthernet0/0/0
      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.0.0/24 is directly connected, GigabitEthernet0/0/0
L      192.168.0.2/32 is directly connected, GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L      192.168.1.1/32 is directly connected, GigabitEthernet0/0/1

```

Router 3:

BGP table version is 7, local router ID is 192.168.2.1
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f
 RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	2.2.2.0/24	192.168.2.2	0			0 3 i
* i	4.4.4.0/24	192.168.0.1	1	100		0 i
*>		192.168.1.1	1		32768	i
*>	192.168.0.0	192.168.1.1	2		32768	i
* i		192.168.1.1	0	100		0 i
*>	192.168.1.0	0.0.0.0	0		32768	i
* i		192.168.1.1	0	100		0 i
*	192.168.2.0	192.168.2.2	0			0 3 i
*>		0.0.0.0	0		32768	i
*>	192.168.3.0	192.168.1.1	1		32768	i
* i		192.168.0.1	1	100		0 i

Gateway of last resort is not set

```

      2.0.0.0/24 is subnetted, 1 subnets
B       2.2.2.0 [20/0] via 192.168.2.2, 00:33:07
      4.0.0.0/24 is subnetted, 1 subnets
O E2    4.4.4.0 [110/1] via 192.168.1.1, 00:35:33,
GigabitEthernet0/0/1
O       192.168.0.0/24 [110/2] via 192.168.1.1, 00:35:33,
GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.1.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0/0
O E2    192.168.3.0/24 [110/1] via 192.168.1.1, 00:35:33,
GigabitEthernet0/0/1

```

Router 4:

BGP table version is 8, local router ID is 2.2.2.2
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f
 RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	2.2.2.0/24	0.0.0.0	0		32768	i
*>	4.4.4.0/24	192.168.2.1	1		0	1 i
*>	192.168.0.0	192.168.2.1	2		0	1 i
*>	192.168.1.0	192.168.2.1	0		0	1 i
*>	192.168.2.0	0.0.0.0	0		32768	i
*		192.168.2.1	0		0	1 i
*>	192.168.3.0	192.168.2.1	1		0	1 i

Gateway of last resort is not set

```

      2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       2.2.2.0/24 is directly connected, Loopback0
L       2.2.2.2/32 is directly connected, Loopback0
      4.0.0.0/24 is subnetted, 1 subnets
B       4.4.4.0 [20/1] via 192.168.2.1, 00:28:53
B       192.168.0.0/24 [20/2] via 192.168.2.1, 00:31:24
B       192.168.1.0/24 [20/0] via 192.168.2.1, 00:31:24
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.2.2/32 is directly connected, GigabitEthernet0/0/0
B       192.168.3.0/24 [20/1] via 192.168.2.1, 00:31:24

```

Problems

- iBGP neighbors do not advertise iBGP routes to other neighbors.

You need to use the update-source command in order to assign the source of a route to the loopback interface if two routers are not directly connected. You can also just manually add in routes to each router on every router.

- Configuring OSPF on the router vs. an interface

The commands are slightly different when configuring OSPF on a router vs just an interface. While OSPF worked when it was configured differently on two different routers in the same AS, it made debugging the router when a different problem occurred more difficult.

Conclusion

iBGP, while offering clear benefits and drawbacks compared to other routing protocols, is a very situational protocol. Most cases in which iBGP can be used are better solved by using another protocol such as OSPF or eBGP. iBGP sort of combines the redundancy/full mesh of IGPs, while retaining some of the scalability of eBGP. iBGP also requires the usage of route-reflectors or loopback interfaces when the network doesn't have physically directly adjacent routers in order to maintain that full mesh capability of a network running iBGP. The redundancy and scalability provided by iBGP makes it a prime option for ISPs to use when distributing internet.

Lab 7: Setting up a Cisco Aironet Access Point with RADIUS Authentication with EAP



Purpose:

The purpose of this lab was to learn how to setup a Cisco Aironet 1040 Series Wireless Access Point with two WPA2-PSK SSIDs and one WPA2-Enterprise SSID. The access point will be connected to a switch and a router for internet connectivity, as well as a RADIUS server for authentication for the Enterprise SSID.

Background:**Wireless Access Point (WAP):**

Wireless Access Points, like their name suggests, allows devices with wireless capabilities to connect to a network without the use of a physical Ethernet connection. APs create a seamless connection between wired and wireless networks and are commonly used in homes and businesses to provide easy internet connectivity.

RADIUS (Remote Authentication Dial-In User Service):

RADIUS is a networking protocol that provides centralized authorization and authentication for users who are trying to access a network. Network administrators are able to set up access credentials, such as a username and password, that allow users to connect to the network.

WPA2-PSK (WPA2-Personal):

WPA2-PSK is a simple security protocol to secure wireless networks. It requires users to use a PSK (Pre-Shared Key), which is like a universal password, in order to connect to the network. WPA2 also encrypts data transmissions between devices on a network, adding another layer of security. WPA2-PSK is typically used in small networks like homes and small businesses.

WPA2-Enterprise:

WPA2-Enterprise is a type of WPA2 that provides “enterprise-grade” security for wireless networks. WPA2-Enterprise uses an authentication server, in this case a RADIUS server, to authenticate devices that are trying to access the network. This allows network administrators to maintain a higher level of security as opposed to

simpler protocols like WPA2-PSK. These benefits make WPA2-Enterprise typically used in large networks, like corporate networks, universities, and government agencies.

Lab Summary:

First, we updated the firmware of the Access Point by creating a TFTP server on the local network. After updating the firmware, we created the three SSIDs and broadcast them on the 5Ghz radio. We then separated each of the SSIDs into different VLANs for security and organization purposes. The ports on the switch were then configured to be trunk ports in order to allow traffic from all VLANs to be sent to the router. After that, we then created three different DHCP pools, one for each of the SSIDs to distribute IP addresses to connected devices, as well as NAT on the router. At this point the two WPA2-PSK SSIDs are fully configured. For the remaining SSID, we need to create the RADIUS server that the SSID will be pointed towards in order to authenticate and authorize any users. In order to create a RADIUS server, we used FreeRadius on a device running with a Ubuntu operating system, assigned it an IP address, and directed the WPA2-Enterprise to use that IP in order to authenticate users.

Lab Commands:

Dot11 ssid [ssid name]: Creates an SSID on the access point with the specified name.

Mbssid guest-mode: Makes the access point start to broadcast the SSID for anyone to detect.

authentication open: Allows any device to try and connect to this SSID

authentication network-eap [list name]: Sets the authentication type for the SSID to Network-EAP

authentication key-management wpa version 2: Sets the authentication type for the SSID to WPA2.

Vlan [vlan number]: Assigns the SSID to the specified VLAN.

Encapsulation Dot1Q [vlan number]: Assigns the interface to the specified VLAN.

IP address [Address]: Sets an ipv4 address.

Radius server [server name]: Lets the access point know there is a RADIUS server to use for authentication.

address ipv4 [RADIUS server IP] auth-port 1812 acct-port 1813:
Specifies the IP of the RADIUS server.

Ip dhcp excluded-address [ip address]: Prevents the DHCP server from assigning this address automatically

Ip dhcp pool [pool name]: Creates a DHCP pool

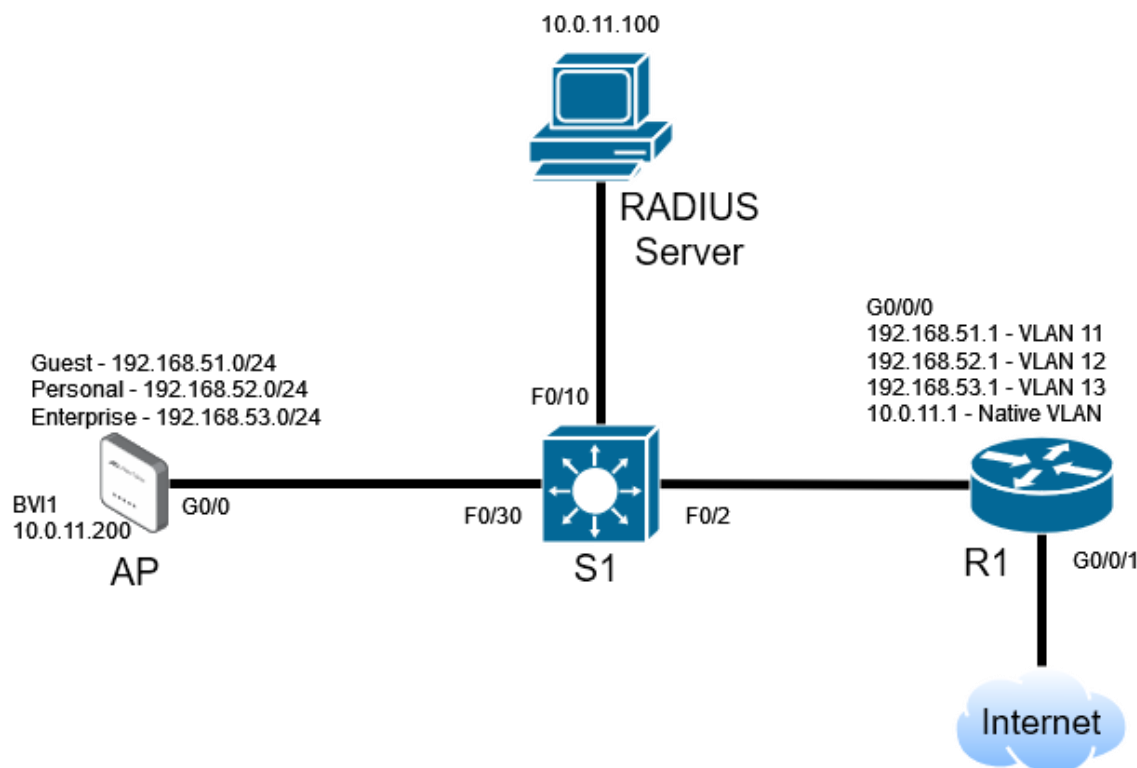
Default-router [ip address]: Tells any device who is using that DHCP pool to make the default gateway the specified IP.

Access-list [list number] permit [ip address] [wildcard number]: Adds a permit statement for a network onto an access list.

Ip nat inside: Makes the router translate the source address using NAT.

Ip nat outside: Makes the router translate the destination address using NAT.

Network Diagram:



Device Configurations:

Router:

```

Current configuration : 2406 bytes
!
! Last configuration change at 21:10:12 UTC Thu Jun 6 2024
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname R1
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
  address-family ipv4
  exit-address-family
!
  address-family ipv6
  exit-address-family
!
!
no aaa new-model
!
!

ip dhcp excluded-address 192.168.51.1
ip dhcp excluded-address 192.168.52.1
ip dhcp excluded-address 192.168.53.1
!
ip dhcp pool Guest
  network 192.168.51.0 255.255.255.0
  dns-server 1.1.1.1
  default-router 192.168.51.1
!
ip dhcp pool Psk
  network 192.168.52.0 255.255.255.0
  dns-server 1.1.1.1
  default-router 192.168.52.1
!

```

```

ip dhcp pool Enterprise
  network 192.168.53.0 255.255.255.0
  dns-server 1.1.1.1
  default-router 192.168.53.1
!
!
!
subscriber templating
multilink bundle-name authenticated
!
!
license udi pid ISR4321/K9 sn FDO211216BL
license accept end user agreement
!
spanning-tree extend system-id
!
!
redundancy
  mode none
!
!
vlan internal allocation policy ascending
!
!
interface GigabitEthernet0/0/0
  no ip address
  ip nat inside
  negotiation auto
!
interface GigabitEthernet0/0/0.1
  encapsulation dot1Q 1 native
  ip address 10.0.11.1 255.255.255.0
!
interface GigabitEthernet0/0/0.11
  encapsulation dot1Q 11
  ip address 192.168.51.1 255.255.255.0
  ip nat inside
!
interface GigabitEthernet0/0/0.12
  encapsulation dot1Q 12
  ip address 192.168.52.1 255.255.255.0
  ip nat inside
!
interface GigabitEthernet0/0/0.13
  encapsulation dot1Q 13
  ip address 192.168.53.1 255.255.255.0
  ip nat inside

```

```

!
interface GigabitEthernet0/0/1
  ip address dhcp
  ip nat outside
  negotiation auto
!
interface Serial0/1/0
  no ip address
  shutdown
!
interface Serial0/1/1
  no ip address
  shutdown
!
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto
!
interface Vlan1
  no ip address
!
ip nat inside source list 1 interface GigabitEthernet0/0/1
  overload
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
access-list 1 permit 192.168.51.0 0.0.0.255
access-list 1 permit 192.168.52.0 0.0.0.255
access-list 1 permit 192.168.53.0 0.0.0.255
!
!
control-plane
!
!
line con 0
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
!
ntp server pool.ntp.org

```

```
!
end
```

Switch:

```
Current configuration : 2381 bytes
!
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname S1
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
system mtu routing 1500
authentication mac-move permit
ip subnet-zero
!
!
spanning-tree mode pvst
spanning-tree etherchannel guard misconfig
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
!
interface FastEthernet0/1
 shutdown
!
interface FastEthernet0/2
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 1,11-13
 switchport mode trunk
!
interface FastEthernet0/3
!
interface FastEthernet0/4
```

```
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!  
interface FastEthernet0/9  
!  
interface FastEthernet0/10  
    switchport mode access  
!  
interface FastEthernet0/11  
!  
interface FastEthernet0/12  
!  
interface FastEthernet0/13  
!  
interface FastEthernet0/14  
!  
interface FastEthernet0/15  
!  
interface FastEthernet0/16  
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24
```

```
!  
interface FastEthernet0/25  
!  
interface FastEthernet0/26  
!  
interface FastEthernet0/27  
!  
interface FastEthernet0/28  
!  
interface FastEthernet0/29  
!  
interface FastEthernet0/30  
    switchport trunk encapsulation dot1q  
    switchport trunk allowed vlan 1,11-13  
    switchport mode trunk  
!  
interface FastEthernet0/31  
!  
interface FastEthernet0/32  
!  
interface FastEthernet0/33  
!  
interface FastEthernet0/34  
!  
interface FastEthernet0/35  
!  
interface FastEthernet0/36  
!  
interface FastEthernet0/37  
!  
interface FastEthernet0/38  
!  
interface FastEthernet0/39  
!  
interface FastEthernet0/40  
!  
interface FastEthernet0/41  
!  
interface FastEthernet0/42  
!  
interface FastEthernet0/43
```



```

!
interface FastEthernet0/44
!
interface FastEthernet0/45
!
interface FastEthernet0/46
!
interface FastEthernet0/47
!
interface FastEthernet0/48
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface GigabitEthernet0/3
!
interface GigabitEthernet0/4
!
interface Vlan1
  no ip address
  shutdown
!
ip classless
ip http server
ip http secure-server
!
!
ip sla enable reaction-alerts
!
!
line con 0
line vty 0 4
  login
line vty 5 15
  login
!
end

```

Access Point:

Current configuration : 4029 bytes

```

!
! Last configuration change at 04:58:51 UTC Fri Mar 1 2002
version 15.3
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname AP
!
!
logging rate-limit console 9
enable secret 5 $1$ySJC$WKGEn6ausDAtFkMop0BIE1
!
aaa new-model
!
!
aaa group server radius AP_RAD_GROUP
    server name Jacob_PC
!
aaa authentication login eap_methods group AP_RAD_GROUP
!
!
aaa session-id common
no ip source-route
no ip cef
no ip domain lookup
!
!
dot11 pause-time 100
dot11 syslog
!
dot11 ssid JacobAaronAiden_67_CCNP_Ent
    vlan 13
    authentication open eap eap_methods
    authentication network-eap eap_methods
    authentication key-management wpa version 2
    mbssid guest-mode
!
dot11 ssid JacobAaronAiden_67_CCNP_Guest
    vlan 11

```

```
    authentication open
    mbssid guest-mode
!
dot11 ssid JacobAaronAiden_67_CCNP_Personal
    vlan 12
    authentication open
    authentication key-management wpa version 2
    mbssid guest-mode
    wpa-psk ascii 7 106D000A061843595F
!
!
eap profile AUTH_METHODS
    method mschapv2
!
no ipv6 cef
!
!
username Cisco password 7 01300F175804
!
!
bridge irb
!
!
interface Dot11Radio0
    no ip address
    shutdown
    antenna gain 0
    station-role root
    bridge-group 1
    bridge-group 1 subscriber-loop-control
    bridge-group 1 spanning-disabled
    bridge-group 1 block-unknown-source
    no bridge-group 1 source-learning
    no bridge-group 1 unicast-flooding
!
interface Dot11Radio1
    no ip address
    !
    encryption vlan 12 mode ciphers aes-ccm
    !
    encryption vlan 13 mode ciphers aes-ccm
```

```

!
ssid JacobAaronAiden_67_CCNP_Ent
!
ssid JacobAaronAiden_67_CCNP_Guest
!
ssid JacobAaronAiden_67_CCNP_Personal
!
antenna gain 0
peakdetect
dfs band 3 block
mbssid
channel dfs
station-role root
!
interface Dot11Radio1.1
 encapsulation dot1Q 1 native
 bridge-group 1
 bridge-group 1 subscriber-loop-control
 bridge-group 1 spanning-disabled
 bridge-group 1 block-unknown-source
 no bridge-group 1 source-learning
 no bridge-group 1 unicast-flooding
!
interface Dot11Radio1.11
 encapsulation dot1Q 11
 bridge-group 11
 bridge-group 11 subscriber-loop-control
 bridge-group 11 spanning-disabled
 bridge-group 11 block-unknown-source
 no bridge-group 11 source-learning
 no bridge-group 11 unicast-flooding
!
interface Dot11Radio1.12
 encapsulation dot1Q 12
 bridge-group 12
 bridge-group 12 subscriber-loop-control
 bridge-group 12 spanning-disabled
 bridge-group 12 block-unknown-source
 no bridge-group 12 source-learning
 no bridge-group 12 unicast-flooding
!

```

```
interface Dot11Radio1.13
  encapsulation dot1Q 13
  bridge-group 13
  bridge-group 13 subscriber-loop-control
  bridge-group 13 spanning-disabled
  bridge-group 13 block-unknown-source
  no bridge-group 13 source-learning
  no bridge-group 13 unicast-flooding
!
interface GigabitEthernet0
  no ip address
  duplex auto
  speed auto
  dot1x pae authenticator
!
interface GigabitEthernet0.1
  encapsulation dot1Q 1 native
  bridge-group 1
  bridge-group 1 spanning-disabled
  no bridge-group 1 source-learning
!
interface GigabitEthernet0.11
  encapsulation dot1Q 11
  bridge-group 11
  bridge-group 11 spanning-disabled
  no bridge-group 11 source-learning
!
interface GigabitEthernet0.12
  encapsulation dot1Q 12
  bridge-group 12
  bridge-group 12 spanning-disabled
  no bridge-group 12 source-learning
!
interface GigabitEthernet0.13
  encapsulation dot1Q 13
  bridge-group 13
  bridge-group 13 spanning-disabled
  no bridge-group 13 source-learning
!
interface BVI1
  mac-address 44d3.ca03.7d9b
```

```

ip address 10.0.11.200 255.255.255.0
ipv6 address dhcp
ipv6 address autoconfig
ipv6 enable
!
ip forward-protocol nd
ip http server
no ip http secure-server
ip http help-path
http://www.cisco.com/warp/public/779/smbiz/prodconfig/help/eag
!
!
!
radius server Jacob_PC
  address ipv4 10.0.11.100 auth-port 1812 acct-port 1813
  timeout 10
  retransmit 3
  key 7 111D1C16031B050B557878
!
bridge 1 route ip
!
!
wlccp ap eap profile AUTH_METHODS
!
line con 0
  logging synchronous
line vty 0 4
  transport input all
!
end

```

IP Routes:

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 192.168.40.1 to network 0.0.0.0

```

S*    0.0.0.0/0 [254/0] via 192.168.40.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.0.11.0/24 is directly connected,
GigabitEthernet0/0/0.1
L      10.0.11.1/32 is directly connected,
GigabitEthernet0/0/0.1
C      192.168.40.0/23 is directly connected,
GigabitEthernet0/0/1
      192.168.40.0/32 is subnetted, 1 subnets
L      192.168.40.236 is directly connected,
GigabitEthernet0/0/1
      192.168.51.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.51.0/24 is directly connected,
GigabitEthernet0/0/0.11
L      192.168.51.1/32 is directly connected,
GigabitEthernet0/0/0.11
      192.168.52.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.52.0/24 is directly connected,
GigabitEthernet0/0/0.12
L      192.168.52.1/32 is directly connected,
GigabitEthernet0/0/0.12
      192.168.53.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.53.0/24 is directly connected,
GigabitEthernet0/0/0.13
L      192.168.53.1/32 is directly connected,
GigabitEthernet0/0/0.13
  
```

DHCP Bindings:

Bindings from all pools not associated with VRF:

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type	State	Interface
192.168.51.3	0114.abc5.36cc.2a	Jun 07 2024 09:39 PM	Automatic	Active	GigabitEthernet0/0/0.11
192.168.52.2	01ac.ed5c.3e56.89	Jun 07 2024 09:08 PM	Automatic	Active	GigabitEthernet0/0/0.12
192.168.52.3	0114.abc5.36cc.2a	Jun 07 2024 09:40 PM	Automatic	Active	GigabitEthernet0/0/0.12
192.168.53.2	01ac.ed5c.3e56.89	Jun 07 2024 05:52 PM	Automatic	Active	GigabitEthernet0/0/0.13

NAT Translations:

```

Pro  Inside global          Inside local          Outside local
-----
Outside global
tcp  192.168.40.236:4603    192.168.53.2:59045
104.16.80.230:443        104.16.80.230:443
tcp  192.168.40.236:4691    192.168.52.2:64828
44.198.23.94:443        44.198.23.94:443
tcp  192.168.40.236:4614    192.168.51.2:52875
104.16.80.230:443        104.16.80.230:443
tcp  192.168.40.236:4613    192.168.51.2:52868
104.16.80.230:443        104.16.80.230:443
tcp  192.168.40.236:4619    192.168.52.2:64709
20.25.227.174:443        20.25.227.174:443
tcp  192.168.40.236:4675    192.168.52.2:64776
52.73.235.185:443        52.73.235.185:443
tcp  192.168.40.236:4604    192.168.51.2:56532
184.28.50.162:443        184.28.50.162:443
tcp  192.168.40.236:4696    192.168.52.2:64842
34.199.53.216:443        34.199.53.216:443
tcp  192.168.40.236:4605    192.168.51.2:56534
35.186.224.39:443        35.186.224.39:443
tcp  192.168.40.236:4608    192.168.53.2:53519
104.17.143.163:443        104.17.143.163:443
tcp  192.168.40.236:4580    192.168.51.2:52838
104.16.80.230:443        104.16.80.230:443

```

Problems:

Accidentally using guest-mode instead of mbssid guest-mode:

Using guest-mode only allows the AP to broadcast one SSID. In order to broadcast more than 1 SSID, you must use the mbssid guest-mode.

WSL and Docker didn't host a RADIUS server properly:

Still don't know why these options for RADIUS servers didn't work properly, so we had to switch to using a PC running Ubuntu as its OS in order to use FreeRadius.

Conclusion:

Wireless Access Points can provide seamless connections between wireless and wired connections, while also bringing along some security risks. In order to improve the security of these wireless connections, different security protocols must be put in place. For a small network, where security may not be as important, a simple WPA2-PSK can be used, requiring just a universal password to connect to the network. In larger networks like universities or large businesses, where security may be more paramount, WPA2-Enterprise can be set up, requiring unique usernames and passwords to be used.

Lab 8: Splitting a Network Using VRF



Purpose:

The purpose of this lab was to learn how to set up VRF on a network with a limited number of routers on GNS3.

Background:

Network Segmenting is the practice of splitting a large network into multiple, smaller networks. This practice provides many benefits, such as minimizing the damage caused by security breaches when any flooding attack is involved, as well as easier overall management.

VRF, or Virtual Routing and Forwarding, is a practice used by network administrators to do so. It creates multiple, smaller routing tables on each router instead of one large one, preventing networks from accessing certain other networks. All of this takes place on one router, which can potentially save resources when deploying a network. Routers running VRF are essentially the layer 3 equivalent of VLANs. VRF is generally used with MPLS VPNs, or Multiprotocol Label Switching VPNs. When not used with these VPNs, VRF turns into VRF-lite. In VRF-lite, the VRF commands must be applied manually on every device in the network. This may be ideal for small to medium sized networks but is not recommended for large scale networks. With MPLS VPNs, VRF is applied without the manual commands of a network administrator, making it very scalable and suitable for large scale networks. For the purposes of this lab, we will be using VRF-lite.

Lab Commands:

Ip vrf [vrf name]: Creates the VRF process on a router.

Ip vrf forwarding [vrf name]: Assigns the interface to route on the specified VRF process.

Ip address [ip address] [subnet mask]: Assigns an IP address to an interface.

Router ospf [process number] vrf [vrf name]: Creates an OSPF process to route only on the specified VRF process.

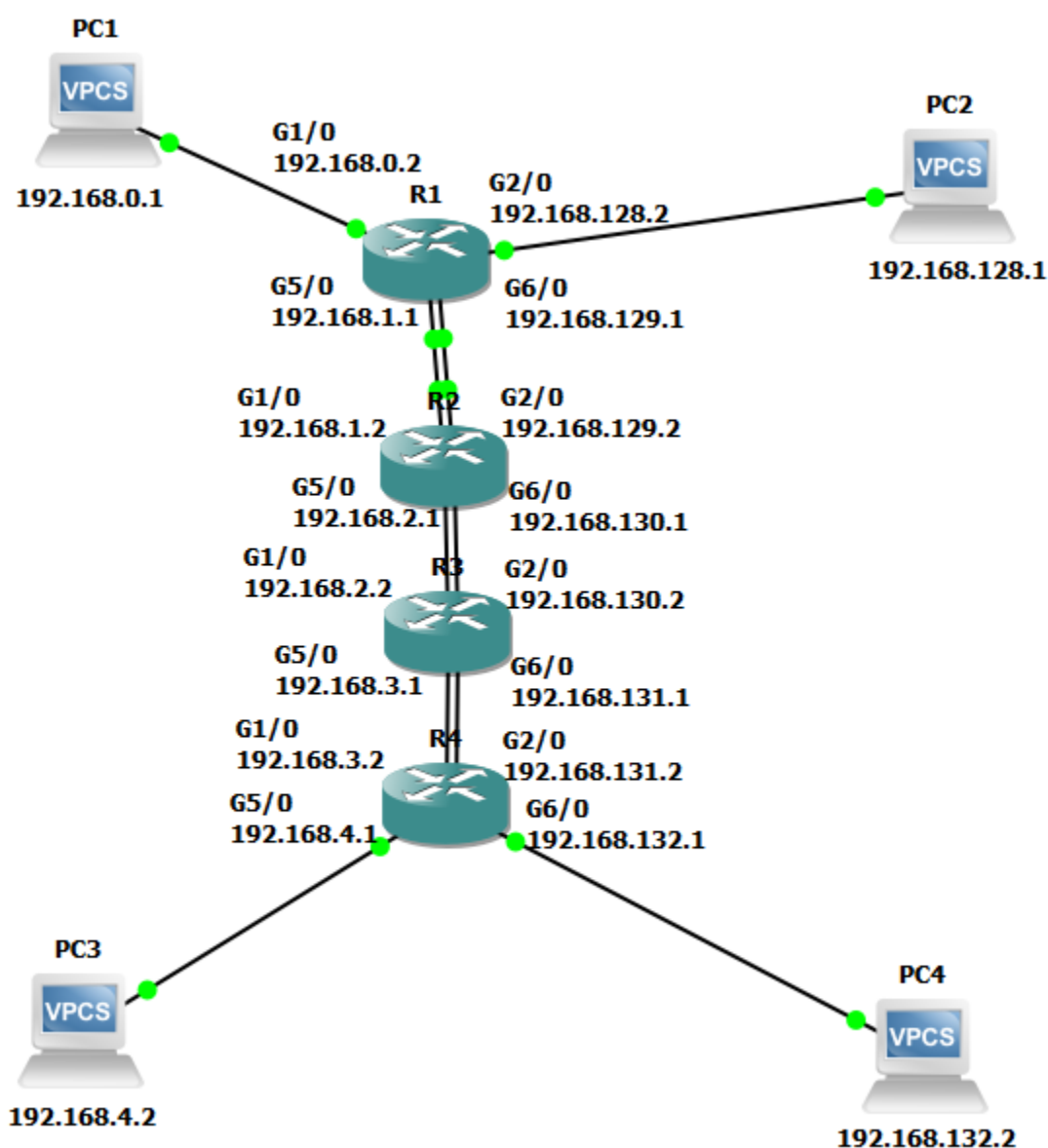
Network [network number] [wildcard number] area [area number]: Adds the specified network to be broadcasted by OSPF.

Capability vrf-lite: Specifies that the VRF process will be running VRF-lite.

Lab Summary:

We first downloaded GNS3 off of the internet, which is like Packet Tracer, a network simulator. We then downloaded a bin file for a suitable router, which in this case was a c7200 series router because it had enough ethernet ports for the lab. After importing the router, we setup the network in the same configuration in the diagram as seen below. After connecting every component, we just entered the commands for VRF, and used OSPF to broadcast networks across the VRF processes to allow other routers to discover the network.

Network Diagram:



Device Configurations:**Router 1:**

```
ip vrf VRF1
ip vrf VRF2

interface g1/0
ip vrf forwarding VRF1
ip address 192.168.0.2 255.255.255.0
no shut

interface g2/0
ip vrf forwarding VRF2
ip address 192.168.128.2 255.255.255.0
no shut

interface g5/0
ip vrf forwarding VRF1
ip address 192.168.1.1 255.255.255.0
no shut

interface g6/0
ip vrf forwarding VRF2
ip address 192.168.129.1 255.255.255.0
no shut

router ospf 1 vrf VRF1
network 192.168.0.0 0.0.0.255 area 0
network 192.168.1.0 0.0.0.255 area 0
capability vrf-lite

router ospf 2 vrf VRF2
network 192.168.128.0 0.0.0.255 area 0
network 192.168.129.0 0.0.0.255 area 0
capability vrf-lite
```

Router 2:

```
ip vrf VRF1
ip vrf VRF2

interface g1/0
ip vrf forwarding VRF1
ip address 192.168.1.2 255.255.255.0
no shut

interface g2/0
ip vrf forwarding VRF2
```

```
ip address 192.168.129.2 255.255.255.0
no shut
```

```
interface g5/0
ip vrf forwarding VRF1
ip address 192.168.2.1 255.255.255.0
no shut
```

```
interface g6/0
ip vrf forwarding VRF2
ip address 192.168.130.1 255.255.255.0
no shut
```

```
router ospf 1 vrf VRF1
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
capability vrf-lite
```

```
router ospf 2 vrf VRF2
network 192.168.129.0 0.0.0.255 area 0
network 192.168.130.0 0.0.0.255 area 0
capability vrf-lite
```

Router 3:

```
ip vrf VRF1
ip vrf VRF2
```

```
interface g1/0
ip vrf forwarding VRF1
ip address 192.168.2.2 255.255.255.0
no shut
```

```
interface g2/0
ip vrf forwarding VRF2
ip address 192.168.130.2 255.255.255.0
no shut
```

```
interface g5/0
ip vrf forwarding VRF1
ip address 192.168.3.1 255.255.255.0
no shut
```

```
interface g6/0
ip vrf forwarding VRF2
ip address 192.168.131.1 255.255.255.0
no shut
```

```
router ospf 1 vrf VRF1
network 192.168.2.0 0.0.0.255 area 0
network 192.168.3.0 0.0.0.255 area 0
capability vrf-lite
```

```
router ospf 2 vrf VRF2
network 192.168.130.0 0.0.0.255 area 0
network 192.168.131.0 0.0.0.255 area 0
capability vrf-lite
```

Router 4:

```
ip vrf VRF1
ip vrf VRF2
```

```
interface g1/0
ip vrf forwarding VRF1
ip address 192.168.3.2 255.255.255.0
no shut
```

```
interface g2/0
ip vrf forwarding VRF2
ip address 192.168.131.2 255.255.255.0
no shut
```

```
interface g5/0
ip vrf forwarding VRF1
ip address 192.168.4.1 255.255.255.0
no shut
```

```
interface g6/0
ip vrf forwarding VRF2
ip address 192.168.132.1 255.255.255.0
no shut
```

```
router ospf 1 vrf VRF1
network 192.168.3.0 0.0.0.255 area 0
network 192.168.4.0 0.0.0.255 area 0
capability vrf-lite
```

```
router ospf 2 vrf VRF2
network 192.168.131.0 0.0.0.255 area 0
network 192.168.132.0 0.0.0.255 area 0
capability vrf-lite
```

VRF IP Routes:

Router 1:

Routing Table: VRF1

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

+ - replicated route, % - next hop override

Gateway of last resort is not set

192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.0.0/24 is directly connected,

GigabitEthernet1/0

L 192.168.0.2/32 is directly connected,

GigabitEthernet1/0

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected,

GigabitEthernet5/0

L 192.168.1.1/32 is directly connected,

GigabitEthernet5/0

O 192.168.2.0/24 [110/2] via 192.168.1.2, 00:01:22,

GigabitEthernet5/0

O 192.168.3.0/24 [110/3] via 192.168.1.2, 00:01:00,

GigabitEthernet5/0

O 192.168.4.0/24 [110/4] via 192.168.1.2, 00:00:50,

GigabitEthernet5/0

Routing Table: VRF2

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP,
 l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```

      192.168.128.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.128.0/24 is directly connected,
GigabitEthernet2/0
L       192.168.128.2/32 is directly connected,
GigabitEthernet2/0
      192.168.129.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.129.0/24 is directly connected,
GigabitEthernet6/0
L       192.168.129.1/32 is directly connected,
GigabitEthernet6/0
O       192.168.130.0/24 [110/2] via 192.168.129.2, 00:03:10,
GigabitEthernet6/0
O       192.168.131.0/24 [110/3] via 192.168.129.2, 00:02:48,
GigabitEthernet6/0
O       192.168.132.0/24 [110/4] via 192.168.129.2, 00:02:38,
GigabitEthernet6/0

```

Router 2:

Routing Table: VRF1

Codes: L - local, C - connected, S - static, R - RIP, M -
 mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
 area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
 type 2

E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
 IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-
 user static route

o - ODR, P - periodic downloaded static route, H - NHRP,
 l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```

O       192.168.0.0/24 [110/2] via 192.168.1.1, 00:04:03,
GigabitEthernet1/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected,
GigabitEthernet1/0

```

```

L      192.168.1.2/32 is directly connected,
GigabitEthernet1/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.2.0/24 is directly connected,
GigabitEthernet5/0
L      192.168.2.1/32 is directly connected,
GigabitEthernet5/0
O      192.168.3.0/24 [110/2] via 192.168.2.2, 00:03:30,
GigabitEthernet5/0
O      192.168.4.0/24 [110/3] via 192.168.2.2, 00:03:20,
GigabitEthernet5/0

```

Routing Table: VRF2

Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-
user static route

o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP

+ - replicated route, % - next hop override

Gateway of last resort is not set

```

O      192.168.128.0/24 [110/2] via 192.168.129.1, 00:04:23,
GigabitEthernet2/0
      192.168.129.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.129.0/24 is directly connected,
GigabitEthernet2/0
L      192.168.129.2/32 is directly connected,
GigabitEthernet2/0
      192.168.130.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.130.0/24 is directly connected,
GigabitEthernet6/0
L      192.168.130.1/32 is directly connected,
GigabitEthernet6/0
O      192.168.131.0/24 [110/2] via 192.168.130.2, 00:03:50,
GigabitEthernet6/0
O      192.168.132.0/24 [110/3] via 192.168.130.2, 00:03:40,
GigabitEthernet6/0

```

Router 3:

Routing Table: VRF1

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

+ - replicated route, % - next hop override

Gateway of last resort is not set

O 192.168.0.0/24 [110/3] via 192.168.2.1, 00:04:41, GigabitEthernet1/0

O 192.168.1.0/24 [110/2] via 192.168.2.1, 00:04:41, GigabitEthernet1/0

192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.2.0/24 is directly connected, GigabitEthernet1/0

L 192.168.2.2/32 is directly connected, GigabitEthernet1/0

192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.3.0/24 is directly connected, GigabitEthernet5/0

L 192.168.3.1/32 is directly connected, GigabitEthernet5/0

O 192.168.4.0/24 [110/2] via 192.168.3.2, 00:04:19, GigabitEthernet5/0

Routing Table: VRF2

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```
O      192.168.128.0/24 [110/3] via 192.168.130.1, 00:05:05,
GigabitEthernet2/0
O      192.168.129.0/24 [110/2] via 192.168.130.1, 00:05:05,
GigabitEthernet2/0
      192.168.130.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.130.0/24 is directly connected,
GigabitEthernet2/0
L      192.168.130.2/32 is directly connected,
GigabitEthernet2/0
      192.168.131.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.131.0/24 is directly connected,
GigabitEthernet6/0
L      192.168.131.1/32 is directly connected,
GigabitEthernet6/0
O      192.168.132.0/24 [110/2] via 192.168.131.2, 00:04:43,
GigabitEthernet6/0
```

Router 4:

Routing Table: VRF1

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

+ - replicated route, % - next hop override

Gateway of last resort is not set

```
O      192.168.0.0/24 [110/4] via 192.168.3.1, 00:05:22,
GigabitEthernet1/0
O      192.168.1.0/24 [110/3] via 192.168.3.1, 00:05:22,
GigabitEthernet1/0
```

```

O      192.168.2.0/24 [110/2] via 192.168.3.1, 00:05:22,
GigabitEthernet1/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.3.0/24 is directly connected,
GigabitEthernet1/0
L      192.168.3.2/32 is directly connected,
GigabitEthernet1/0
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.4.0/24 is directly connected,
GigabitEthernet5/0
L      192.168.4.1/32 is directly connected,
GigabitEthernet5/0

```

```

O      192.168.0.0/24 [110/4] via 192.168.3.1, 00:05:22,
GigabitEthernet1/0
O      192.168.1.0/24 [110/3] via 192.168.3.1, 00:05:22,
GigabitEthernet1/0
O      192.168.2.0/24 [110/2] via 192.168.3.1, 00:05:22,
GigabitEthernet1/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.3.0/24 is directly connected,
GigabitEthernet1/0
L      192.168.3.2/32 is directly connected,
GigabitEthernet1/0
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.4.0/24 is directly connected,
GigabitEthernet5/0
L      192.168.4.1/32 is directly connected,
GigabitEthernet5/0

```

Routing Table: VRF2

Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-
user static route

o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP

+ - replicated route, % - next hop override

Gateway of last resort is not set

```

O      192.168.128.0/24 [110/4] via 192.168.131.1, 00:05:41,
GigabitEthernet2/0
O      192.168.129.0/24 [110/3] via 192.168.131.1, 00:05:41,
GigabitEthernet2/0
O      192.168.130.0/24 [110/2] via 192.168.131.1, 00:05:41,
GigabitEthernet2/0
      192.168.131.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.131.0/24 is directly connected,
GigabitEthernet2/0
L      192.168.131.2/32 is directly connected,
GigabitEthernet2/0
      192.168.132.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.132.0/24 is directly connected,
GigabitEthernet6/0
L      192.168.132.1/32 is directly connected,
GigabitEthernet6/0

```

Problems:

No problems were encountered during the creation of this lab.

Conclusion:

Overall, VRF is pretty simple to setup. You only need to adjust a few commands by adding “vrf” in them. Network segmentation is essential to basic network security, and network administrators can do so by using things like VRF or VLANs. VRF without the usage of MPLS VPNs isn’t highly scalable and is only suitable for small to medium scale networks. Using VRF on large-scale networks necessitates the usage of MPLS VPNs in order to make VRF scalable and applied on every router without direct configuration by the network administrator.

Factory Resetting a PA-220 Firewall

By
Aiden Yin



Purpose:

When a device starts to malfunction, sometimes the last option is to factory reset it to start again on a clean slate. The purpose of this lab was to learn how to factory reset a Palo Alto PA-220 firewall without knowing the password.

Background:

Palo Alto is a cybersecurity company based in North America that sells firewalls and other cloud-based solutions for commercial and consumer use. A firewall is a network security device that monitors traffic to and from your own network. Based on a network administrator's set of security rules, the firewall will then proceed to block or permit traffic to and from a destination. Installing a firewall brings a multitude of benefits, such as promoting privacy and improving network security, all while being relatively cheap.

There are many different types of firewalls that network administrators can choose to deploy on their network. For example, there are Host-Based firewalls that are installed on individual hosts or computers that monitor inbound and outbound traffic from that specific device. The opposite of Host-Based firewalls are network firewalls which, like its name suggests, monitors the inbound and outbound traffic of entire networks. Firewalls can come as both hardware or software. A hardware firewall is a network device that needs to be plugged in with a cable to a physical network, while a software firewall is usually deployed on servers or virtual machines to secure cloud networks.

When powering on and booting a Palo Alto firewall, one can choose to boot into a maintenance partition or the default partition. The default partition is what most people will use, in order to normally utilize their firewall. The maintenance mode, however, offers functions not found in the default partition for network admins to utilize. By entering the maintenance partition, network admins can do a variety of tasks, such as factory resetting the firewall, setting the FIPS (Federal Information Processing Standards), or performing a firewall disk check. In this lab, maintenance mode was utilized to factory reset the firewall.

Lab Summary:

We first plugged in the console cable from the firewall to the computer, and then plugged and screwed in the power cable for the firewall. After plugging in the power

cable, the firewall proceeded to boot up, allowing us to enter the maintenance partition and then to factory reset the firewall.

Walkthrough:

After booting up the firewall, you will be greeted with the following bootup screen.

 COM1 - PuTTY

```
SPI stage 1 bootloader
SPI ID: ef:40:18:00:00
header found at offset 0x1d80
Image 1.2: address: 0xffffffff81000000, header length: 192, data length: 8160
Validating data...
Starting next bootloader at 0xffffffff81000000
SPI stage 1.5 bootloader
SPI ID: ef:40:18:00:00
Header 1 found at offset 0x10000
Header 2 found at offset 0x10000
Found bootloaders, booting bootloader 1 of 2 at offset 0x10000.
Starting next bootloader at 0xffffffff81000000
Board type: KINGFISHER
Welcome to the PanOS Bootloader.

U-Boot 11.0.0.0-63 (Build time: May 10 2023 - 21:55:50)
```

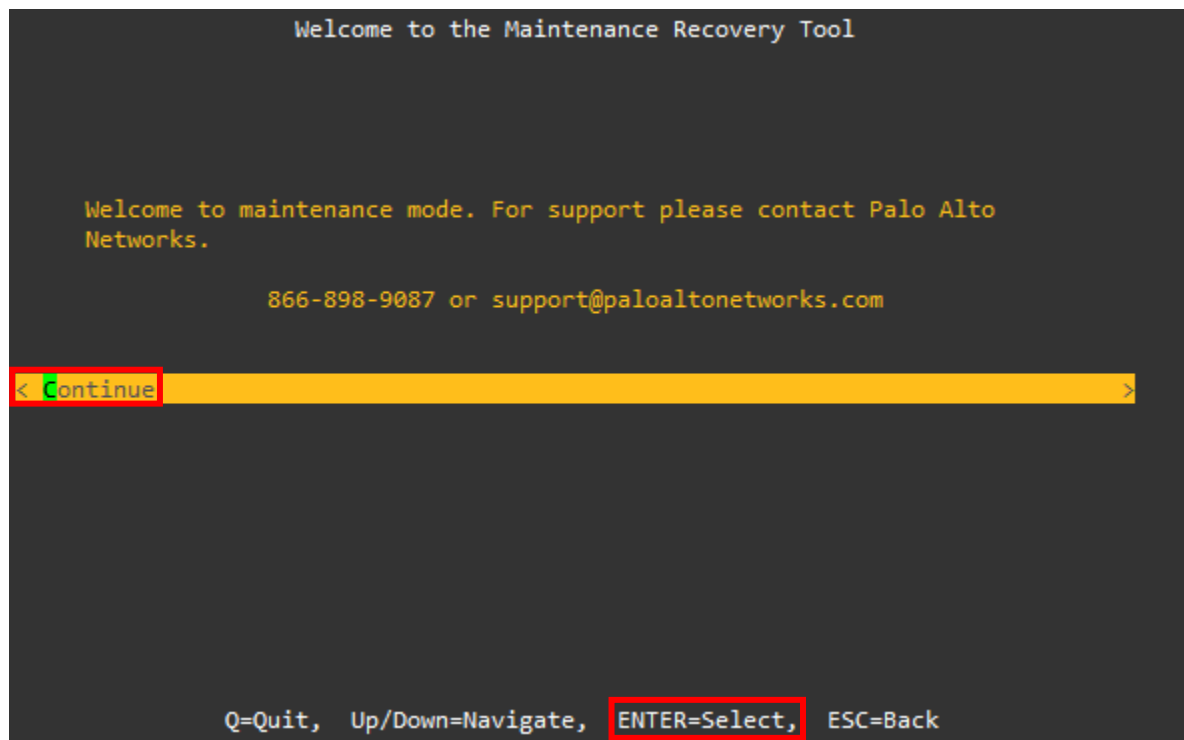
Around 30 seconds later, the firewall will ask you if you want to enter the default or maintenance partition. If you don't type in anything, the firewall will automatically boot into the default partition. In order to boot into the maintenance partition, type in "maint" when prompted.

```
Autoboot to default partition in 5 seconds.
Enter 'maint' to boot to maint partition.

Entry: maint

Booting to maint mode.
```

After entering maint, you will be greeted with the following screen, and you need to press ENTER to continue to the maintenance tools.



When selecting the maintenance tools, you will need to use the arrow keys to navigate up and down the list, and use ENTER to select the maintenance tool you want to use. We will be using the Factory Reset tool for this lab.

```
Welcome to the Maintenance Recovery Tool

< Maintenance Entry Reason >
< Get System Info >
< Factory Reset >
< Set FIPS-CC Mode >
< FSCK (Disk Check) >
< Log Files >
< Bootloader Recovery >
< Disk Image >
< Select Running Config >
< Content Rollback >
< Set IP Address >
< Diagnostics >
< Debug Reboot >
< Reboot >

Q=Quit, Up/Down=Navigate, ENTER=Select, ESC=Back
```

After selecting the factory reset tool, you will then be asked if you want to scrub the firewall while factory resetting it. We will not be scrubbing the firewall for this lab, so make sure it remains unselected before navigating to the Factory Reset button using the arrow keys and pressing ENTER.

```
Factory Reset

WARNING: Performing a factory reset will remove all logs and configuration.

Using Image:
  (X) panos-10.2.6

WARNING: Scrubbing will iteratively write patterns on pancfg, panlogs, and any
extra disks to make retrieving the data more difficult.
NOTE: This could take several hours to several days if selected. Scrubbing is
not recommended unless explicitly required.

  [ ] Scrub

If scrubbing, select scrub type:
  (X) nnsa ( ) dod

< Factory Reset >
< Advanced >

Q=Quit, Up/Down=Navigate, ENTER=Select, ESC=Back
```

Problems:

Overall, the lab was pretty straightforward, and no mistakes were made while resetting the firewall. The only obstacle was knowing when the firewall booted into the maintenance partition after entering maint, as it took more time than expected (around 10 minutes). This isn't really a problem, and it just stemmed from me being unfamiliar with the usage of Palo Alto Firewalls.

Conclusion:

Firewalls provide a variety of security features to the networks they are installed on. However, when a network administrator forgets the password to a firewall, they can factory reset the firewall to set it up again through the maintenance partition. The maintenance partition can be used to do a variety of tasks, such as factory resetting the firewall, setting the FIPS (Federal Information Processing Standards), or performing a firewall disk check.

Setting up a Firewall for a SOHO Network.

By
Aiden Yin



Purpose:

The purpose of this lab was to learn how to setup and configure a PA-220 firewall for deployment in a SOHO network. The firewall would cover the entire network, acting as a DHCP server for all the connected end devices. We also created security zones on the firewall in order to tag and classify incoming traffic coming from various ports.

Background:SOHO Network:

Small Office Home Office, or SOHO for short, is a size classification of networks, usually consisting of around 10 or less devices in the network. Depending on the size of the network, different routing protocols may be used to connect all the devices. A SOHO network brings a variety of benefits such as being easy to set-up and allowing the connected devices to share information seamlessly.

DHCP:

Dynamic Host Configuration Protocol, or DHCP for short, is a protocol used to automate the assignment of IP addresses to devices on a network through the usage of a DHCP server. This allows the servers to dynamically lease IP addresses to ensure a network administrator doesn't need statically assign IP addresses to one device, which allows a network to essentially function with fewer IPs depending on how active certain devices are. A DHCP server is given a pool, or a range, of IP addresses that it can lease out to DHCP clients. DHCP relays are any TCP/IP devices that forward messages between DHCP clients and servers, acting as a guide.

NAT:

IPv4 utilizes 32-bit addresses, which means that there are roughly 4.3 billion possible unique addresses. This ultimately means that it is possible for us to run out of IPv4 addresses, which has already happened. In order to circumvent this, Network Address Translation, or NAT for short, aims to solve this by assigning entire networks using private IP addresses to one public IP address, allowing those networks to take up only one IP address. The router facing the internet will be the device using the one public IP address. It then translates the source address of packets coming from devices in its network to use its own IP address, tracks the basic data like destination address and

port number, before forwarding it into the Internet. When the router receives Internet traffic, it uses the stored data in order to determine which private address to forward the packet to.

Security Zones:

Zones, at the most basic level, segment your network into smaller, more easily manageable sections. Palo Alto firewalls require interfaces to be assigned to zones in order to start processing packets in the first place. Zones are also classified based on the type of traffic it should be processing (i.e. Layer 2 or Layer 3). The zone type and interface type on the firewall should match in order for the interface to process traffic. Traffic destined for different zones are also unable to use the same interface to prevent packets from accidentally entering a zone it shouldn't be in.

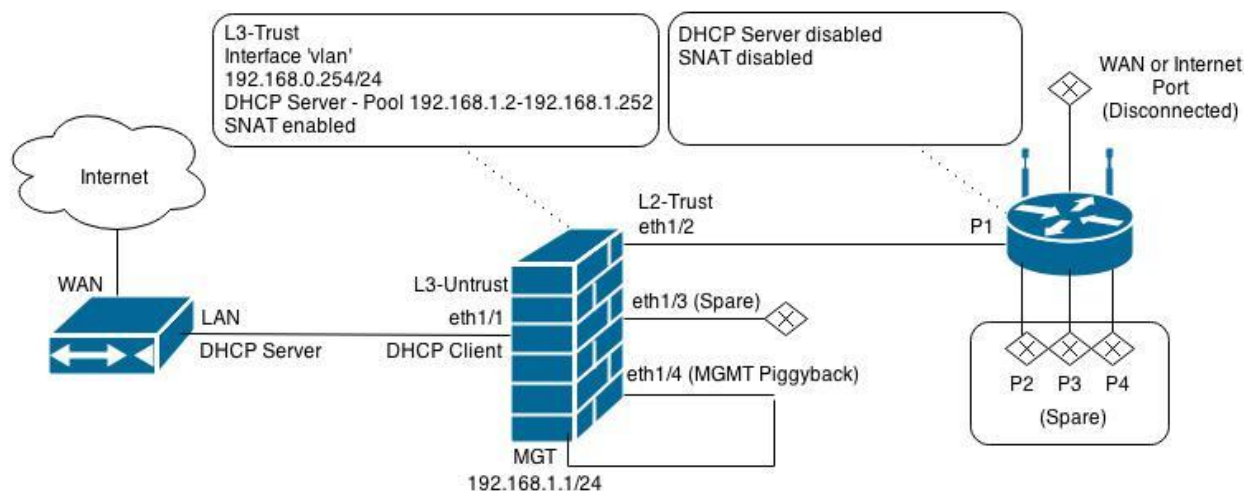
VLAN:

Virtual Local Area Networks, or VLANs for short, are a way of splitting up a network into individual network segments. VLANs take already existing hardware and splits them up into more networks, eliminating the need to physically adjust the network topology when a new network is needed to be added. VLANs allow you to separate and organize traffic from multiple devices. In a SOHO network, they can be used to separate printers, VOIP, security cameras, etc. While VLANs do provide a variety of segmentation benefits, it can make the network more complicated to manage,

Lab Summary:

After logging into the firewall through the use of a web browser, we followed the Palo Alto setup guide for the PA-200 for Home and Small Office guide step by step. After finishing the guide, we disconnected the PC we were using for the lab from WIFI, connected the switch and firewall to the internet through Layer 2, and used the web browser to test that we indeed were connected to the internet through the firewall.

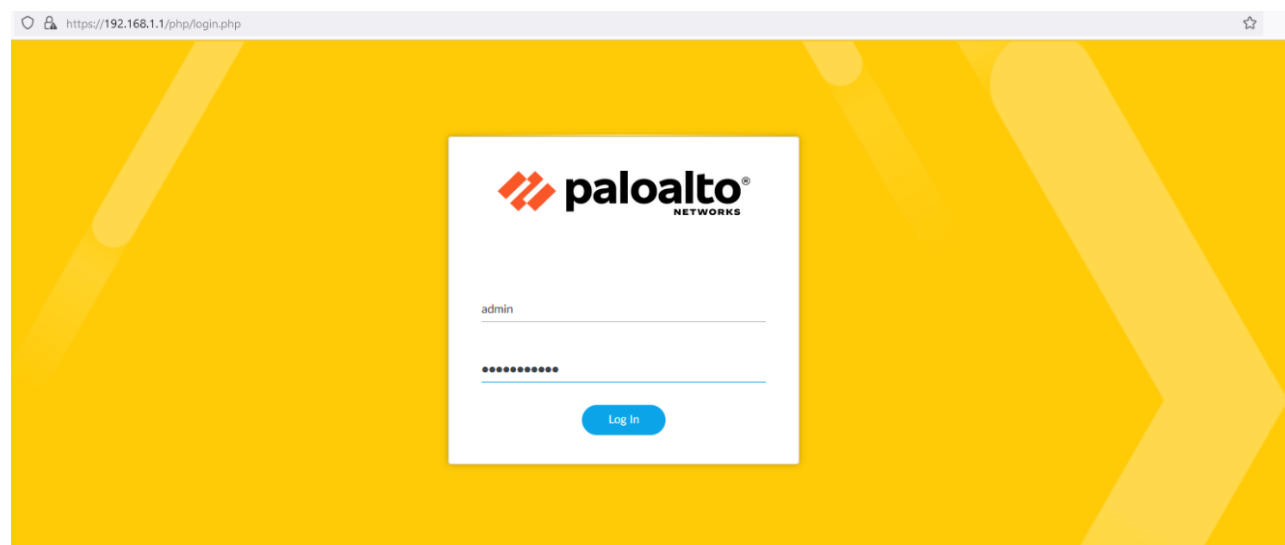
Topology:



Walkthrough:

Accessing the GUI:

1. Set your connected device's IP address to 192.168.1.2/24, and use any web browser of your choice to connect to the firewall through <https://192.168.1.1>. The default user should be admin and the password is whatever you set after factory resetting the firewall through the CLI.



Setting up the Security Zones:

1. Click on Network at the top toolbar, navigate to Zones, and then click add.

The screenshot displays the Palo Alto Networks PA-220 web interface. The top navigation bar includes tabs for DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, NETWORK, and DEVICE. The NETWORK tab is selected. Below the navigation bar, there's a section for the selected device (PA-220) showing its name, address, netmask, and gateway. A sidebar on the left lists various configuration options, with 'Zones' highlighted. The main content area shows a table of configured zones. At the bottom, there are buttons for '+ Add', '- Delete', and 'PDF/CSV'.

NAME	TYPE	INTERFACE
trust	virtual-wire	ethernet1/24
untrust	virtual-wire	ethernet1/24

admin | Logout | Last Login Time: 09/17/2024 12:35:23 | Session Expire Time:

2. Create 3 zones as such: Untrust-L3 - Type Layer3, Trust-L3 - Type Layer3, Trust-L2 - Type Layer2.

Zone ⓘ

Name: **Untrust-L3**
 Log Setting: **None** ▾
 Type: **Layer3** ▾

INTERFACES ▾

+ Add - Delete

Zone Protection
 Zone Protection Profile: **None** ▾
☒ Enable Packet Buffer Protection

User Identification ACL
☐ Enable User Identification
☒ **INCLUDE LIST** ▾
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Users from these addresses/subnets will be identified.
☐ **EXCLUDE LIST** ▾
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Users from these addresses/subnets will not be identified.

Device-ID ACL
☐ Enable Device Identification
☒ **INCLUDE LIST** ▾
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Devices from these addresses/subnets will be identified.
☐ **EXCLUDE LIST** ▾
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Devices from these addresses/subnets will not be identified.

OK Cancel

Zone ⓘ

Name: **Trust-L3**
 Log Setting: **None** ▾
 Type: **Layer3** ▾

INTERFACES ▾

+ Add - Delete

Zone Protection
 Zone Protection Profile: **None** ▾
☒ Enable Packet Buffer Protection

User Identification ACL
☐ Enable User Identification
☒ **INCLUDE LIST** ▾
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Users from these addresses/subnets will be identified.
☐ **EXCLUDE LIST** ▾
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Users from these addresses/subnets will not be identified.

Device-ID ACL
☐ Enable Device Identification
☒ **INCLUDE LIST** ▾
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Devices from these addresses/subnets will be identified.
☐ **EXCLUDE LIST** ▾
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Devices from these addresses/subnets will not be identified.

OK Cancel

Zone ?

Name: **Trust-L2**
 Log Setting: **None**
 Type: **Layer2**

INTERFACES ^
 + Add - Delete

Zone Protection
 Zone Protection Profile: **None**
☒ Enable Packet Buffer Protection

User Identification ACL
☐ Enable User Identification
☒ **INCLUDE LIST** ^
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Users from these addresses/subnets will be identified.
☒ **EXCLUDE LIST** ^
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Users from these addresses/subnets will not be identified.

Device-ID ACL
☐ Enable Device Identification
☒ **INCLUDE LIST** ^
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Devices from these addresses/subnets will be identified.
☒ **EXCLUDE LIST** ^
 Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
 + Add - Delete
 Devices from these addresses/subnets will not be identified.

OK **Cancel**

Connecting the ISP to the firewall

1. Connect a cable from the ISP modem to eth1/1 on the firewall.
2. Click on Network on the top toolbar, then navigate to interfaces, and click on eth1/1
3. Set the interface type as L3, set the virtual router to default and set the security zone to untrust-L3

Ethernet Interface ?

Interface Name: ethernet1/1
 Comment:
 Interface Type: **Layer3**
 Netflow Profile: **None**

Config | IPv4 | IPv6 | SD-WAN | Advanced

Assign Interface To
 Virtual Router: **default**
 Security Zone: **Untrust-L3**

OK **Cancel**

- Next, click the IPv4 tab and set the type as DHCP if the ISP provides a modem which the configuration can be obtained automatically (provides DHCP)

Ethernet Interface ⓘ

Interface Name: ethernet1/1
 Comment:
 Interface Type: Layer3
 Netflow Profile: None

Config **IPv4** | IPv6 | SD-WAN | Advanced

☐ Enable SD-WAN ☐ Enable Bonjour Reflector

Type ☐ Static ☐ PPPoE **☒ DHCP Client**

☒ Enable
☒ Automatically create default route pointing to default gateway provided by server
☐ Send Hostname: system-hostname

Default Route Metric: 10

[Show DHCP Client Runtime Info](#)

OK Cancel

- To set up a VLAN object, click on Network at the top toolbar, then navigate to VLANs and click Add at the bottom of the screen.

VLAN ⓘ

Name: **vlan1**

VLAN Interface: **vlan**

Static MAC Configuration

		MAC ADDRESS	INTERFACE
☐	INTERFACES ^		
☐	ethernet1/2		
☐	ethernet1/3		

OK Cancel

Configuring Layer 2 Ports and VLAN Objects

1. Go to network, then interface and be under the ethernet tab and configure eth1/2-4 with the following.

Interface type: layer 2

VLAN: (vlan name)

Security zone: Trust-L2

Ethernet Interface ⓘ

Interface Name

Comment

Interface Type

Netflow Profile

Config | Advanced

Assign Interface To

VLAN

Security Zone

Ethernet Interface ⓘ

Interface Name

Comment

Interface Type

Netflow Profile

Config | Advanced

Assign Interface To

VLAN

Security Zone

Ethernet Interface ⓘ

Interface Name

Comment

Interface Type

Netflow Profile

Config | Advanced

Assign Interface To

VLAN

Security Zone

Configuring the VLAN interface

1. Go to network from the top toolbar, then navigate to interfaces and click on the VLAN tab and change the following settings.
 VLAN: (VLAN name)
 Virtual Router: default
 Security Zone: Trust-L3

VLAN Interface ⓘ

Interface Name:

Comment:

Netflow Profile:

Config | IPv4 | IPv6 | Advanced

Assign Interface To

VLAN:

Virtual Router:

Security Zone:

OK Cancel

2. Next, Under the IPv4 tab Enter the IP address 192.168.1.254/24

VLAN Interface ⓘ

Interface Name:

Comment:

Netflow Profile:

Config | **IPv4** | IPv6 | Advanced

Type: ☒ Static ☐ DHCP Client

☐	IP
☐	192.168.1.254/24

+ Add - Delete ↑ Move Up ↓ Move Down

IP address/netmask. Ex. 192.168.2.254/24

Configuring the DHCP Server

1. Click on Network in the top toolbar, navigate to DHCP and click DHCP server.
2. Add the DHCP server then edit the configurations as shown.

DHCP Server

Interface: vlan

Mode: enabled

☐ Ping IP when allocating new IP

Options

Lease: ☐ Unlimited ☒ Timeout

0 Days 1 Hours 0 Minutes

Inheritance Source: ethernet1/1

[Check inheritance source status](#)

Gateway: 192.168.1.254

Ippool Subnet: 255.255.255.0

Primary DNS: inherited

Secondary DNS: inherited

Primary WINS: inherited

Secondary WINS: inherited

Primary NIS: inherited

Secondary NIS: inherited

Primary NTP: inherited

Secondary NTP: inherited

POP3 Server: inherited

SMTP Server: inherited

DNS Suffix: None

IP Pools

192.168.1.2-192.168.1.252

[+ Add](#) [- Delete](#)

Reserved Address	MAC Address
192.168.1.20	xx:xx:xx:xx:xx:xx (Optional MAC Address)

[+ Add](#) [- Delete](#)

OK **Cancel**

Define a Security Profile Group

1. Go to Objects in the top toolbar, then navigate to Security profile groups then click add.
2. Configure as shown.

Security Profile Group ⓘ

Name	internet
Antivirus Profile	default
Anti-Spyware Profile	strict
Vulnerability Protection Profile	strict
URL Filtering Profile	default
File Blocking Profile	None
Data Filtering Profile	None
WildFire Analysis Profile	None

OK Cancel

Configure an outbound internet security policy

1. Go to policies in the top toolbar, then navigate to security and click Add.
2. Add a name, rule type, and description.

Security Policy Rule ⓘ

General | Source | Destination | Application | Service/URL Category | Actions

Name	internet outgoing
Rule Type	universal (default)
Description	All traffic to the internet
Tags	
Group Rules By Tag	None
Audit Comment	

[Audit Comment Archive](#)

OK Cancel

3. Next, click the source tab and add a source zone.

Security Policy Rule

General | **Source** | Destination | Application

☐	Any
☒	SOURCE ZONE ▾
☐	 Trust-L3

4. Then, Click the destination tab and add a destination zone.

Security Policy Rule

General | Source | **Destination** | Application

select ▾

☐	DESTINATION ZONE ▲
☐	 Untrust-L3

5. Finally, go under the “actions” tab and configure the profile settings.

The screenshot shows the 'Security Policy Rule' configuration window with the 'Actions' tab selected. The 'Action Setting' section has 'Action' set to 'Allow' and 'Send ICMP Unreachable' unchecked. The 'Profile Setting' section has 'Profile Type' set to 'Group' and 'Group Profile' set to 'internet'. The 'Log Setting' section has 'Log at Session Start' unchecked, 'Log at Session End' checked, and 'Log Forwarding' set to 'None'. The 'Other Settings' section has 'Schedule' set to 'None', 'QoS Marking' set to 'None', and 'Disable Server Response Inspection' unchecked. The 'OK' button is highlighted in blue.

Configuring the outbound internet NAT policy

1. Go to policies, then NAT and click Add.
2. Add a name and set NAT type as IPV4.

The screenshot shows the 'NAT Policy Rule' configuration window with the 'General' tab selected. The 'Name' field is 'Internet Outgoing'. The 'Description' field is empty. The 'Tags' field is empty. The 'Group Rules By Tag' field is 'None'. The 'NAT Type' field is 'ipv4'. The 'Audit Comment' field is empty. The 'Audit Comment Archive' link is visible. The 'OK' button is highlighted in blue.

3. Click Original Packet and select trust-l3 as the source zone.

The screenshot shows the 'NAT Policy Rule' configuration window with the 'Original Packet' tab selected. The 'SOURCE ZONE' dropdown is open, showing 'trust-L3' selected. The 'Destination Zone' dropdown is empty. The 'Destination Interface' dropdown is 'any'. The 'Service' dropdown is 'any'. The 'Any' checkbox is checked. The 'Add' and 'Delete' buttons are visible at the bottom.

4. Select Untrust-L3 as destination zone and eth1/1 for destination interface.

NAT Policy Rule ?

General | **Original Packet** | Translated Packet

☐ Any ☐ SOURCE ZONE ^ ☒ Trust-L3	Destination Zone Untrust-L3 Destination Interface ethernet1/1 Service any	☒ Any ☐ SOURCE ADDRESS ^	☒ Any ☐ DESTINATION ADDRESS ^
--	---	--	---

+ Add - Delete

OK Cancel

5. Click the translated packets and configure as following.

NAT Policy Rule ?

General | Original Packet | **Translated Packet**

Source Address Translation Translation Type: Dynamic IP And Port Address Type: Interface Address Interface: ethernet1/1 IP Address: None	Destination Address Translation Translation Type: None
---	---

OK Cancel

Configuring the Management IP

1. Go to Device, then setup, then go to management and configure the IP address, netmask, and default gateway.

Management Interface Settings ?

IP Address	192.168.1.1
Netmask	255.255.255.0
Default Gateway	192.168.1.254
IPv6 Address/Prefix Length	
Default IPv6 Gateway	
Speed	auto-negotiate
MTU	1500

Services

- ☐ HTTP
- ☐ HTTP OSCP
- ☒ HTTPS
- ☐ Telnet
- ☒ SSH
- ☒ Ping
- ☐ SNMP
- ☐ User-ID
- ☐ User-ID Syslog Listener-SSL
- ☐ User-ID Syslog Listener-UDP

Permitted IP Addresses

+ Add - Delete

OK Cancel

Setting DNS for Management

1. Go to device, then setup, then services
2. Enter the DNS server and secondary if needed.

Services

Services | NTP

Update Server

☒ Verify Update Server Identity

DNS Settings

DNS ☒ Servers ☐ DNS Proxy Object

Primary DNS Server

Secondary DNS Server

Minimum FQDN Refresh Time (sec)

FQDN Stale Entry Timeout (min)

Proxy Server

Server

Port

User

Password

Confirm Password

☐ Use proxy to send logs to Cortex Data Lake

OK

Cancel

Saving/Committing the Changes

1. Click the “commit button” at the top right, you may need to delete the default wire to commit.

Commit

Doing a commit will overwrite the running configuration with the commit scope.

☒ Commit All Changes
 ☐ Commit Changes Made By: (1) admin

COMMIT SCOPE	LOCATION TYPE	OBJECT TYPE	ENTITIES	ADMINS
policy-and-objects	Policy and Objects			
device-and-network	Device and Network Configuration			

Preview Changes

Change Summary

Validate Commit

Note: This shows all the changes in login admin's accessible domain.

Description

Commit

Cancel

Problems:

We misconfigured the VLAN when entering the IP address. When setting the VLAN, you have to include the subnet mask in the IP address or else the VLAN breaks. This will cause anything VLAN-related to break. To solve this, simply add a “/24” to the end of your IP address.

Otherwise, there weren't any other problems, as we were following the Palo Alto Documentation for setting up a PA-200 firewall for Home and Small Office Networks.

Conclusion:

Firewalls play an essential role in network security, filtering all traffic that enters and exits a network. There are a multitude of firewalls of different sizes, features, and form factors. They utilize a variety of security features, like VLANs and Security Zones, as well as basic networking features like DHCP and NAT to effectively secure its network. A SOHO network is just one of many types of common network sizes, and learning how to setup a firewall on these common networks enhances our proficiency at basic network security.

Configuring URL Filtering with Override

By
Aiden Yin



Purpose:

The purpose of the lab was to set up URL filtering for a simulated elementary school network. We needed to decide which websites students would be able to access, and which would be restricted. Some websites would also need to be accessed by teachers, but not students.

Background:URL Addresses:

A Uniform Resource Locator, or URL for short, is a reference to something's location on a network, and the method of getting there. URLs most commonly reference web pages through HTTP/HTTPS but can also be used for file transfers through FTP, emails through mailto, among others. A standard format for URLs is

“http://www.example.com:80”

The “http” part is called the scheme of the URL, which indicates the protocol used to access the resource. If the URL was like a shipping address, the scheme would be the shipping service you use to deliver the package. “www.example.com:80” is called the authority. www.example.com is called the domain, which can be a string of text or an IP address, while the :80 is the port number used. If no port number is given, the default port used is 80 for http and 443 for https.

Palo Alto URL Filters:

When a URL filter is set up through a Palo Alto firewall, a network admin can choose from a variety of actions that would occur when a URL is passing through the filter. Websites are categorized by Palo Alto into groups in which admins can choose actions to take when a URL from one of these groups passes through the filter.

Allow: The website is allowed to be accessed, and no log entry is generated

Block: The website is unable to be accessed, and a log entry is generated

Alert: The website is allowed to be accessed, but a log entry is generated

Continue: The website will initially be blocked, but the user is given an option to continue onto the site. A log entry is generated.

Override: The user is required to enter user credentials to access the site. A log entry is also generated.

URL Override Types:

When prompting a user for user credentials to override a filter, there are two behaviors that a URL filter can follow.

Redirect: Forces the user's traffic to a specified URL on the firewall to prompt for authentication.

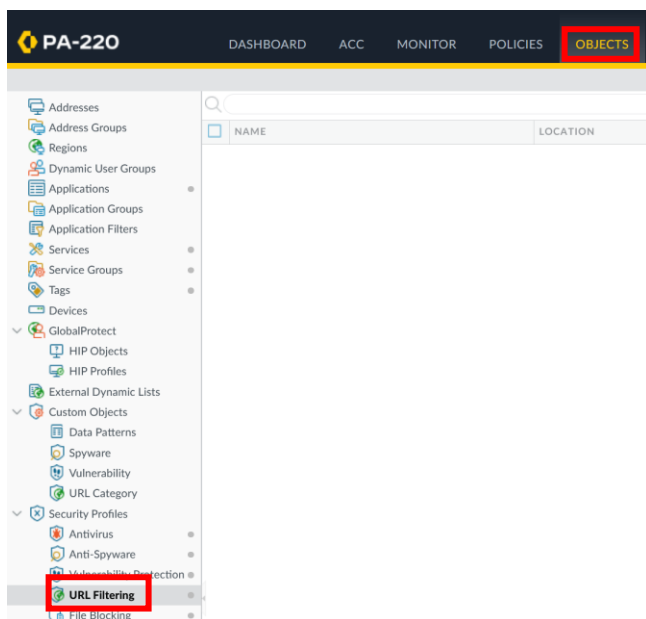
Transparent: “Intercepts” traffic from the original URL and authentication appears to originate from the original site, essentially mimicking the site.

Lab Summary:

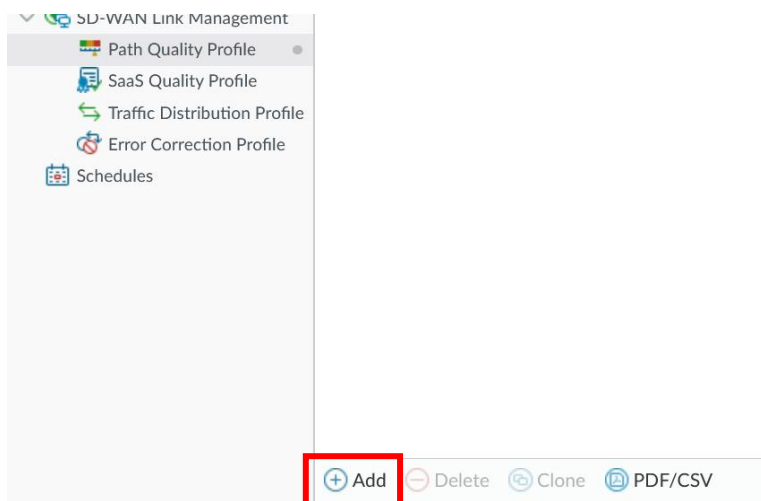
We took the firewall that we previously setup for SOHO networking and created a new URL Filtering Profile for a fake elementary school. Categories like abused-drugs and games would be blocked, while shopping was set to override.

Instructions:

1. Navigate to the objects tab and click on URL Filtering.



2. At the bottom of the page, click on “Add”



3. Create a name for your new filtering profile, select the actions for each category, and click OK.

URL Filtering Profile

Name: **URL Block**

Description:

Categories | URL Filtering Settings | User Credential Detection | HTTP Header Insertion | Inline Categorization

74 items → ×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
Pre-defined Categories		
☐ abortion	block	block
☐ abused-drugs	block	block
☐ adult	block	block
☐ alcohol-and-tobacco	block	block
☐ auctions	block	block
☐ business-and-economy	allow	allow

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK Cancel

4. Navigate to the Policies tab and click on “internet outgoing”.

The screenshot shows the PA-220 dashboard with the **POLICIES** tab selected. On the left, the **Security** menu is highlighted. The main area displays a table of Security Policy Rules:

	NAME	TAGS	TYPE	ZONE
1	rule1	none	universal	tr
2	internet outgoing	none	universal	Tr

- Click on the Actions tab. Under URL filtering, choose the new filtering profile you created.

Security Policy Rule

General | Source | Destination | Application | Service/URL Category | **Actions**

Action Setting

Action: Allow ☐ Send ICMP Unreachable

Profile Setting

Profile Type: Profiles

Antivirus: default

Vulnerability Protection: default

Anti-Spyware: default

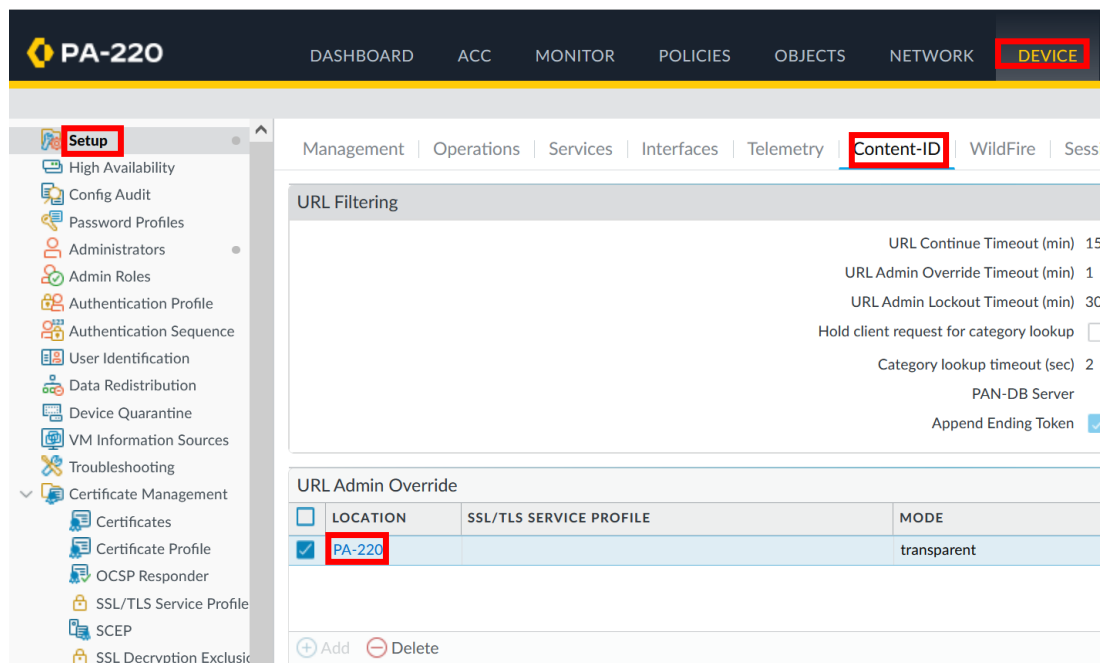
URL Filtering: URL Block

File Blocking: None

Data Filtering: None

WildFire Analysis: None

6. Navigate to the Device tab. Under Setup, click on the Content-ID tab. Click on the firewall you want to configure overrides on.



7. Enter a password that will be used for credential-based overrides. Select transparent mode for the override. Click OK.

The screenshot shows the 'URL Admin Override' dialog box. It contains the following fields and options:

- Password:** A text field with a red box around it, containing ten black dots.
- Confirm Password:** A text field with a red box around it, containing ten black dots.
- SSL/TLS Service Profile:** A dropdown menu with 'None' selected.
- Mode:** Two radio buttons. The 'Transparent' radio button is selected and highlighted with a red box. The 'Redirect' radio button is unselected.
- Buttons:** At the bottom right, there are two buttons: 'OK' (highlighted with a red box) and 'Cancel'.

Problems:

No problems with the GUI were encountered while setting up the filter itself. There were faulty ethernet cables that didn't work and needed to be replaced.

Conclusion:

While internet access is important for places such as schools, it is not ideal for students and teachers to be able to access every part of the internet. Blocking entertainment and other unrelated sites is highly beneficial to the school. Through a URL filter, network admins can block internet traffic going and coming from these URLs, while allowing access to acceptable sites like educational ones. Palo Alto provides grouping for URLs within their firewalls so that admins do not need to manually configure access to each URL. If a site needs to be accessed by teachers but not students, network admins can configure URL override access to the site, where upon accessing a URL, the user is prompted for login credentials in order to access what would otherwise be a blocked URL.

Setting up Palo Alto GlobalProtect VPN

By
Aiden Yin





The purpose of the lab was to learn how to set up Palo Alto's GlobalProtect VPN on a PA-220 firewall. We used Remote Desktop to ensure that the VPN was setup correctly and devices connecting to the VPN could send data.

Background:

A virtual private network, or VPN for short, is an encrypted connection between two devices over the internet, essentially privatizing all data transmitted between them. A VPN creates a secured connection between a network and the user device called a tunnel, and data from devices is encrypted and then sent through this tunnel.

After connecting to a VPN, it will hide the device's original IP address from other devices by routing traffic through a different server. This makes the device seem like it is located where the VPN's server is, instead of the device's actual location, adding a layer of privacy to connected devices.

Encryption adds a layer of security to all encrypted data, taking the data from a device and scrambling it according into an unreadable format. In order to read the data, the recipient of the encrypted data needs to have a specific key to unscramble the data. Without the proper key, any data intercepted by an unauthorized individual is generally unusable and therefore useless. Common encryption algorithms DES, RSA, and AES.

In this lab, certificates were utilized to establish trust between devices. Certificates are like identification for devices; a device with the correct certificate presents it to the VPN like a proof of identity, allowing it to establish an encrypted connection with the VPN. Besides the encryption key, a certificate usually also includes things like expiration dates as well as the issuer of the certificate, who is usually a trusted third party called a Certificate Authority, or CA for short. In this lab, the certificates are self-signed, meaning that the network admin is the third party.

There is generally one root certificate, which is essentially "original" certificate. The root certificate can then sign multiple intermediate certificates, which are like

weaker versions of the root certificate. This way, when you have multiple network administrators distributing intermediate certificates and one accidentally gets leaked, you can just reconfigure that one intermediate certificate instead of the root certificate, saving a lot of trouble and adding another layer of security, almost like an airlock.

Instructions:

1. Firstly generate 3 certificates, Root, Intermediate and Server.
Device -> certificate -> generate

The screenshot shows the 'Generate Certificate' dialog box. The 'Certificate Type' is set to 'Local'. The 'Certificate Name' and 'Common Name' are both 'RootCert'. The 'Signed By' dropdown is set to 'Certificate Authority'. The 'OCSP Responder' is empty. Under 'Cryptographic Settings', the 'Algorithm' is 'RSA', 'Number of Bits' is '2048', 'Digest' is 'sha256', and 'Expiration (days)' is '365'. Under 'Certificate Attributes', the 'Country = "C" from "Subject" field' is checked with a value of 'US'. There are 'Add' and 'Delete' buttons at the bottom left. 'Generate' and 'Cancel' buttons are at the bottom right.

Have the certificate name and common name be the same.

Make sure the “Certificate authority” is checked

Certificate attributes just give more info about the certificate but not required to function

The screenshot shows the 'Generate Certificate' dialog box for an intermediate certificate. The 'Certificate Type' is 'Local'. The 'Certificate Name' and 'Common Name' are both 'IntermediateCert'. The 'Signed By' dropdown is set to 'RootCert'. The 'OCSP Responder' is empty. Under 'Cryptographic Settings', the 'Algorithm' is 'RSA', 'Number of Bits' is '2048', 'Digest' is 'sha256', and 'Expiration (days)' is '365'. Under 'Certificate Attributes', there are no attributes listed. There are 'Add' and 'Delete' buttons at the bottom left. 'Generate' and 'Cancel' buttons are at the bottom right.

Similarly, keep the same certificate name and common name and checked certificate authority. Although in the “signed by” click the previously created RootCert

Generate Certificate ⓘ

Certificate Type ☒ Local ☐ SCEP

Certificate Name

Common Name

Signed By

☐ Certificate Authority

☐ Block Private Key Export

OCSP Responder

▼ Cryptographic Settings

Certificate Attributes

☐	TYPE	VALUE
+ Add - Delete		

Next for the server root, put the common name as the gateway IP and have it signed by the intermediate certificate. Make sure to uncheck Certificate Authority

- After creating the certificates, select and send each one to your user device

delete Revoke Renew

- Next make an SSL/TLS Service profile referencing the Server Certificate
Device -> SSL/TLS service profile

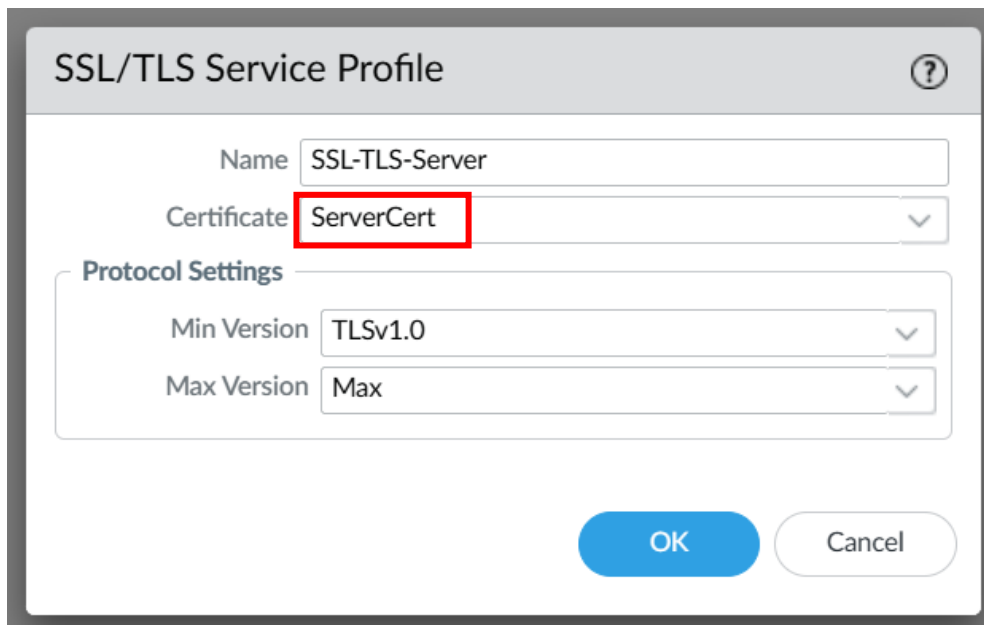
PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK **DEVICE**

Setup High Availability Config Audit Password Profiles Administrators Admin Roles Authentication Profile Authentication Sequence User Identification Data Redistribution Device Quarantine VM Information Sources Troubleshooting

▼ Certificate Management

- Certificates
- Certificate Profile
- OCSP Responder
- SSL/TLS Service Profile**
- SCEP
- SSL Decryption Exclusion

NAME	LOCATION	CERTIFICATE
------	----------	-------------



The image shows a dialog box titled "SSL/TLS Service Profile". It has a header bar with a question mark icon. The main area contains several fields: "Name" with the value "SSL-TLS-Server", "Certificate" with a dropdown menu showing "ServerCert" (highlighted with a red box), and a "Protocol Settings" section. The "Protocol Settings" section has two dropdowns: "Min Version" with "TLSv1.0" and "Max Version" with "Max". At the bottom right, there are two buttons: "OK" (blue) and "Cancel" (white with a grey border).

SSL/TLS Service Profile

Name SSL-TLS-Server

Certificate ServerCert

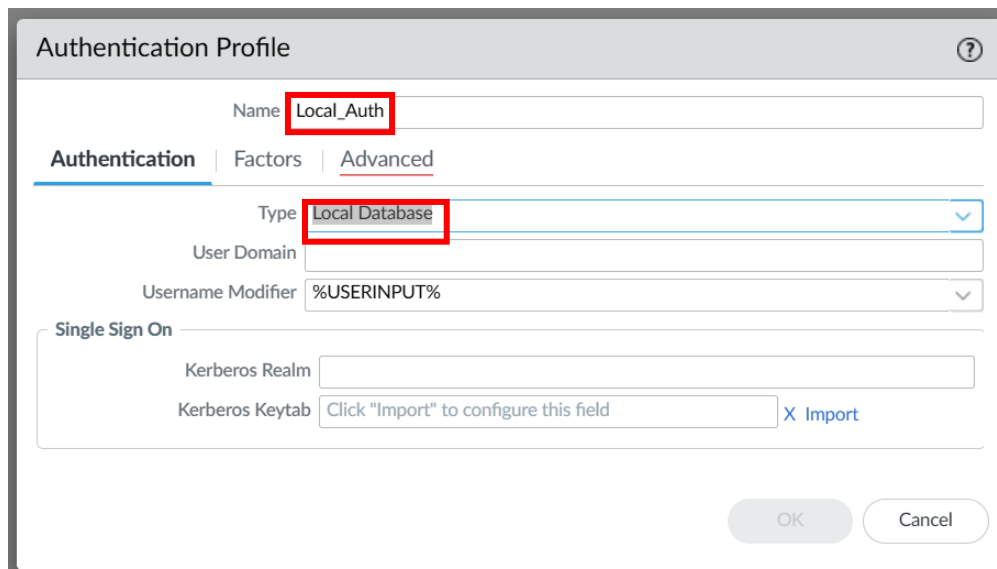
Protocol Settings

Min Version TLSv1.0

Max Version Max

OK Cancel

4. Next make an authentication profile
Device -> authentication profile



The image shows a dialog box titled "Authentication Profile". It has a header bar with a question mark icon. The main area has three tabs: "Authentication" (selected), "Factors", and "Advanced". Under the "Authentication" tab, there are several fields: "Name" with the value "Local_Auth" (highlighted with a red box), "Type" with a dropdown menu showing "Local Database" (highlighted with a red box), "User Domain", "Username Modifier" with the value "%USERINPUT%", and a "Single Sign On" section. The "Single Sign On" section has two fields: "Kerberos Realm" and "Kerberos Keytab" with the text "Click 'Import' to configure this field" and a blue "X Import" link. At the bottom right, there are two buttons: "OK" (grey) and "Cancel" (white with a grey border).

Authentication Profile

Name Local_Auth

Authentication Factors Advanced

Type Local Database

User Domain

Username Modifier %USERINPUT%

Single Sign On

Kerberos Realm

Kerberos Keytab Click "Import" to configure this field X Import

OK Cancel

Authentication Profile

Name

Local_Auth

Authentication

Factors

Advanced

Allow List

☐ ALLOW LIST ^

☒ all

+ Add

- Delete

Account Lockout

Failed Attempts

[0 - 10]

Lockout Time (min)

0

OK

Cancel

5. Next create a tunnel interface in the L3 security zone
Network -> interface -> tunnel

Ethernet VLAN Loopback <u>Tunnel</u> SD-WAN						
INTERFACE	MANAGEMENT PROFILE	IP ADDRESS	VIRTUAL ROUTER	SECURITY ZONE	FEATURES	COMMENT
tunnel		none	none	none		
tunnel.10		none	default	Trust-L3		

6. Create the GlobalProtect portal on the outbound interface of the firewall.

Network -> Portals -> add

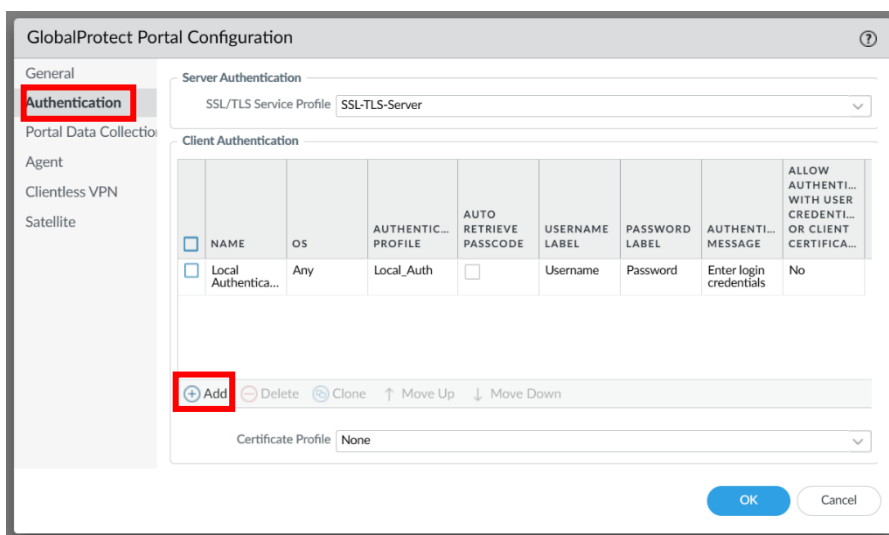
The screenshot shows the Palo Alto Networks PA-220 configuration interface. The top navigation bar includes DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, NETWORK (highlighted in red), and DEVICE. The left sidebar lists various configuration categories, with GlobalProtect and Portals (highlighted in red) selected. The main area displays a table with columns: NAME, LOCATION, INTERFACE, IP, and SSL/TLS SERV PROFILE. Below this, the GlobalProtect Portal Configuration dialog is open, showing the following settings:

- General**: Name: GP-portal
- Network Settings**: Interface: ethernet1/1 (highlighted in red), IP Address Type: IPv4 Only, IPv4 Address: None (highlighted in red)
- Appearance**: Portal Login Page: factory-default, Portal Landing Page: factory-default, App Help Page: None
- Log Settings**: Log Successful SSL Handshake: ☐, Log Unsuccessful SSL Handshake: ☒, Log Forwarding: None

At the bottom right of the dialog are OK and Cancel buttons.

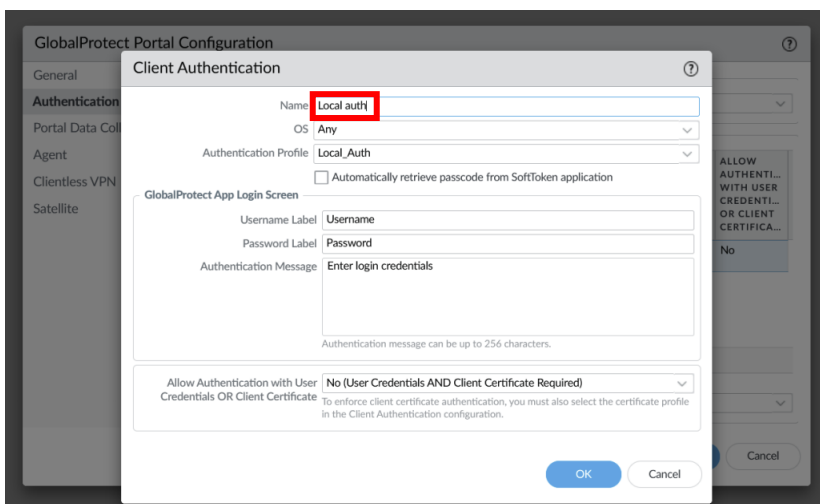
7. Navigate to the Authentication tab.

Under client authentication, Click add and create an authentication



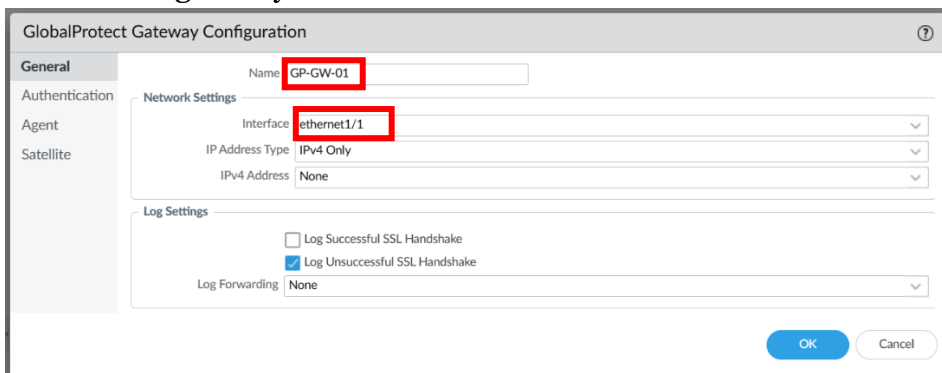
The screenshot shows the 'GlobalProtect Portal Configuration' window with the 'Authentication' tab selected. The 'Server Authentication' section is set to 'SSL-TLS-Server'. The 'Client Authentication' section contains a table with columns: NAME, OS, AUTHENTIC... PROFILE, AUTO RETRIEVE PASSCODE, USERNAME LABEL, PASSWORD LABEL, AUTHENTI... MESSAGE, and ALLOW AUTHENTI... WITH USER CREDENTIAL... OR CLIENT CERTIFICA... The table has one row: Local Authentica..., Any, Local_Auth, [checkbox], Username, Password, Enter login credentials, No. Below the table is an 'Add' button (circled in red), a 'Delete' button, a 'Clone' button, and 'Move Up'/'Move Down' buttons. A 'Certificate Profile' dropdown is set to 'None'. 'OK' and 'Cancel' buttons are at the bottom right.

NAME	OS	AUTHENTIC... PROFILE	AUTO RETRIEVE PASSCODE	USERNAME LABEL	PASSWORD LABEL	AUTHENTI... MESSAGE	ALLOW AUTHENTI... WITH USER CREDENTIAL... OR CLIENT CERTIFICA...
Local Authentica...	Any	Local_Auth	☐	Username	Password	Enter login credentials	No



The screenshot shows the 'Client Authentication' dialog box. The 'Name' field is 'Local auth' (circled in red). The 'OS' dropdown is 'Any'. The 'Authentication Profile' dropdown is 'Local_Auth'. There is a checkbox for 'Automatically retrieve passcode from SoftToken application' which is unchecked. The 'GlobalProtect App Login Screen' section has fields for 'Username Label' (Username), 'Password Label' (Password), and 'Authentication Message' (Enter login credentials). A note states 'Authentication message can be up to 256 characters.' The 'Allow Authentication with User Credentials OR Client Certificate' dropdown is set to 'No (User Credentials AND Client Certificate Required)'. A note states 'To enforce client certificate authentication, you must also select the certificate profile in the Client Authentication configuration.' 'OK' and 'Cancel' buttons are at the bottom.

8. Next, create and configure the gateway
Network -> gateway -> add



GlobalProtect Gateway Configuration

General

Name: GP-GW-01

Network Settings

Interface: ethernet1/1

IP Address Type: IPv4 Only

IPv4 Address: None

Log Settings

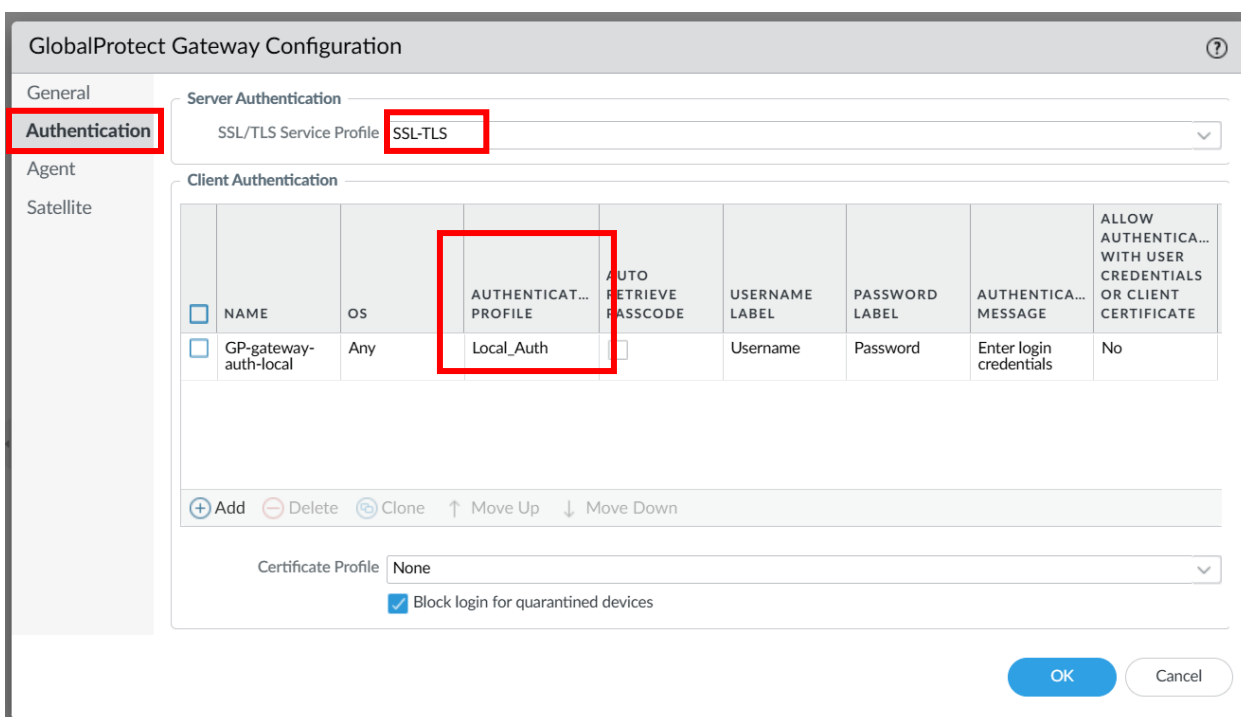
☐ Log Successful SSL Handshake

☒ Log Unsuccessful SSL Handshake

Log Forwarding: None

OK Cancel

Next go under authentication and click add, select the authentication profile/create a new one. Select the proper SSL/TLS service profile



GlobalProtect Gateway Configuration

Authentication

Server Authentication

SSL/TLS Service Profile: SSL-TLS

Client Authentication

☐	NAME	OS	AUTHENTICAT... PROFILE	AUTO RETRIEVE PASSCODE	USERNAME LABEL	PASSWORD LABEL	AUTHENTICA... MESSAGE	ALLOW AUTHENTICA... WITH USER CREDENTIALS OR CLIENT CERTIFICATE
☐	GP-gateway-auth-local	Any	Local_Auth	☐	Username	Password	Enter login credentials	No

+ Add - Delete Clone ↑ Move Up ↓ Move Down

Certificate Profile: None

☒ Block login for quarantined devices

OK Cancel

Configs ?

Authentication | Config Selection Criteria | Internal | External | App | HIP Data Collection

Name **GP-client-config-1**

Client Certificate **None**

The selected client certificate including its private key will be installed on client machines.

Save User Credentials **Yes**

Authentication Override

☒ Generate cookie for authentication override

☒ Accept cookie for authentication override

Cookie Lifetime **Hours** **24**

Certificate to Encrypt/Decrypt Cookie **RootCer**

Components that Require Dynamic Passwords (Two-Factor Authentication)

☐ Portal ☐ External gateways-manual only

☐ Internal gateways-all ☐ External gateways-auto discovery

Select the options that will use dynamic passwords like one-time password (OTP) to authenticate users as opposed to using saved credentials. As a result, the user will always be prompted to enter new credentials for each selected option.

OK **Cancel**

Next go under agent and click on the client settings tab and click add.

GlobalProtect Gateway Configuration ?

General | **Tunnel Settings** | **Client Settings** | Client IP Pool | Network Services | Connection Settings | Video Traffic | HIP Notific

Authentication

Agent

Satellite

☒ **Tunnel Mode**

Tunnel Interface **tunnel.10**

Max User **[1 - 250]**

☒ **Enable IPSec**

GlobalProtect IPSec Crypto **default**

☐ **Enable X-Auth Support**

Group Name

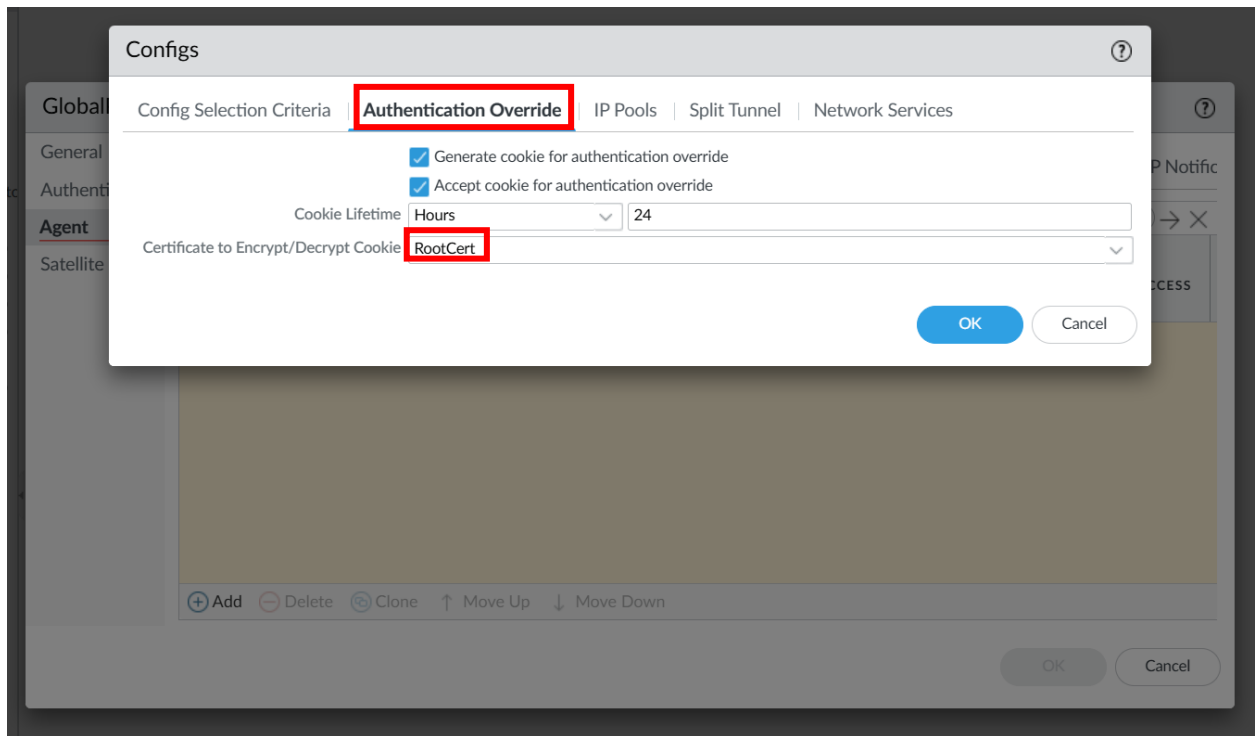
Group Password

Confirm Group Password

☒ **Skip Auth on IKE Rekey**

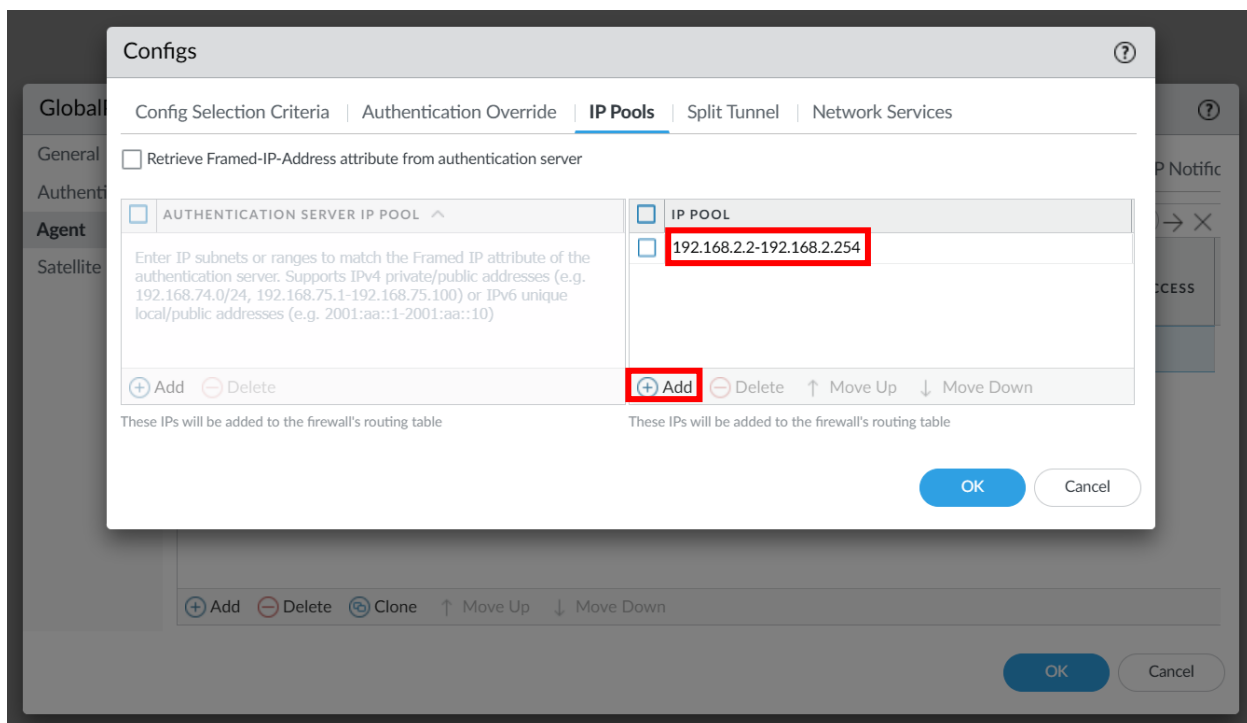
OK **Cancel**

Under the Authentication Override tab, make sure the Root Certificate is selected

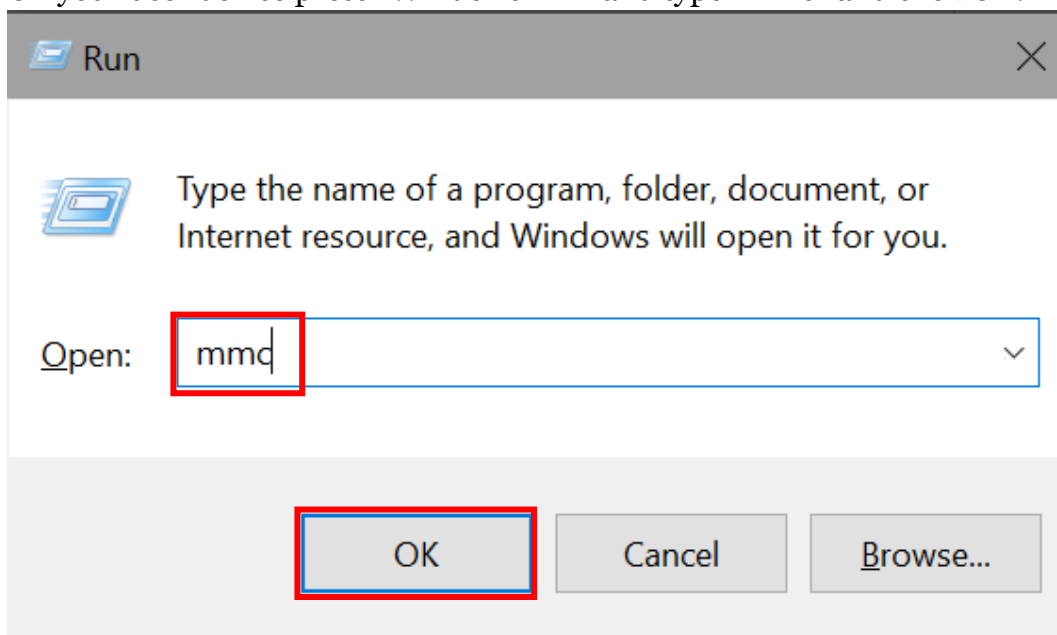


Under the IP Pools tab, designate an IP pool that will be used to assign IP addresses to devices connected to the VPN.

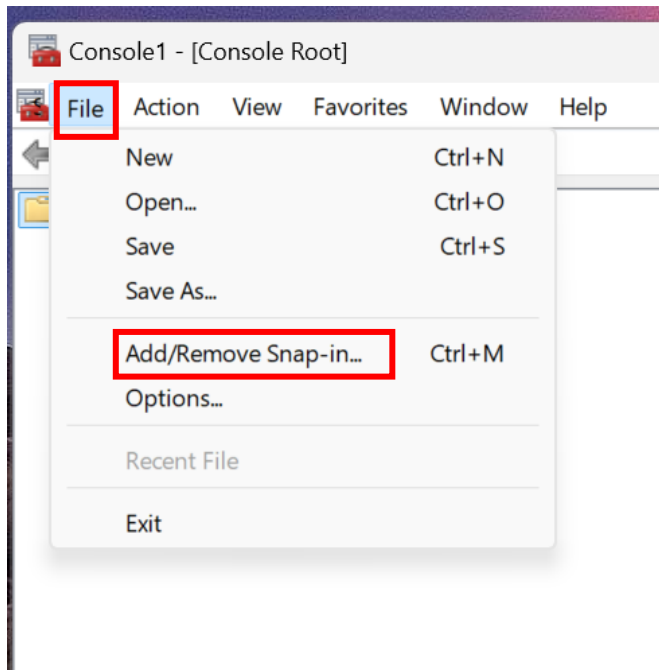
* Important, make sure the Ip pool is not in the same network as any IP pools in the internal firewall network



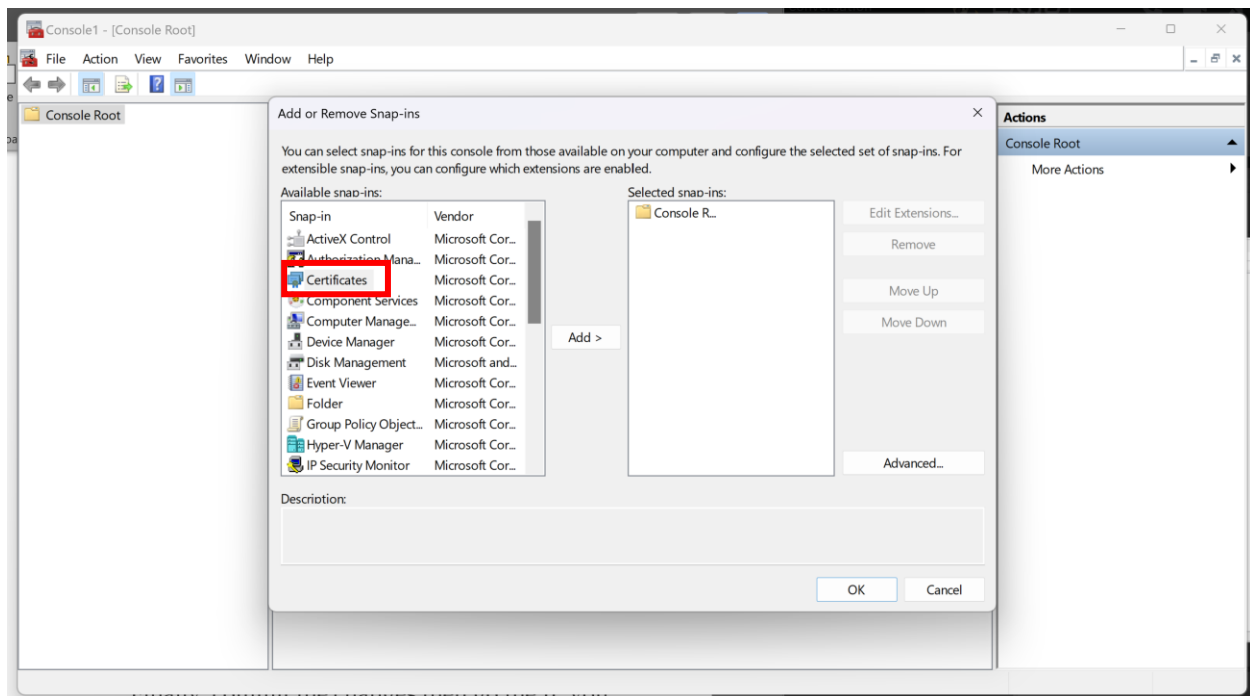
9. On your user device press “Windows + R” and type “mmc” and click OK.

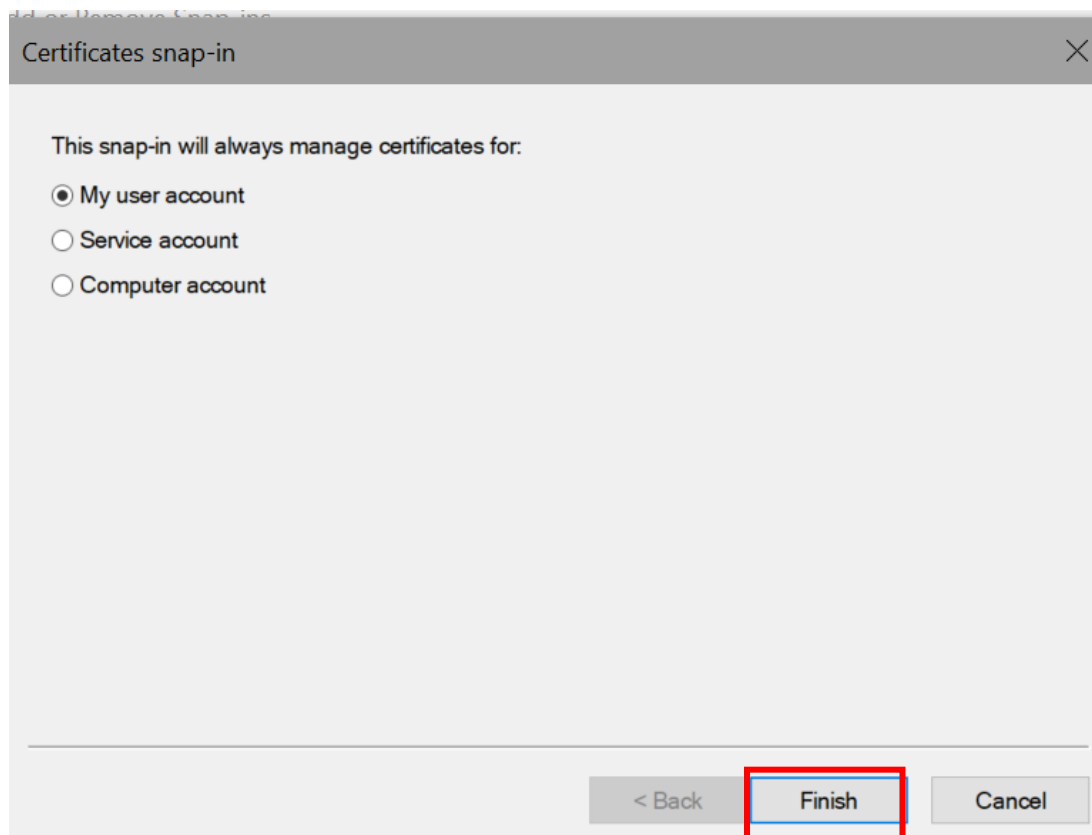


10. At the top left, click file and then Add/remove snap-in

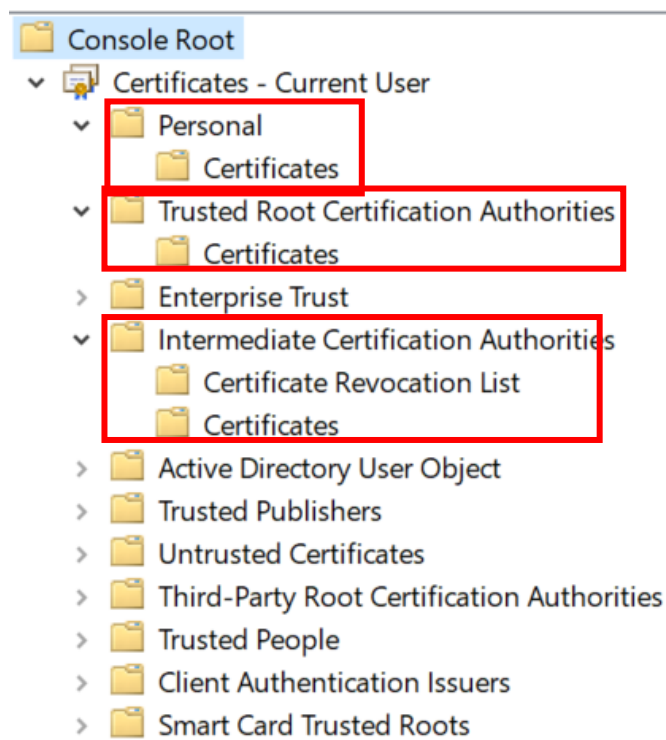


11. Double click Certificates then click through the prompts.

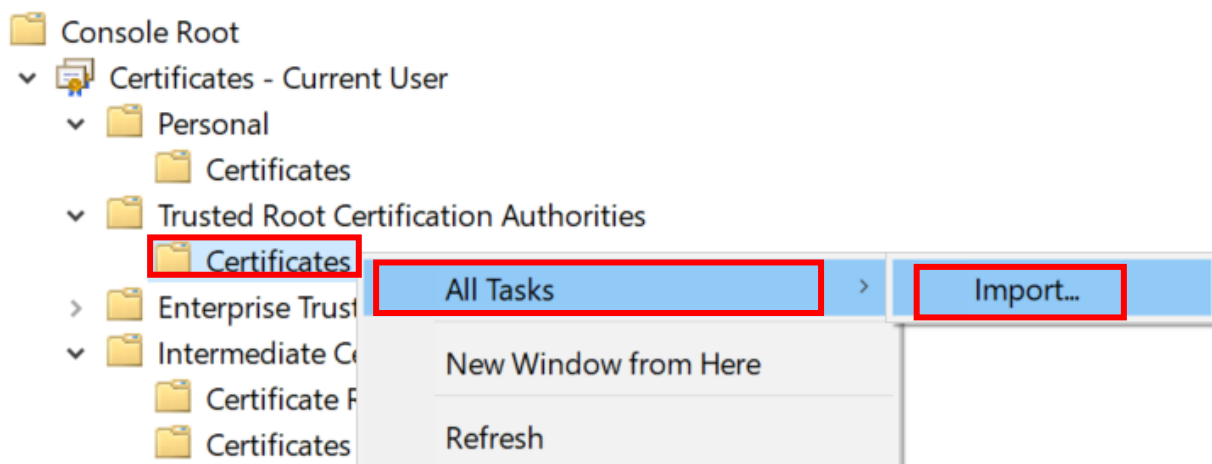




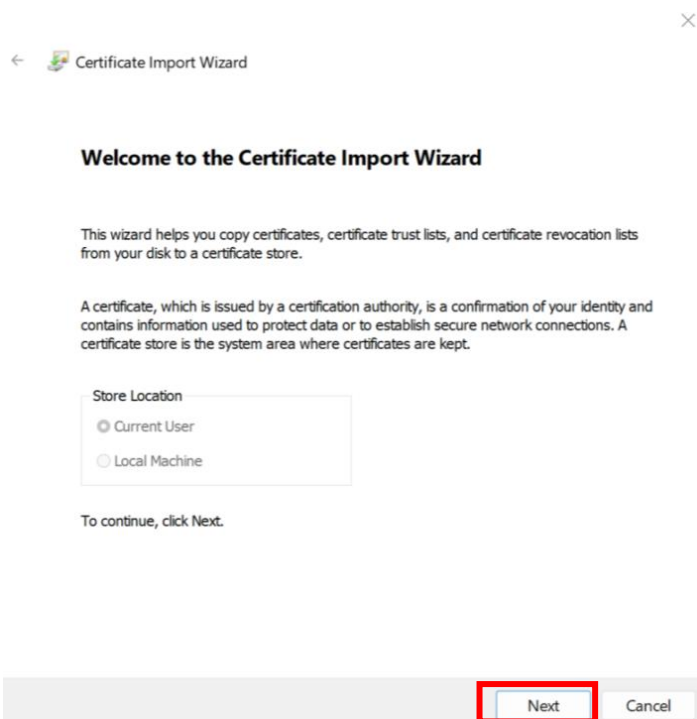
Navigate to the certificates folder under Personal, Trusted Root, and Intermediate.

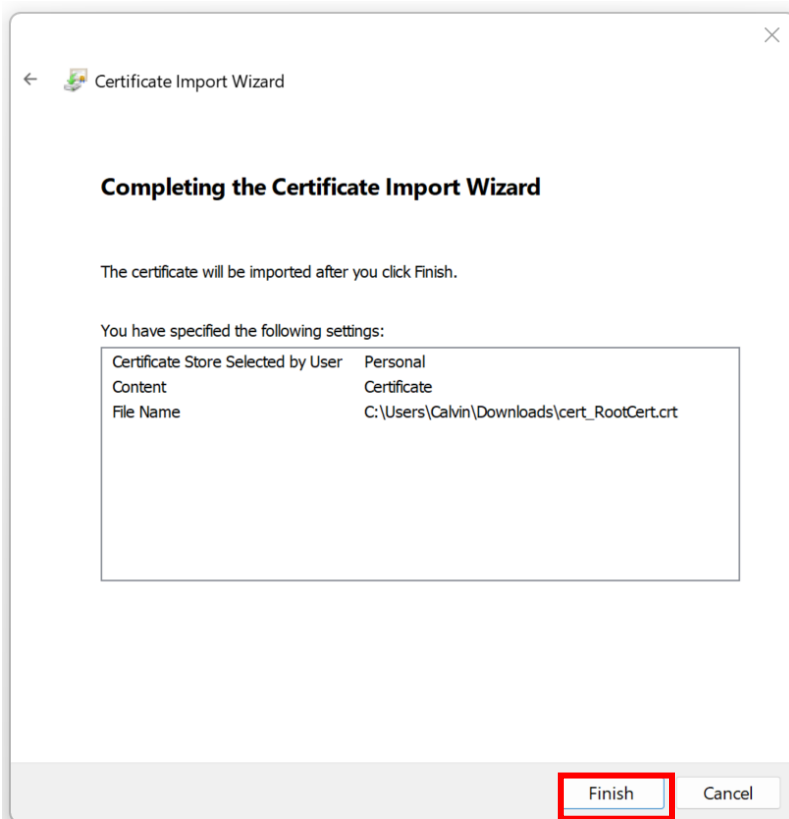
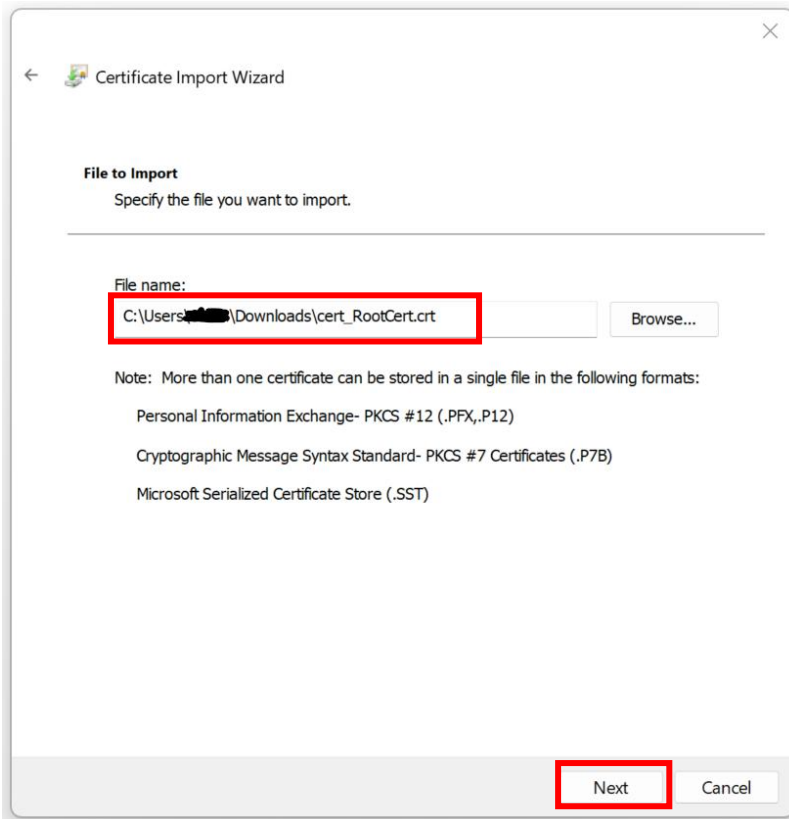


Right click on each Certificates folder, hover over All Tasks, and import the Server Certificate under Personal, the Root Certificate under Trusted Root, and the Intermediate Certificate under Intermediate.



Navigate through the prompts and select the correct certificate.



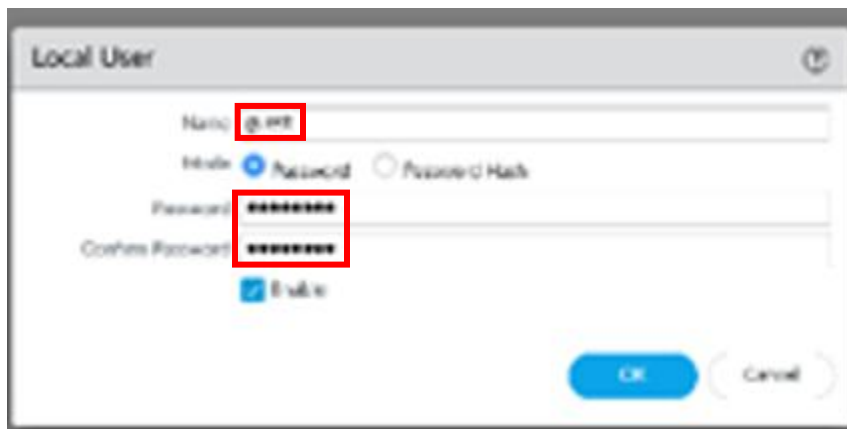


Exit out and save when prompted.

12. Lastly, create a user

Device -> Users -> add

Enter the username and password for user authentication



Local User

Name:

Mode: ☒ Password ☐ Password Hash

Password:

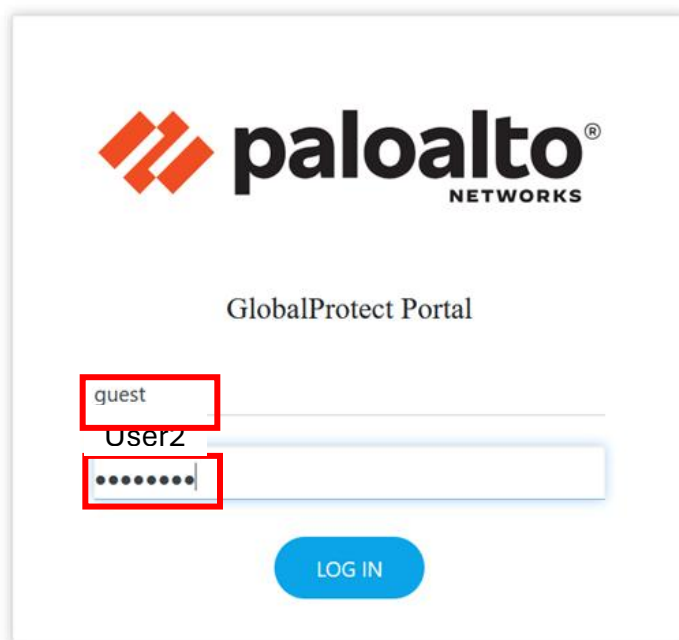
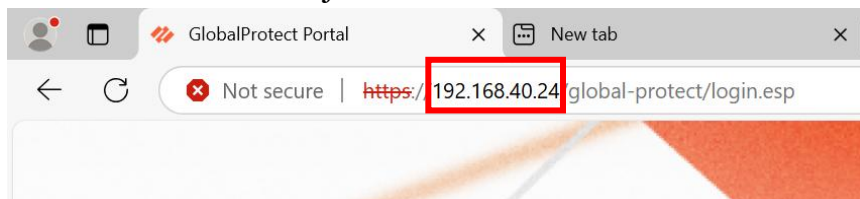
Confirm Password:

☒ Enable

OK Cancel

13. Now to connect to the GlobalProtect VPN.

In a separate web browser, enter in the IP address of the GP gateway and login with the user that was just created.



paloalto
NETWORKS

GlobalProtect Portal

LOG IN

14. Download the appropriate GlobalProtect Agent



GlobalProtect Portal

[Download Windows 32 bit GlobalProtect agent](#)

[Download Windows 64 bit GlobalProtect agent](#)

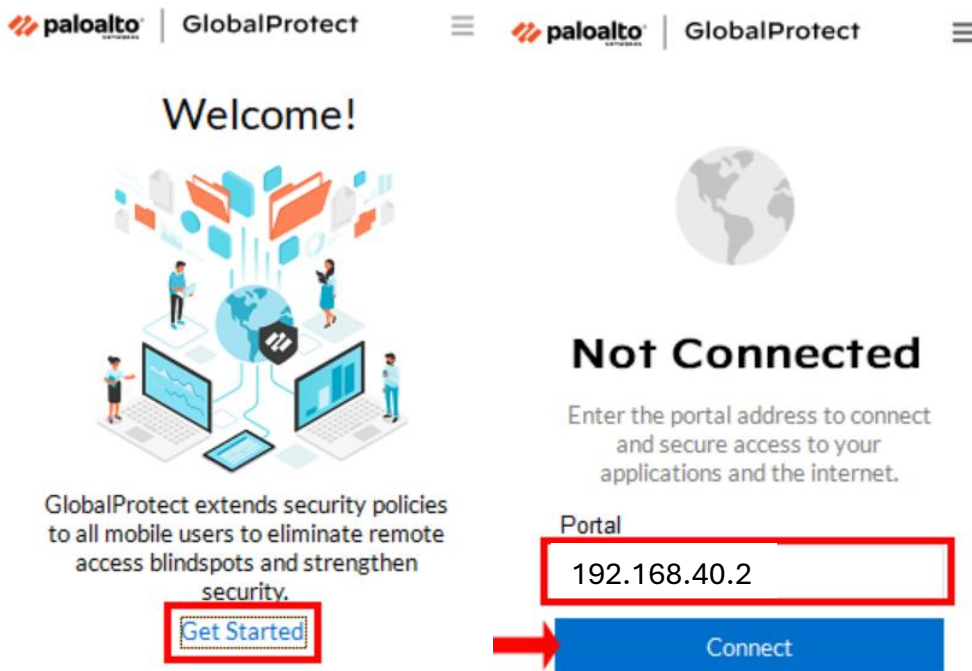
[Download Mac 32/64 bit GlobalProtect agent](#)

Windows 32 bit OS needs to download and install Windows 32 bit GlobalProtect agent.

Windows 64 bit OS needs to download and install Windows 64 bit GlobalProtect agent.

Mac OS needs to download and install Mac 32/64 bit GlobalProtect agent.

Once downloaded, Click Get started and input the gateway IP and click Connect.



Enter the credentials for the user you created and hit connect. You should now be connected to your VPN



Enter login credentials

Username

Password

Connect

Cancel

15. To check if our VPN is setup properly, we will be using remote desktop to connect from the external network device to an internal network device.

To turn on remote desktop, in the windows search bar look up “remote desktop settings” and turn on remote desktop.

Make sure both computers have remote desktop on.

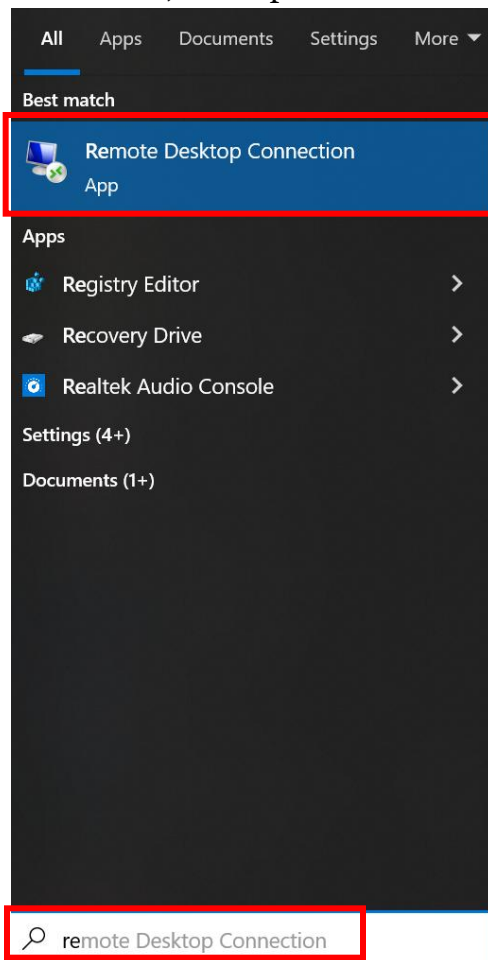
Remote Desktop

Remote Desktop lets you connect to and control this PC from a remote device by using a Remote Desktop client (available for Windows, Android, iOS and macOS). You'll be able to work from another device as if you were working directly on this PC.

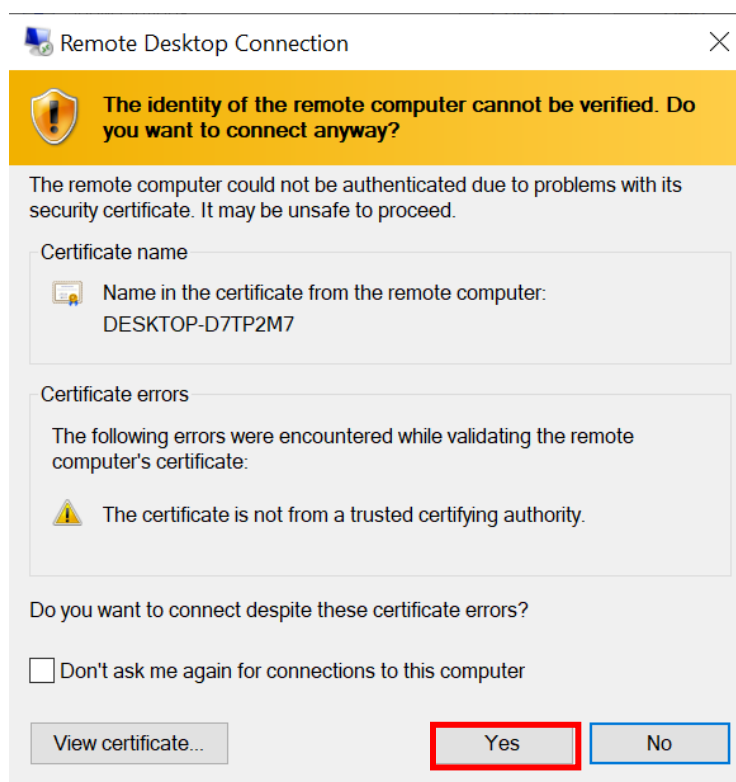
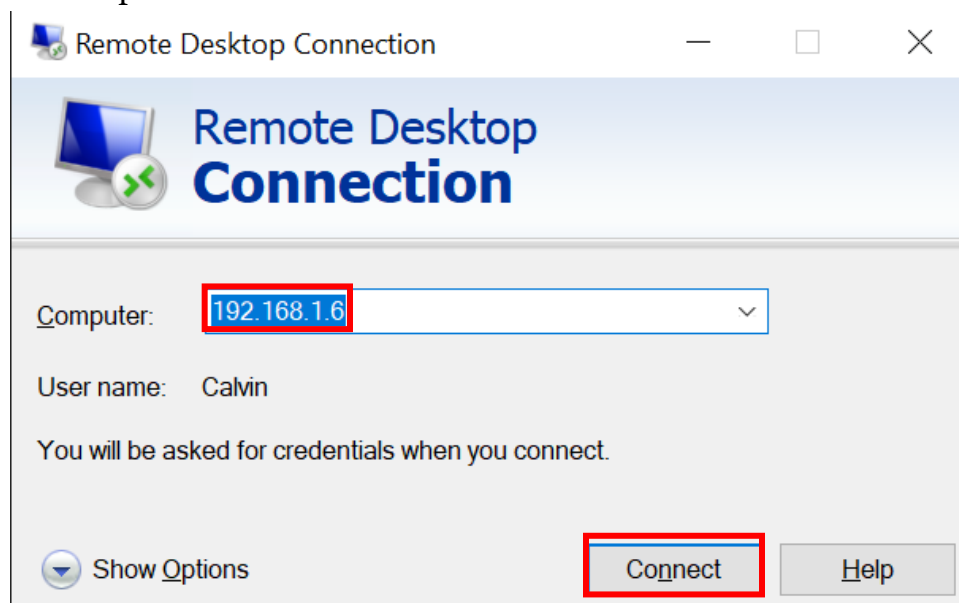
Enable Remote Desktop



To connect, look up remote desktop connection on the external network device.



Then input the IP address of the internal network device and click connect



You may see this warning, click Yes to continue.

You should now have successfully connected to the internal network device through the VPN.

Problems:

- Our original firewall didn't seem to want to update to the latest PAN-OS version
 - o Needed to replace firewall
- When connected to the VPN, the outside device couldn't reach any of the internal devices
 - o When configuring the GP Gateway's IP address pool, make sure it doesn't overlap with any preexisting internal network IP pools.

Conclusion:

Virtual Private Networks play a crucial role in securing online communications by creating an encrypted tunnel that protects and encrypts data. The encryption process ensures that intercepted data remains unreadable without the proper key, adding an essential layer of security. Certificates, which act as digital IDs, help establish trust between devices and the use of root and intermediate certificates further strengthens the system by allowing easier management and added security, ensuring that the VPN connection remains reliable and protected from potential breaches.

Fortigate 40F SOHO Network Setup

By
Aiden Yin



Purpose:

The purpose of this lab was to learn how to set up a SOHO network on a Fortinet 40F firewall as well as wireless networks through the usage of Fortinet wireless access points.

Background:

Fortinet is a cybersecurity company based in North America that sells firewalls and other cloud-based solutions for commercial and consumer use. A firewall is a network security device that monitors traffic to and from your own network. Based on a network administrator's set of security rules, the firewall will then proceed to block or permit traffic to and from a destination. Installing a firewall brings a multitude of benefits, such as promoting privacy and improving network security, all while being relatively cheap. Compared to Palo Alto, Fortinet networking solutions are cheaper and easier to use, making them more suitable for SOHO networks, whereas Palo Alto solutions are more feature-rich, making those more suitable for large commercial use.

There are many different types of firewalls that network administrators can choose to deploy on their network. For example, there are Host-Based firewalls that are installed on individual hosts or computers that monitor inbound and outbound traffic from that specific device. The opposite of Host-Based firewalls are network firewalls which, like its name suggests, monitors the inbound and outbound traffic of entire networks. Firewalls can come as both hardware or software. A hardware firewall is a network device that needs to be plugged in with a cable to a physical network, while a software firewall is usually deployed on servers or virtual machines to secure cloud networks. For the purposes of this lab, we will be configuring a FortiGate 40F, a hardware network firewall.

Wireless Access Points, sometimes referred to just as Access Points (APs), allow for user devices to wirelessly connect to a network. Access Points are then physically wired to the rest of the network, sending all network traffic it receives to whatever device it is connected to, which is typically a switch or a firewall. On the AP for this lab, we set up one WPA2-PSK SSID and one WPA2-Enterprise SSID.

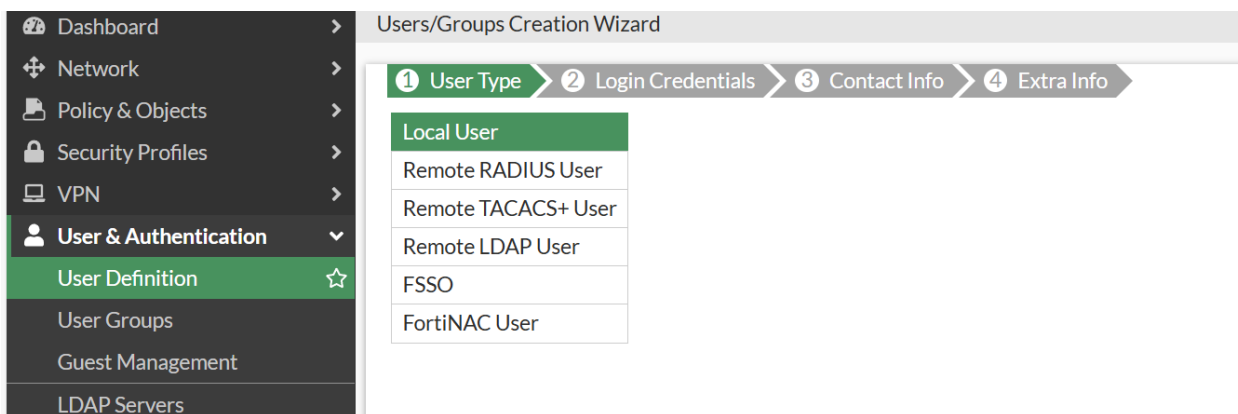
WPA2-PSK is a simple security protocol to secure wireless networks. It requires users to use a PSK (Pre-Shared Key), which is like a universal password, in order to

connect to the network. WPA2 also encrypts data transmissions between devices on a network, adding another layer of security. WPA2-PSK is typically used in small networks like homes and small businesses.

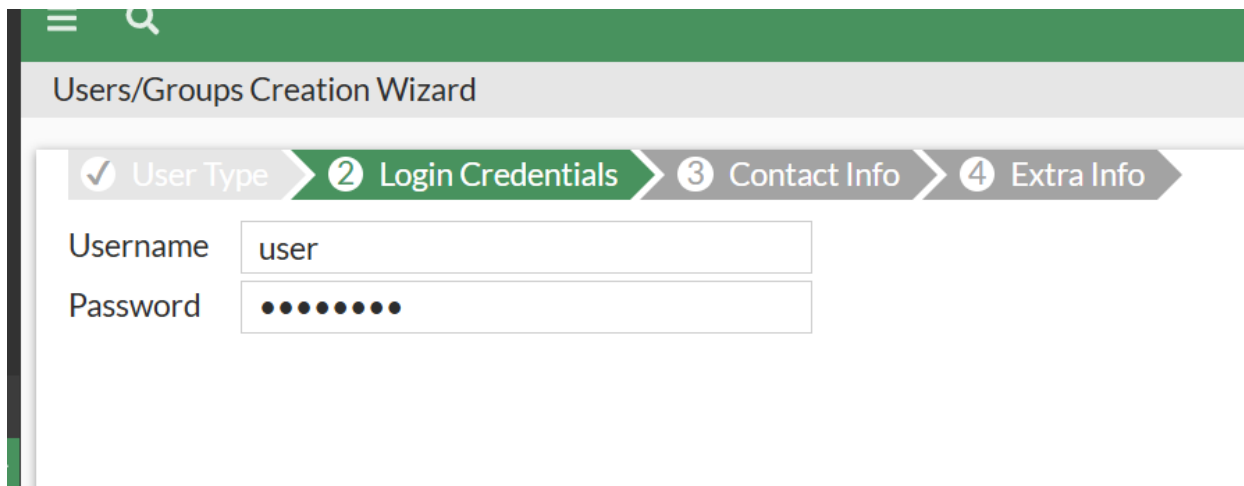
WPA2-Enterprise is a type of WPA2 that provides “enterprise-grade” security for wireless networks. WPA2-Enterprise uses an authentication server to authenticate devices that are trying to access the network. This allows network administrators to maintain a higher level of security as opposed to simpler protocols like WPA2-PSK. These benefits make WPA2-Enterprise typically used in large networks, like corporate networks, universities, and government agencies.

Lab Walkthrough:

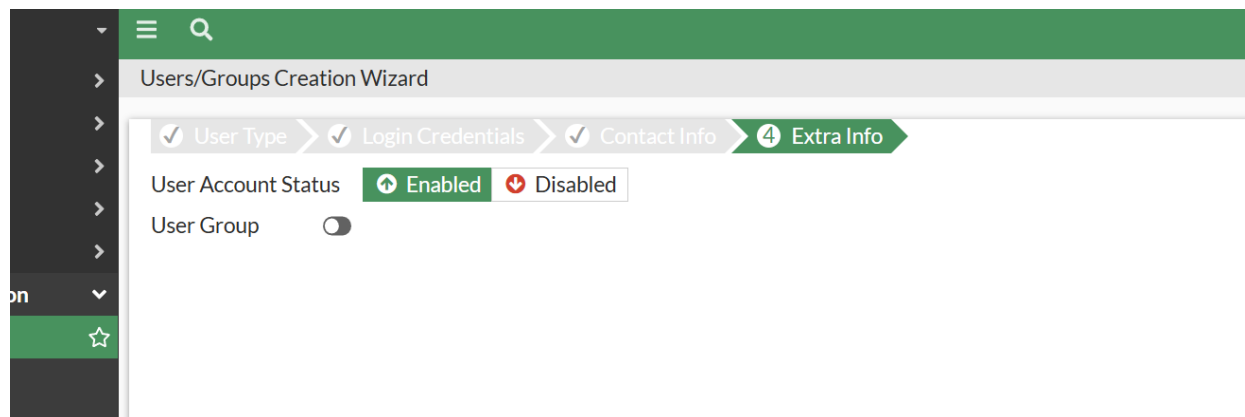
1. Firstly, to get into the firewall. connect your computer into port 1 & Set your IP to 192.168.1.1 with a subnet of 255.255.255.0
2. Next, go to your browser and look up 192.168.1.99
3. Once you are in, we must create a user. Go to User & authentication, then User Definition and click create new



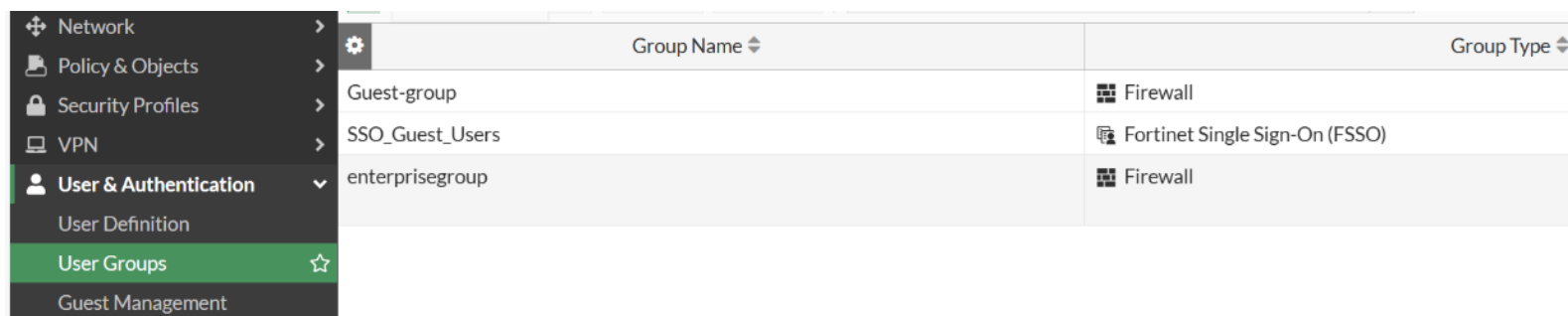
4. Select Local user and click next. Then create the user and password



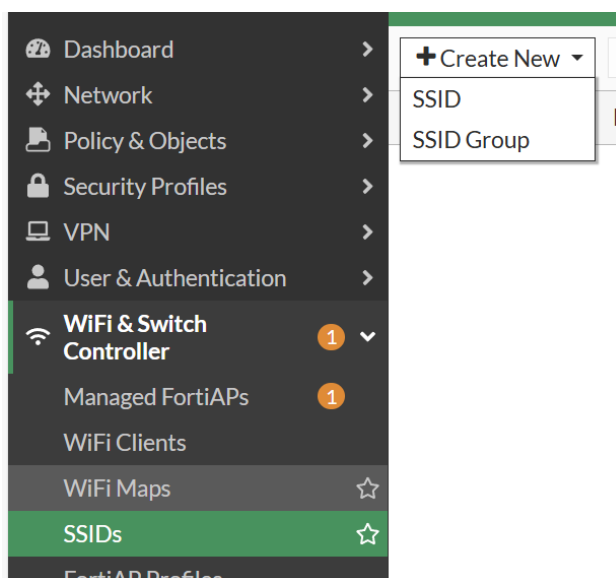
- Click next through contact info, and in extra info enable the account status and add to user group if needed.



- Create another user if needed for different sign in method
- Go to User Groups and create 2 new user groups, one for the group (PSK) and the enterprise.



- Next, go to WiFi and Switch controller and go to SSIDs. Click create new



9. Create the name for the SSID, choose which IP you want and the DHCP addresses. Keep traffic mode as normal. Here this is for the PSK

Name  ACguest (ACguest)

Alias

Type  WiFi SSID

Traffic mode   Tunnel

Address

IP/Netmask

Create address object matching subnet ☒

 The interface-subnet address associated to this interface is currently in use and will not be deleted.

Name  ACguest address

Destination 10.1.2.0/24

Secondary IP address ☐

Administrative Access

IPv4

☐ HTTPS	☐ HTTP 	☒ PING
☐ FMG-Access	☐ SSH	☐ SNMP
☐ FTM	☐ RADIUS Accounting	☐ Security Fabric Connection 
☐ Speed Test		

☒ DHCP Server

DHCP status Enabled Disabled

Address range

Netmask

Default gateway Same as Interface IP Specify

DNS server Same as System DNS Same as Interface IP Specify

Lease time  ☒ second(s)

 Advanced

Then scroll down and name the SSID, Pick the Security mode and pick the password

Network

Device detection  

WiFi Settings

SSID

Client limit  

Broadcast SSID 

Beacon advertising ☐ Name ☐ Model ☐ Serial number

Security Mode Settings

Security mode

Pre-shared Key

Mode 

Passphrase  

Client MAC Address Filtering

RADIUS server 

Address group policy

Additional Settings

Schedule  

Block intra-SSID traffic 

Optional VLAN ID

Broadcast suppression  
 
 

Quarantine host 

VLAN pooling 

NAC profile 

Traffic Shaping

Outbound shaping profile 

Miscellaneous

Comments 0/255

Status

1. Next Create another SSID for the Enterprise WIFI, it is the same steps although change the address range so the SSIDs are not overlapping
2. Under wifi settings in the SSID, select WPA2 Enterprise and have the authentication be set to local. Set the group to the previously created enterprise group with the correct users

WiFi Settings

SSID

Client limit ☐

Broadcast SSID ☒

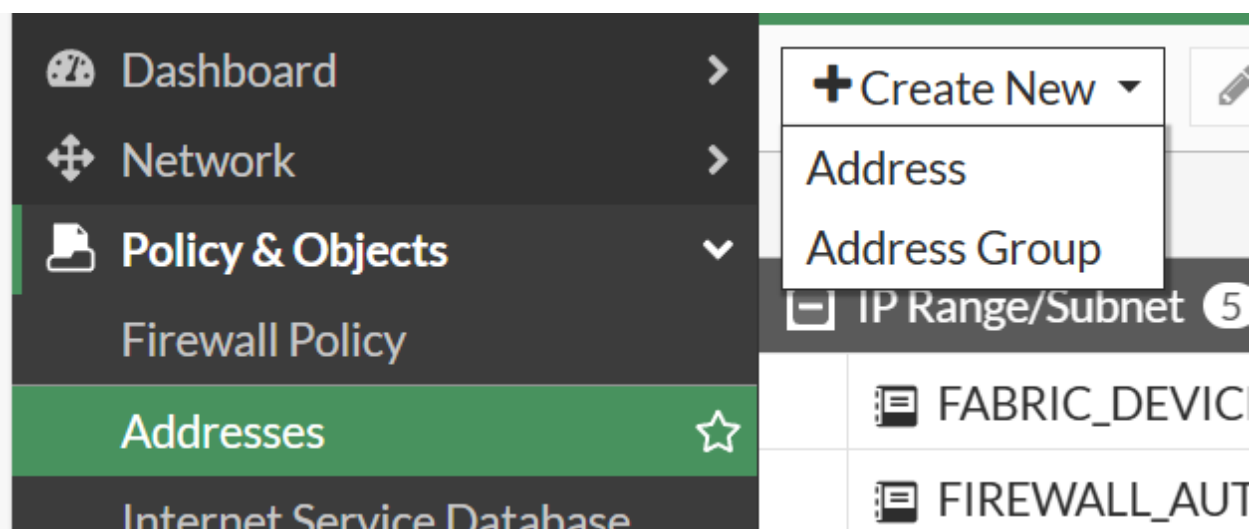
Beacon advertising ☐ Name ☐ Model

Security Mode Settings

Security mode

Authentication ☒ Local ☐ RADIUS Server

3. Next go to Policy & Objects and create new



4. Create 2 new addresses for the corresponding WPA2s.

Edit Address

Name	ACenterprise address	
Color		Change
Type	Interface Subnet	
IP/Netmask	10.1.1.0 255.255.255.0	
Interface	 ACenterprise (ACenterprise)	
Comments	Write a comment... 0/255	

5. Next go to Firewall policy, Click Create New. Make one for the corresponding policies
6. Have the Incoming Interface be the SSID, (that is where the users are connecting from) and have the WAN port be the outgoing interface.
7. Have the source as the corresponding Address. Set the Destination as all.

Name ⓘ

EnterpriseIN

Incoming Interface

ACenterprise (ACenterprise) ▾

Outgoing Interface

wan ▾

Source

ACenterprise address ×

Destination

all ×

Schedule

always ▾

Service

ALL ×

Action

✓ ACCEPT

✗ DENY

Firewall/Network Options

NAT

☒

IP Pool Configuration

Use Outgoing Interface Address Use Dynamic IP Pool

Preserve Source Port

☐

Protocol Options

PROT default ▾ ✎

Security Profiles

AntiVirus

☐

Web Filter

☐

DNS Filter

☐

Application Control

☐

IPS

☐

SSL Inspection

SSL no-inspection ▾ ✎

Logging Options

Log Allowed Traffic

☒ Security Events All Sessions

Comments

Write a comment... 0/1023

Enable this policy

☒

Name ⓘ

GuestIN

Incoming Interface

ACguest (ACguest) ▾

Outgoing Interface

wan ▾

Source

ACguest address ×

Destination

all ×

Schedule

always ▾

Service

ALL ×

Action

✓ ACCEPT

✗ DENY

Firewall/Network Options

NAT

☒

IP Pool Configuration

Use Outgoing Interface Address Use Dynamic IP Pool

Preserve Source Port

☐

Protocol Options

PROT default ▾ ✎

Security Profiles

AntiVirus

☐

Web Filter

☐

DNS Filter

☐

Application Control

☐

IPS

☐

SSL Inspection

SSL no-inspection ▾ ✎

Logging Options

Log Allowed Traffic

☒ Security Events All Sessions

Comments

Write a comment... 0/1023

Enable this policy

☒

8. Finally, Under Wi-Fi Switch Controller, Click the FortiAP Profile and select the preplaced one which came with the AP
9. Here you can add the SSID's to the radios 2.4GHz and 5GHz along with change setting for broadcasting.

Radio 1

Mode: Disabled **Access Point** Dedicated Monitor

WIDS profile: ☐

Radio resource provision: ☐

Band: 2.4 GHz 802.11n/g

Channel width: 20MHz

Channel plan: **Three Channels** Four Channels Custom

Channels: ☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 6 ☐ 7 ☐ 8

Short guard interval: ☐

Transmit power mode: ☒ **Percent**
Transmit power is determined by multiplying set percentage with maximum available power determined by region and FortiA

☐ dBm
Power is setting using a dBm value.

☐ Auto
Set a range of dBm values and the power is set automatically.

Transmit power: 100 %

SSIDs: **Tunnel** **Bridge** **Manual**

ACenterprise (ACenterprise) ✕

ACguest (ACguest) ✕

+

Monitor channel utilization: ☒

Radio 2

Mode: Disabled **Access Point** Dedicated Monitor

WIDS profile: ☐

Radio resource provision: ☐

Band: 5 GHz 802.11ac/n/a

Channel width: **20MHz** 40MHz 80MHz

Channels: Set Channels

Short guard interval: ☐

Transmit power mode: ☐ Percent
Transmit power is determined by multiplying set percentage with maximum available power determined by region and FortiA

☐ dBm
Power is setting using a dBm value.

☒ **Auto**
Set a range of dBm values and the power is set automatically.

Transmit power: 10 - 17 dBm

SSIDs: **Tunnel** **Bridge** **Manual**

ACenterprise (ACenterprise) ✕

ACguest (ACguest) ✕

+

Monitor channel utilization: ☒

10. Once created, with SSID's added to FortiAP profile. Go to Managed FortiAPs and click on the pre-existing firewall.
11. Under status, select authorize
12. Under wireless setting, select the previously configured FortiAP. Then Add the SSIDs to the radios

Serial number FP221C3X15011168

Name ACfirewall

Comments Aiden Calvin # 12/35

Status Authorize Deauthorize Reject

Wireless Settings

FortiAP profile test

Override Radio 1

Band 802.11n/g (2.4 GHz Band)

2.4 GHz 802.11n/g

Channels 1, 6, 11

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 ☒ 11

Transmit power mode 100%

SSIDs ACenterprise (ACenterprise) ACguest (ACguest)

Tunnel Bridge Manual

ACenterprise (ACenterprise)

ACguest (ACguest)

+

Override Radio 2

Band 802.11ac/n/a (5 GHz Band)

5 GHz 802.11ac/n/a

Channels 36, 40, 44, 48, 52*, 56*, 60*, 64*, 100*, 104*, 108*, 112*, 116*, 132*, 136*, 140*, 149, 153, 157, 161, 165

Set Channels

36 40 44 48 52 56 60 64

100 104 108 112 116 132 136 140

149 153 157 161 165

Transmit power mode 100%

SSIDs ACenterprise (ACenterprise) ACguest (ACguest)

Tunnel Bridge Manual

ACenterprise (ACenterprise)

ACguest (ACguest)

+

13. Once configured, the AP should be broadcasting the SSIDs and should be able to be connected to. If you are able to connect but WiFi isn't, look back at the Firewall policy to make sure you have the correct Inbound and Outbound ports.

1

total

Status

Online

1

total

2.4 GHz Radio Channel Utilization

Good

1

total

5 GHz Radio Channel Utilization

Good

Create New

Edit

Delete

Authorization

Register All

Search

Access Point	Status	SSIDs	Channel	Clients	OS Version	LLDP	FortiAP Profile	Connected Via
ACFirewall	Online	All Tunnel Mode SSIDs All Tunnel Mode SSIDs	1 140	1	FP221C-v6.0-build0044	No LLDP neighbors found.	FAP221C-default	lan

Problems:

The only error in the lab was setting the wrong incoming and outgoing ports in the GUI, which was a pretty simple fix of just switching the ports around in the GUI.

Conclusion:

Firewalls are a cheap and effective method of adding another layer of security to networks. Compared to Palo Alto firewalls, Fortinet solutions are easier to use and setup in exchange for being less feature-rich. WPA2-PSK is a sufficient security protocol for small offices where ease of access is prioritized. WPA2-Enterprise is a more secure and robust network that should be used when security is of high importance due to the server-based authentication it uses.

Fortigate IPsec Site-to-Site VPN Setup

By
Aiden Yin



Purpose:

The purpose of this lab was to learn how to set up a site-to-site VPN, which provides the benefits of a VPN while restricting access to user devices on a predefined network.

Background:

Encryption protocols are what privatize and make data unreadable to unauthorized users. These protocols are the backbone of all VPNs, providing rules on how to encrypt and decrypt data at both ends of the VPN tunnel. Two of the most common encryption protocols used in VPNs are IPsec and SSL/TLS.

Internet Protocol Security, or IPsec for short, is an encryption protocol that operates at the network layer, which is Layer 3 of the OSI model. This means that IPsec encrypts everything coming from your device, no matter what application or port number you are using. Due to how wide the protocol spans, it is usually implemented in scenarios where the user should be able to remotely access all company resources through the VPN, as if they were in the office. IPsec also requires more setup on both the company and user devices to get going.

Secure Sockets Layer and Transport Layer Security, or SSL/TLS for short, is an encryption protocol that operates at the transport layer, which is Layer 4 of the OSI model. Instead of encrypting all traffic coming from your device, it instead focuses on encrypting specific applications, especially web-based applications. SSL has been deprecated as a whole, and TLS is the direct successor to SSL. SSL/TLS VPNs are usually more simple to set up, only requiring a connection through a secure web browser or application without needing super complex configuration. This makes SSL/TLS VPNs better choices for when a user only needs to access certain services or company resources on the internal network.

Encryption protocols play a vital role in securing digital communication. The variety of properties different encryption protocols contain ensures that there is an optimal choice for every scenario, allowing VPNs to be an extremely versatile solution for users to remotely access specific networks.

Lab Walkthrough:

1. Click on the IPsec Wizard tab from the left dropdown menu and create a custom VPN and name it.

Dashboard > VPN Creation Wizard

1 VPN Setup

Name

Template type ☐ Site to Site ☐ Hub-and-Spoke ☐ Remote Access ☒ Custom

VPN

- Overlay Controller VPN
- IPsec Tunnels
- IPsec Wizard**
- IPsec Tunnel Template

2. Set the Remote Gateway IP address to the outward IP address of the partner firewall. Set the Interface to the personal outwards port.

Network

IP Version IPv4

Remote Gateway Static IP Address

IP Address 192.168.40.8

Interface wan

Local Gateway ☐

Mode Config ☐

NAT Traversal ☒ Enable ☐ Disable ☐ Forced

Keepalive Frequency 10

Dead Peer Detection ☐ Disable ☐ On Idle ☒ On Demand

DPD retry count 3

DPD retry interval 20 s

Forward Error Correction Egress ☐ Ingress ☐

+ Advanced...

3. Enter a pre-shared key password for the connection and leave everything else to their default values.

Authentication
✓ ↺

Method Pre-shared Key ▼

Pre-shared Key ••••••••

IKE

Version 1 2

Mode Aggressive Main (ID protection)

4. Set the Local Address to your personal network, and the Remote Address as the partner firewall's network with their respective subnets.

Phase 2 Selectors

Name	Local Address	Remote Address	
fort	192.168.1.0/255.255.255.0	192.168.19.0/255.255.255.0	

Edit Phase 2
✓ ↺

Name fort

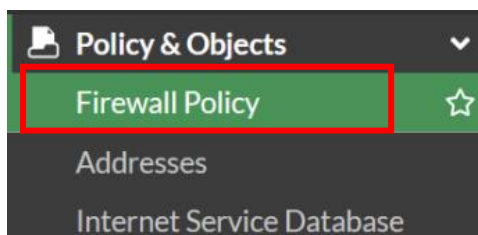
Comments Comments

Local Address Subnet ▼ 192.168.1.0/255.255.25

Remote Address Subnet ▼ 192.168.19.0/255.255.2

+ **Advanced...**

5. Then, navigate to the Firewall Policy tab on the left dropdown menu, and create a new policy with your inwards facing port as the Incoming Interface and outwards facing port as the Outgoing Interface. Then proceed to select the Source as your own network and the Destination as your partner firewall's network.



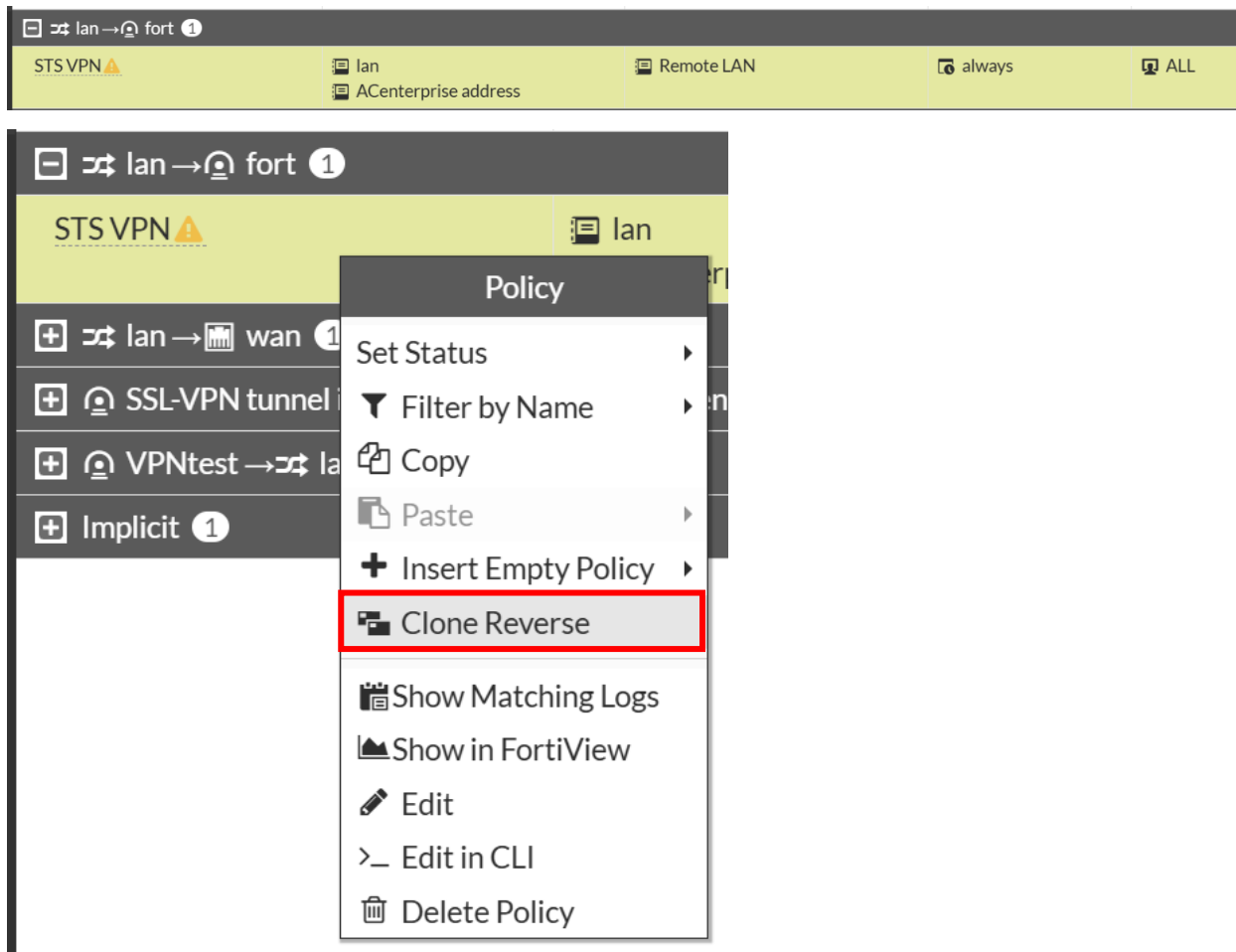
Edit Policy

Name 	STS VPN
Incoming Interface	 lan
Outgoing Interface	 fort
Source	 lan
	+
Destination	 Remote LAN
	+
Schedule	 always
Service	 ALL
	+
Action	☒ ACCEPT ☐ DENY

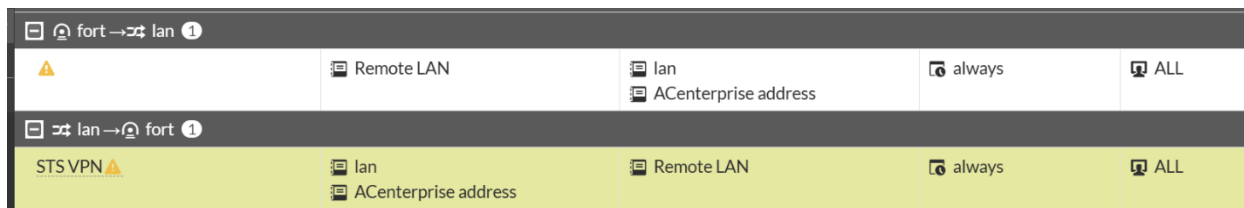
Firewall/Network Options

NAT	☒
IP Pool Configuration	☒ Use Outgoing Interface Address ☐ Use Dynamic IP Pool
Preserve Source Port	☐
Protocol Options	☒ PROT default

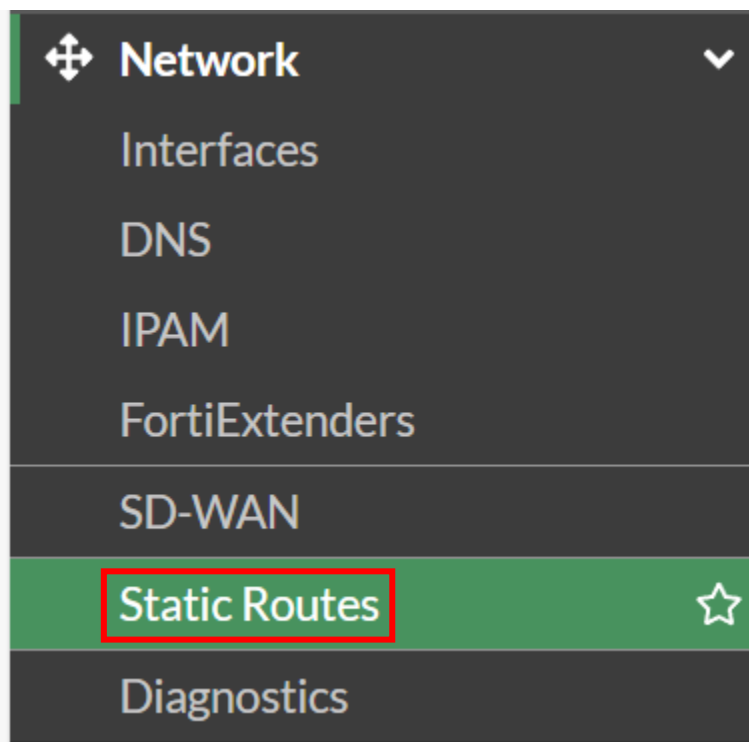
6. You should now have the policy appear in the table. Right click on the policy you just created and click Clone Reverse.



7. Now, both policies should appear in the table.



8. Navigate to the Static Routes tab on the left dropdown menu.



9. Enter in the partner firewall's network and subnet mask and choose the VPN connection we just created as the interface.

 A screenshot of the 'Static Routes' configuration page in FortiGate. The page has a light grey header with three tabs: 'Subnet' (selected), 'Named Address', and 'Internet Service'. Below the header, there are several fields:

- Destination:** A text box containing '192.168.19.0/255.255.255.0', which is highlighted with a red box.
- Interface:** A dropdown menu showing 'fort' with a red icon to its left. This dropdown is also highlighted with a red box. To the right of the dropdown is a close button (X).
- Administrative Distance:** A text box containing '10'.
- Comments:** A text box with the placeholder 'Write a comment...' and a character count '0/255'.
- Status:** Two buttons: 'Enabled' (green with an up arrow) and 'Disabled' (red with a down arrow). The 'Enabled' button is selected.

 At the bottom of the page, there is a grey bar with a plus icon and the text 'Advanced Options'.

10. Complete steps 1-9 again but with the partner firewall and local firewall switched.

11. Your IPsec site-to-site VPN should be up and running.

Problems:

No problems were encountered during the setup and execution of this lab.

Conclusion:

Site-to-site VPNs are a simple and secure way of connecting two designated networks together. They are used in corporate scenarios when connecting the network between a branch office and a main office, allowing for users to share files, access company resources, and communicate as if they were all in the same building. Different types of encryption protocols may also be used within these VPNs. IPsec functions on the network layer, allowing for all traffic from a device to be encrypted, while SSL/TLS functions on the transport layer, allowing for traffic from designated apps or websites to be encrypted. When setting up a VPN, the type of VPN as well as the encryption protocol it uses should be considered on a case-by-case basis in order to promote best security practices.

Fortigate SSL Remote Access VPN Setup

By
Aiden Yin



Purpose:

The purpose of this lab was to learn how to set up a remote access VPN on Fortinet firewalls, allowing secure access to a private network through a public one.

Background:

Virtual Private Networks, or VPNs for short, are encrypted connections that secure data transmission between devices over the Internet. This protects sensitive data from unauthorized access, eavesdropping, and other security threats. VPNs help ensure confidentiality, integrity, and authenticity when sending data across potentially untrusted networks such as public Wi-Fi. For personal devices, connecting to a VPN routes all internet traffic through the VPN server first, before forwarding it to the personal device. This masks the IP address of the device, enhances user privacy, and can also help bypass geographic restrictions on content.

In corporate environments, a VPN typically connects remote users to the internal network, providing secure access to company resources such as file servers, internal websites, or proprietary applications. This setup enables employees to work remotely while still operating within the company's secure digital infrastructure. For this lab, we will be setting up a remote access VPN. This type of VPN uses a VPN client installed on the remote device to connect securely to the company network. Once connected, it's as if the device is physically located within the office network, with full access to internal systems.

To establish this connection, a VPN portal or concentrator needs to be installed and configured on a central server or firewall. A VPN tunnel is then created, which forms a secure channel between the client and the company server using encryption protocols such as IPsec or SSL/TLS. These tunnels protect all data passing through from tampering or interception.

Another type of VPN is the site-to-site VPN, which connects entire private networks to one another, often between different office locations. This type of VPN operates at the network level, rather than the individual user level. In a site-to-site configuration, users on either side can access resources on the other network, but only while they are within their designated local network. There is no support for remote, roaming connections like with remote access VPNs. Site-to-site VPNs are commonly

used by organizations with multiple branches to securely integrate their local area networks over the internet without exposing them to public access.

Overall, VPNs are a critical component in modern cybersecurity architecture, offering scalable and secure ways to connect users and networks across geographical boundaries.

Lab Walkthrough:

1. Navigate to SSL-VPN Settings on the left dropdown menu.
 - a. Choose the outside-facing port as the listening interface
 - b. Set the listening port to port 4433
 - c. You can leave the listening port, server certificate, access restriction, address range, and DNS server all as default.
 - d. Create a new Authentication/Portal mapping to your user group and choose the correct portal. Tunnel-access only allows VPN client connections, web-access only allows web client connections, and full-access allows for both.

SSL-VPN Settings

Connection Settings

- Enable SSL-VPN: ☒
- Listen on Interface(s): wan
- Listen on Port: 4433
- Web mode access will be listening at <https://192.168.40.167:4433>
- Server Certificate: Fortinet_Factory
- You are using a default built-in certificate, which will not be able to verify your server's domain name (your users will see a warning). Let's Encrypt can be used to easily generate a trusted certificate if you do not have one. [Create Certificate](#)

Redirect HTTP to SSL-VPN ☐

Restrict Access ☒ Allow access from any host ☐ Limit access to specific hosts

Negate Source ☐

Idle Logout ☒

Inactive For: 300 Seconds

Require Client Certificate ☐

Tunnel Mode Client Settings

Address Range ☒ Automatically assign addresses ☐ Specify custom IP ranges

Tunnel users will receive IPs in the range of 10.212.134.200 - 10.212.134.210

DNS Server ☒ Same as client system DNS ☐ Specify

Specify WINS Servers ☐

Web Mode Settings

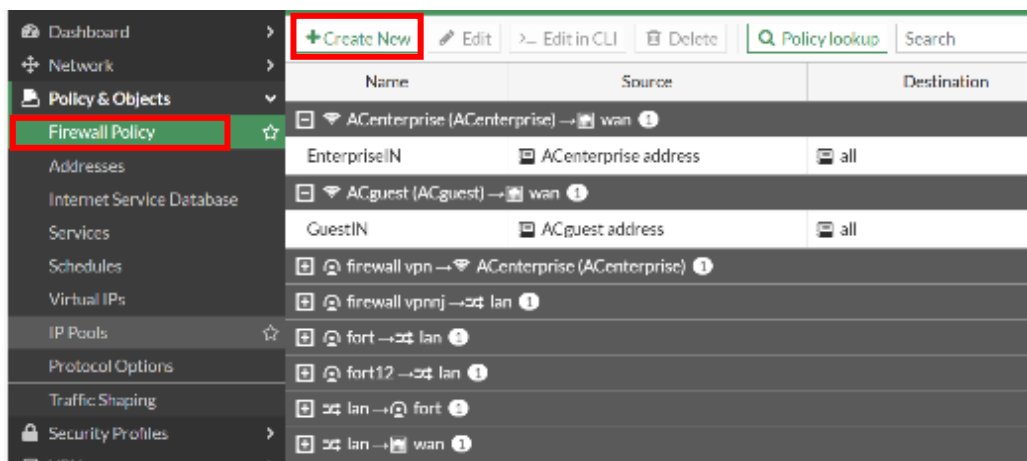
Language ☒ Browser preference ☐ System

Authentication/Portal Mapping

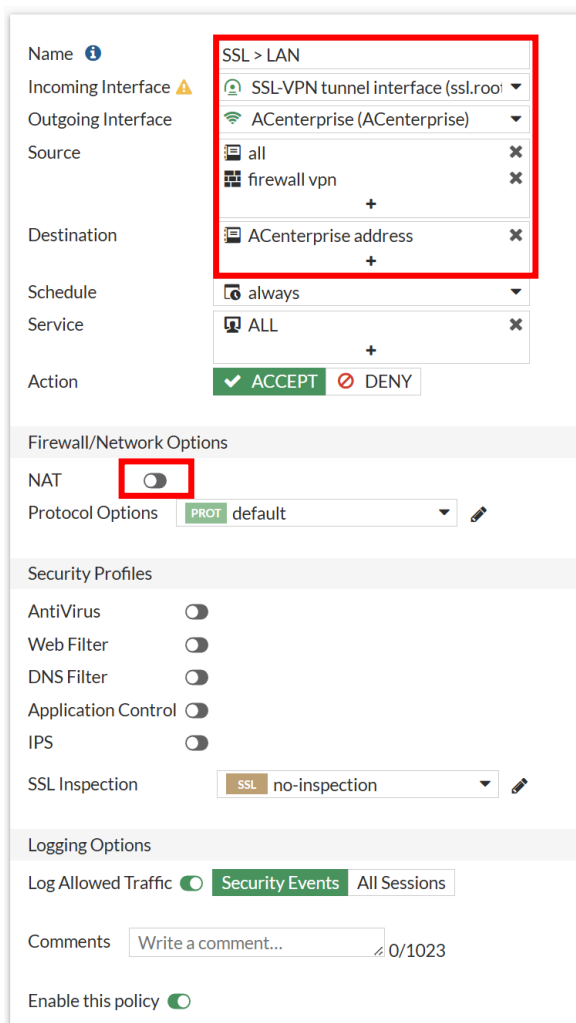
[+ Create New](#) [Edit](#) [Delete](#) [Send SSL-VPN Configuration](#)

Users/Groups	Portal
firewall vpn	tunnel-access
All Other Users/Groups	vpn portal

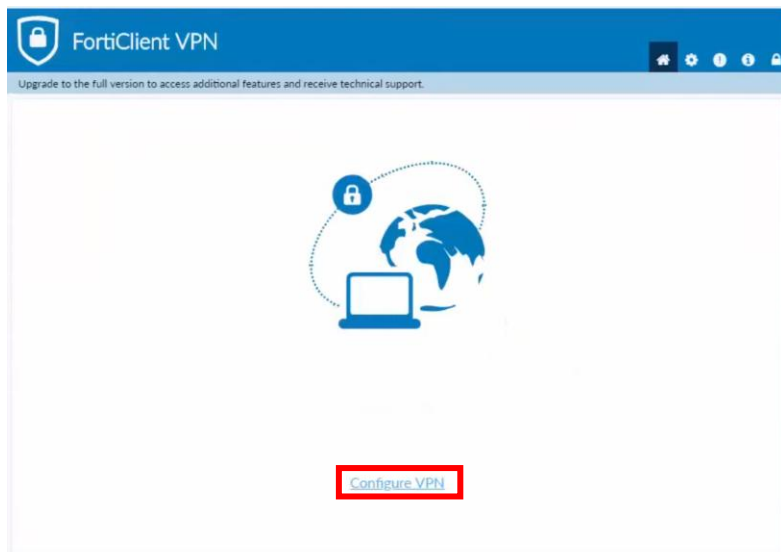
2. Navigate to Firewall Policy on the left dropdown menu and click Create New.



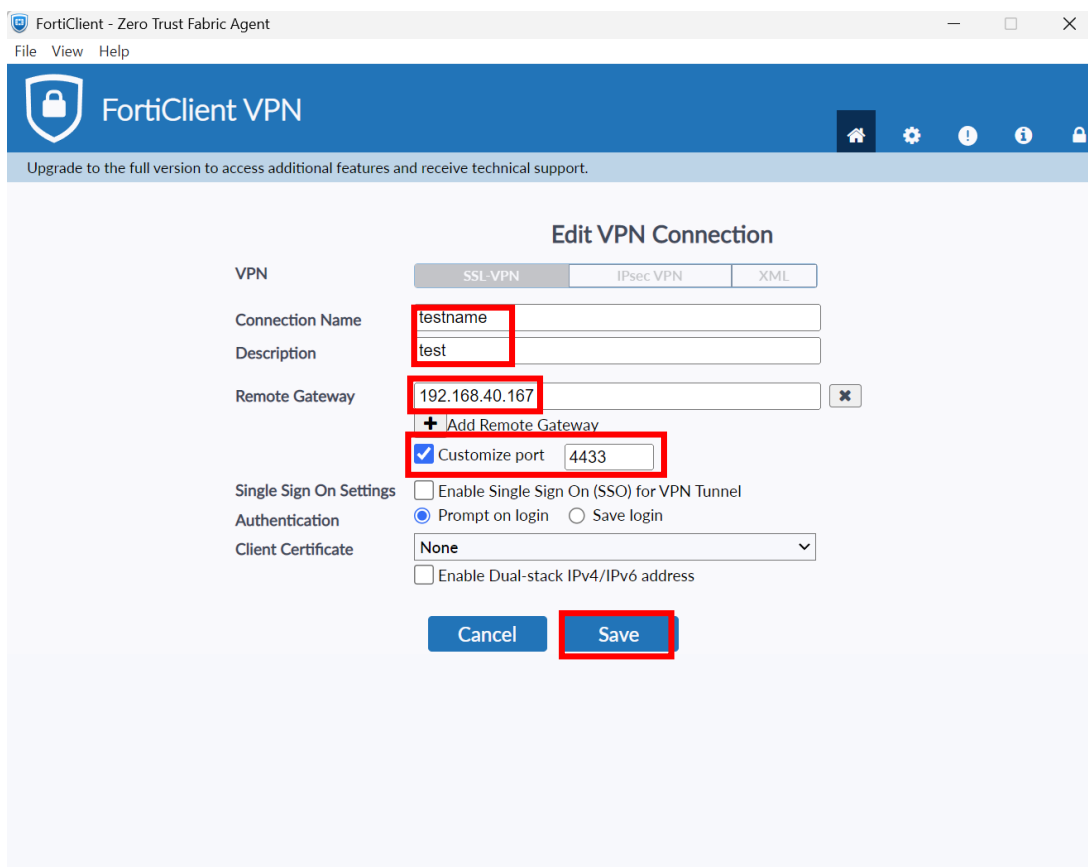
3. Create a name for the policy and select the VPN tunnel interface as the incoming interface, the company network as the outgoing interface, and the corresponding source and destinations. Leave NAT disabled and you can finish the policy.



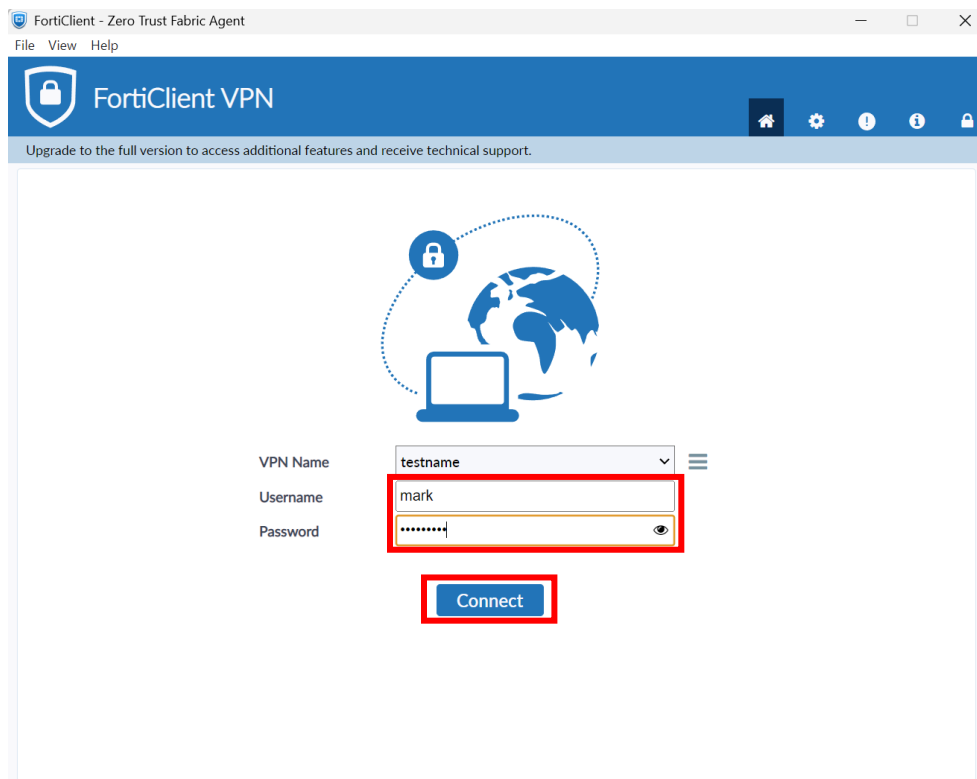
4. Your VPN should now be set up properly. Now we are going to connect to it using the Fortinet VPN Client. Open the VPN Client and hit Configure VPN.



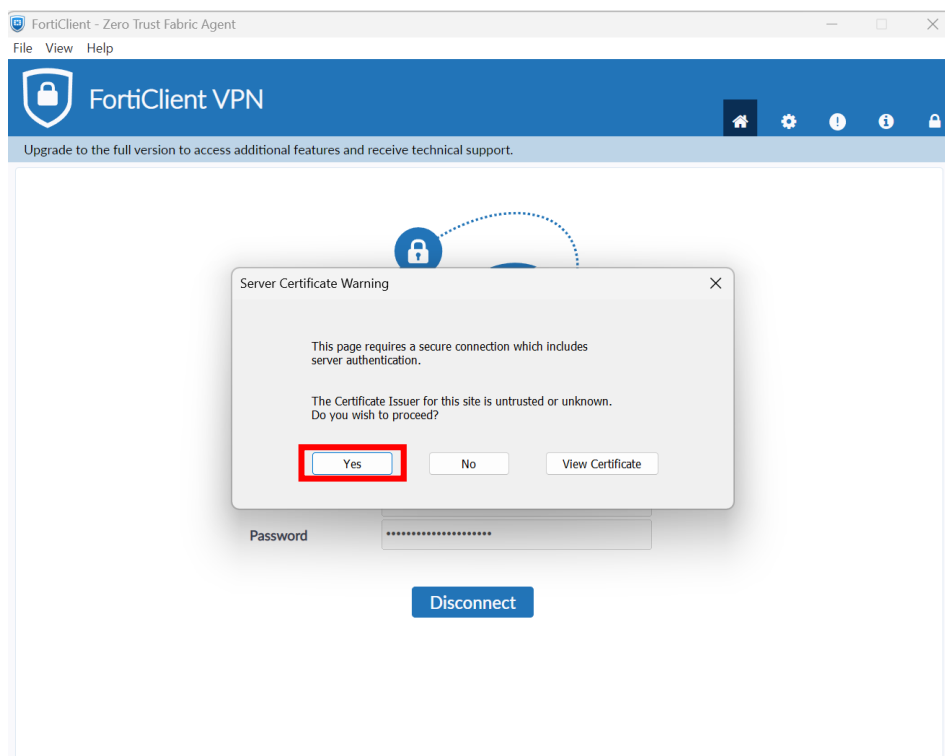
5. Name the connection and description whatever you want. Set the remote gateway to the public IP of the firewall. Set the port to the listening port that was previously set and click save.



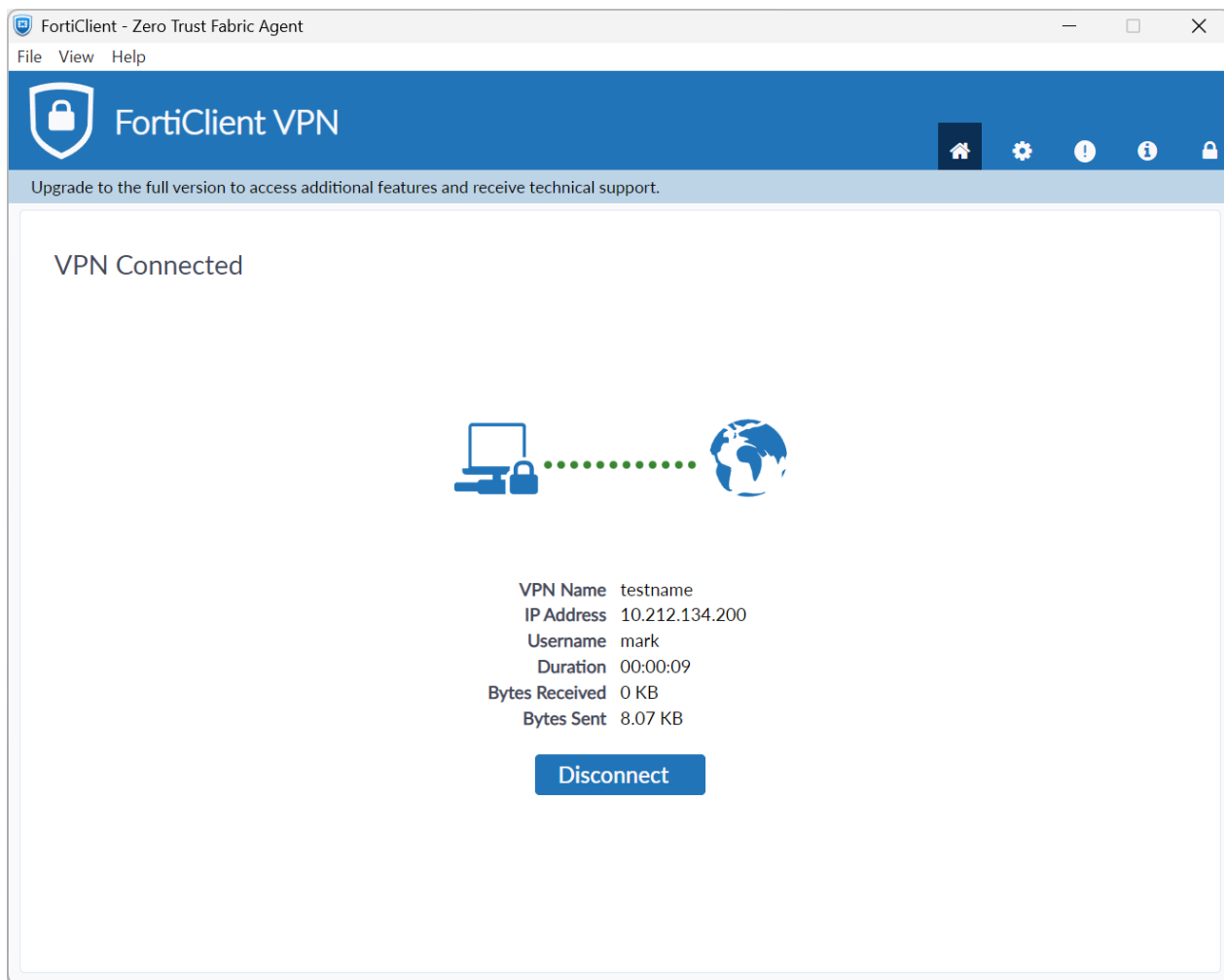
6. Login with a user account that is in your previous Authentication/Portal Mapping group and click Connect.



7. Click through the warning window.



8. You should now be connected to the VPN and be able to access the network on the other side of the VPN.



Problems:

Overall, the lab was relatively simple and no problems were encountered during the setup of this lab.

Conclusion:

Remote access VPNs allow users to connect to a private network from anywhere on the Internet using either a web client or a VPN client. Remote access VPNs offer the same core benefits of encryption, authentication, and secure tunneling as other VPNs, but are specifically designed for individual users. Like other VPNs, they protect sensitive data from eavesdropping, interception, and unauthorized access by routing traffic through a secure and encrypted tunnel using protocols like IPsec or SSL/TLS. This enables remote users to access internal company resources such as file servers or

internal web applications as if they were physically present in the office. By ensuring confidentiality, integrity, and availability of data over untrusted networks like public Wi-Fi, remote access VPNs play a vital role in securing enterprise infrastructure.